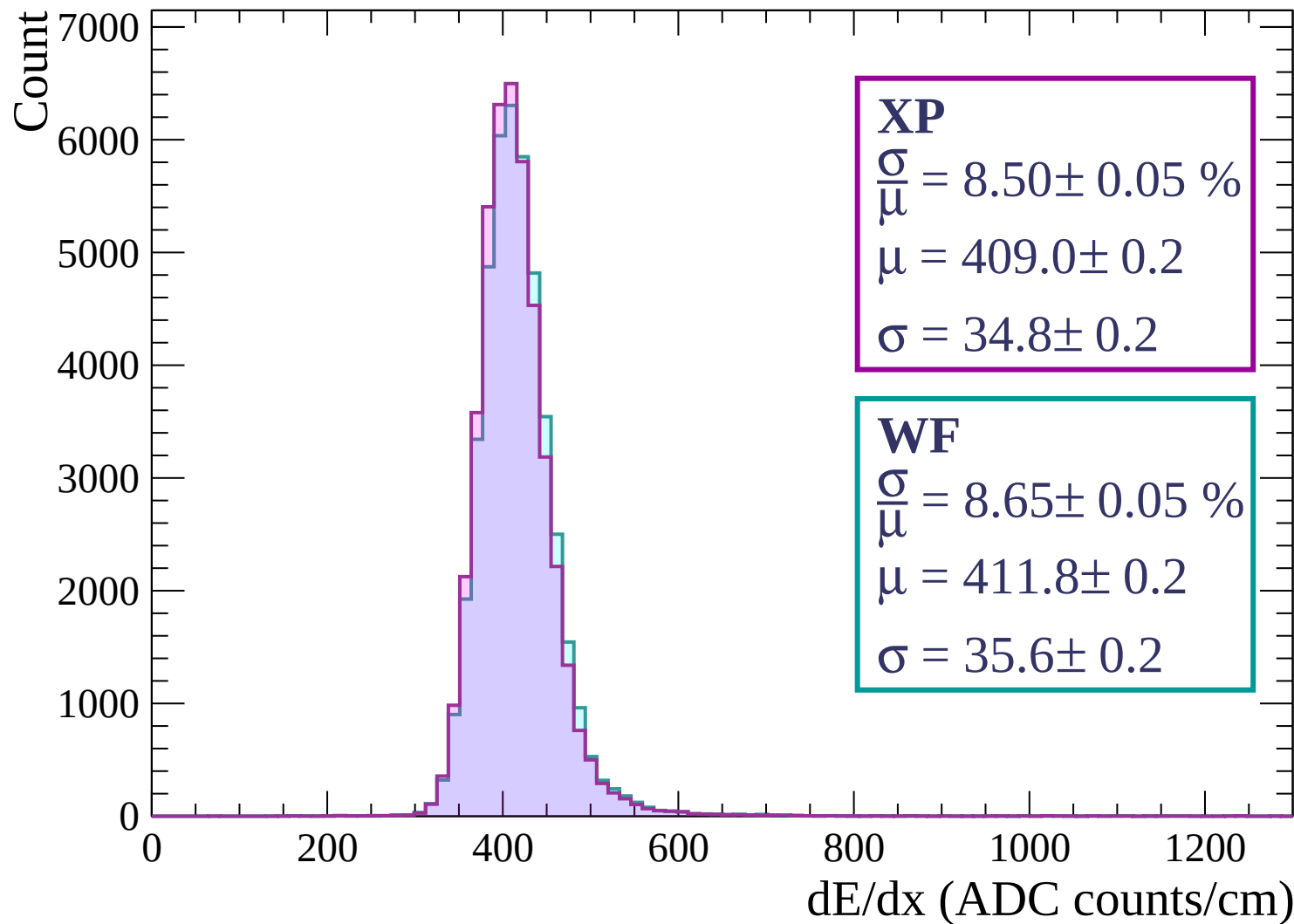
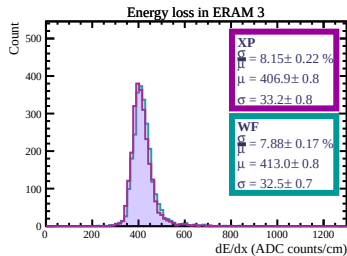
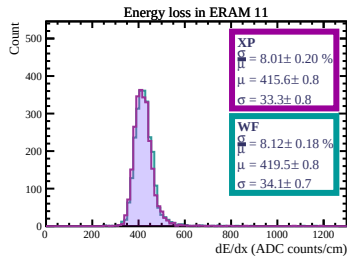
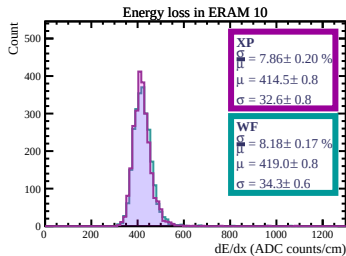
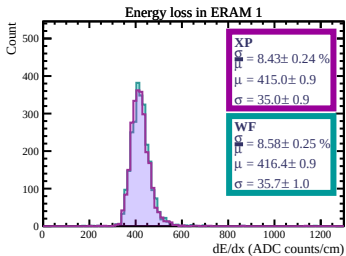
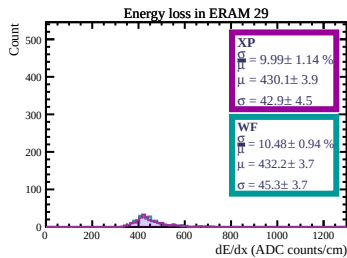
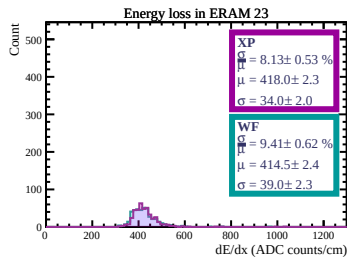
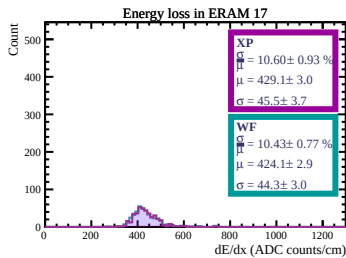
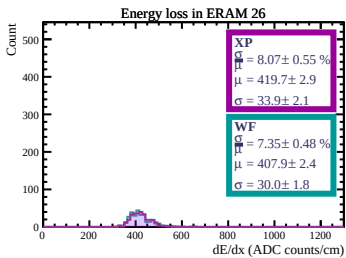
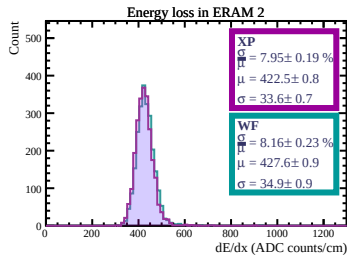
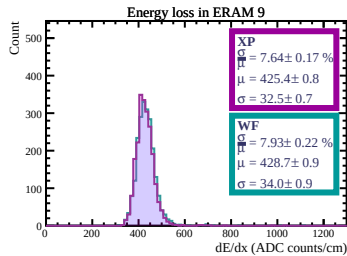
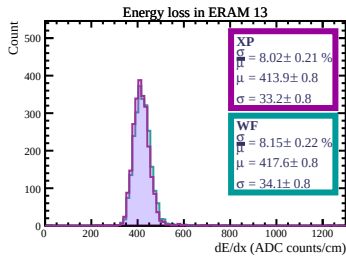
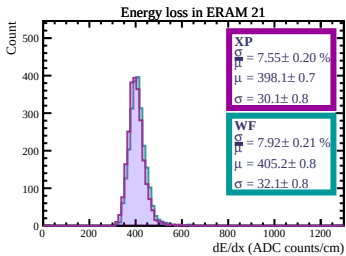
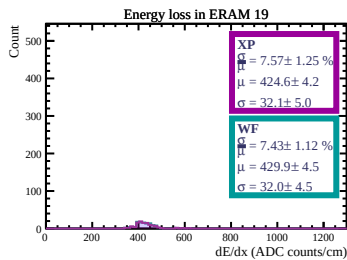
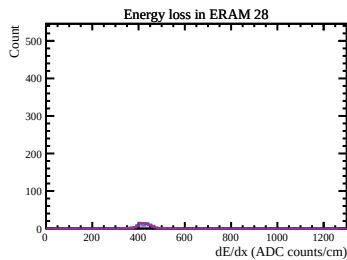
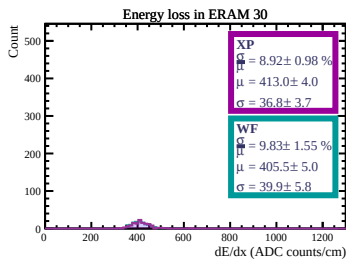
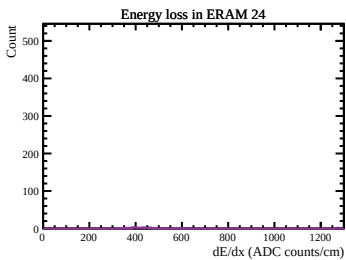
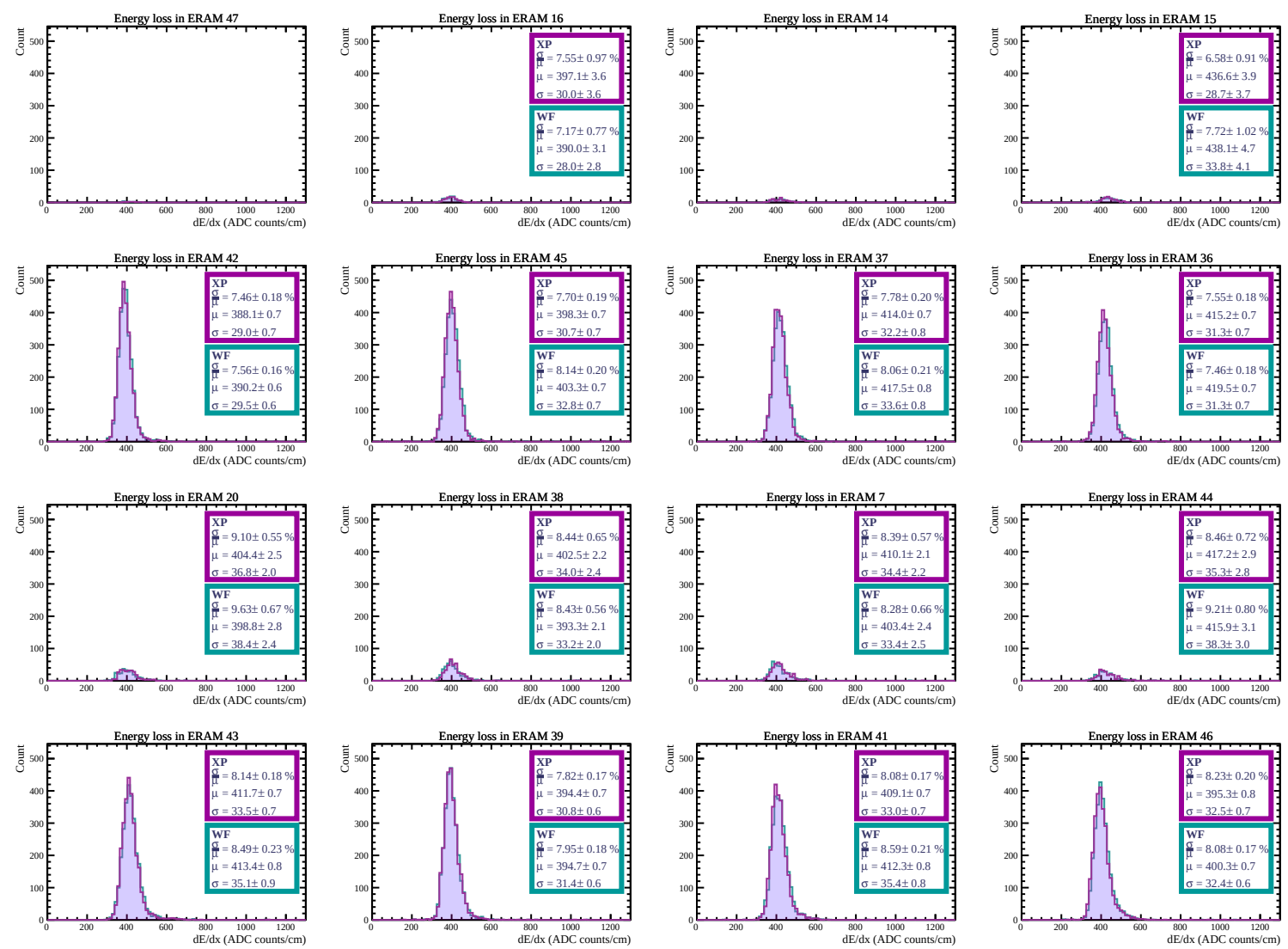
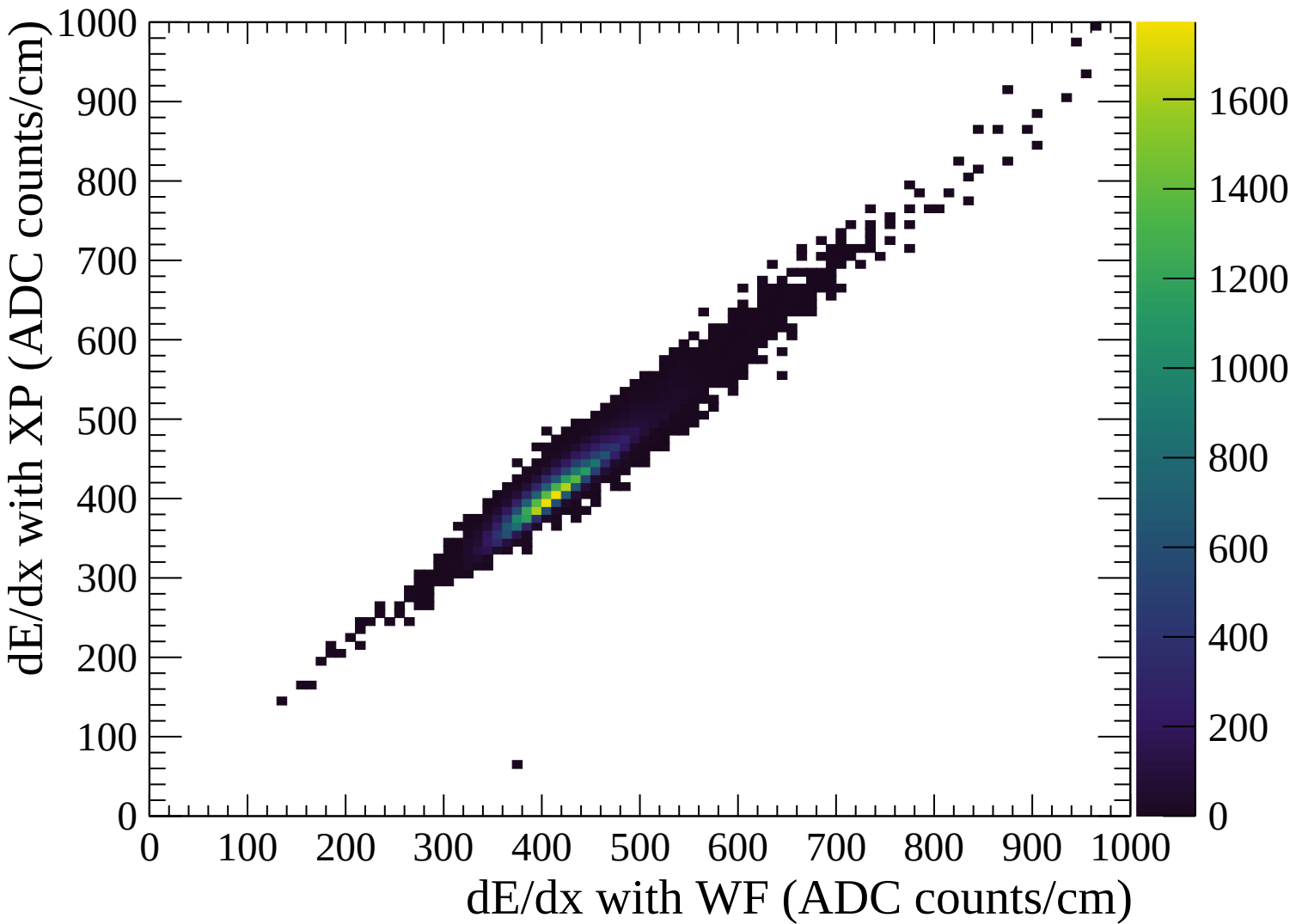
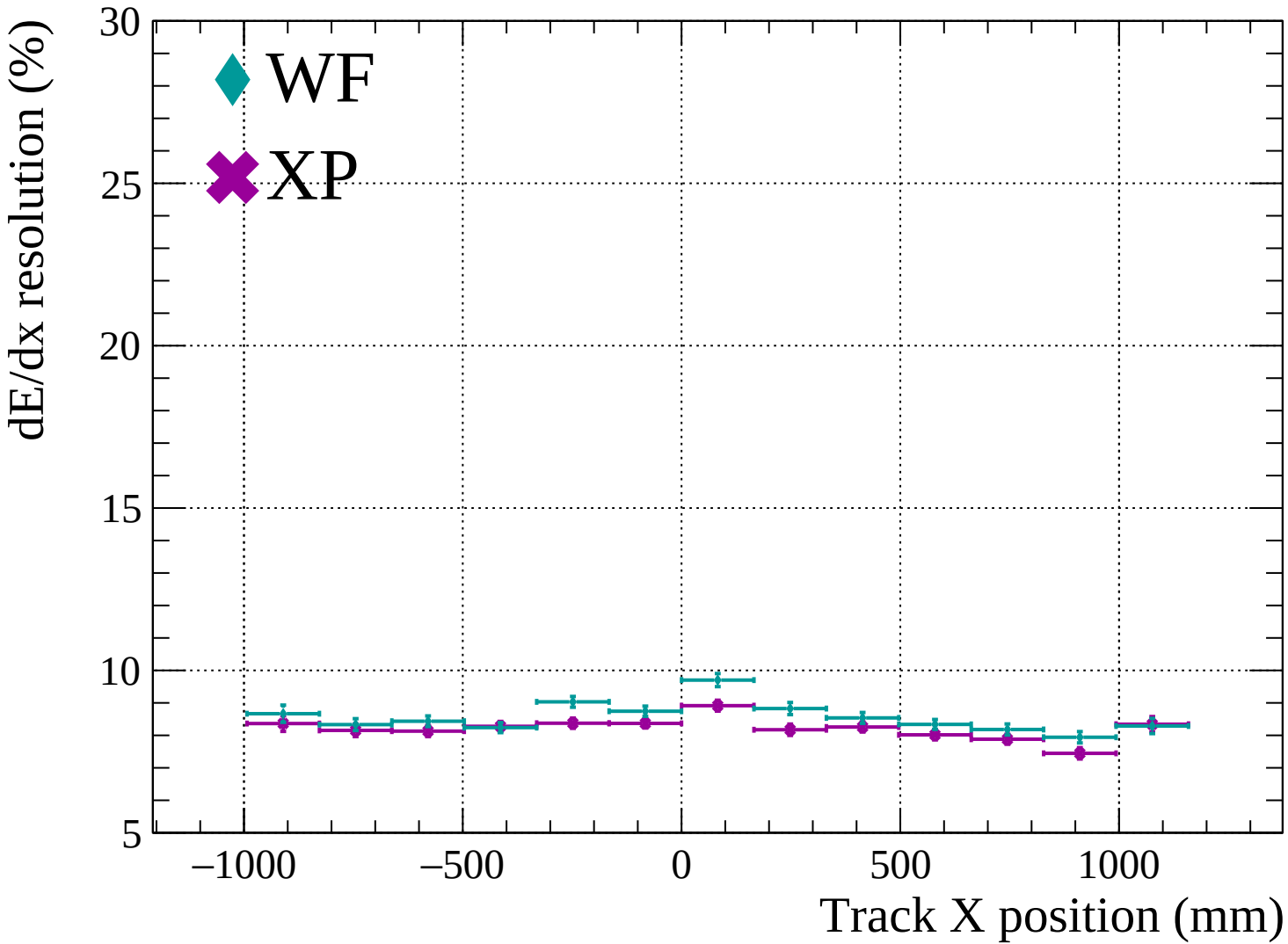


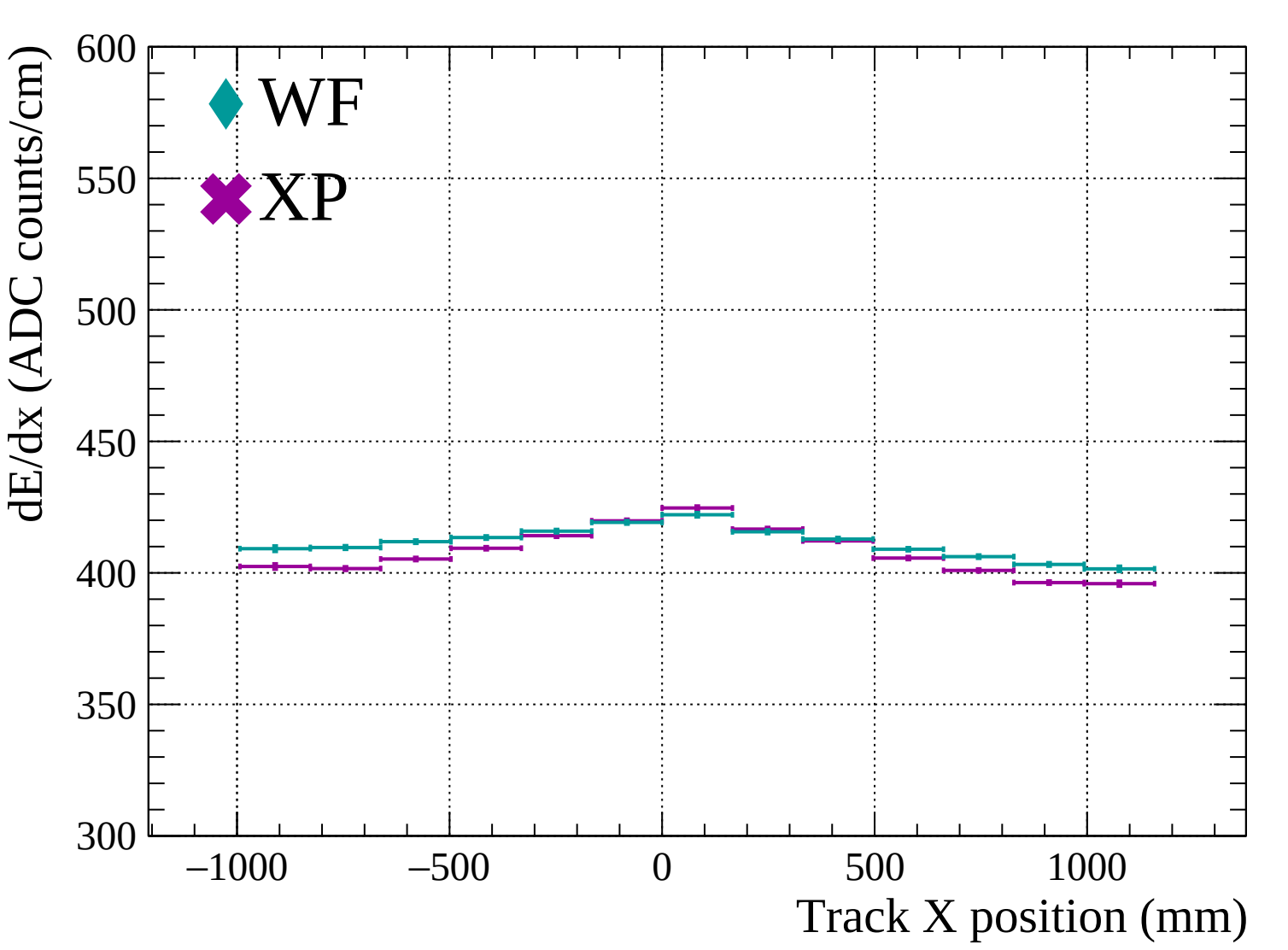


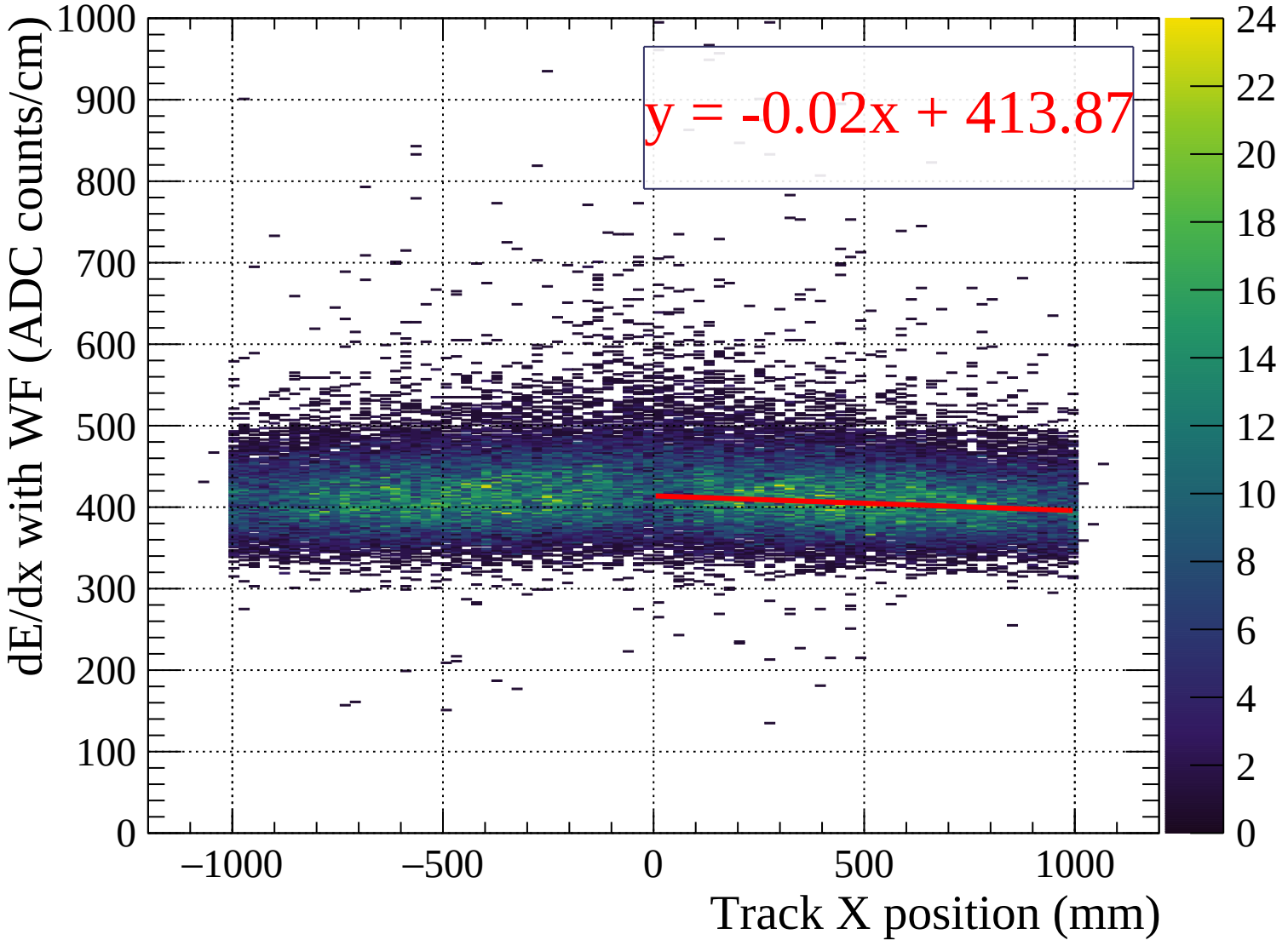


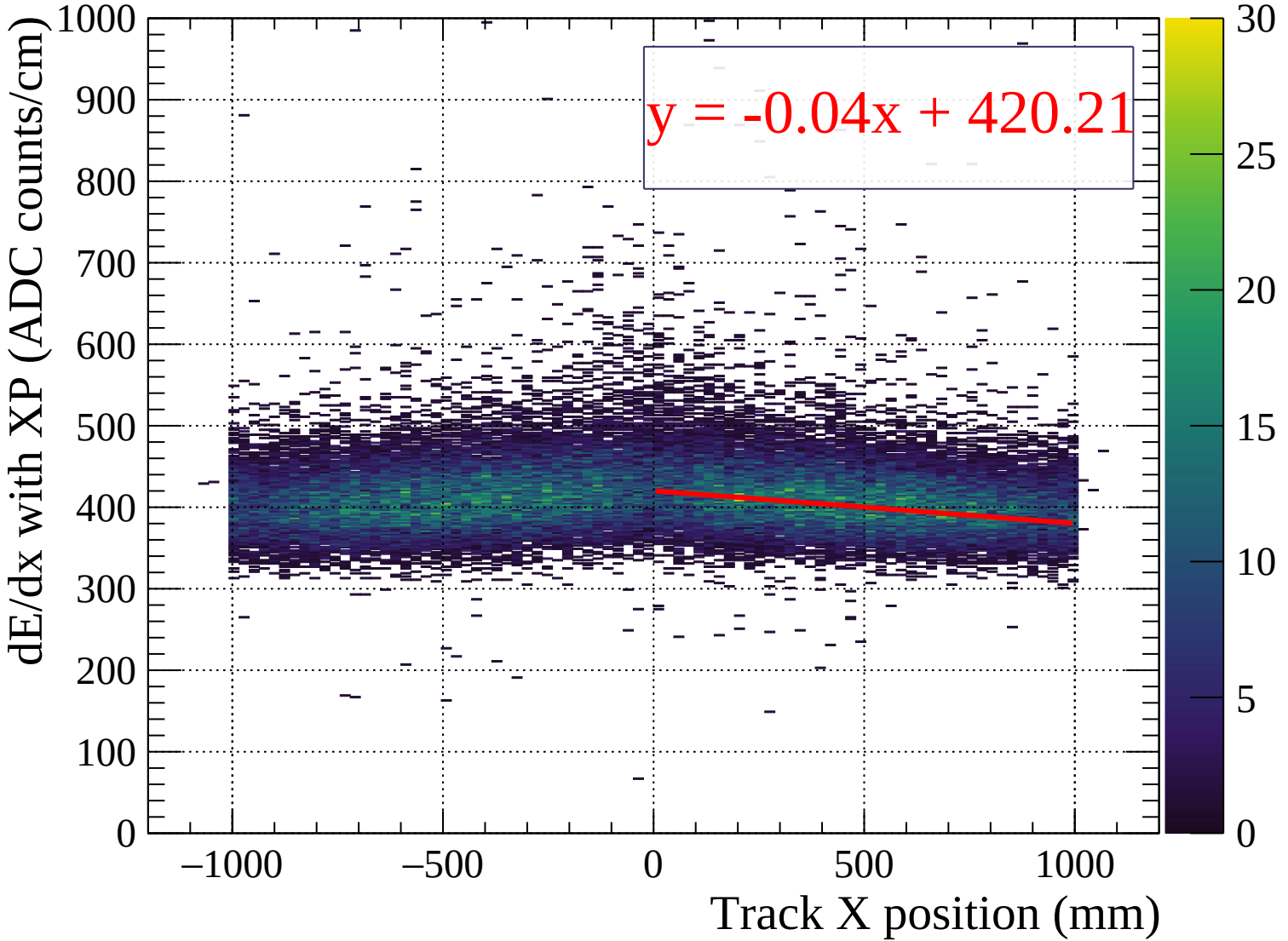


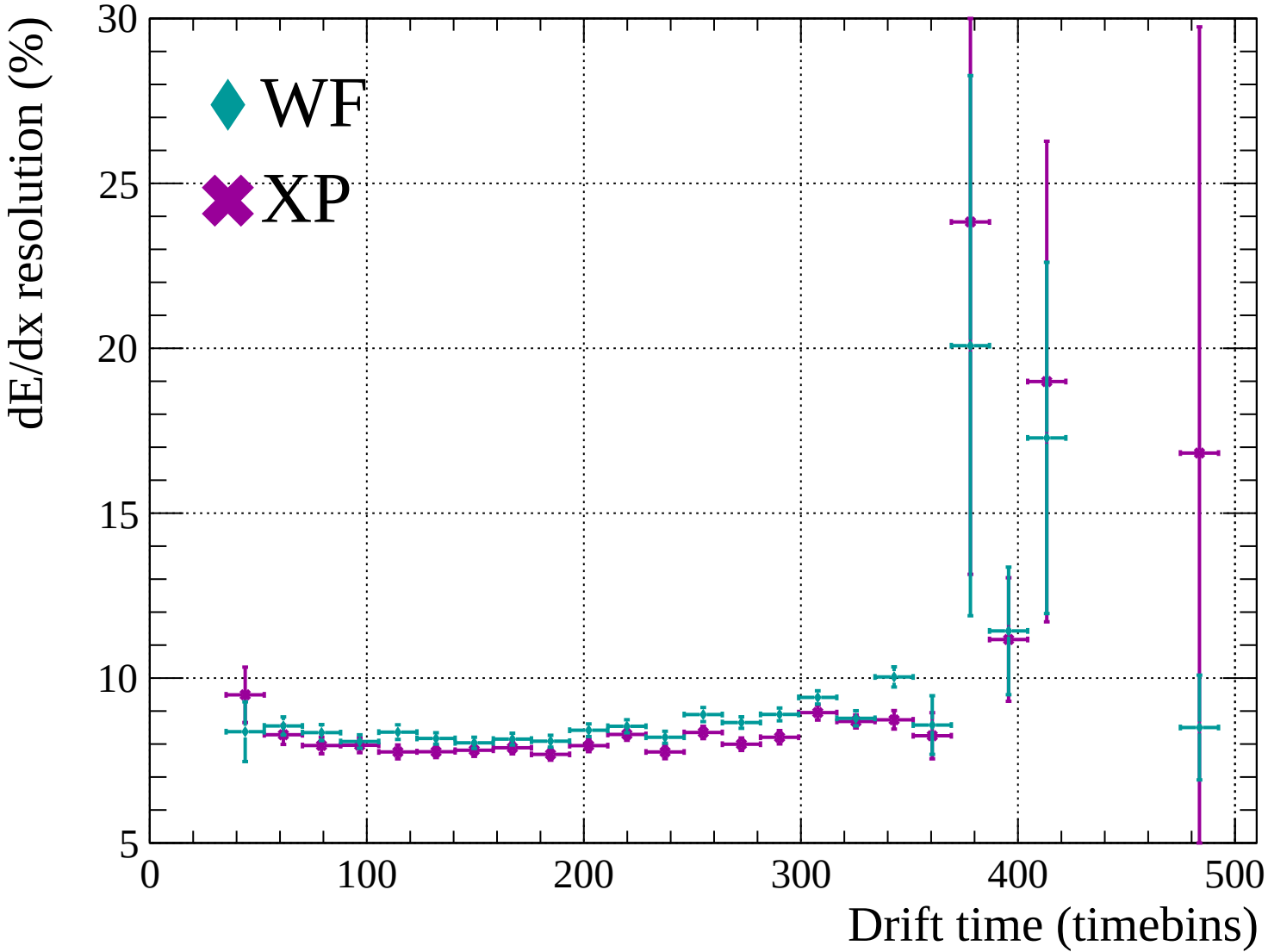


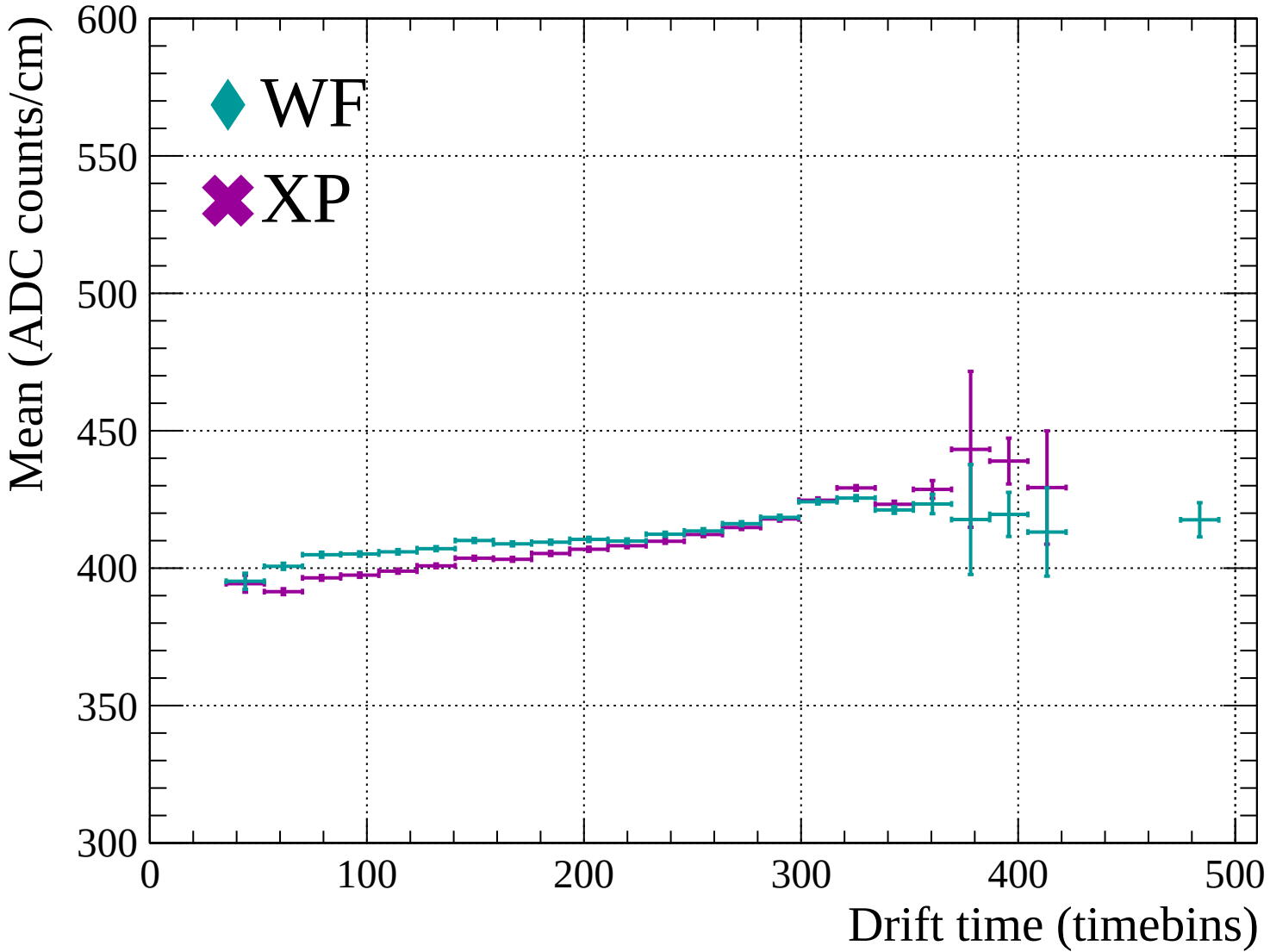


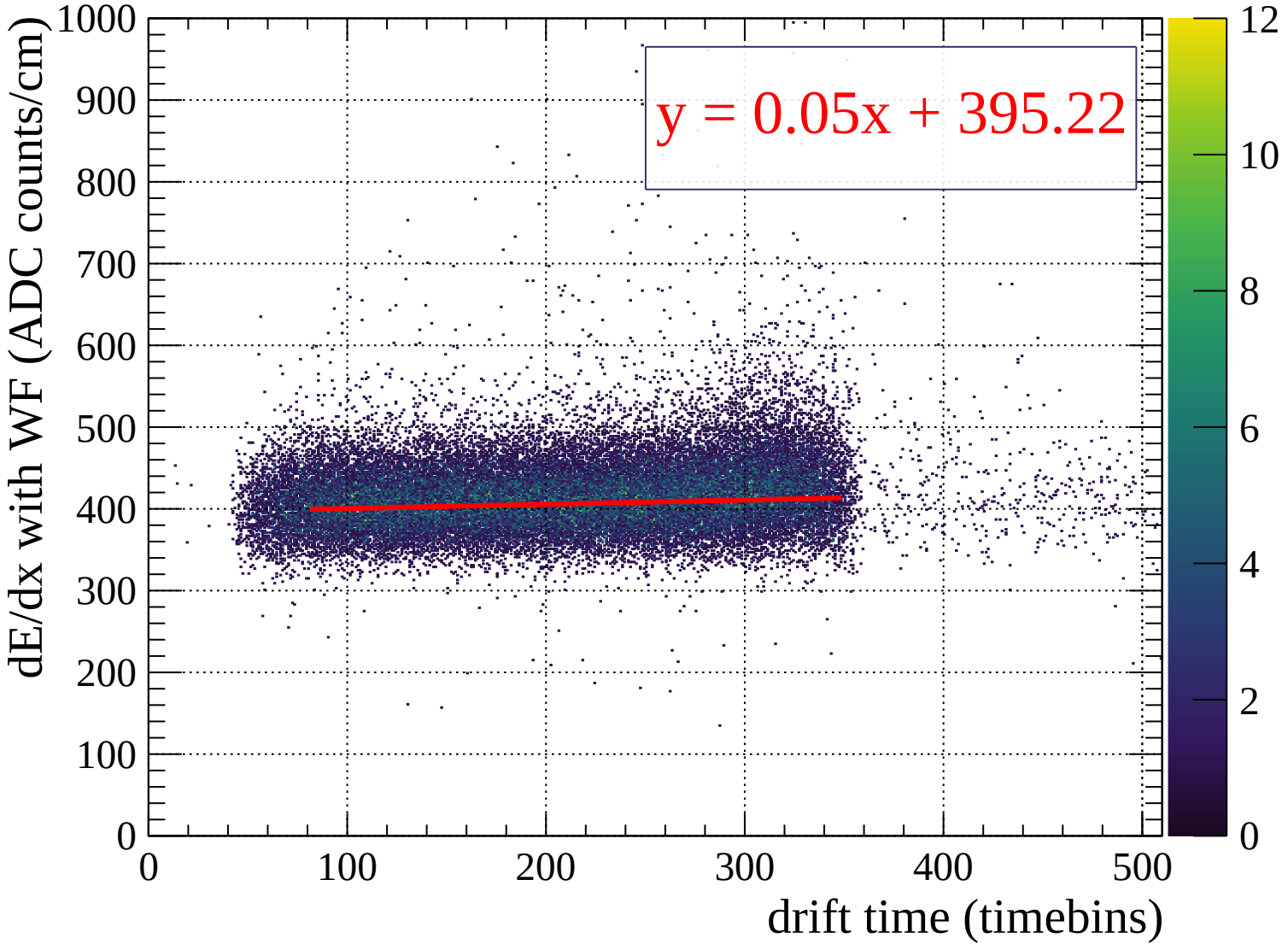


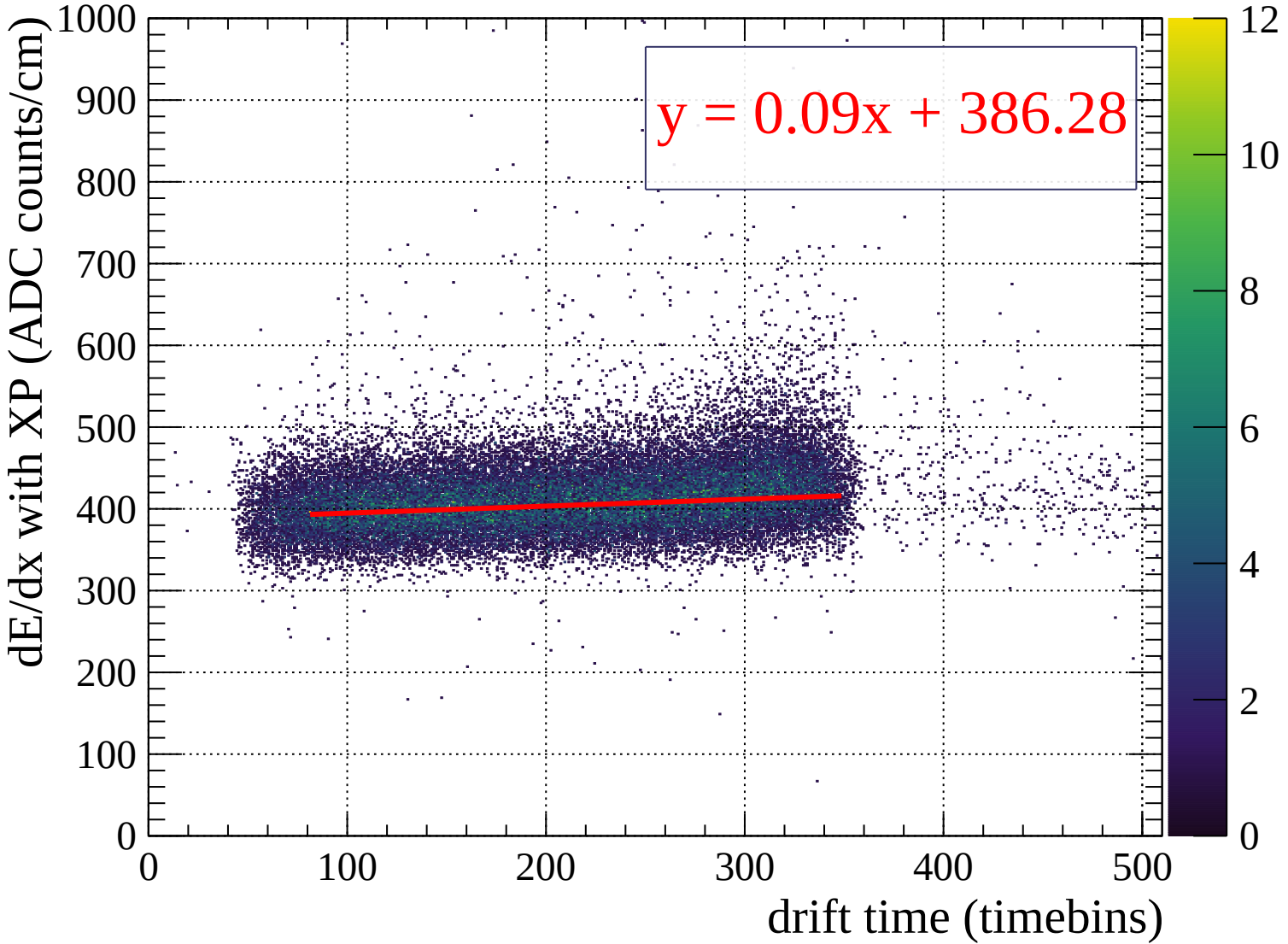


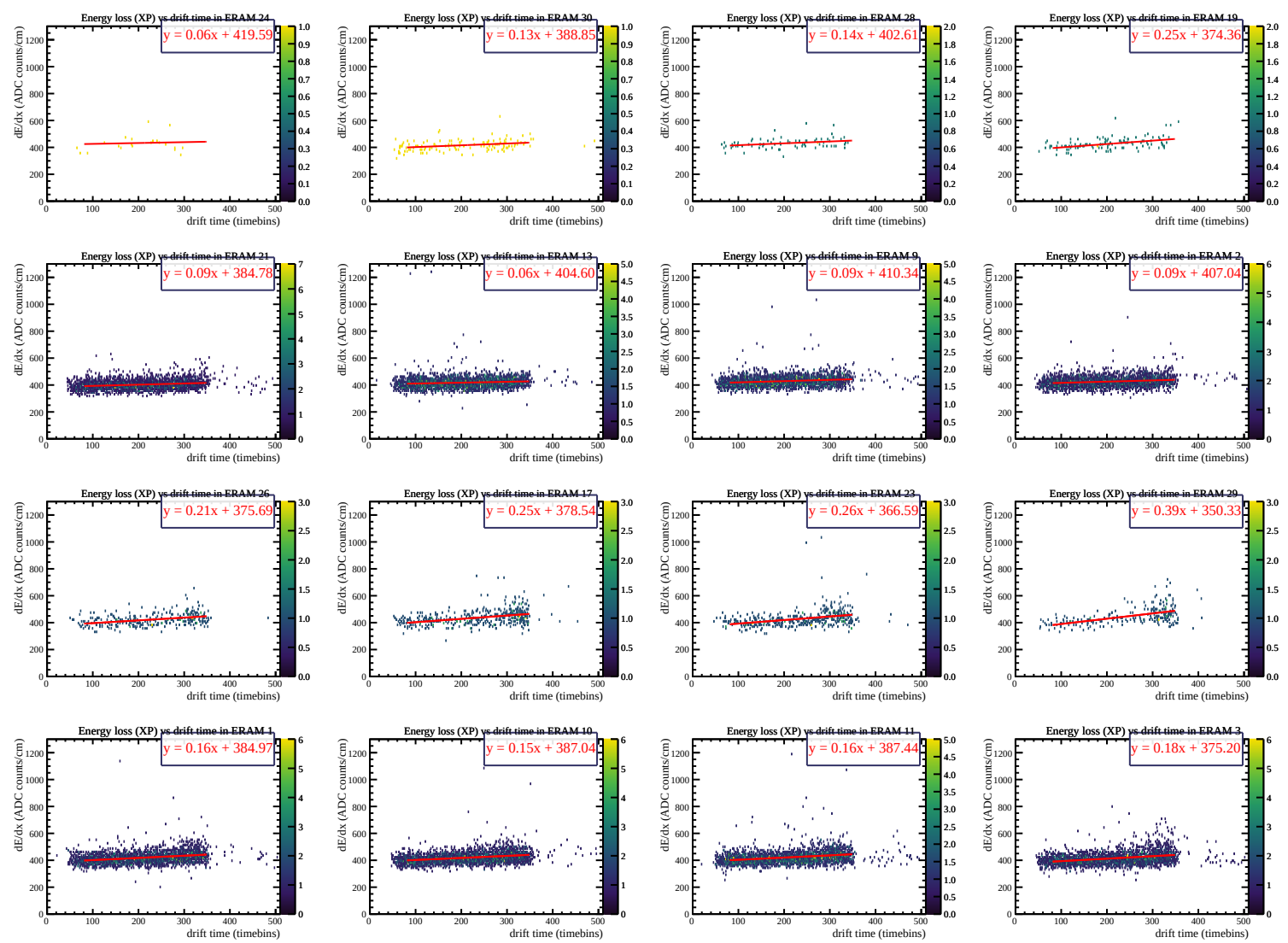


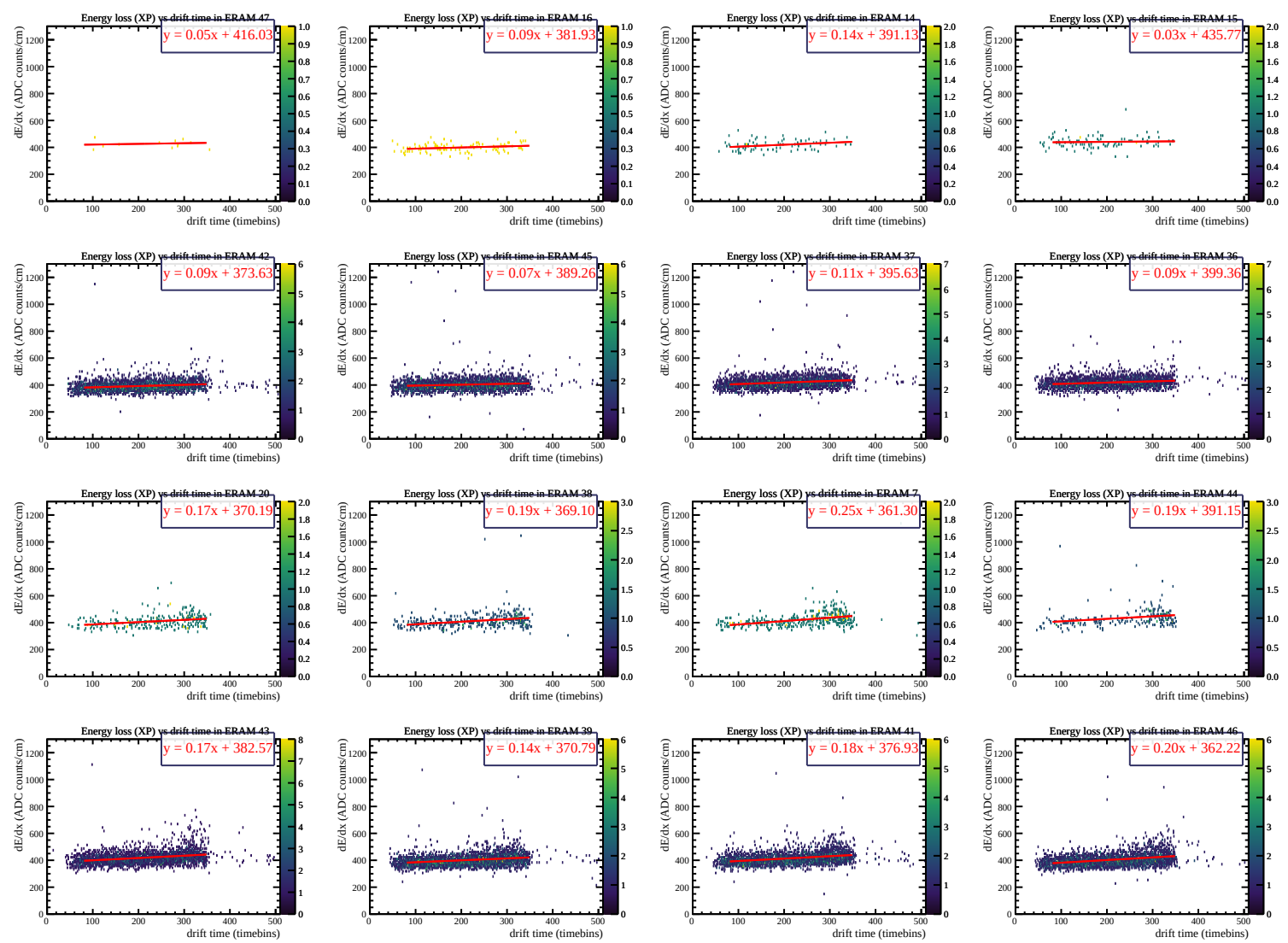


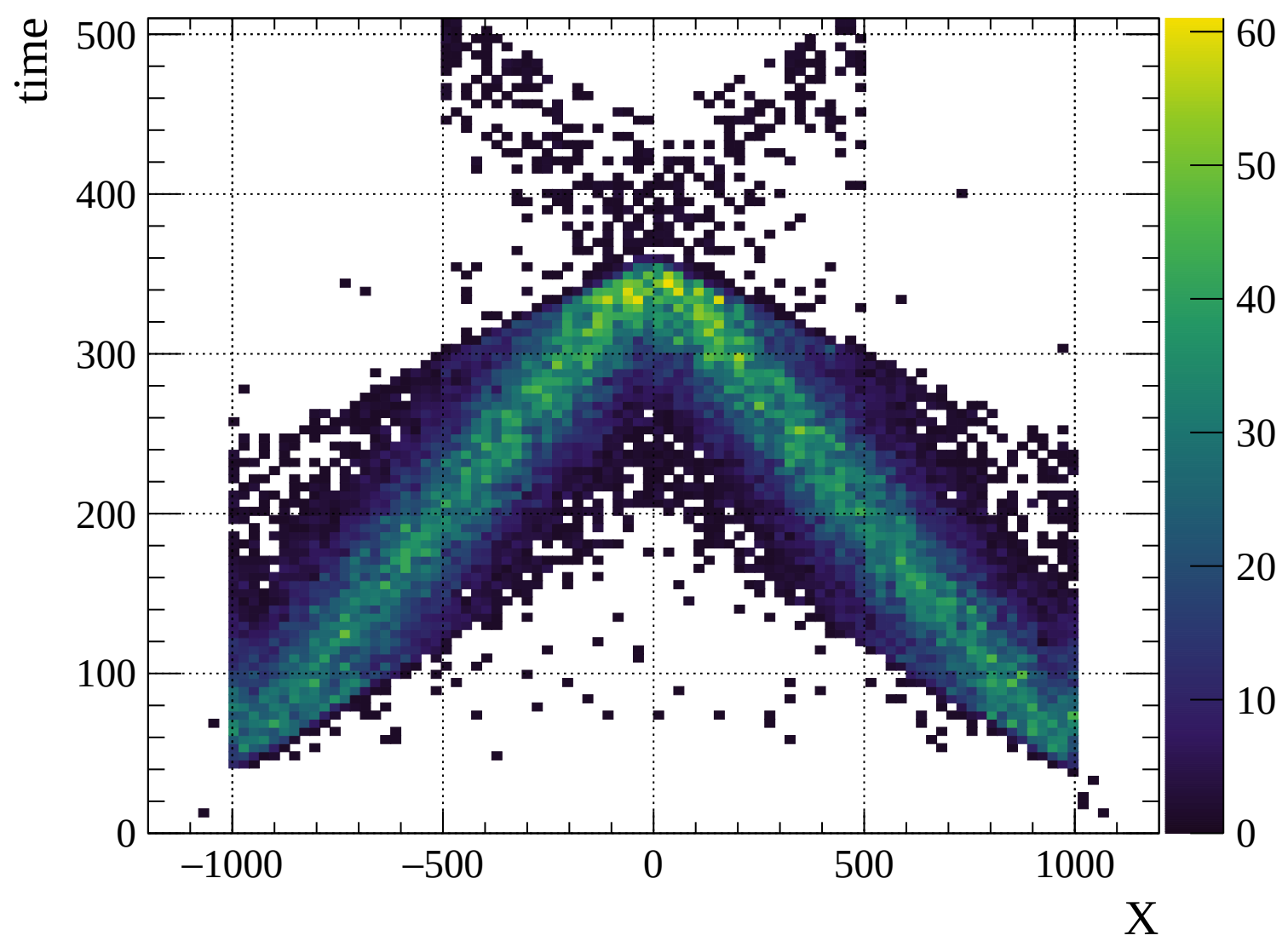


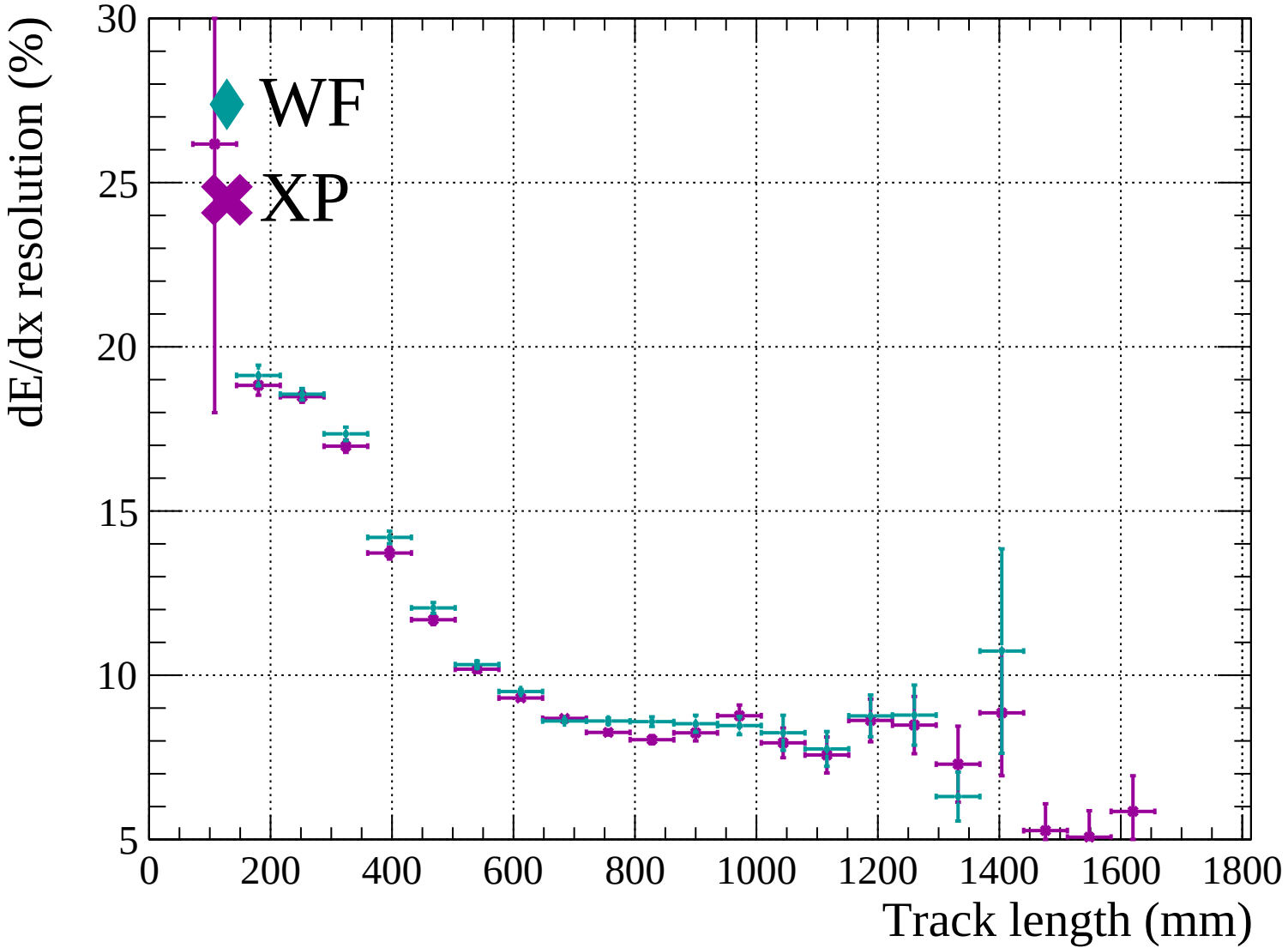


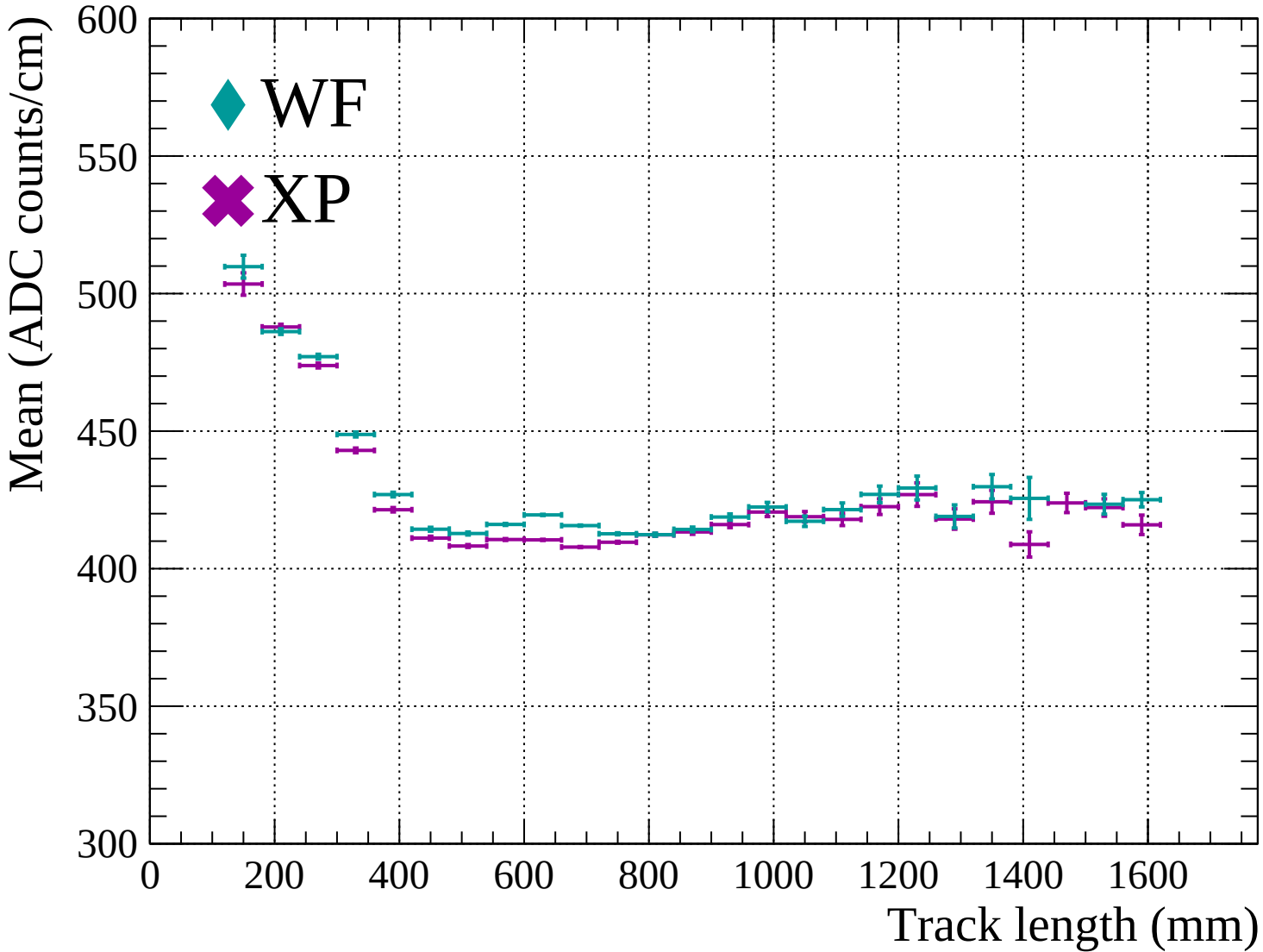


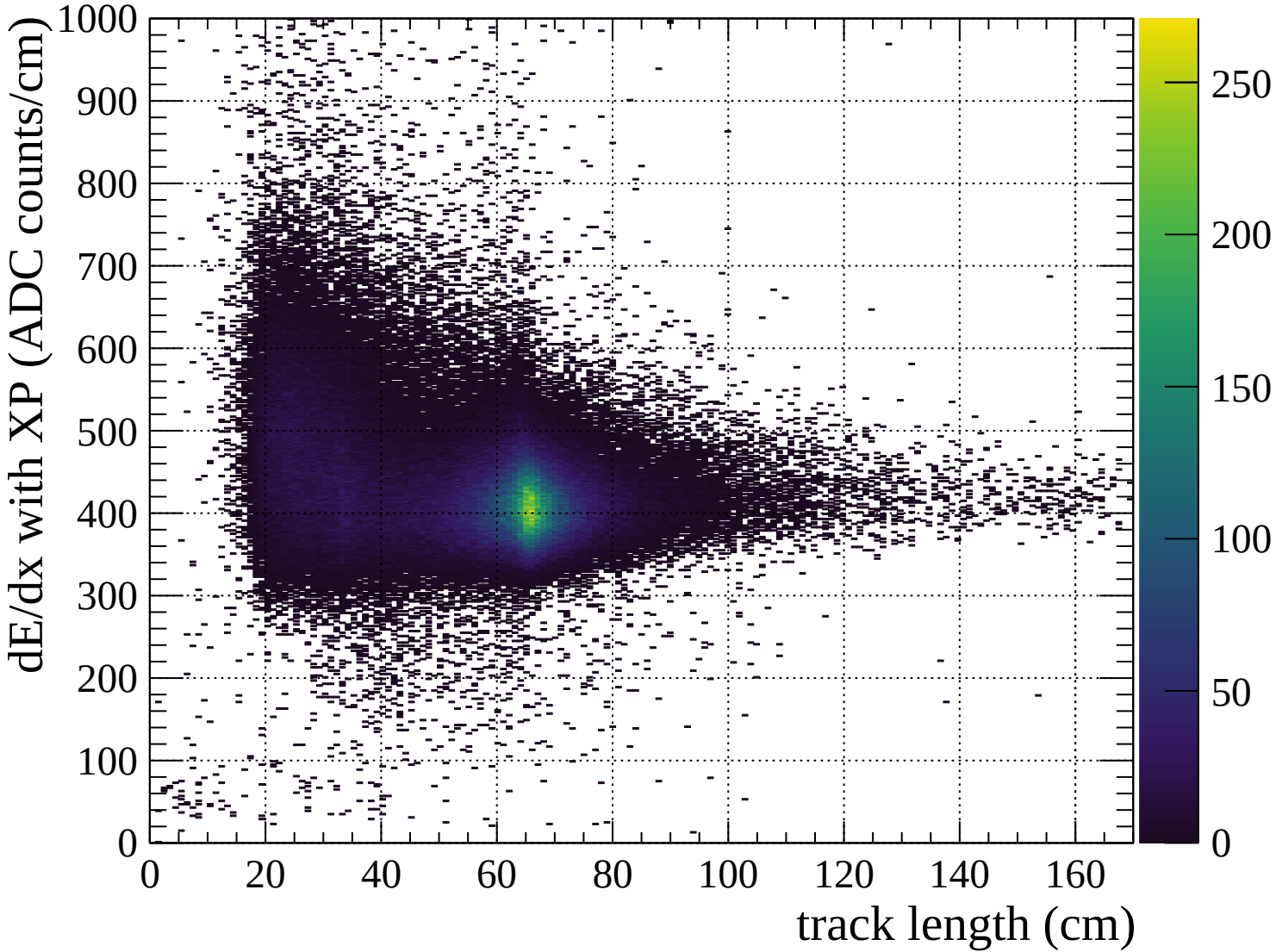


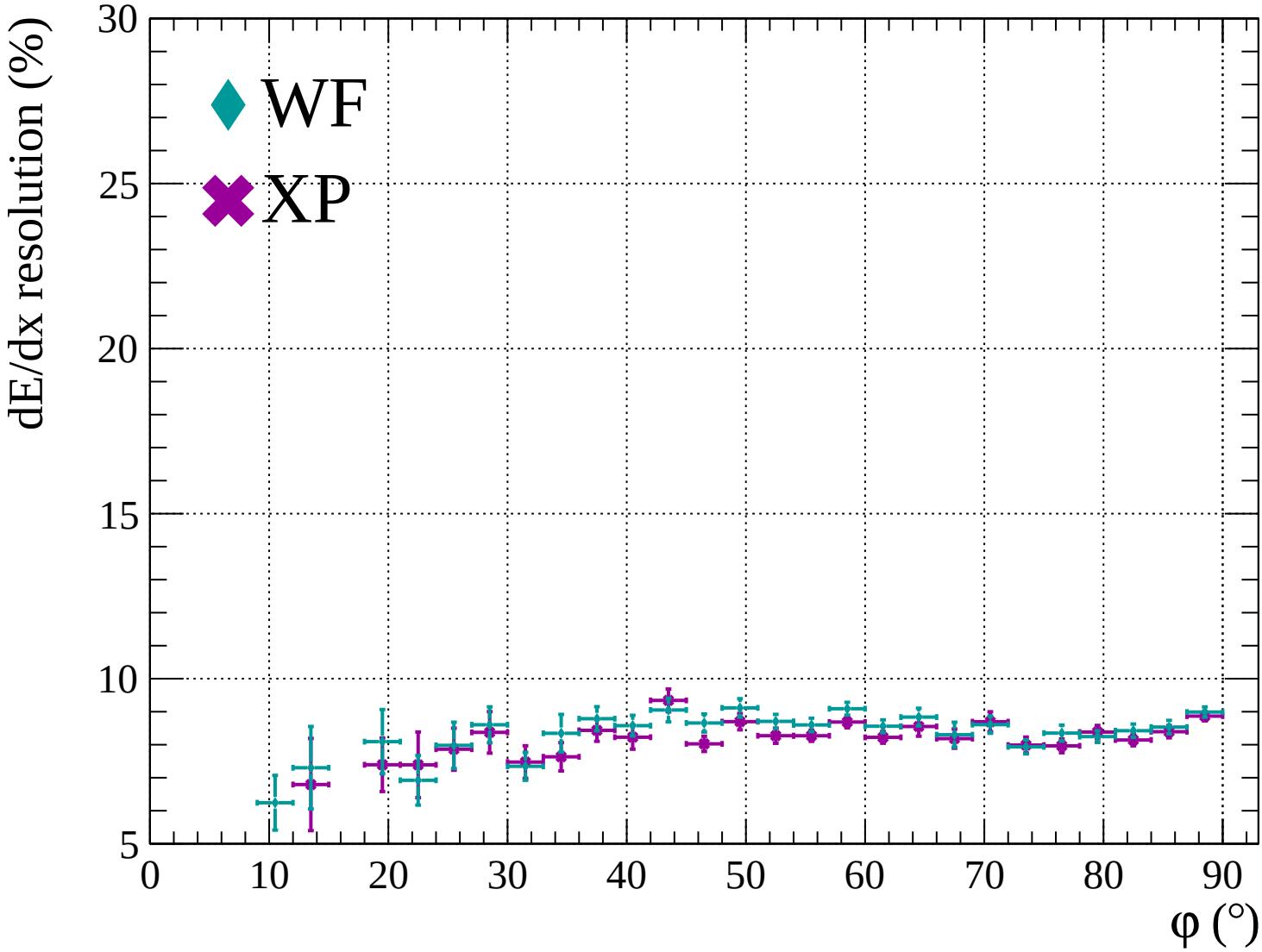


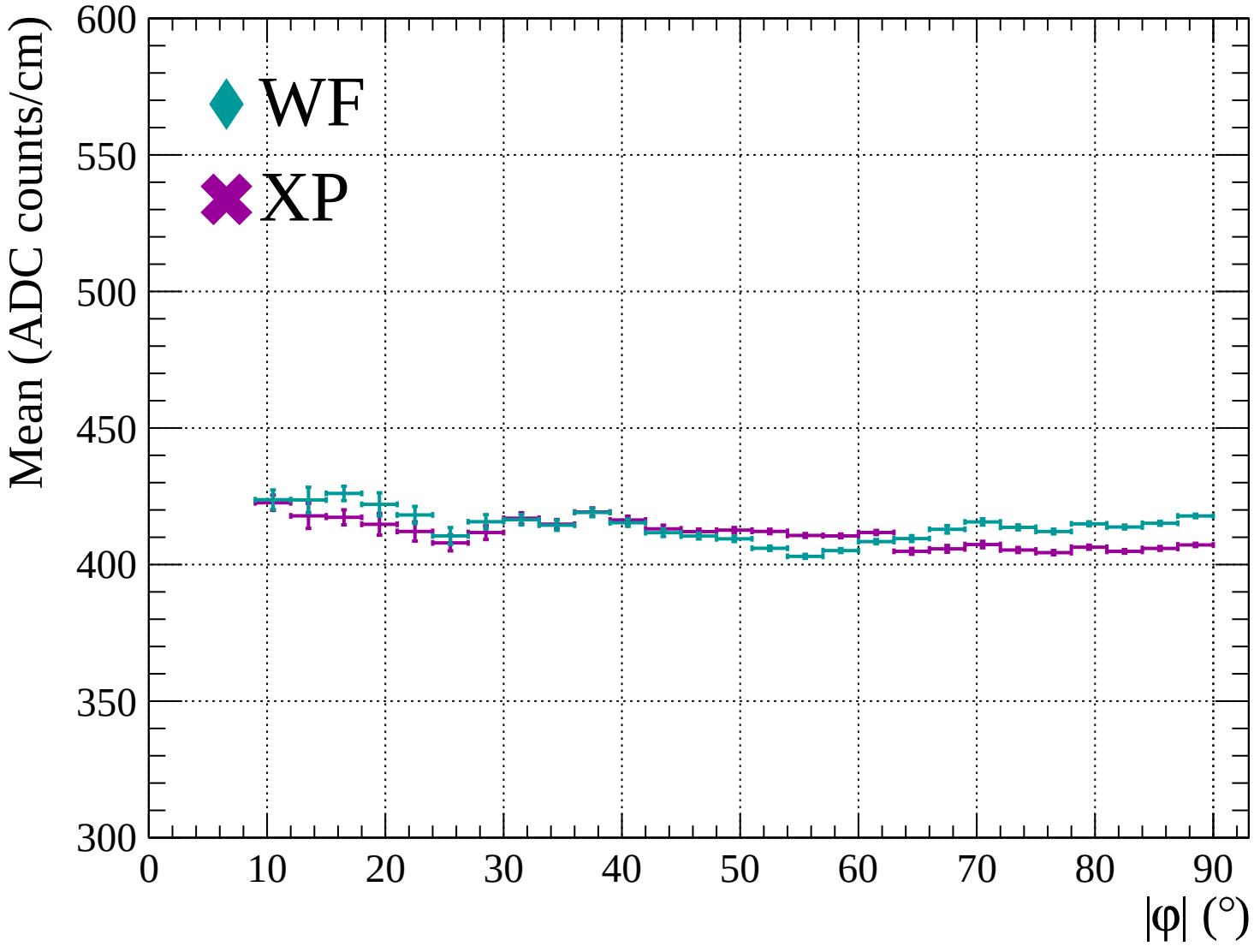


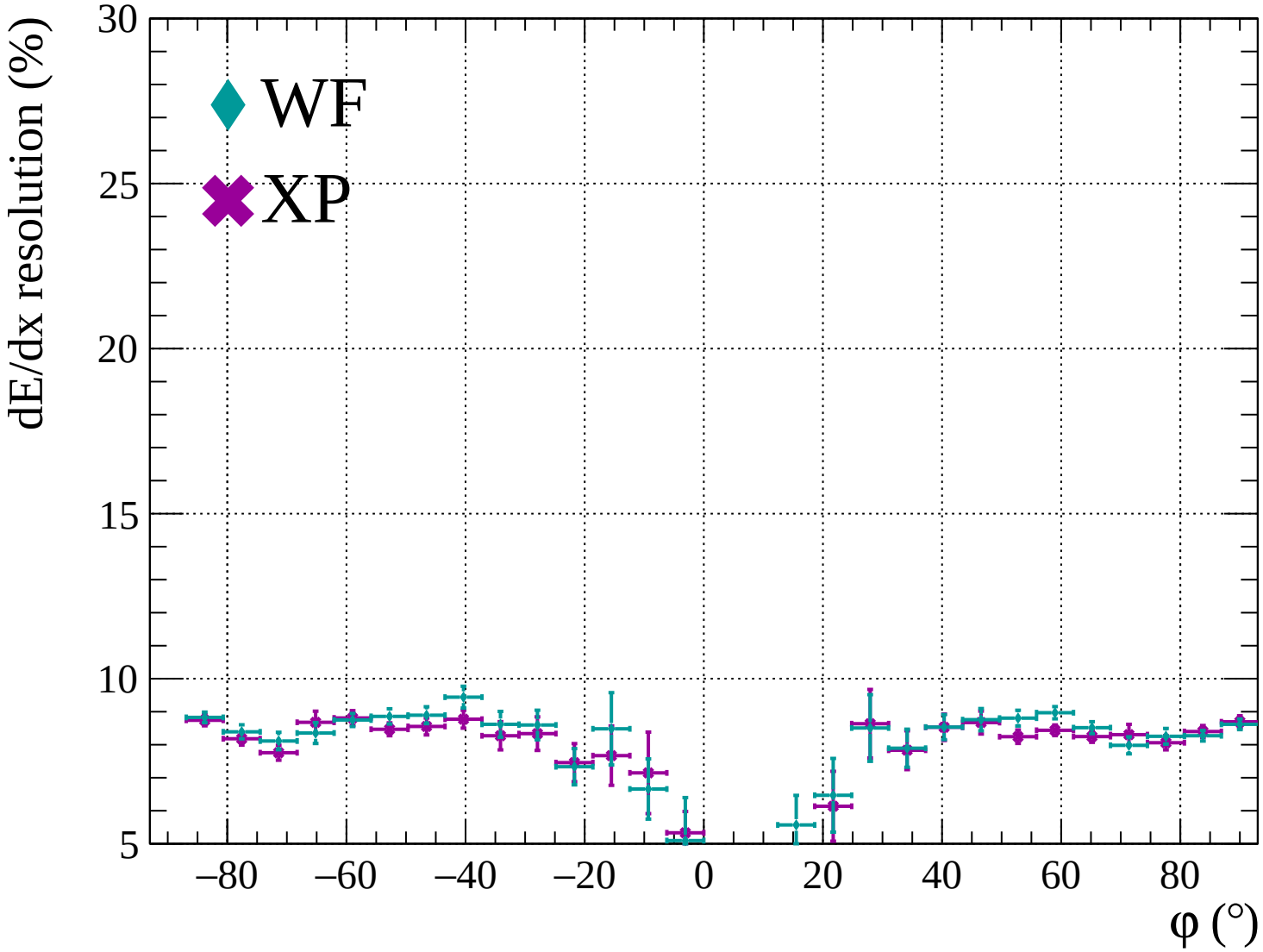


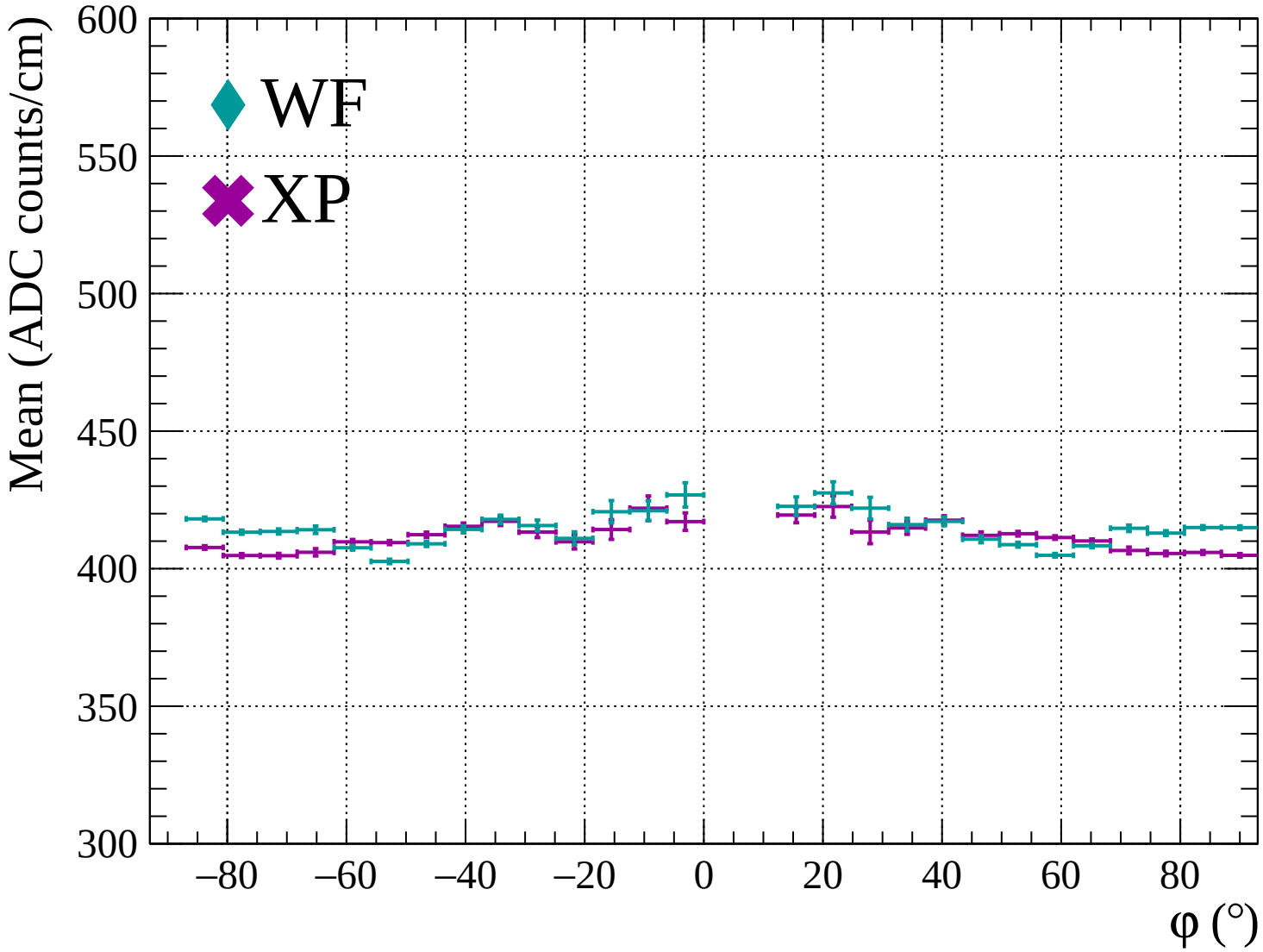


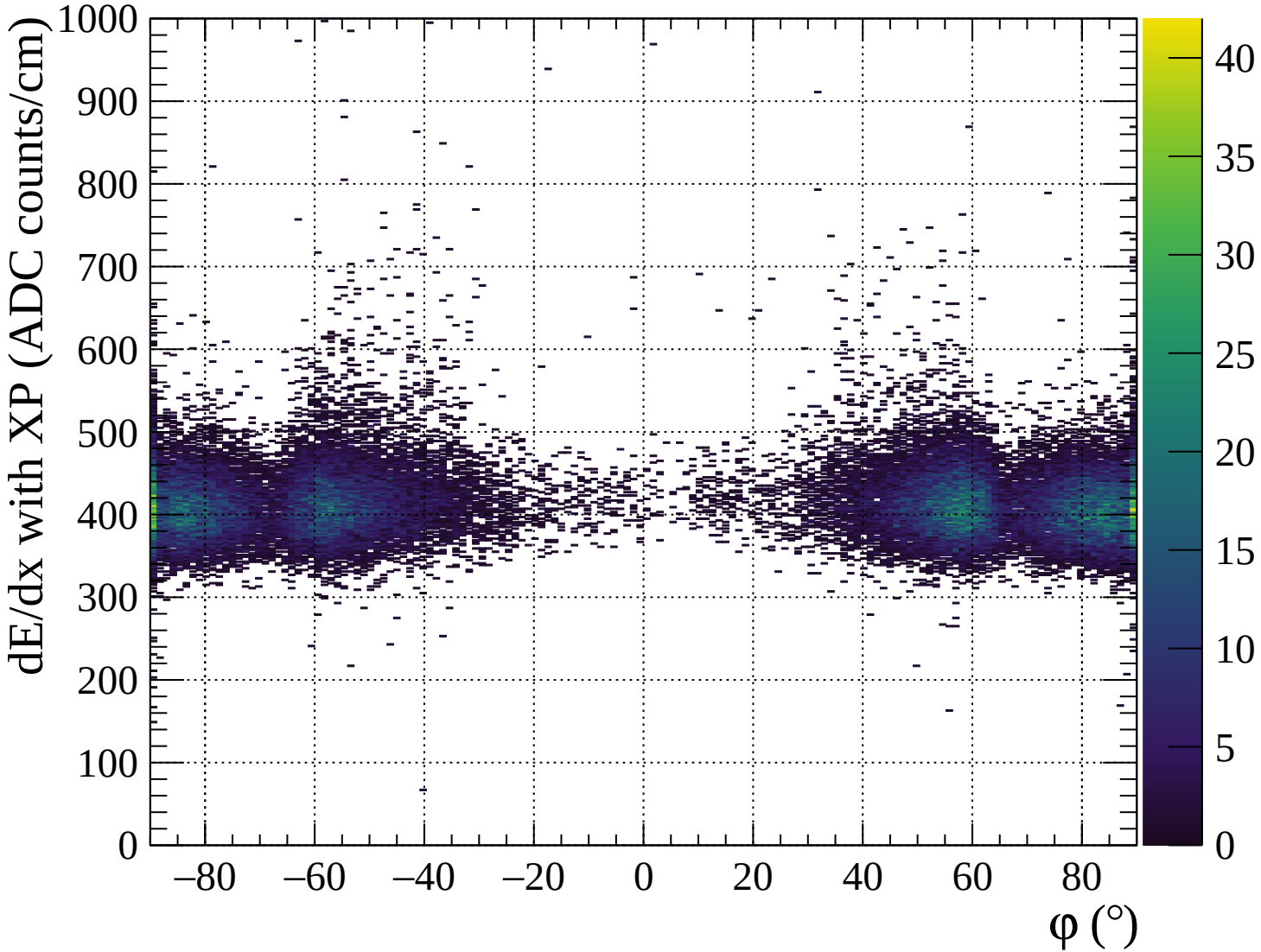


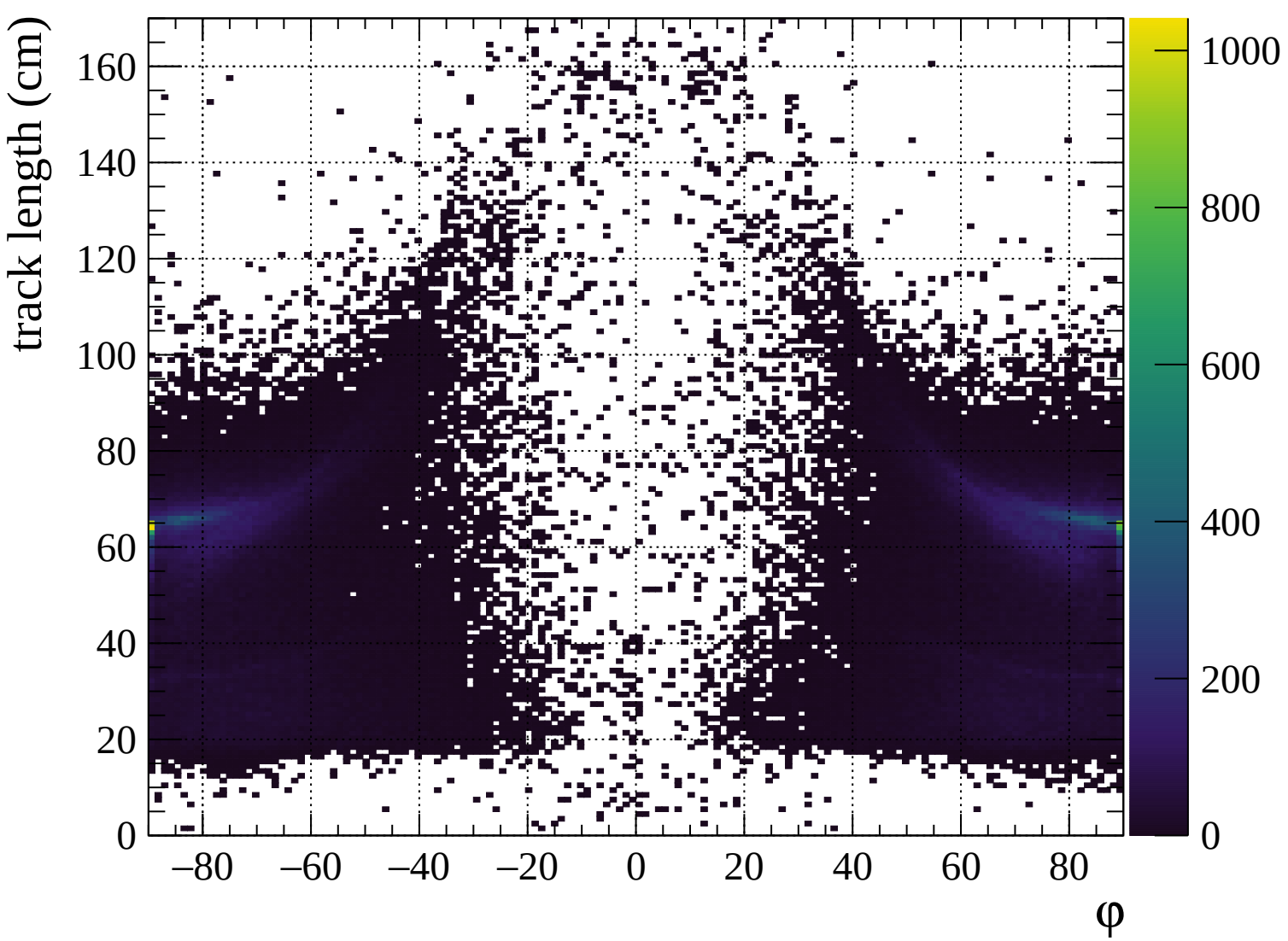


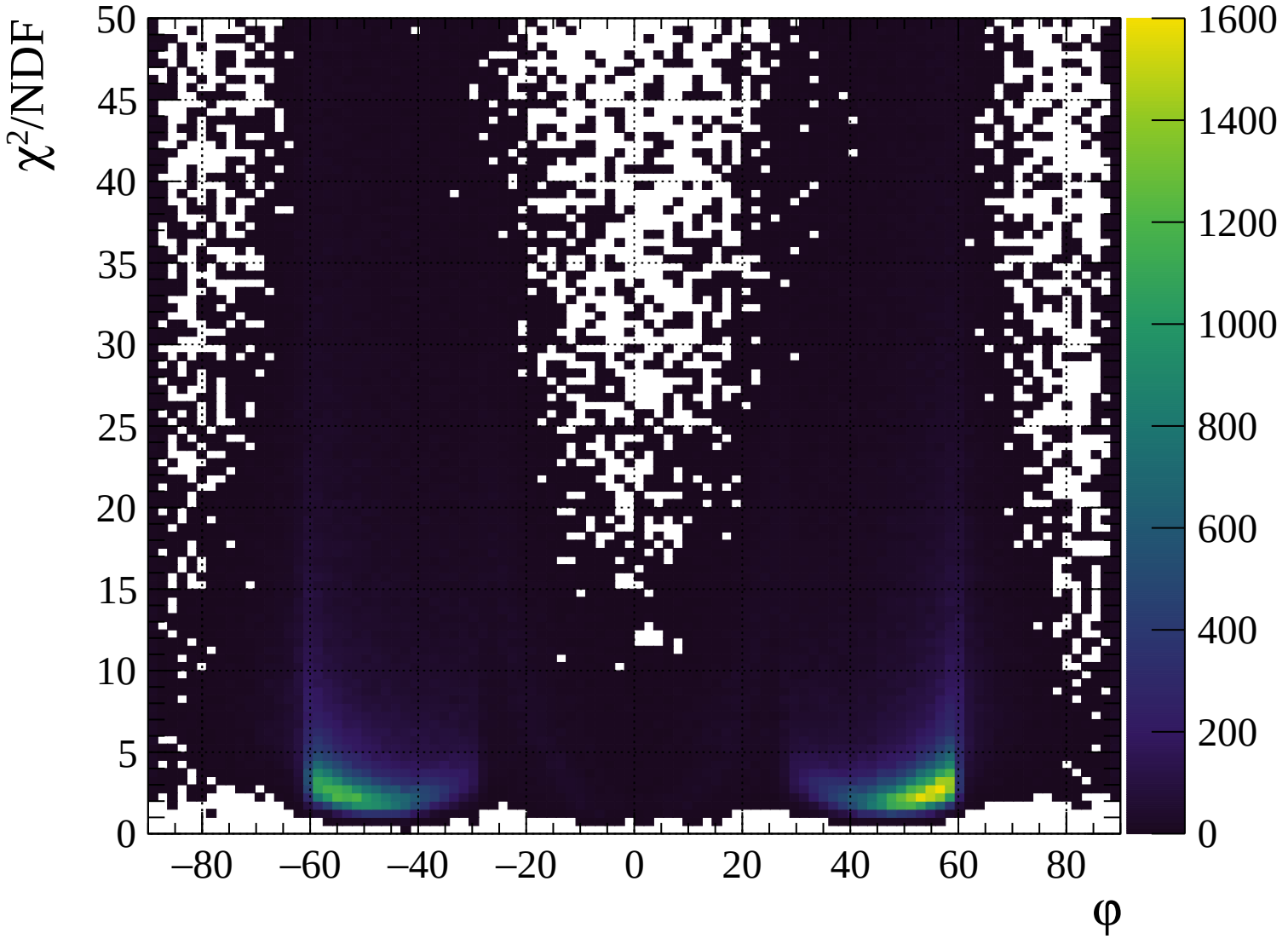


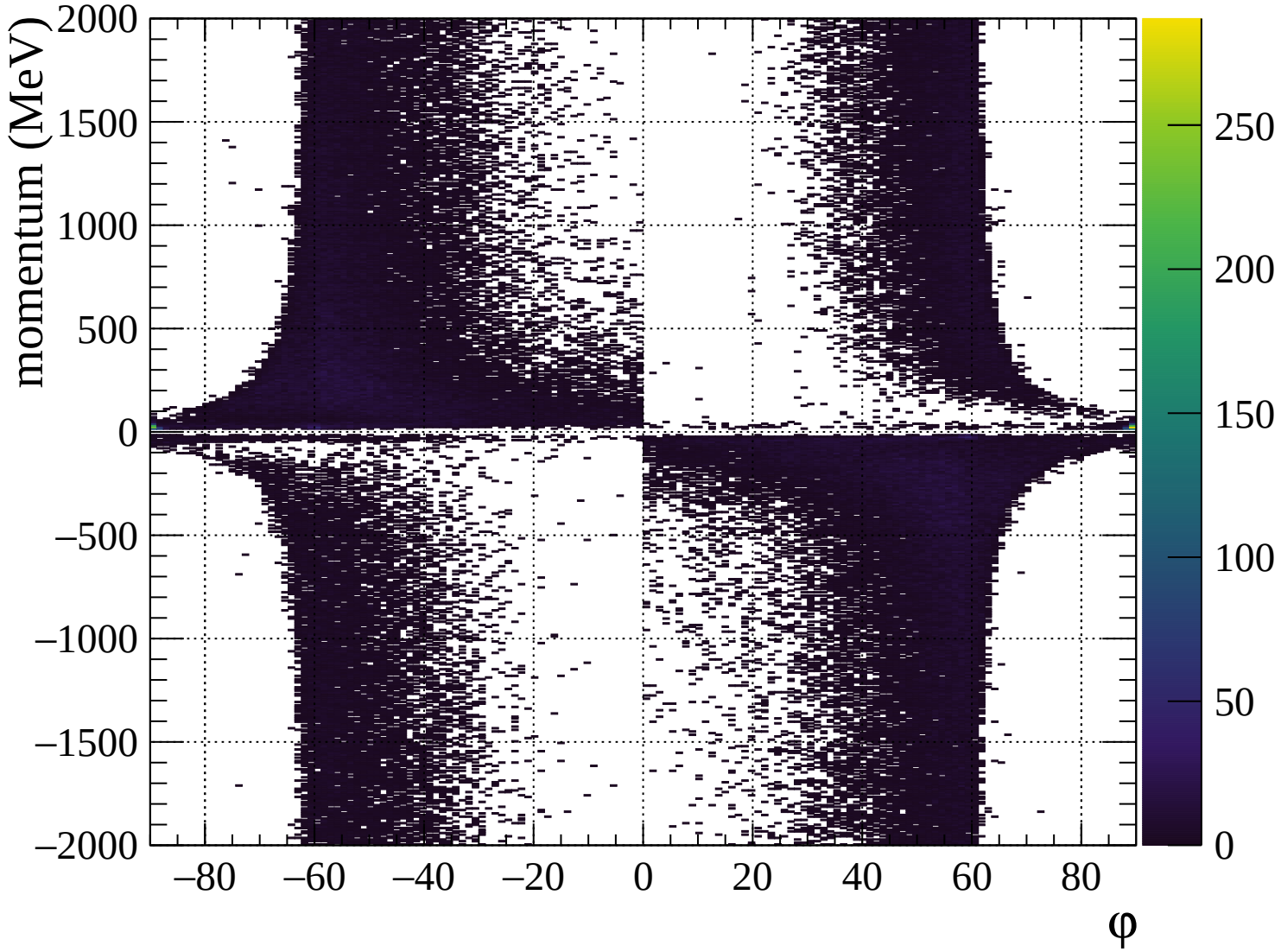


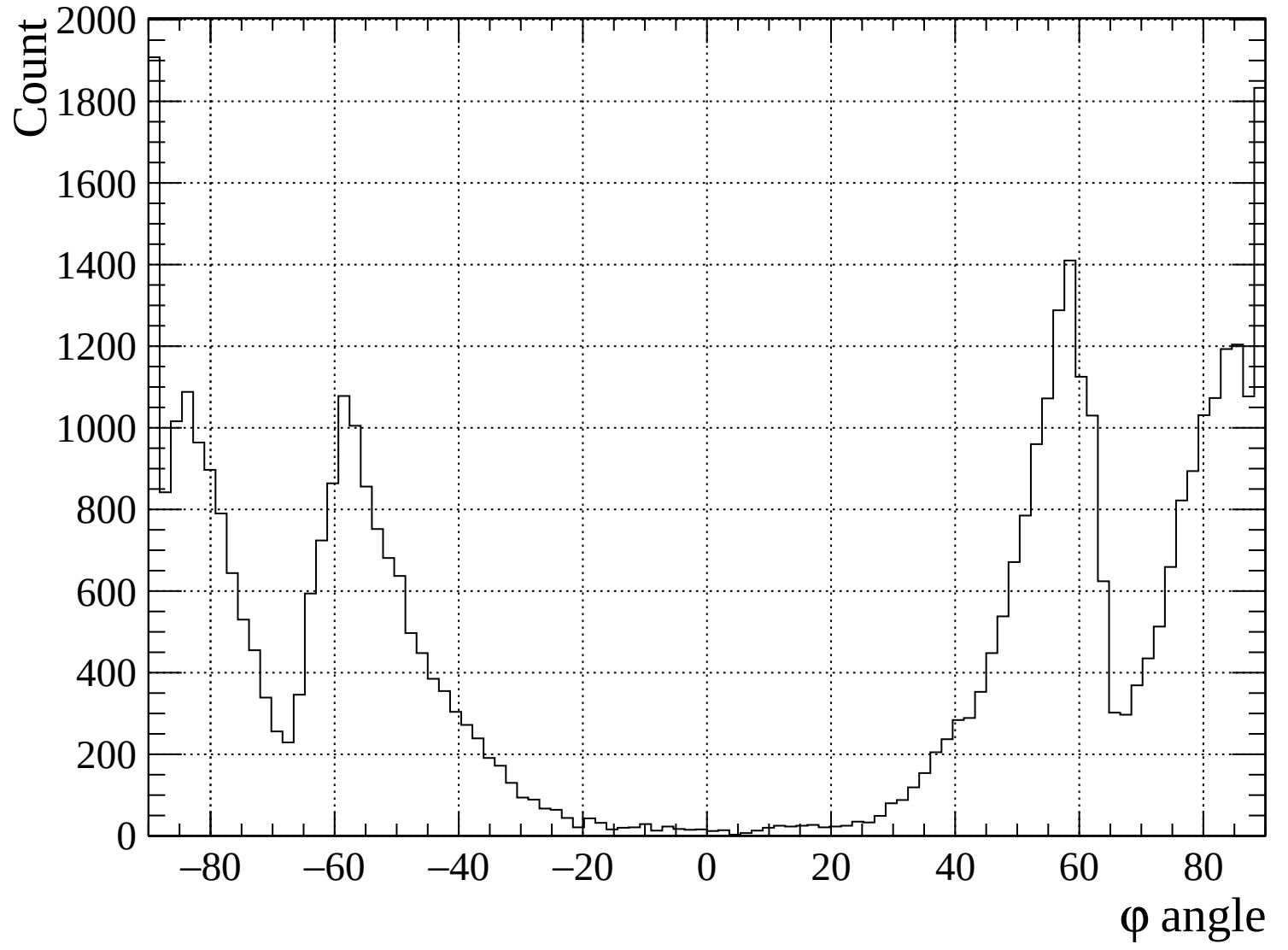


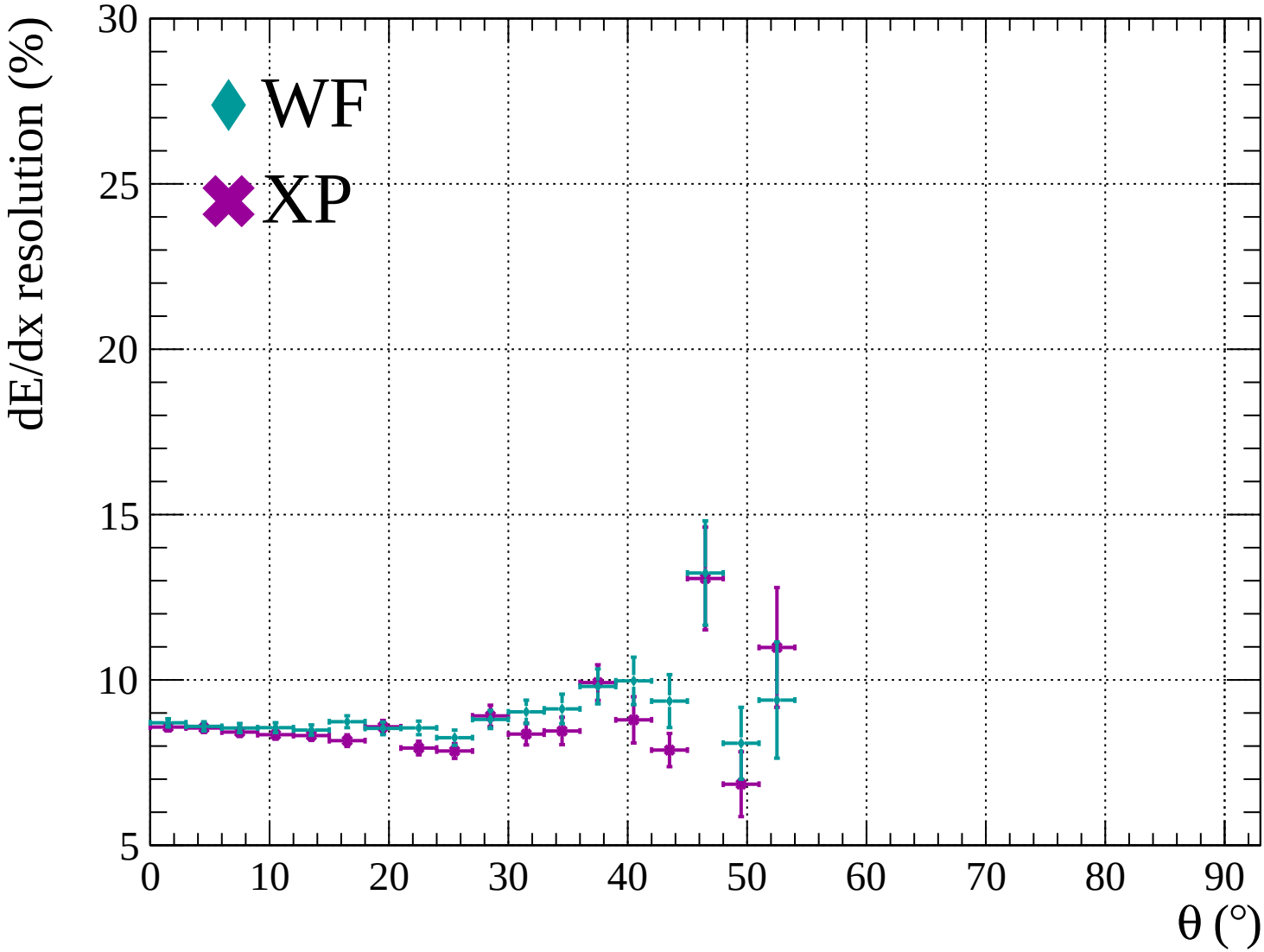


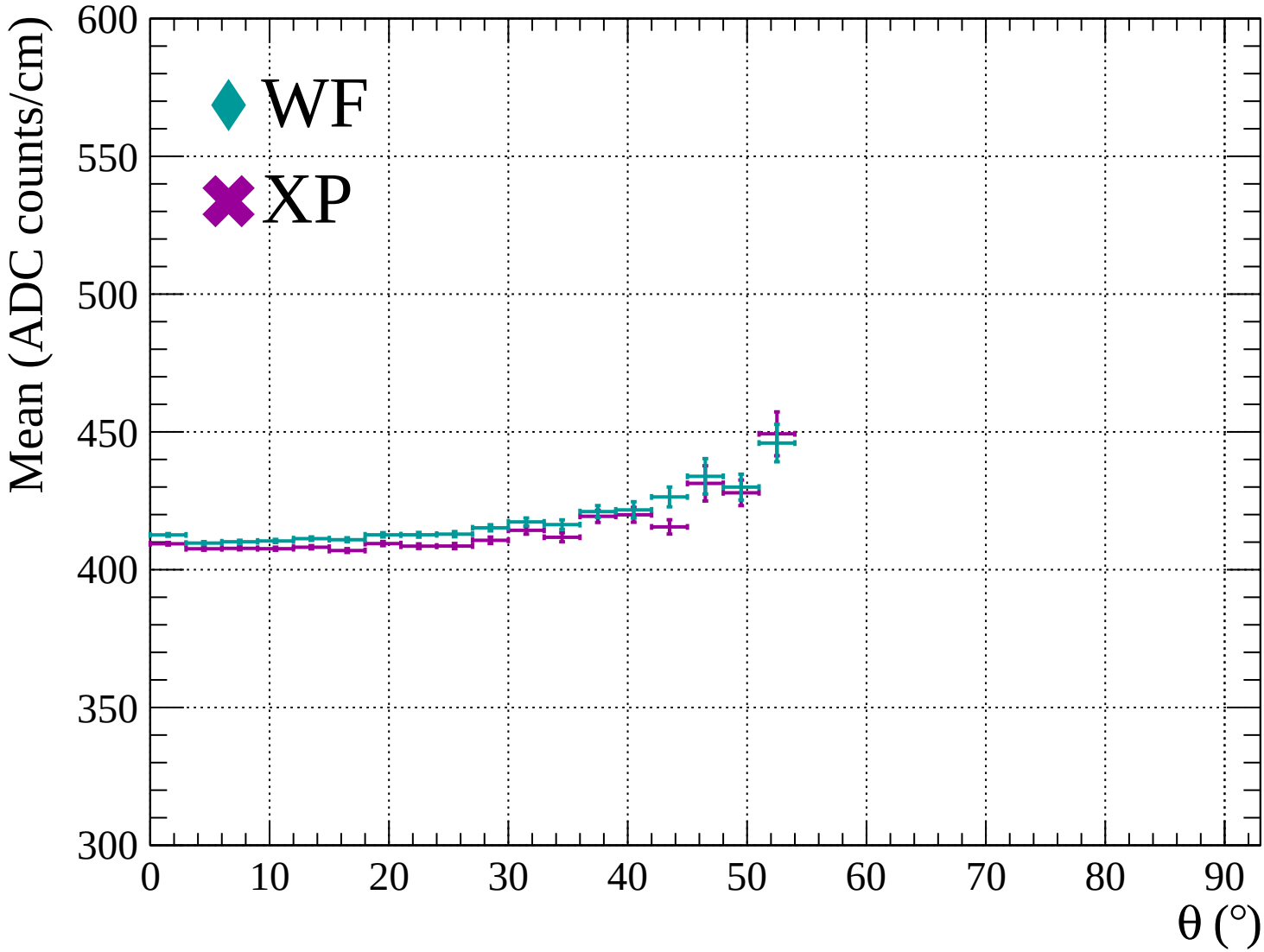


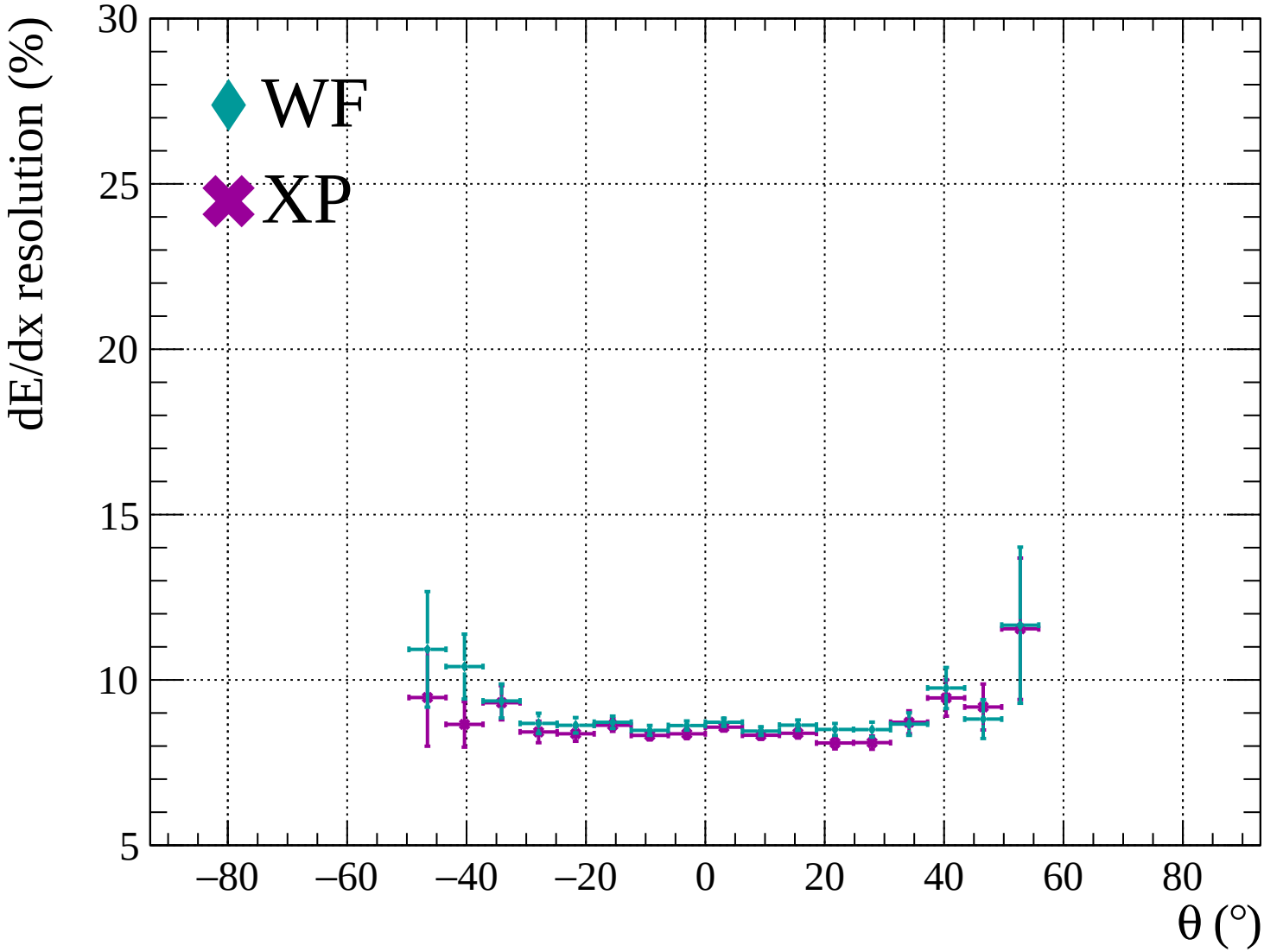


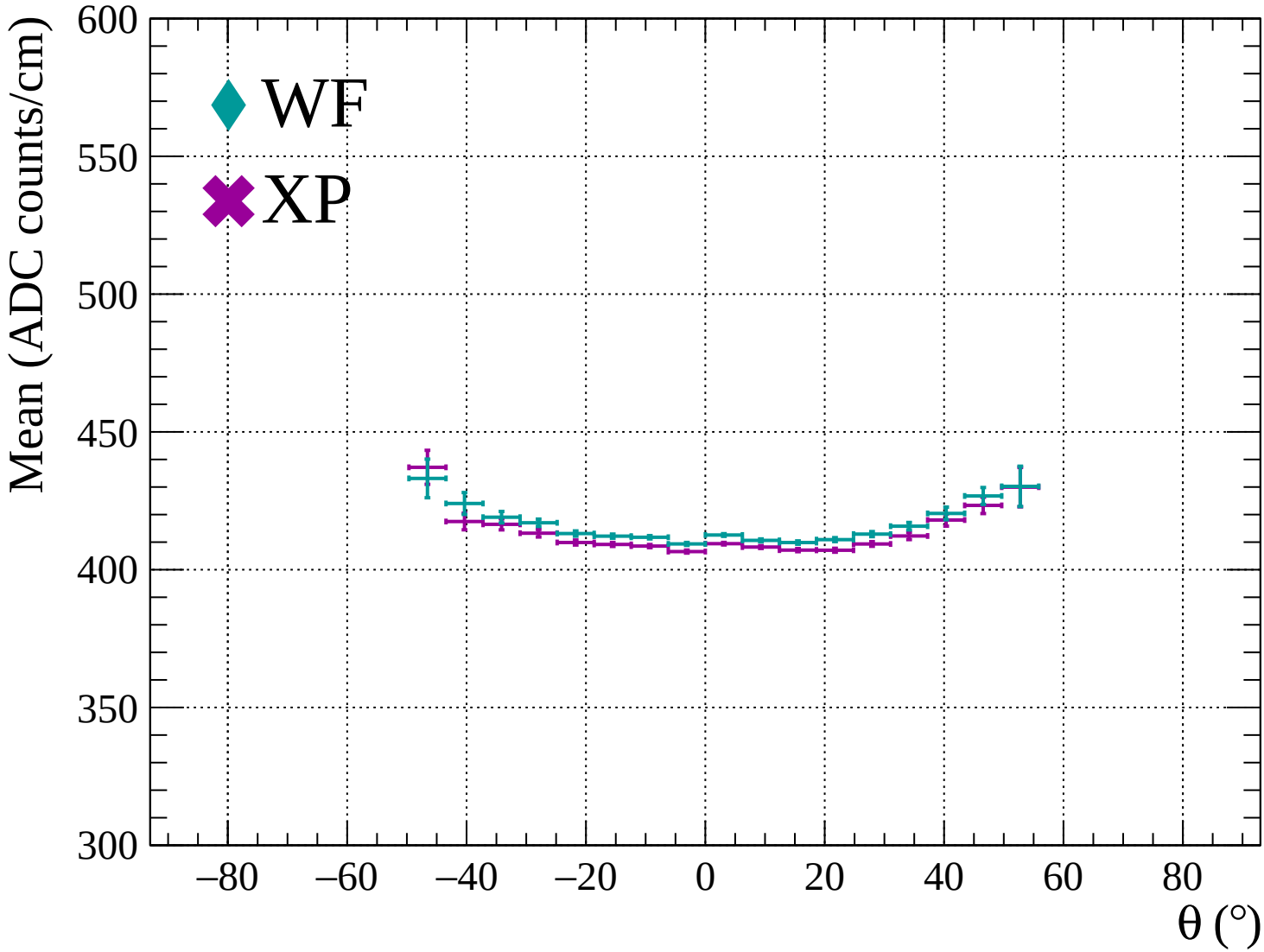


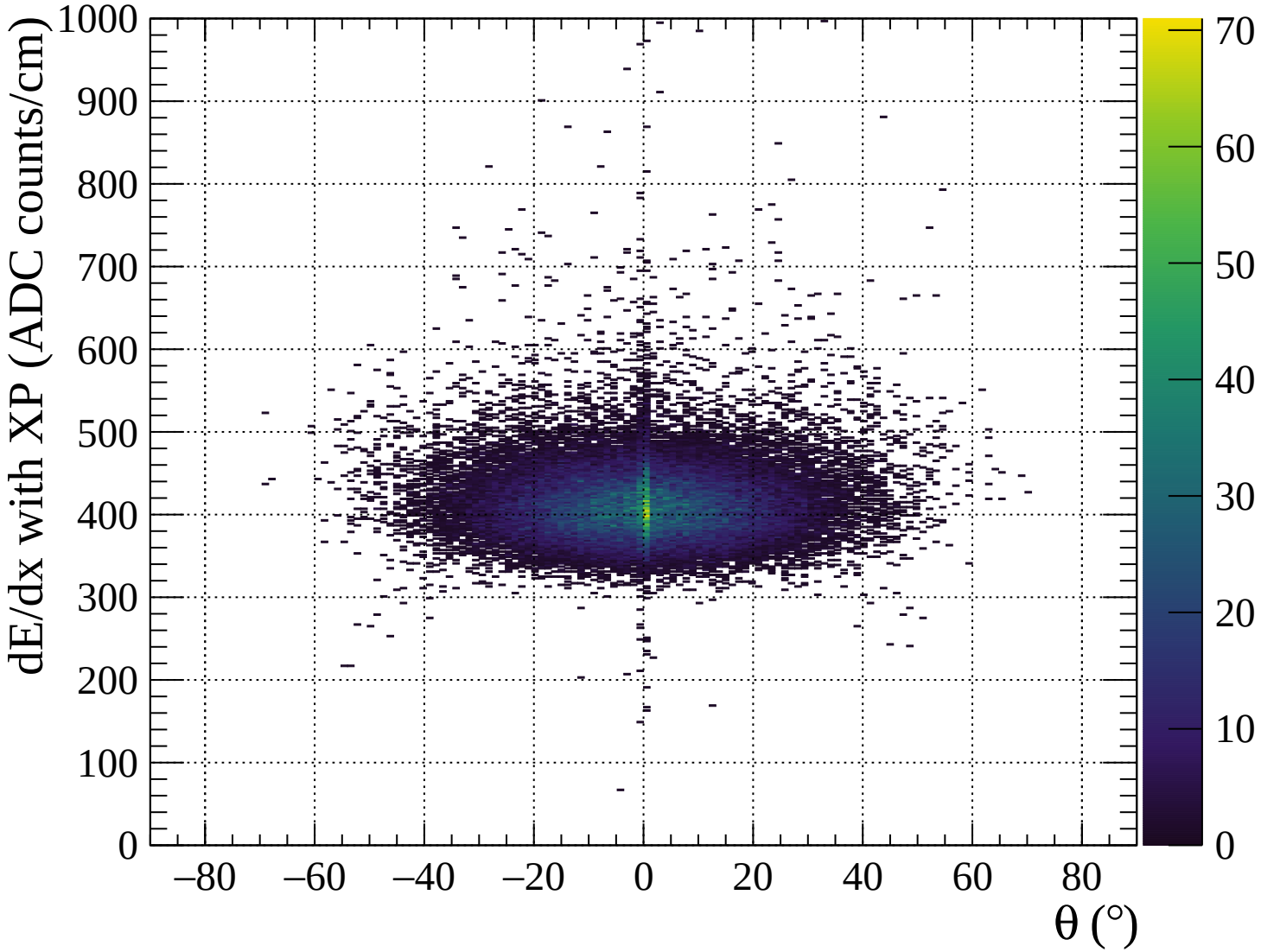


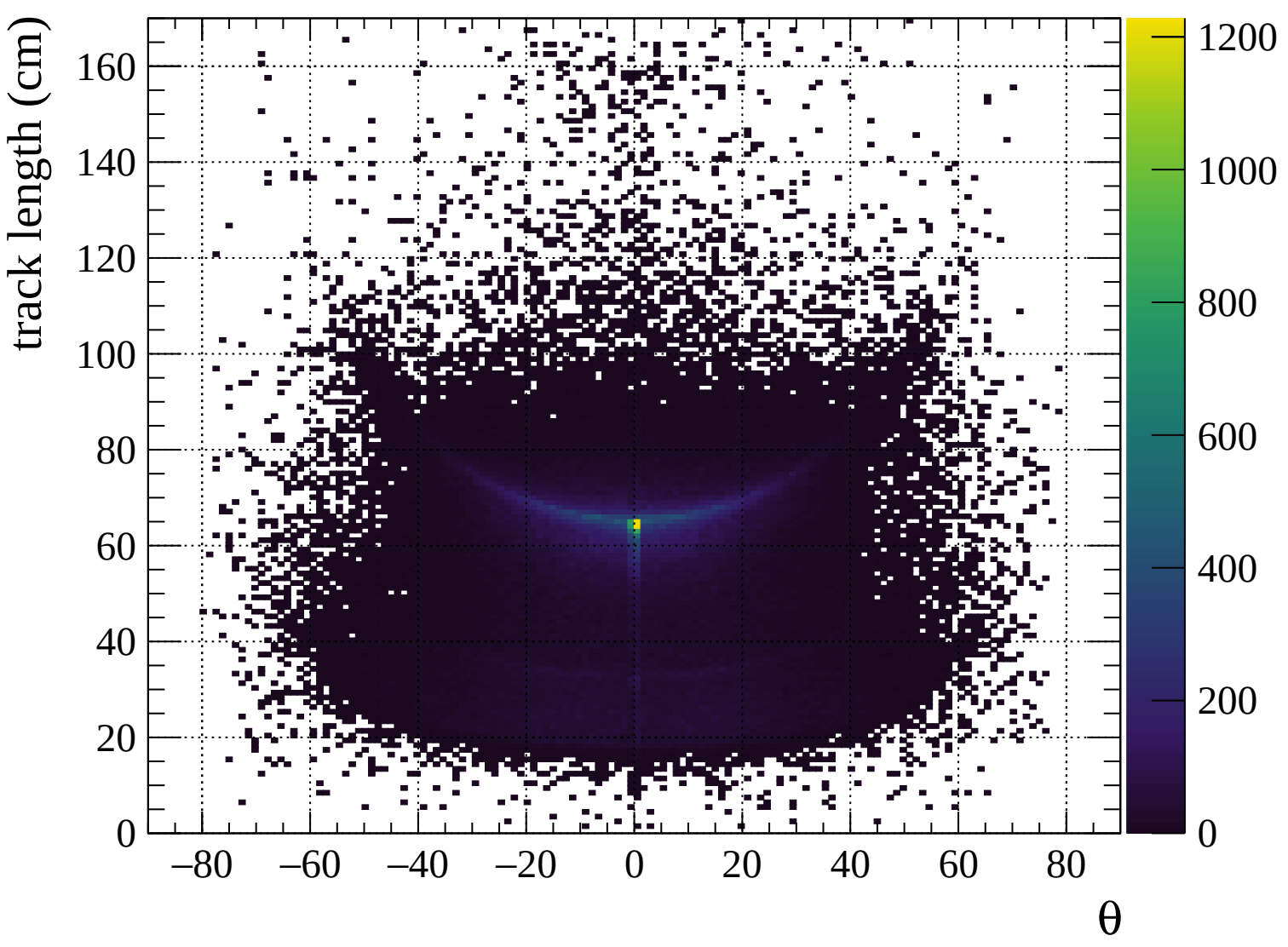


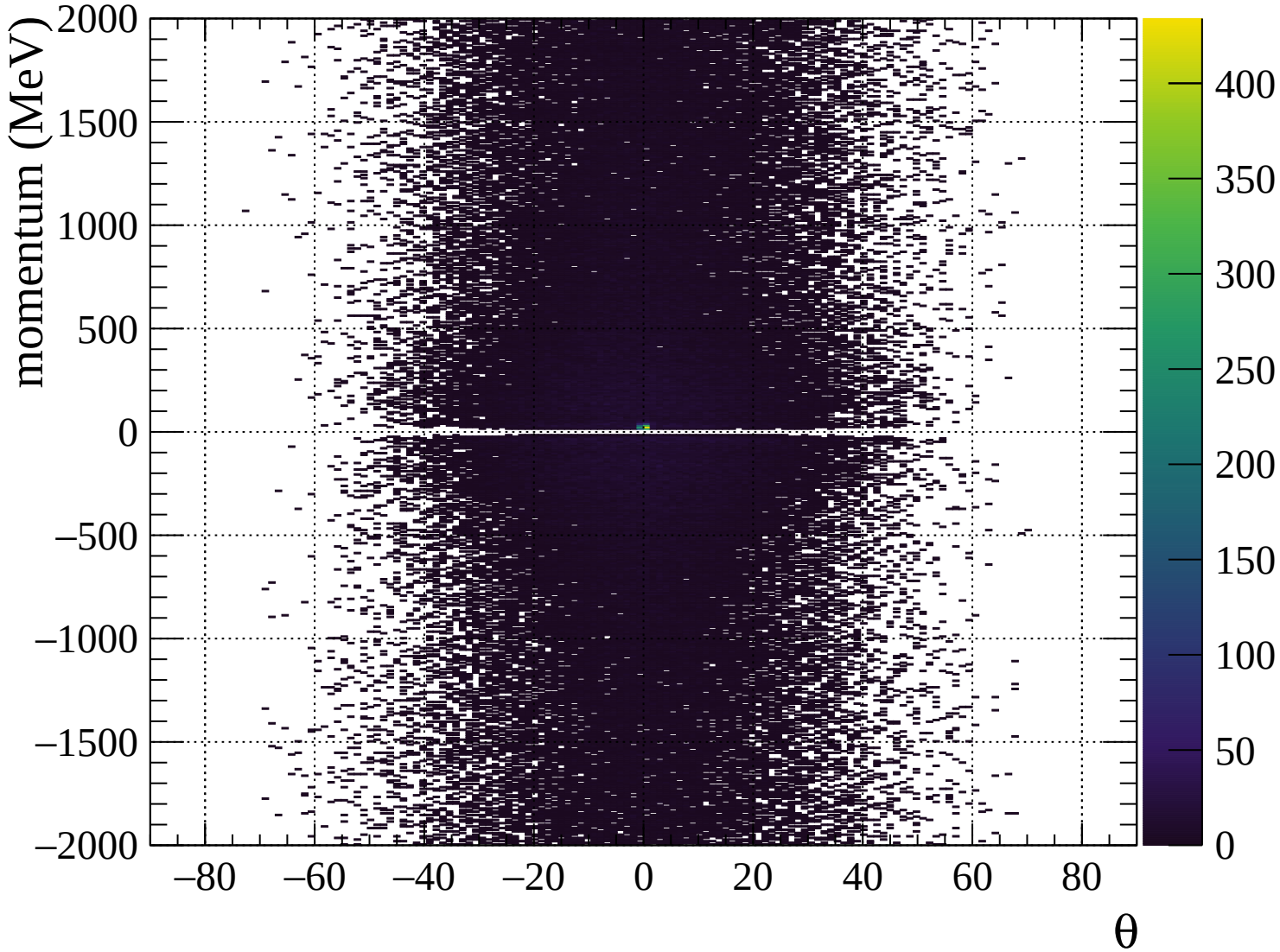


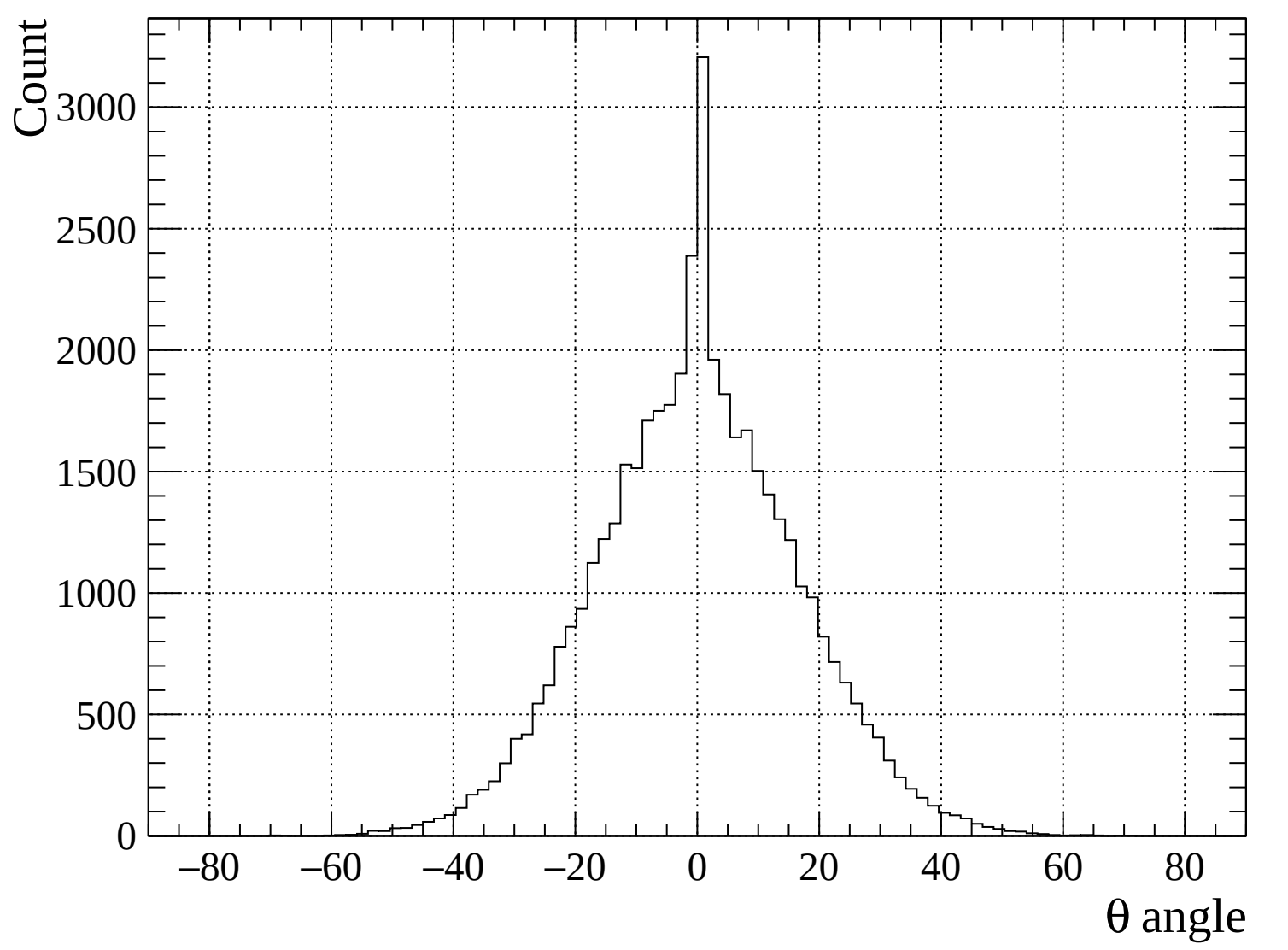


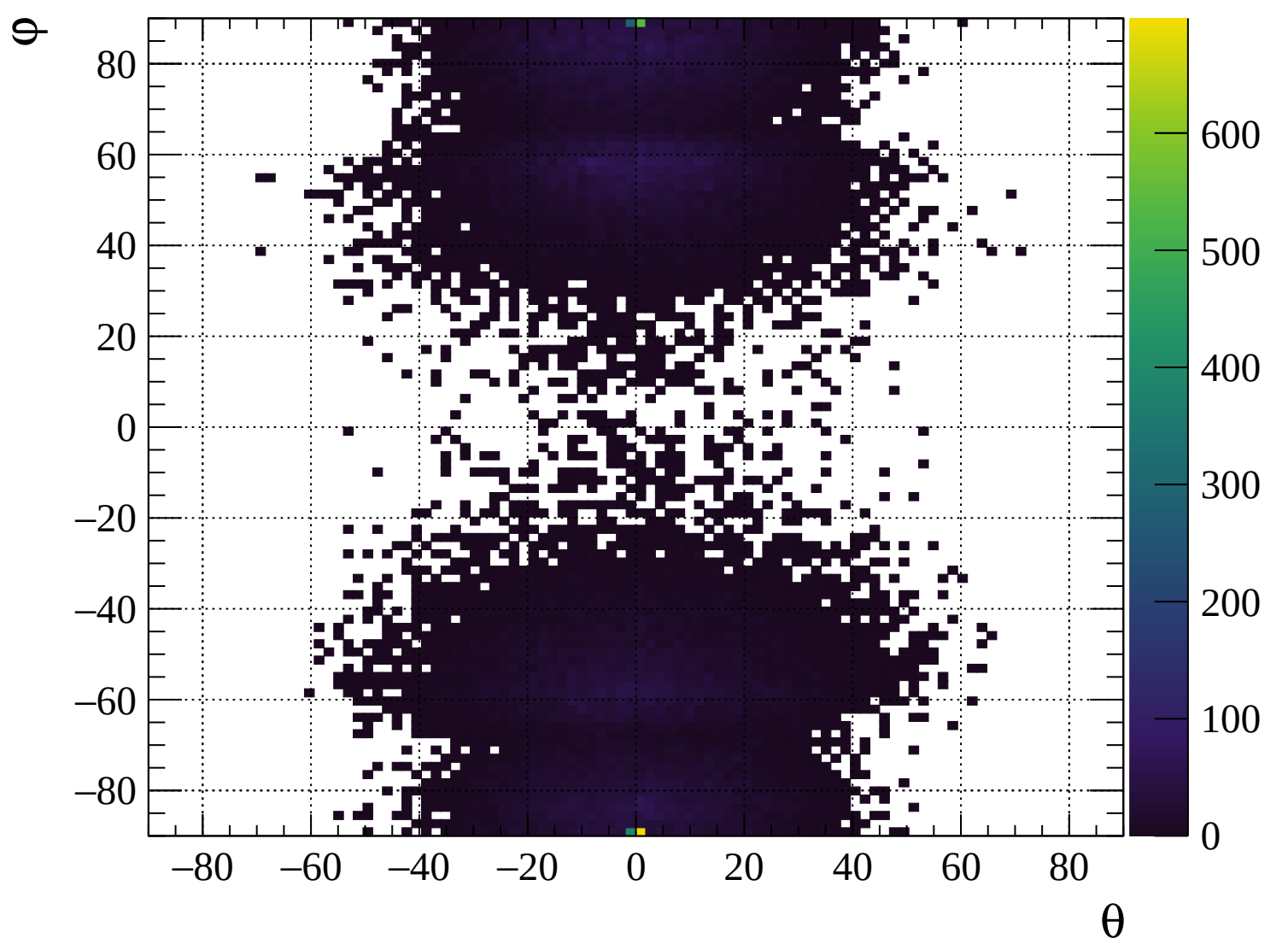


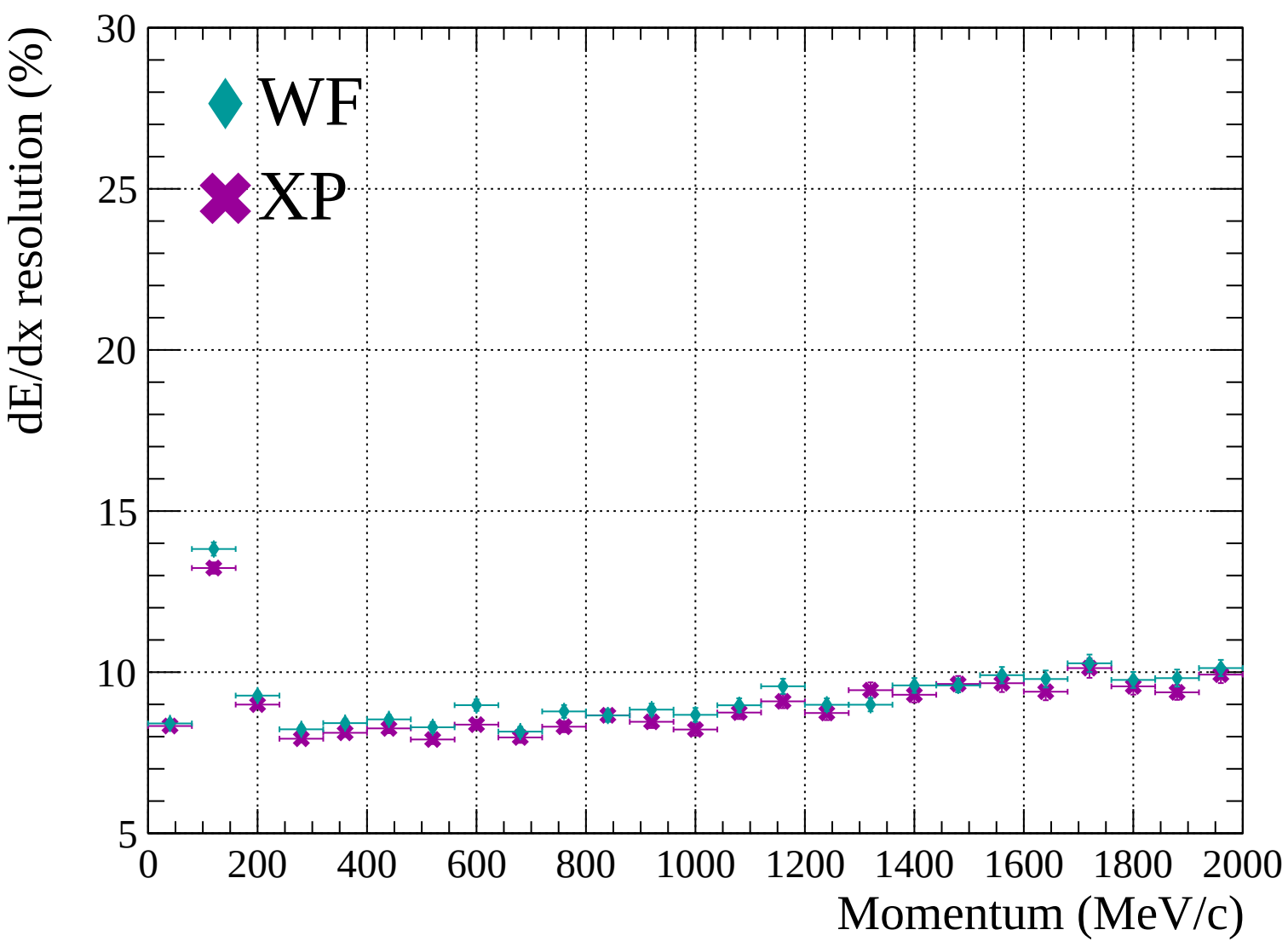


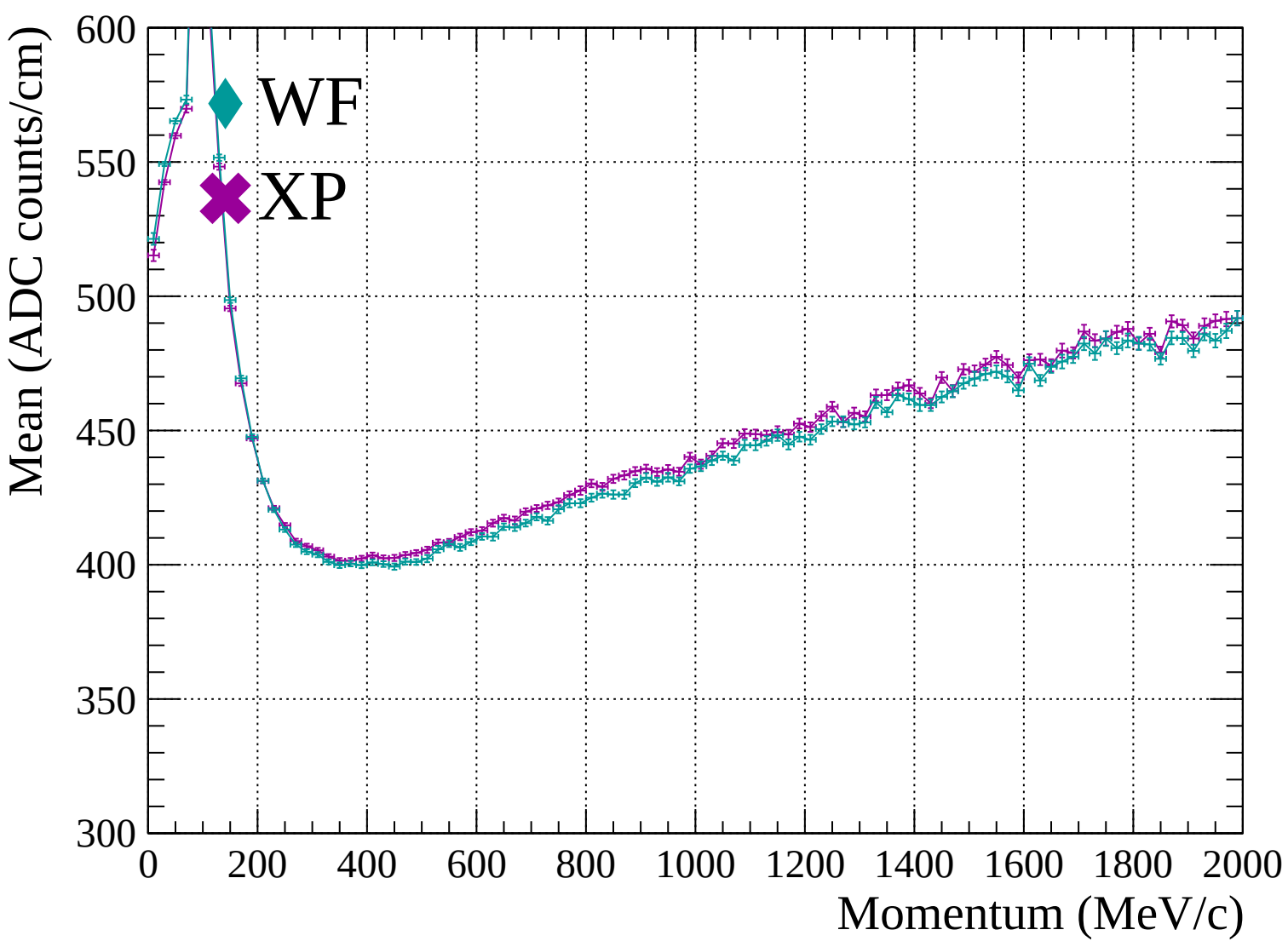


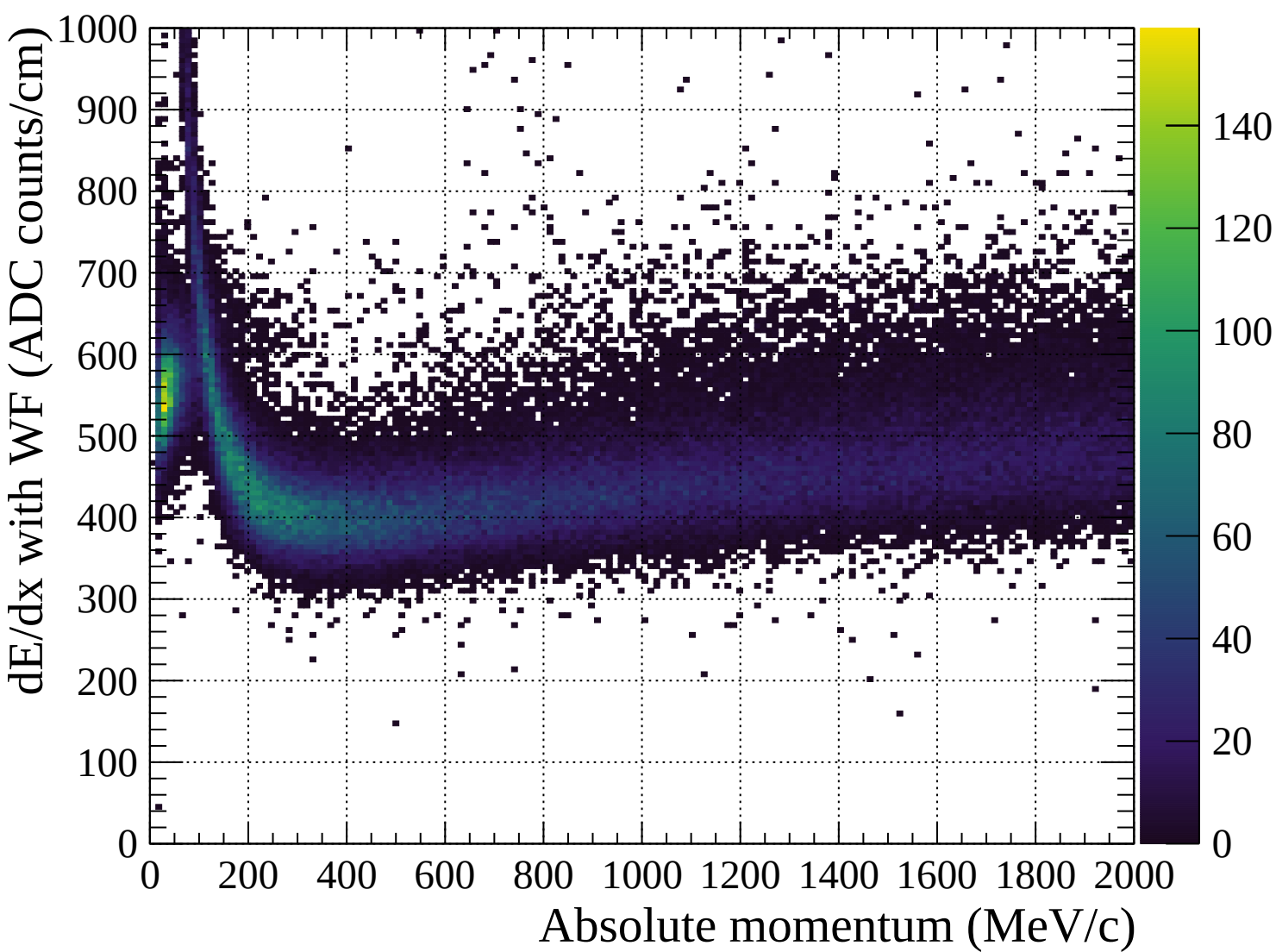


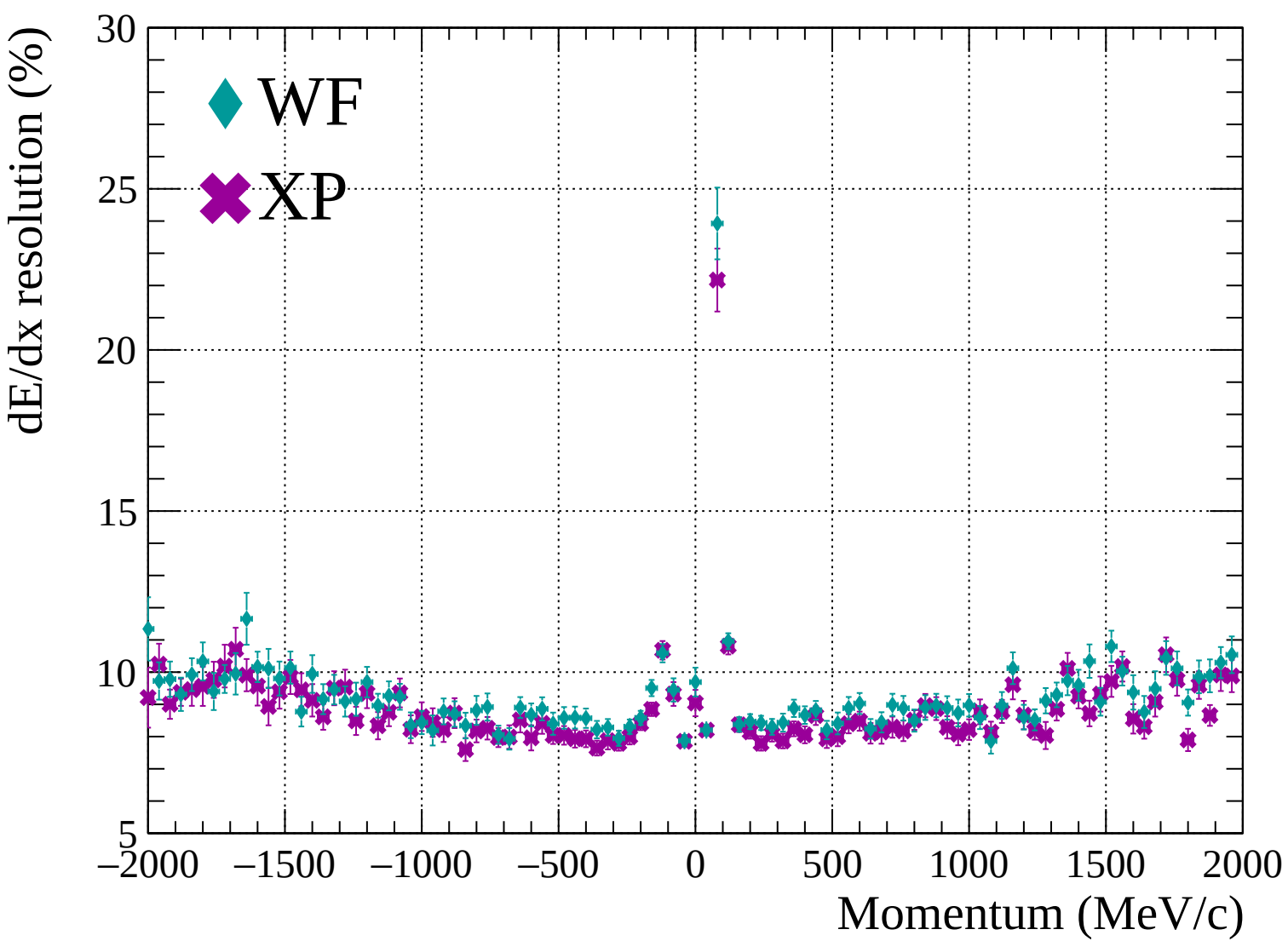


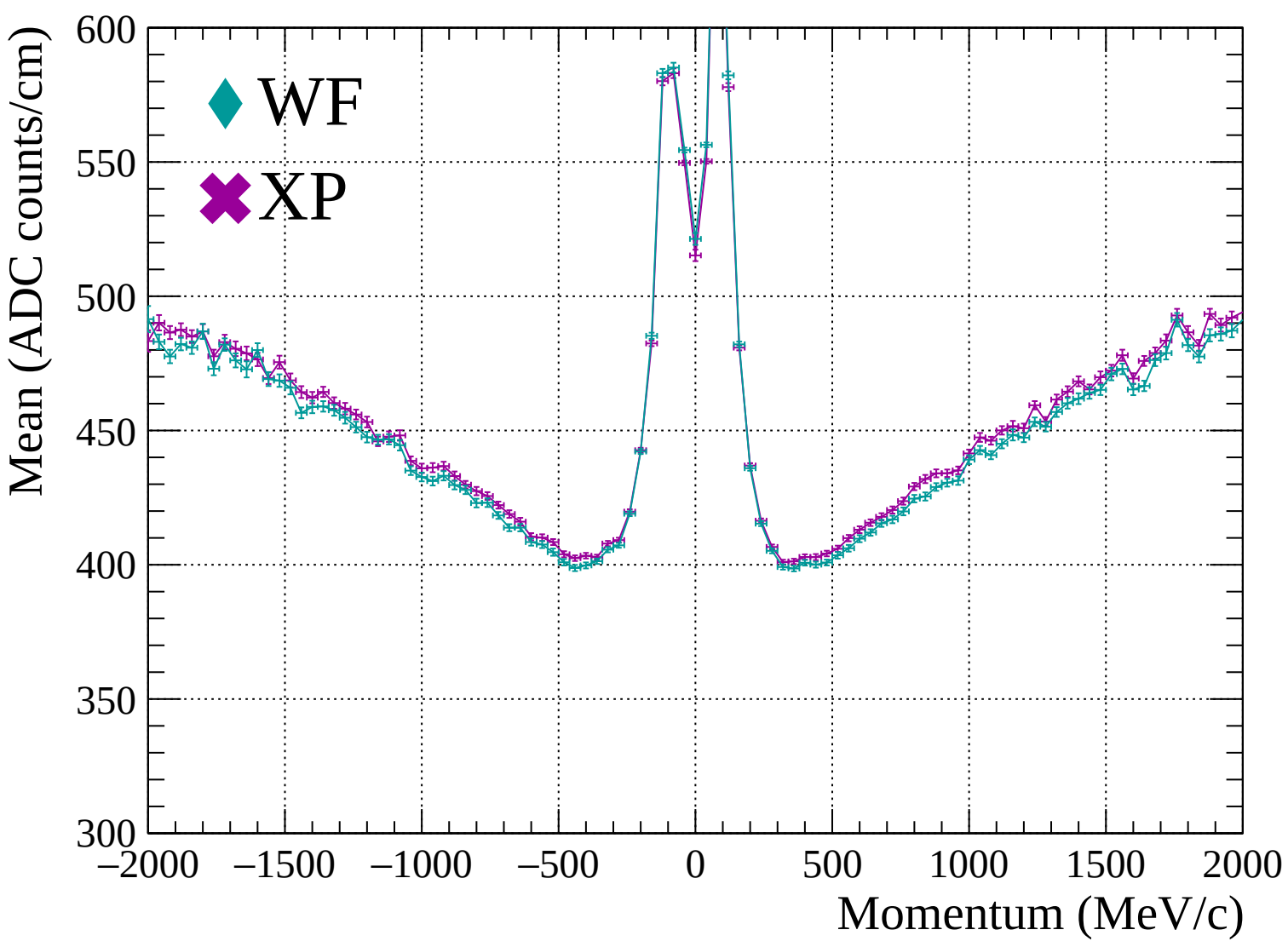


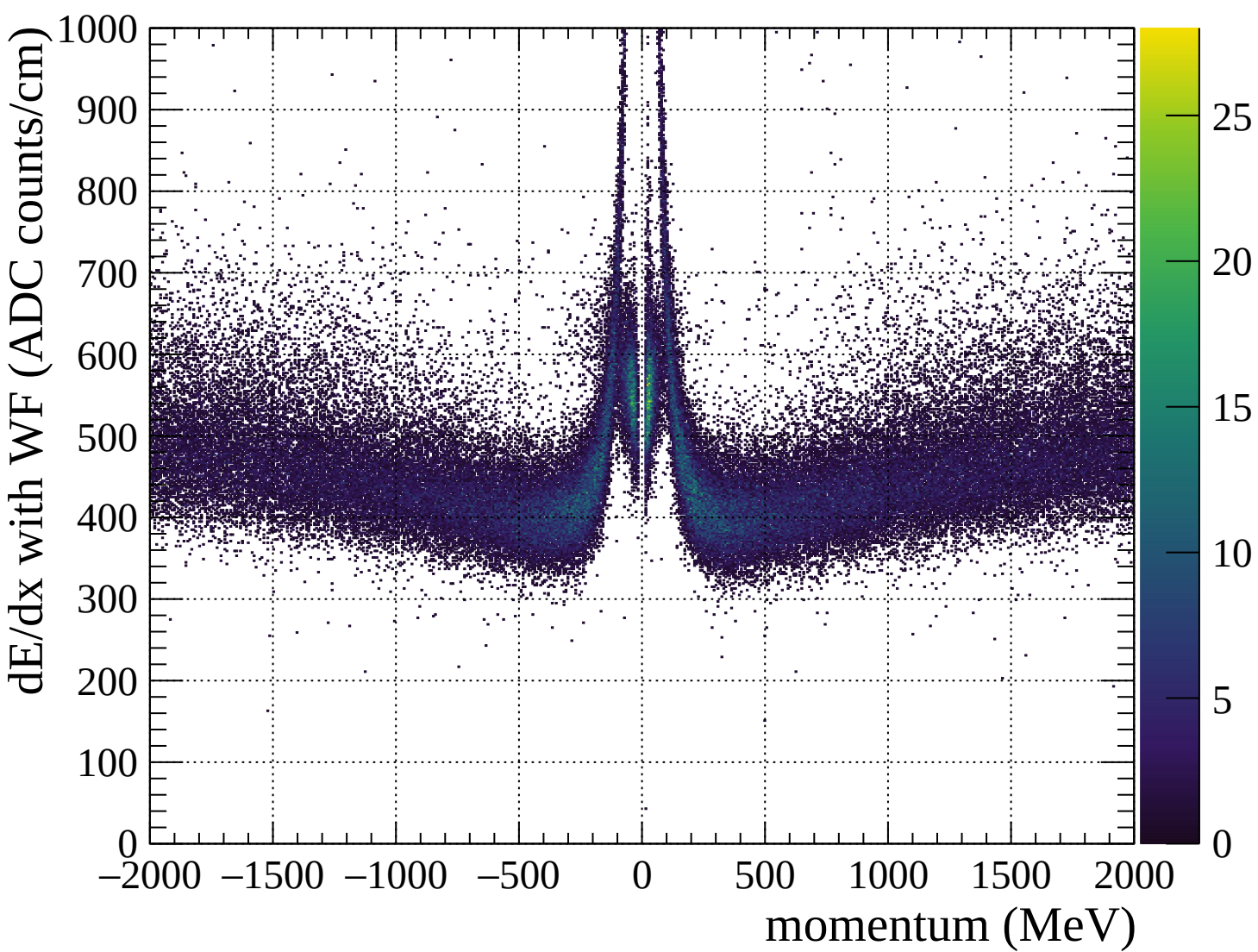


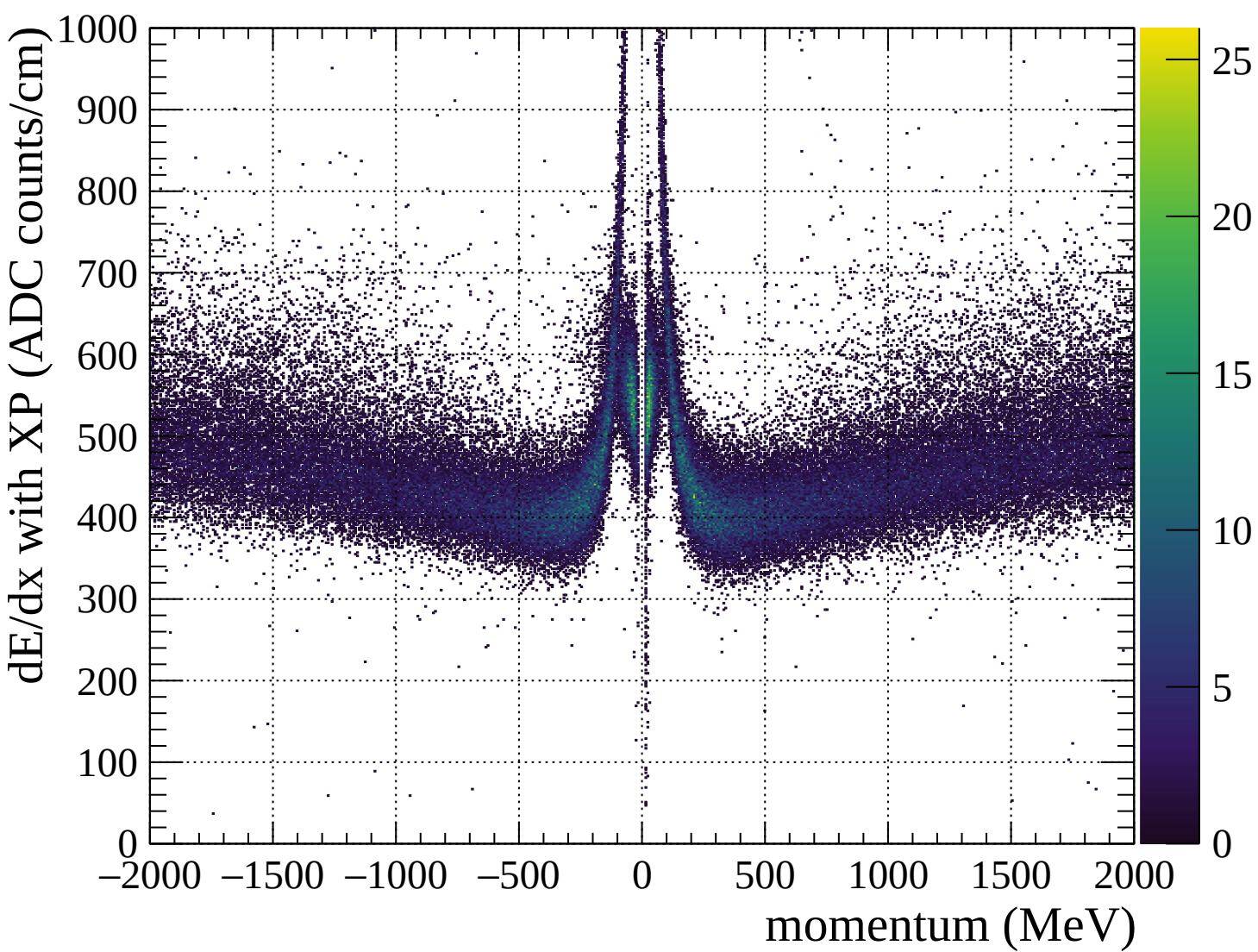


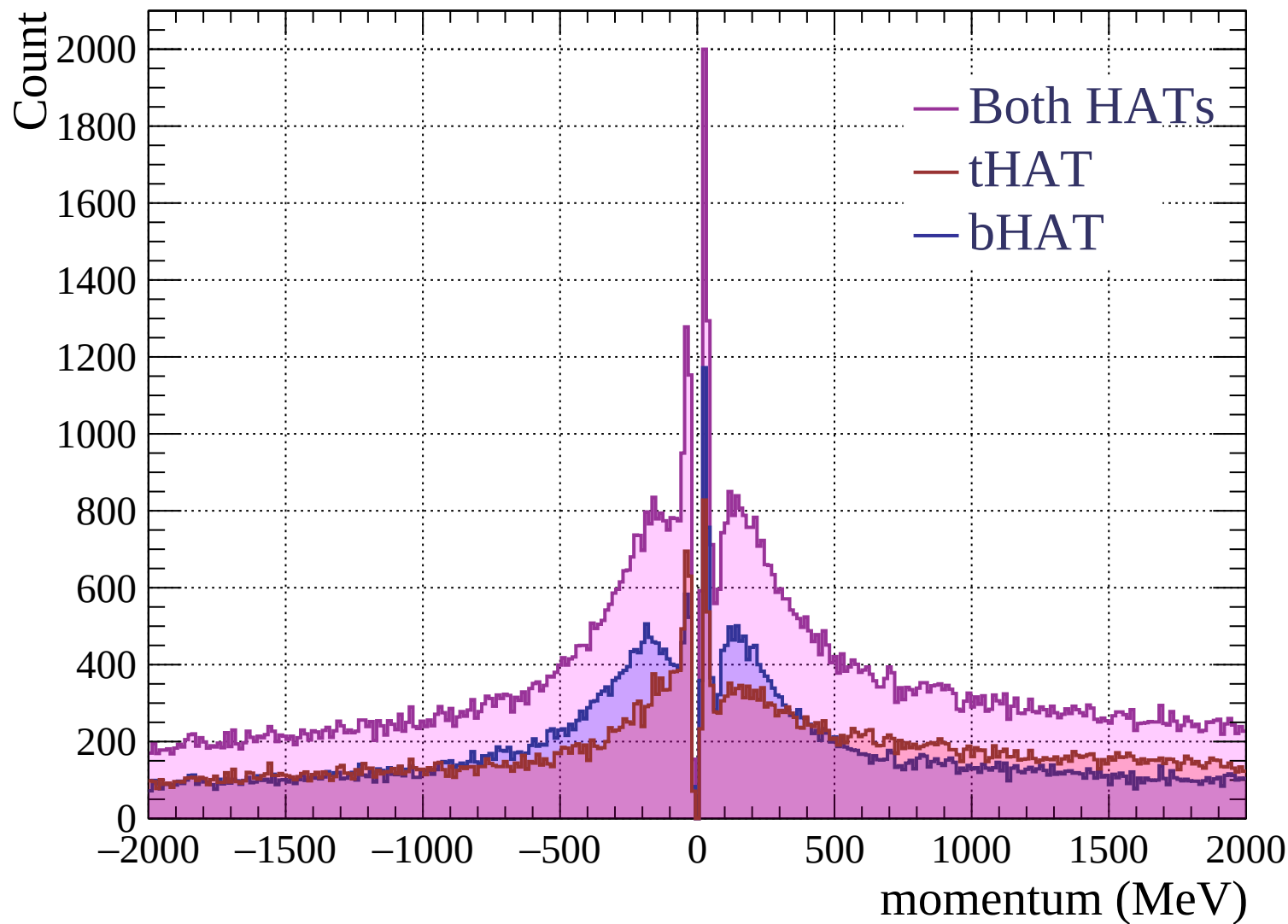


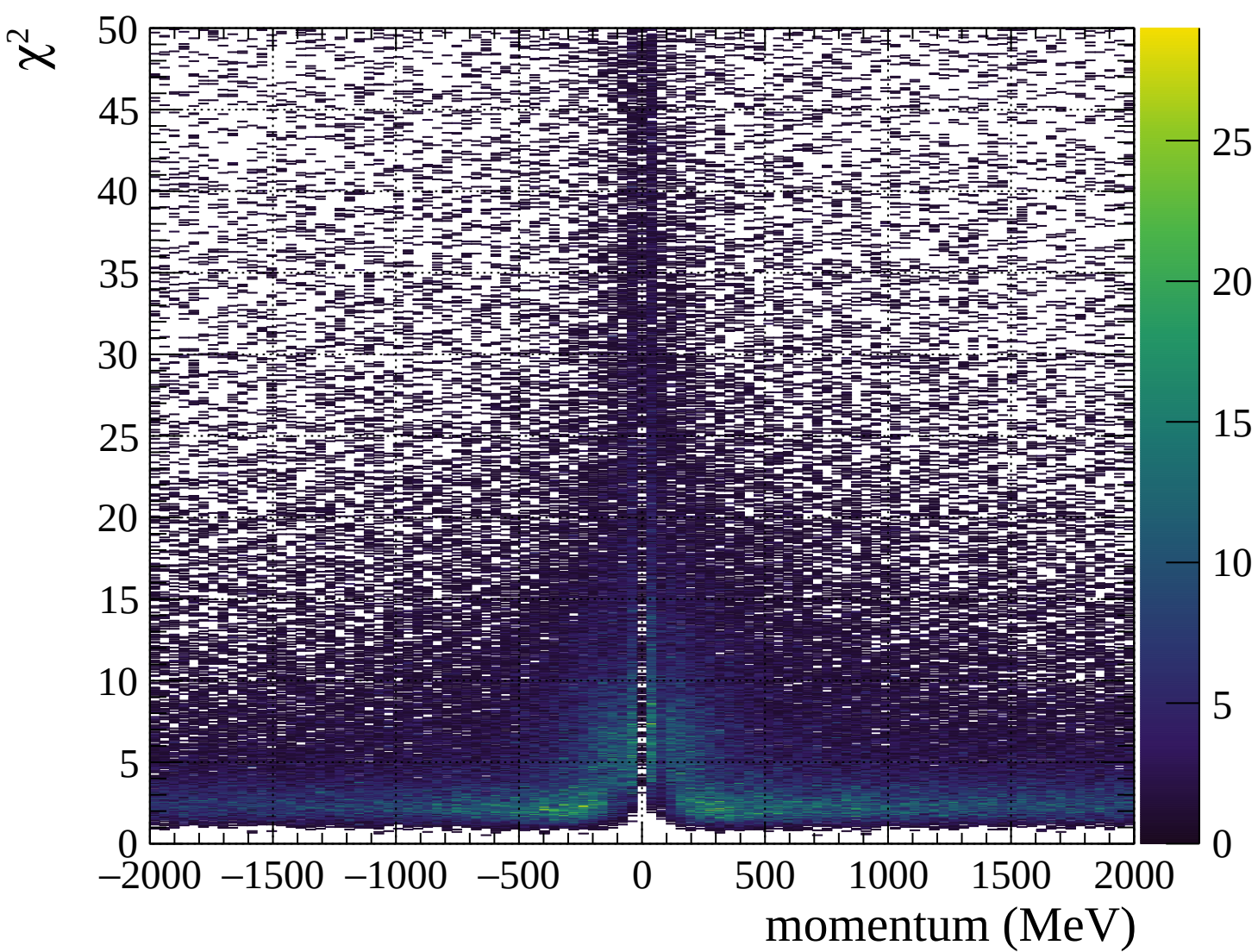




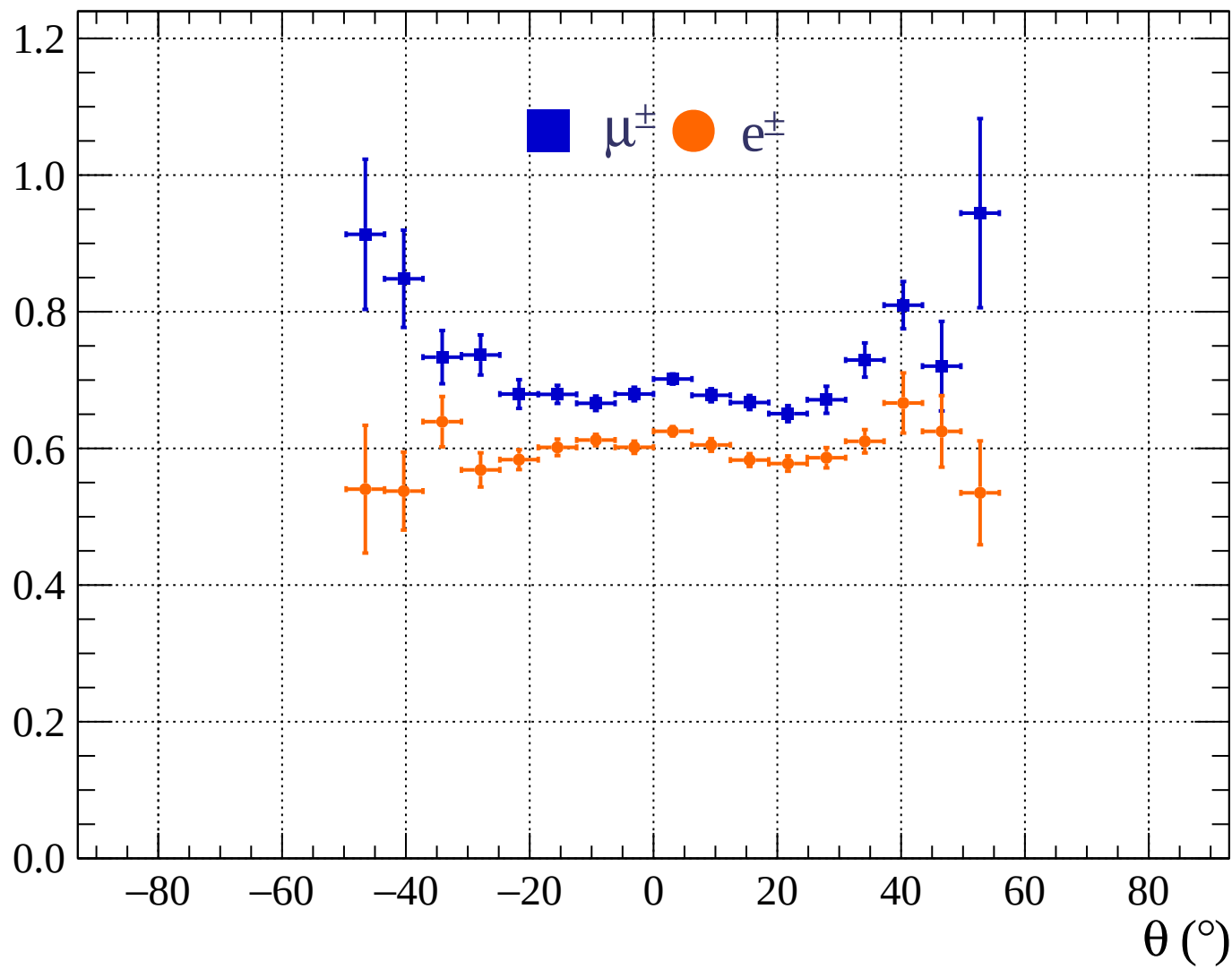


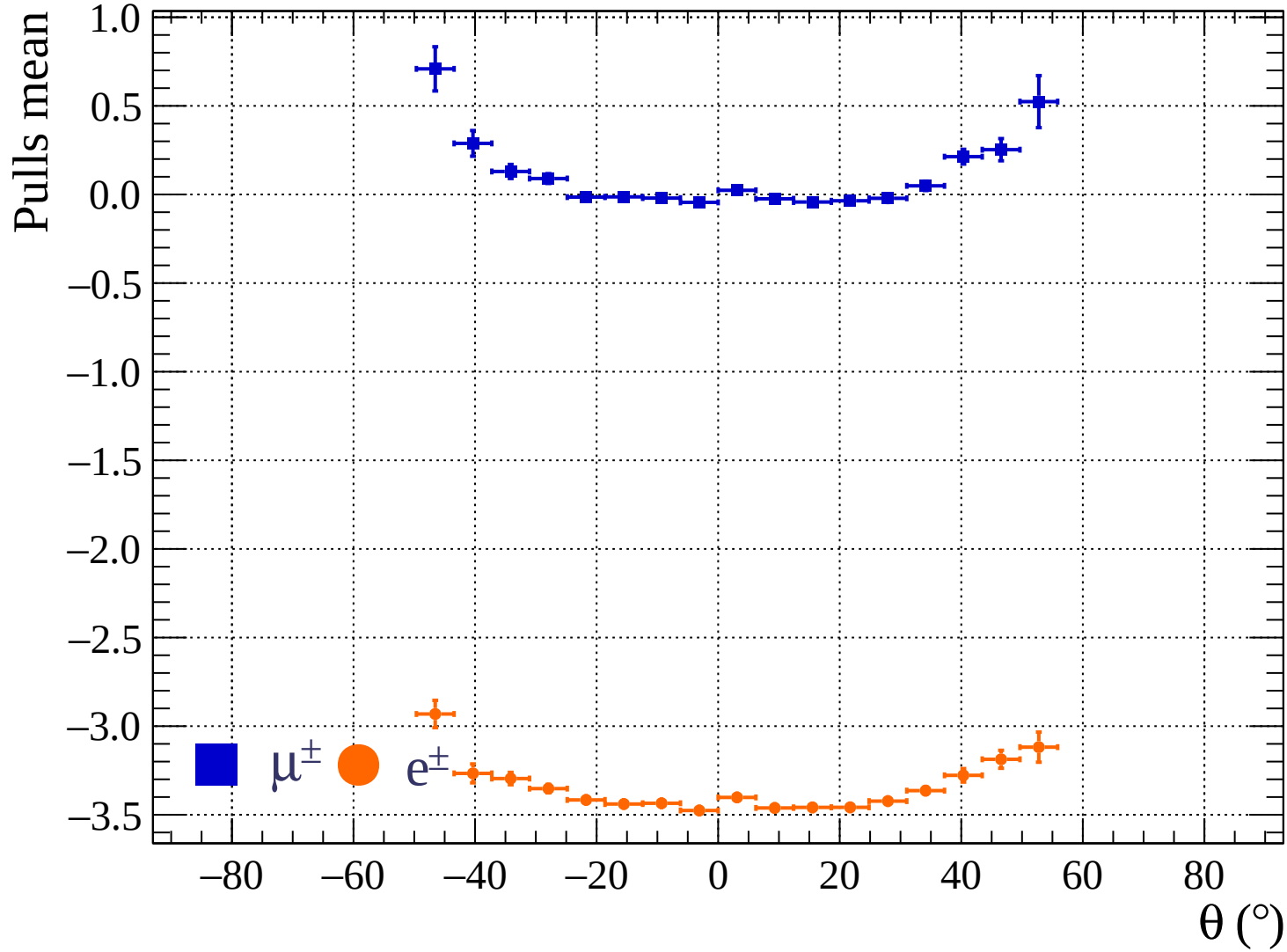




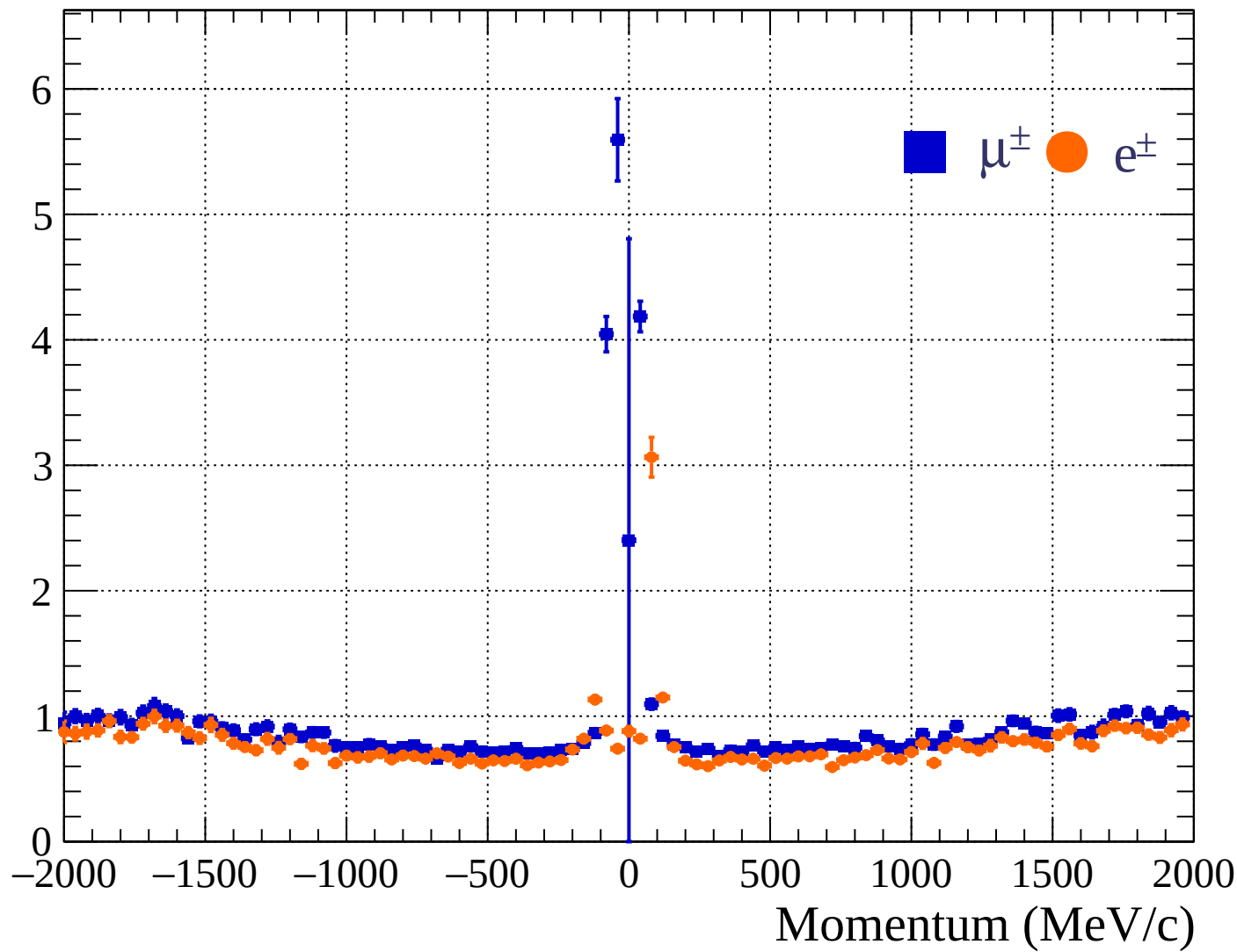


Pulls standard deviation

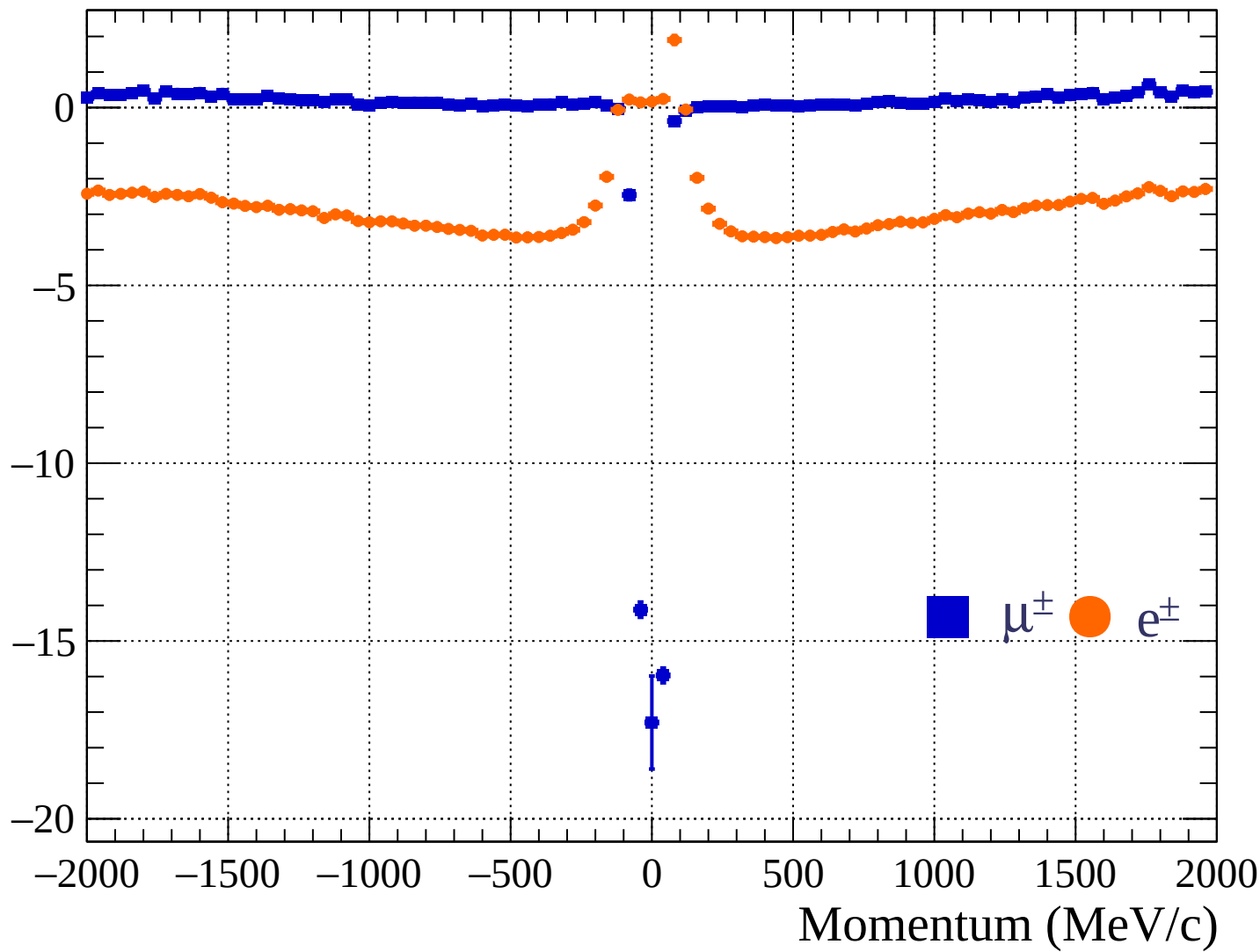


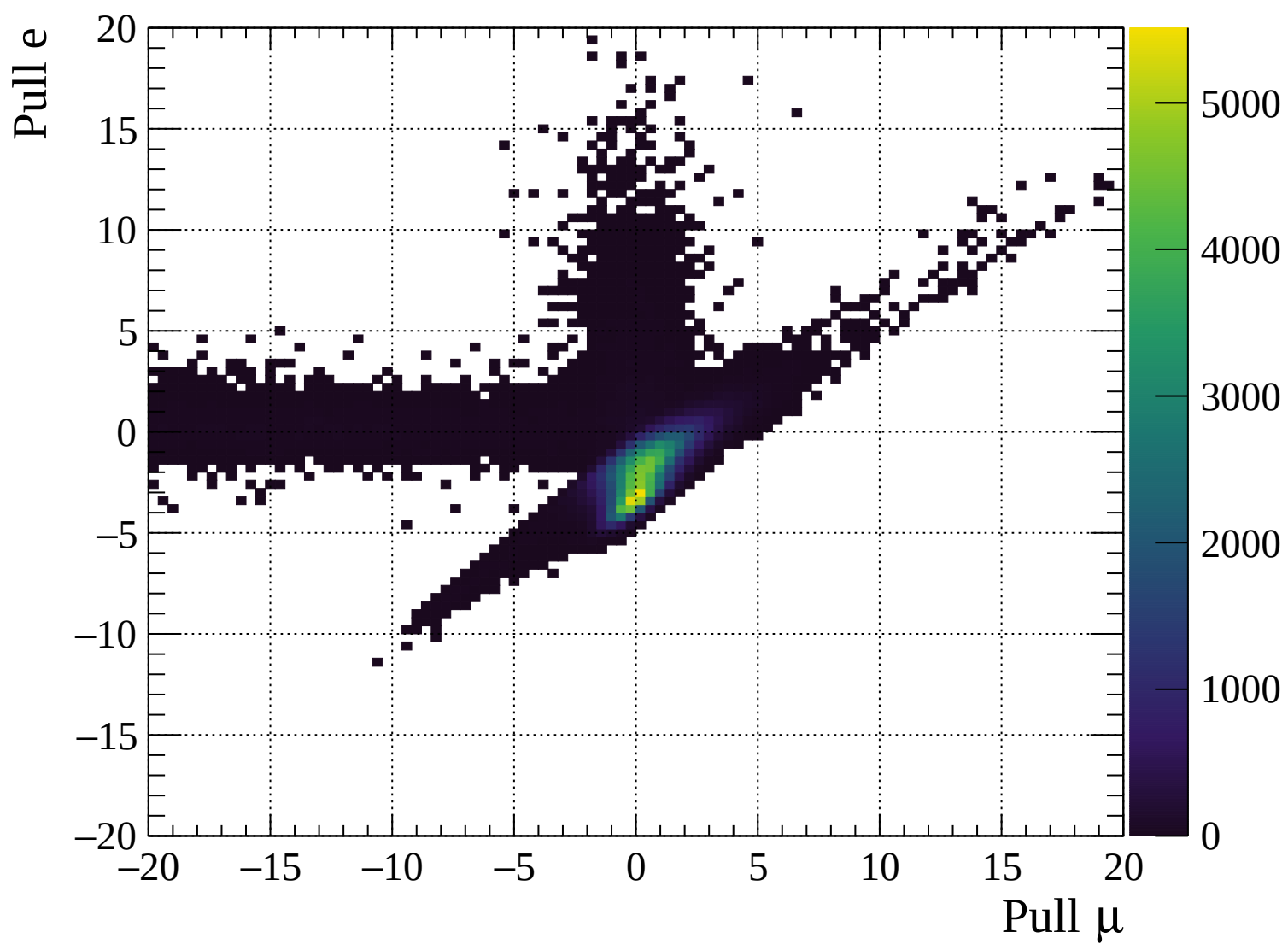


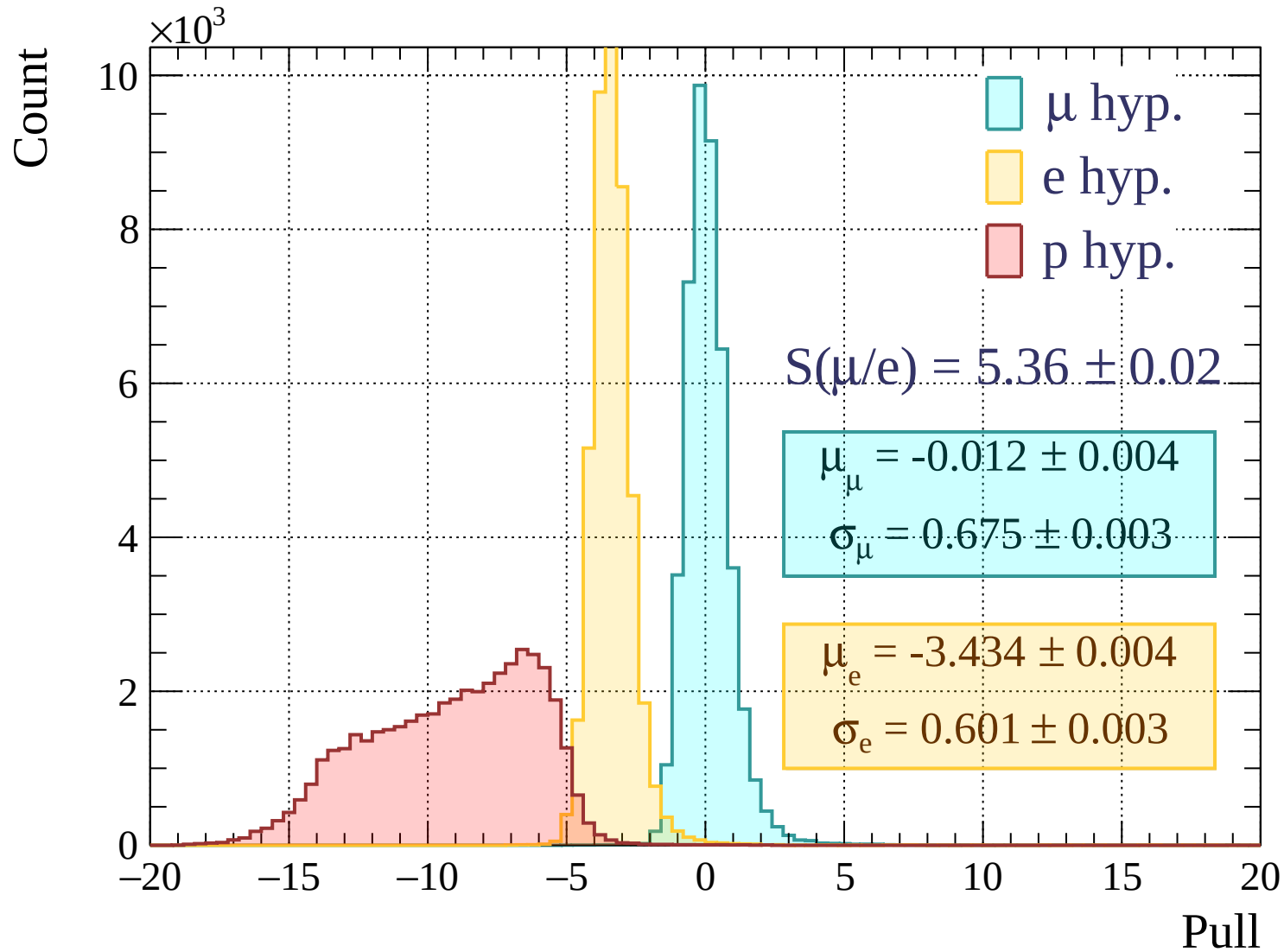
Pulls standard deviation

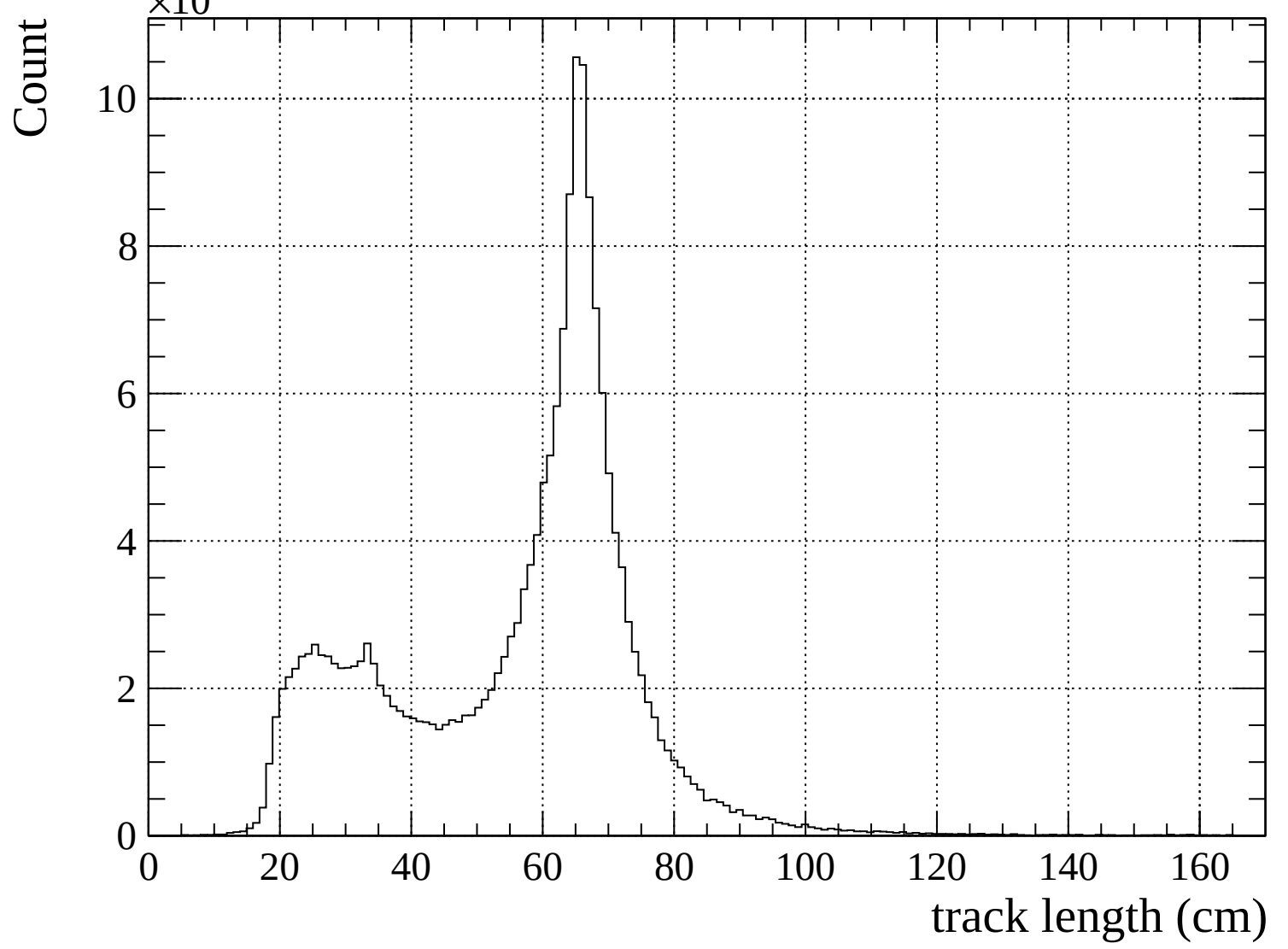


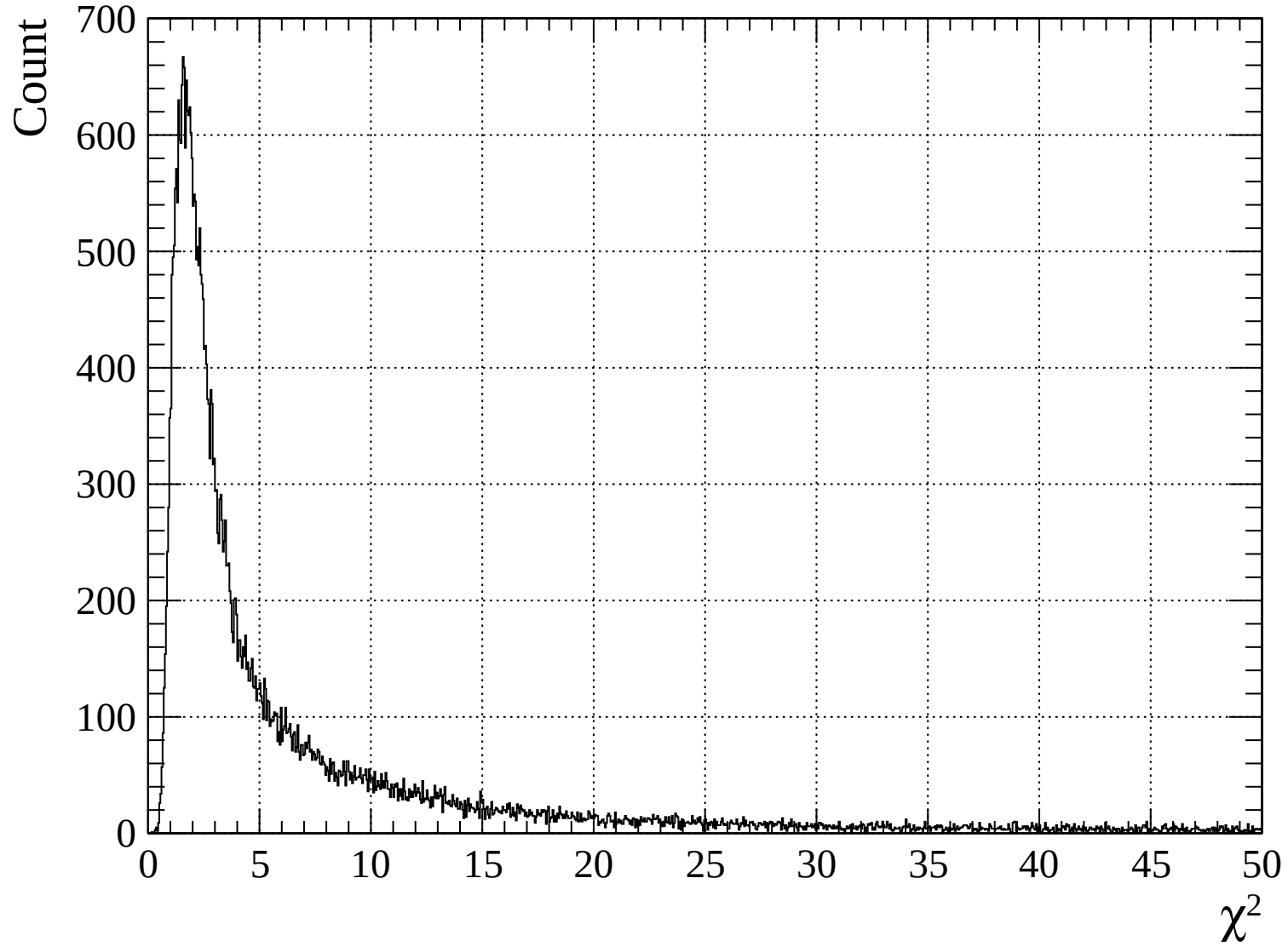
Pulls mean

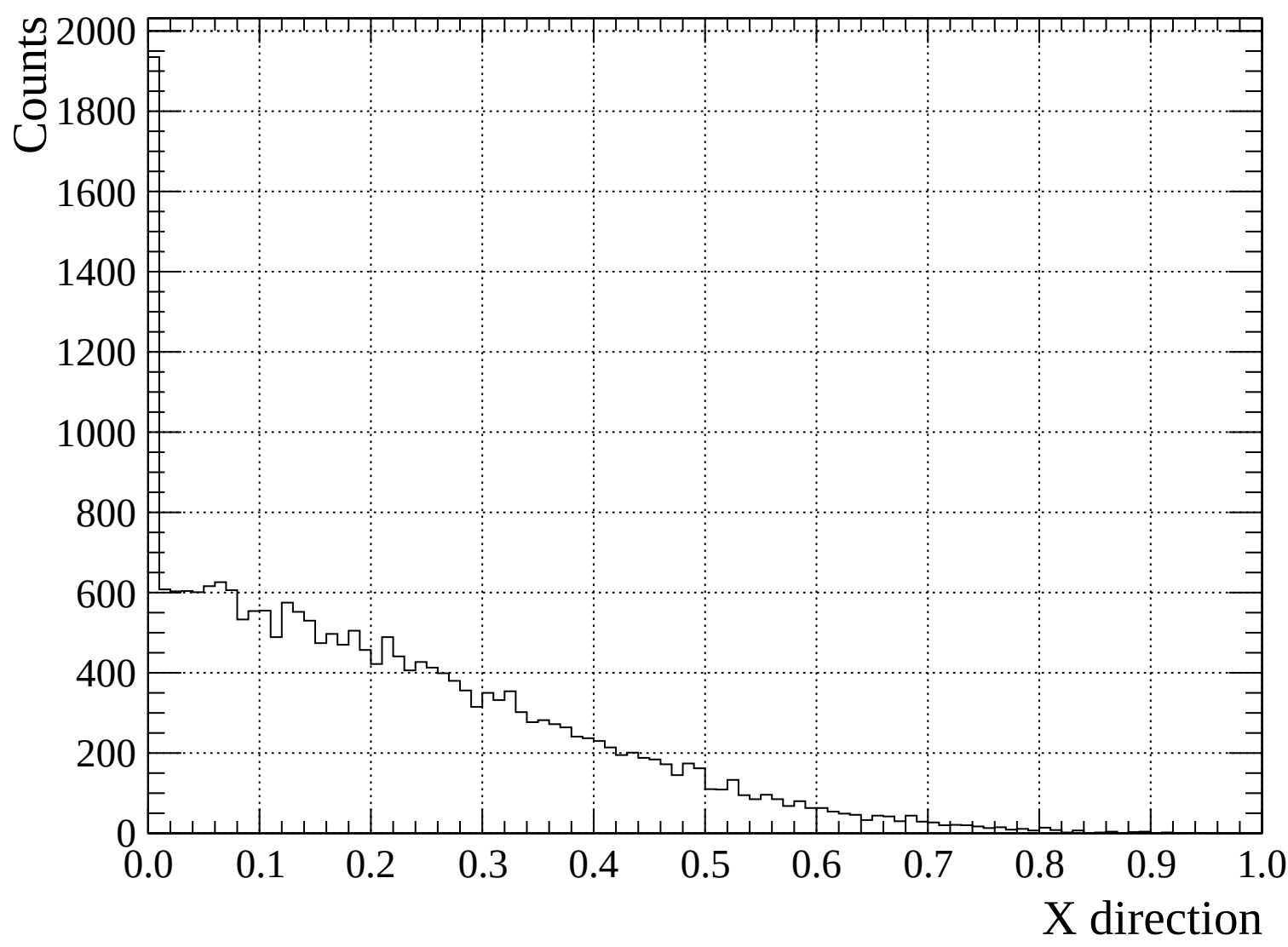


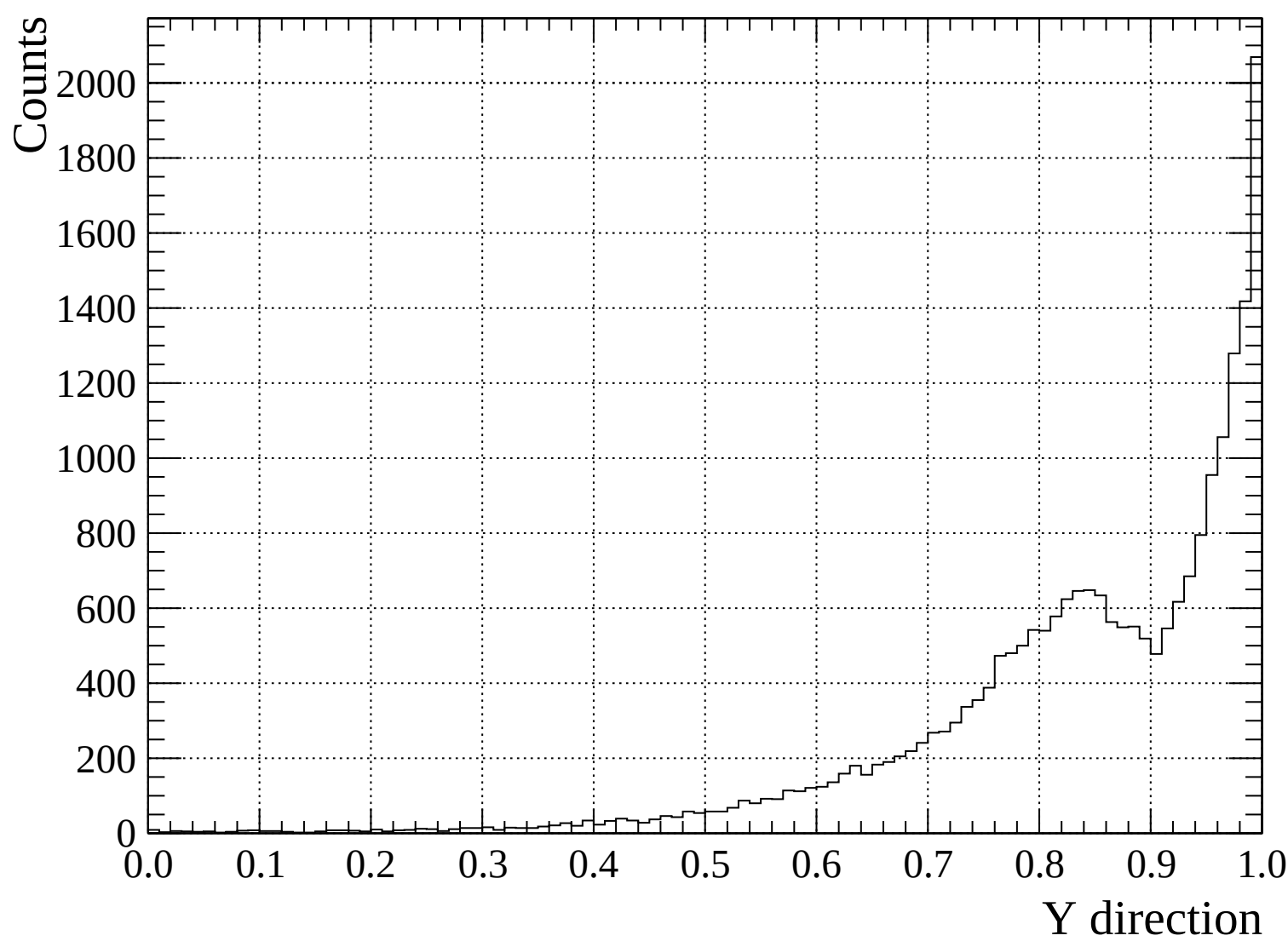


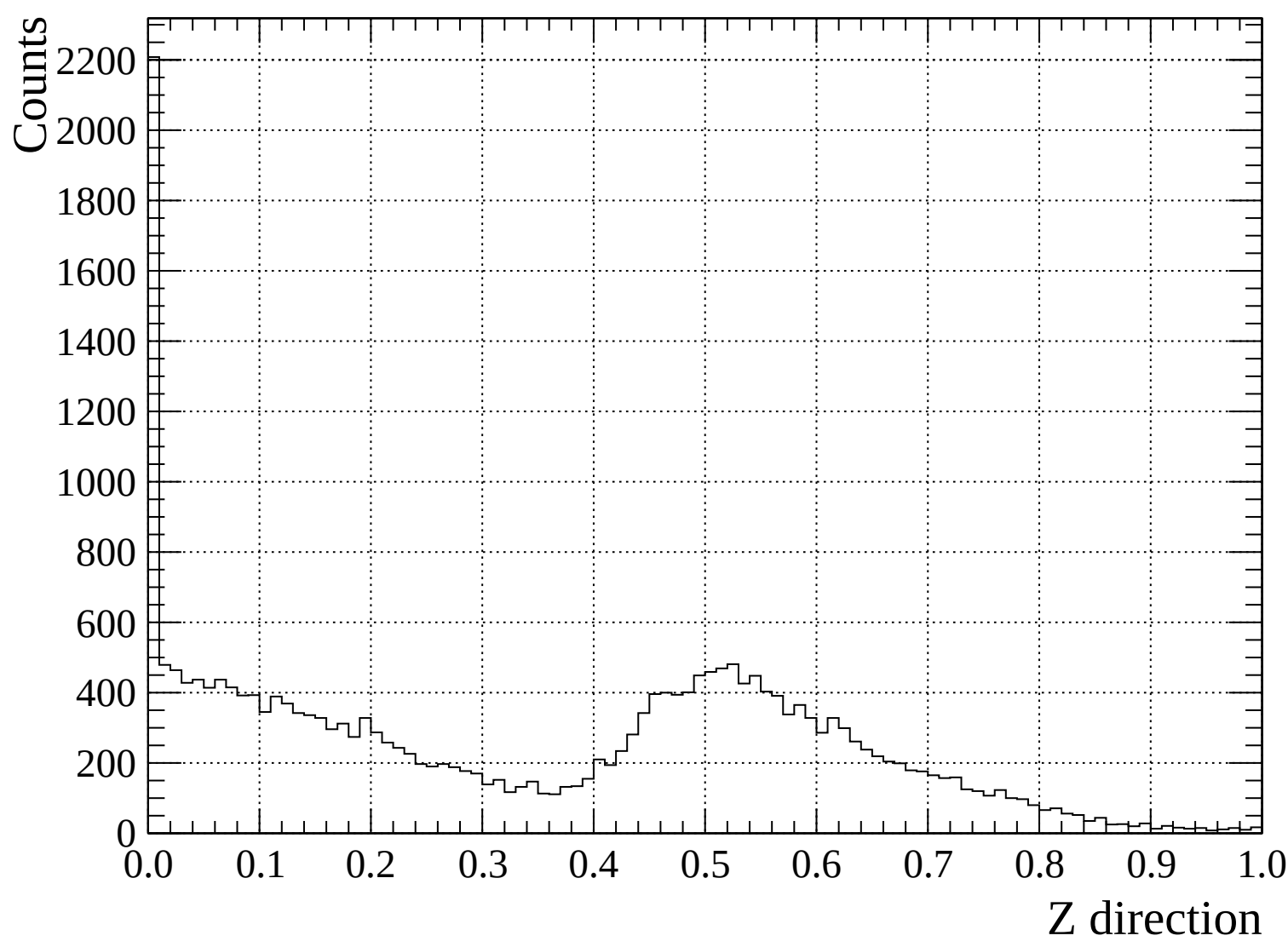


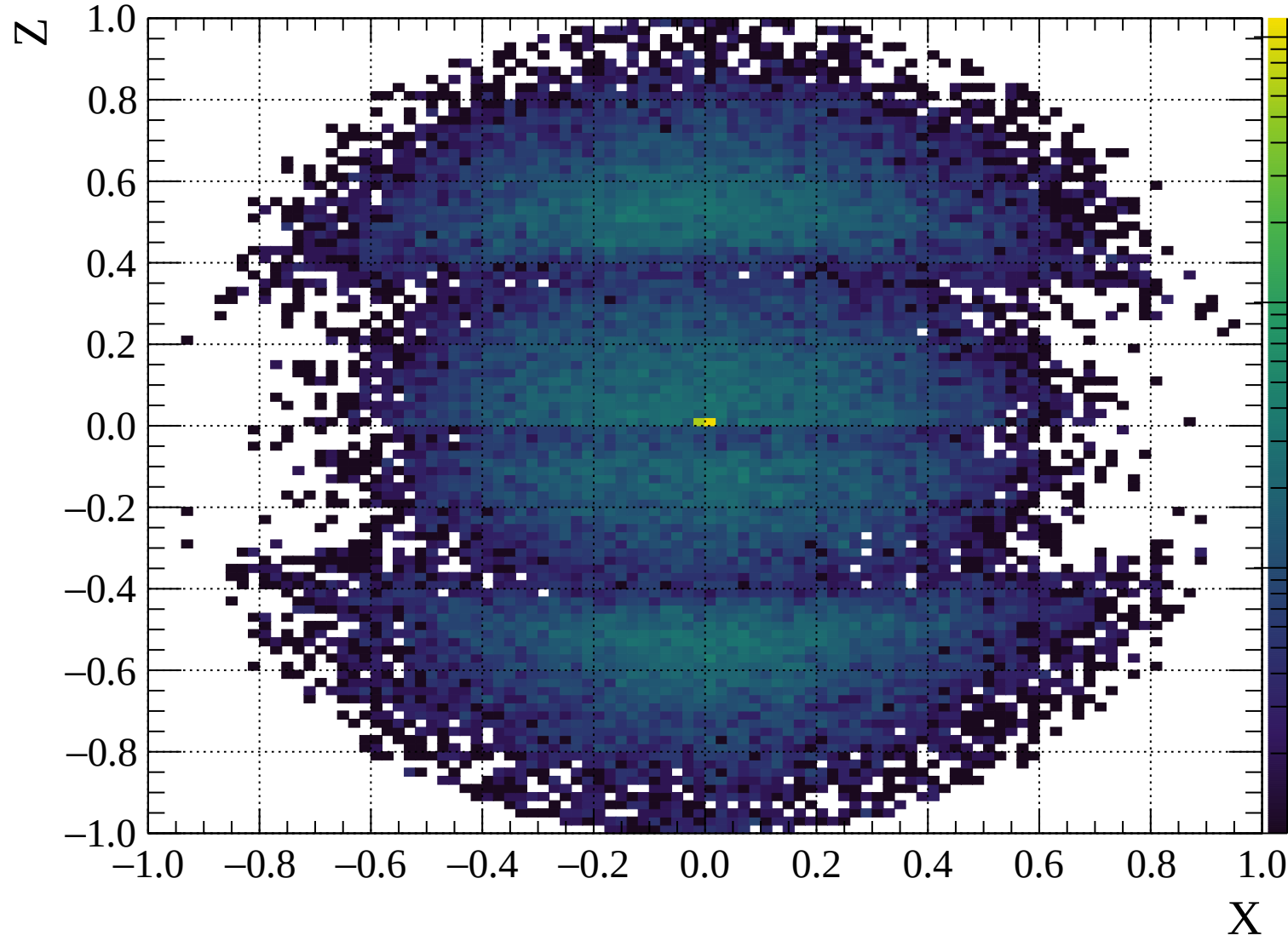


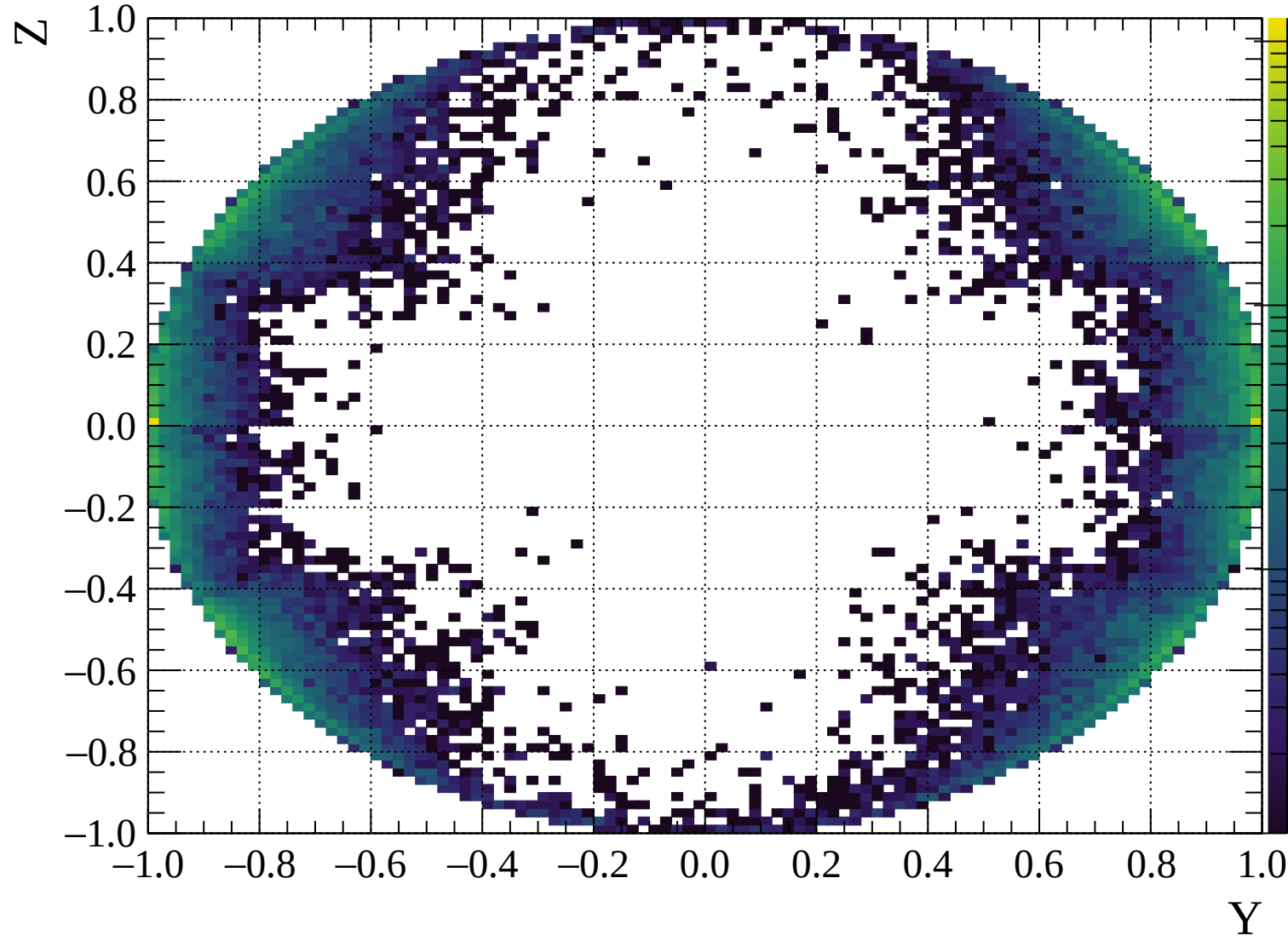


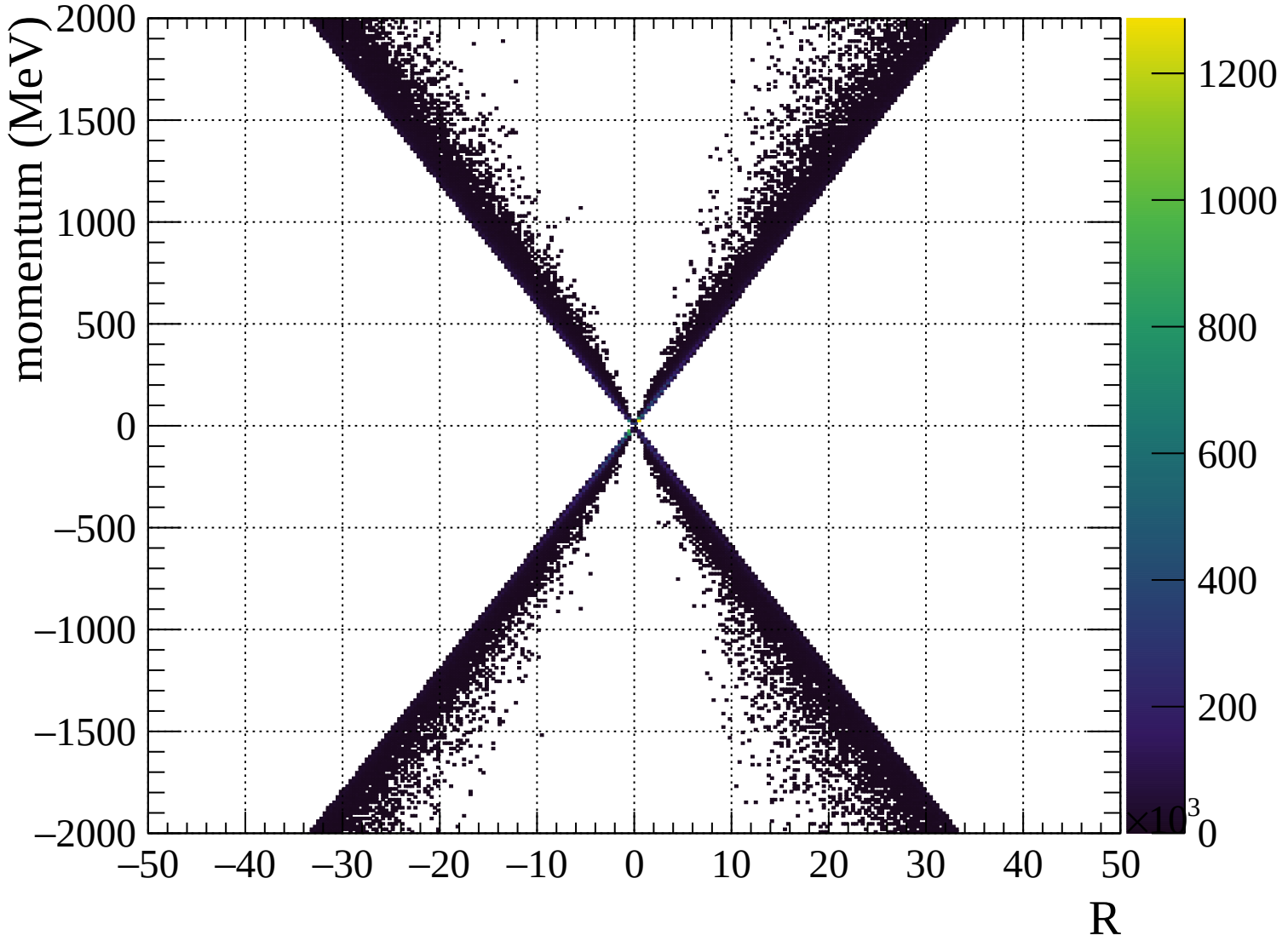


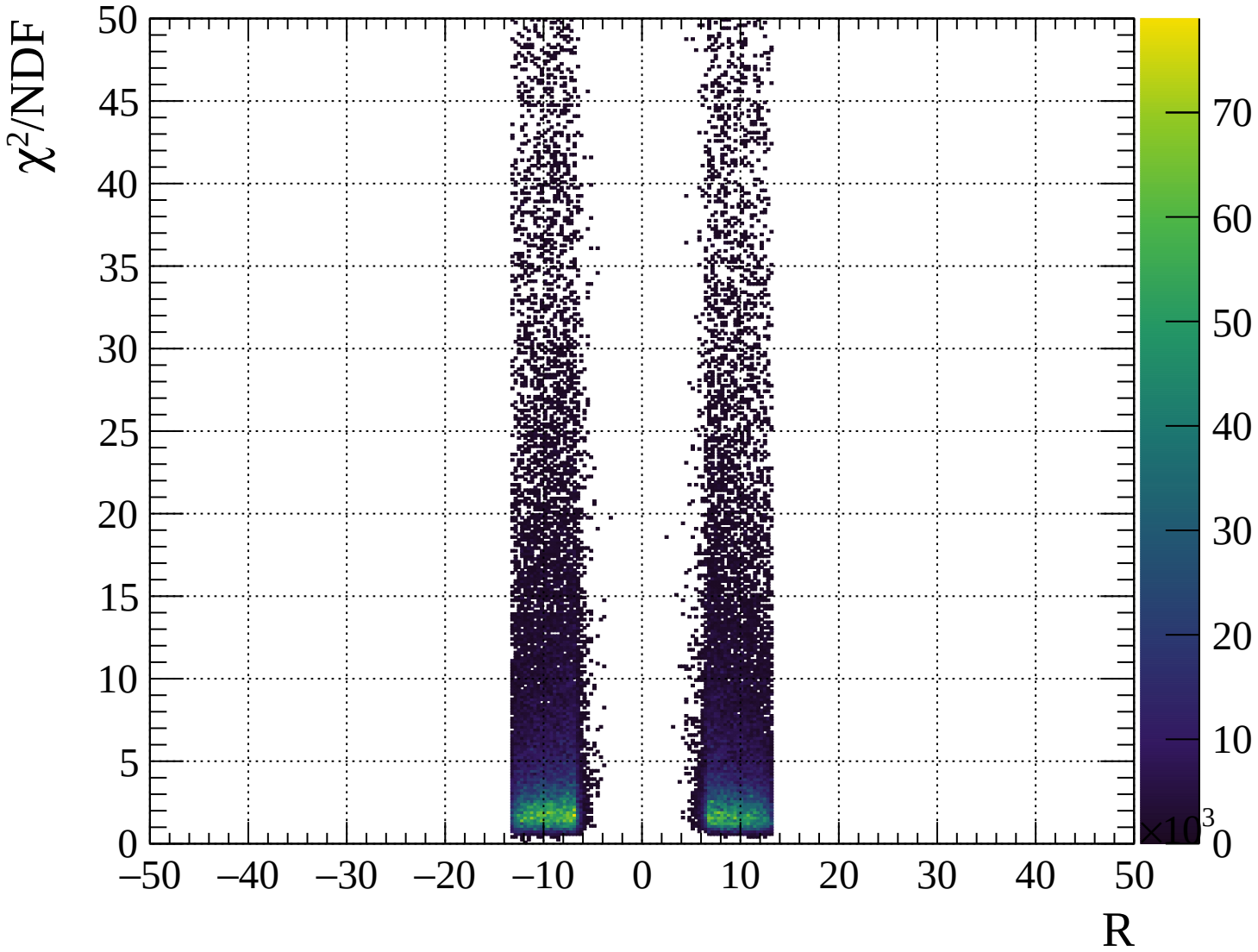




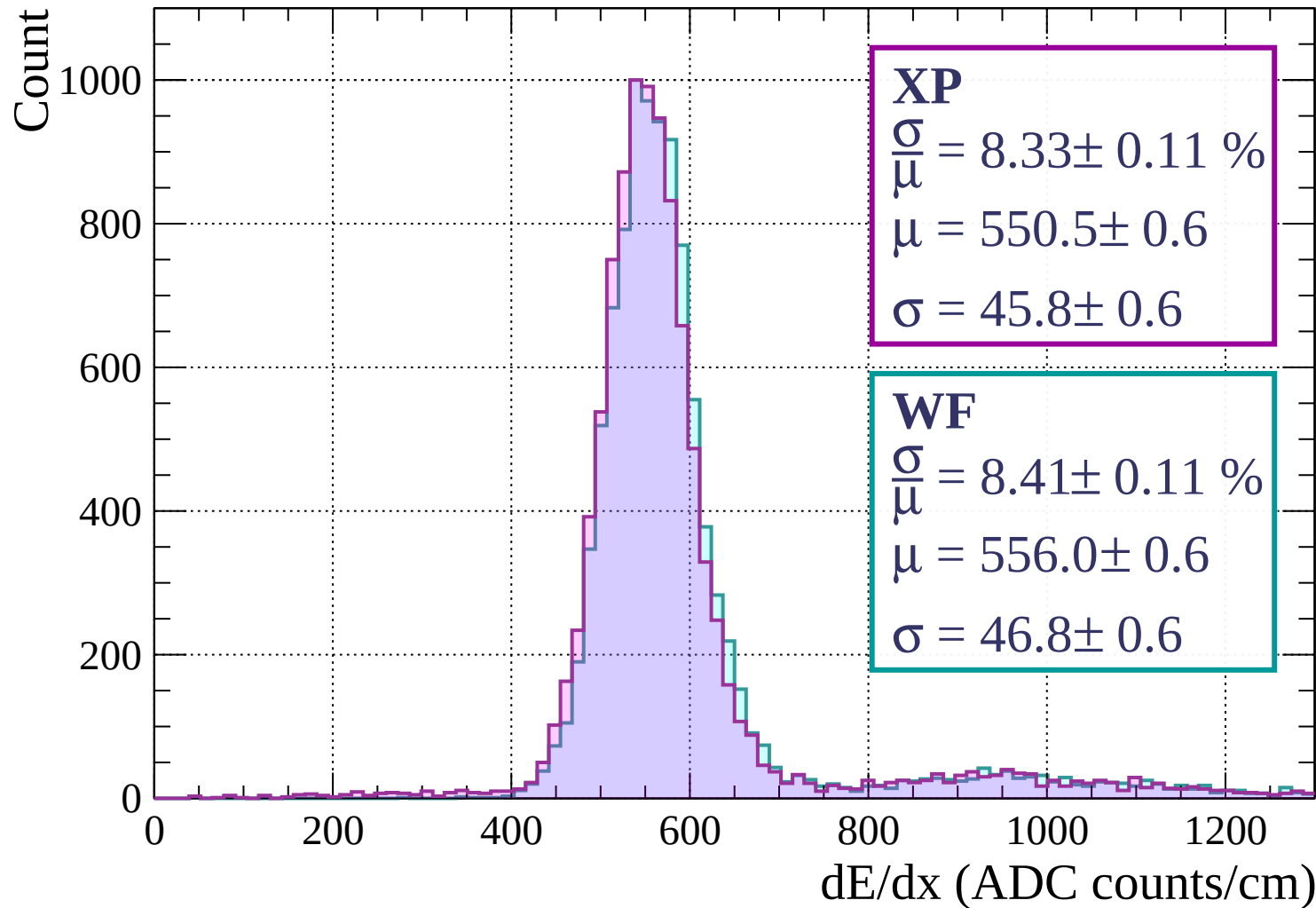




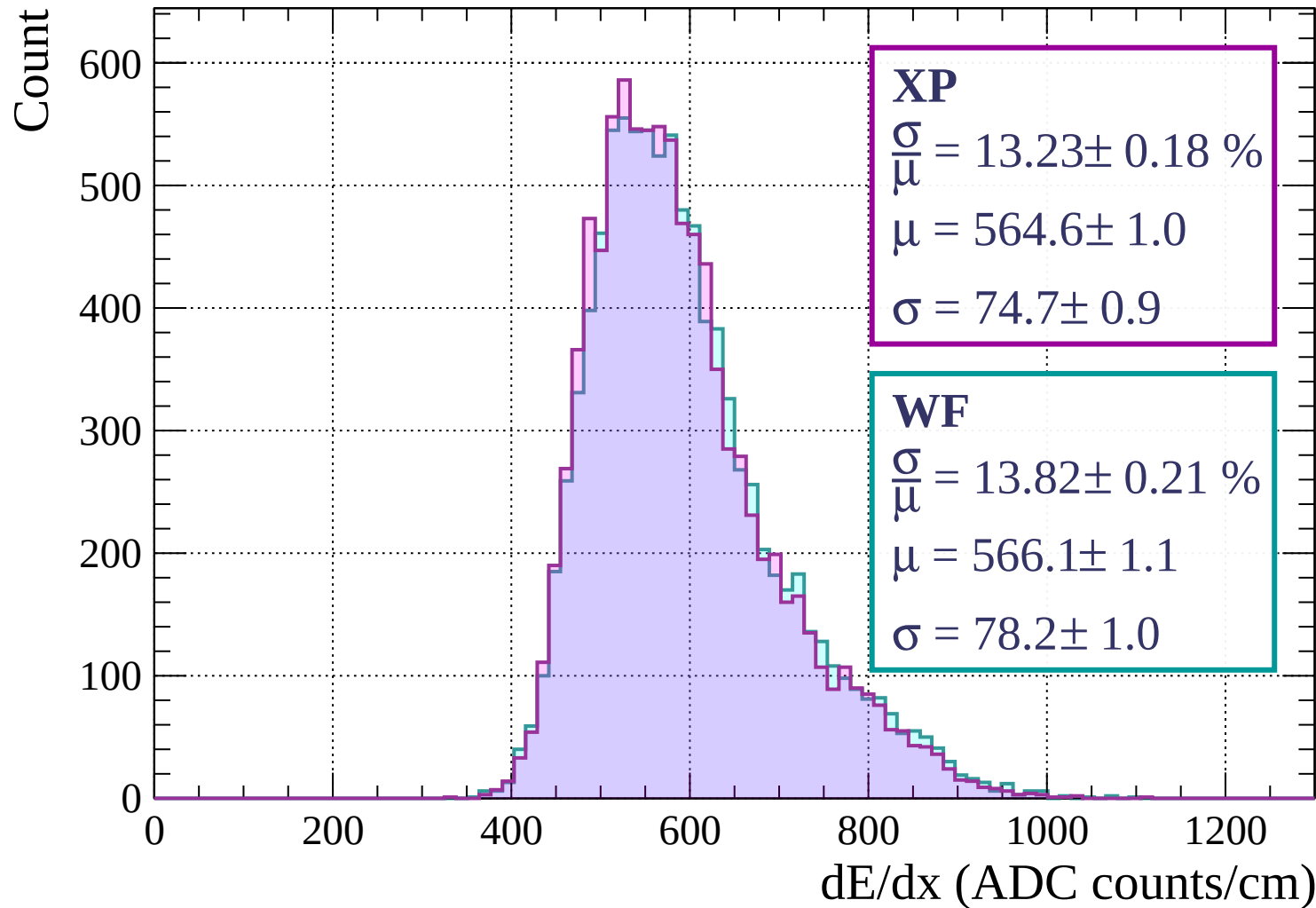




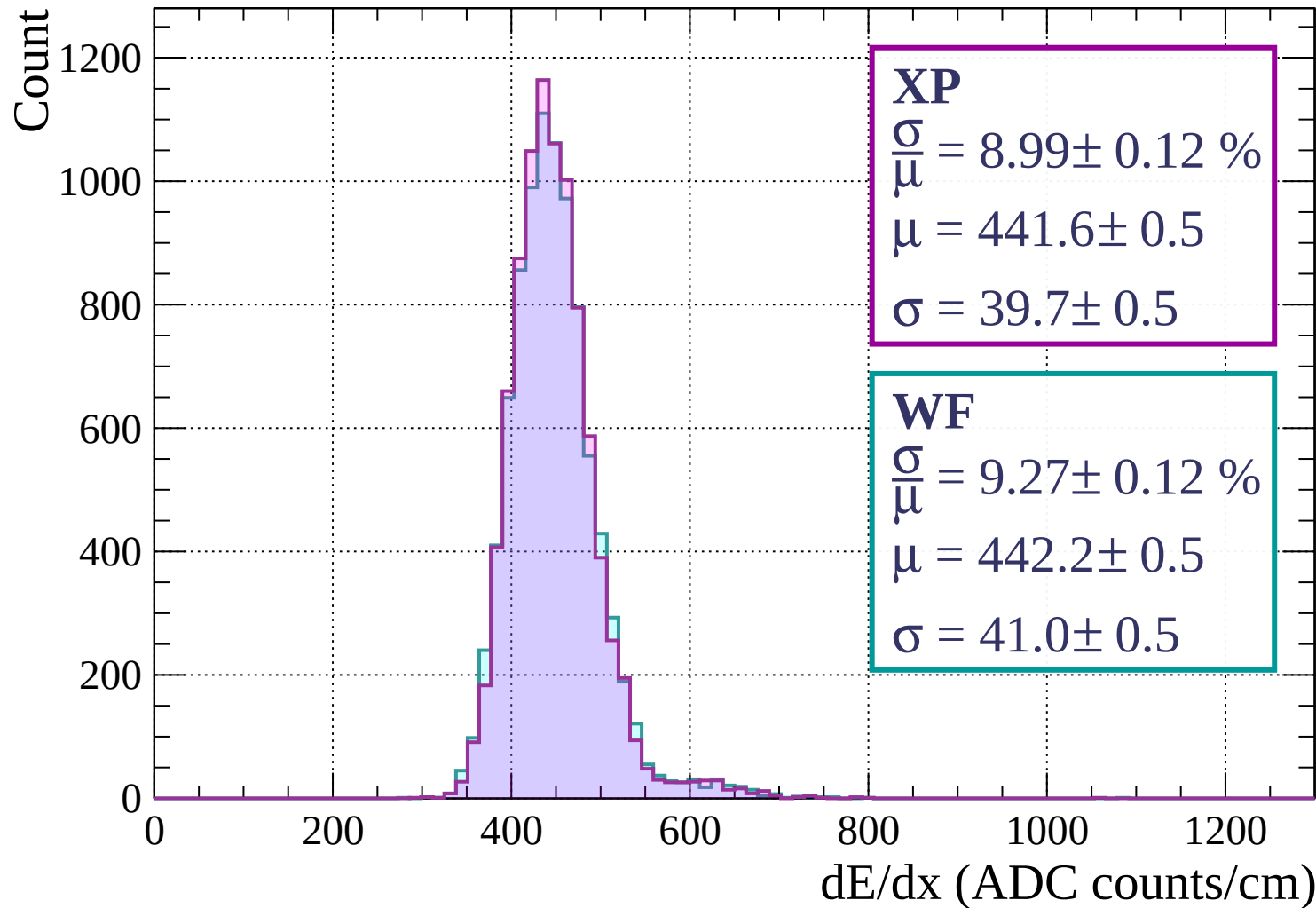
Energy loss | $0 < p < 80$



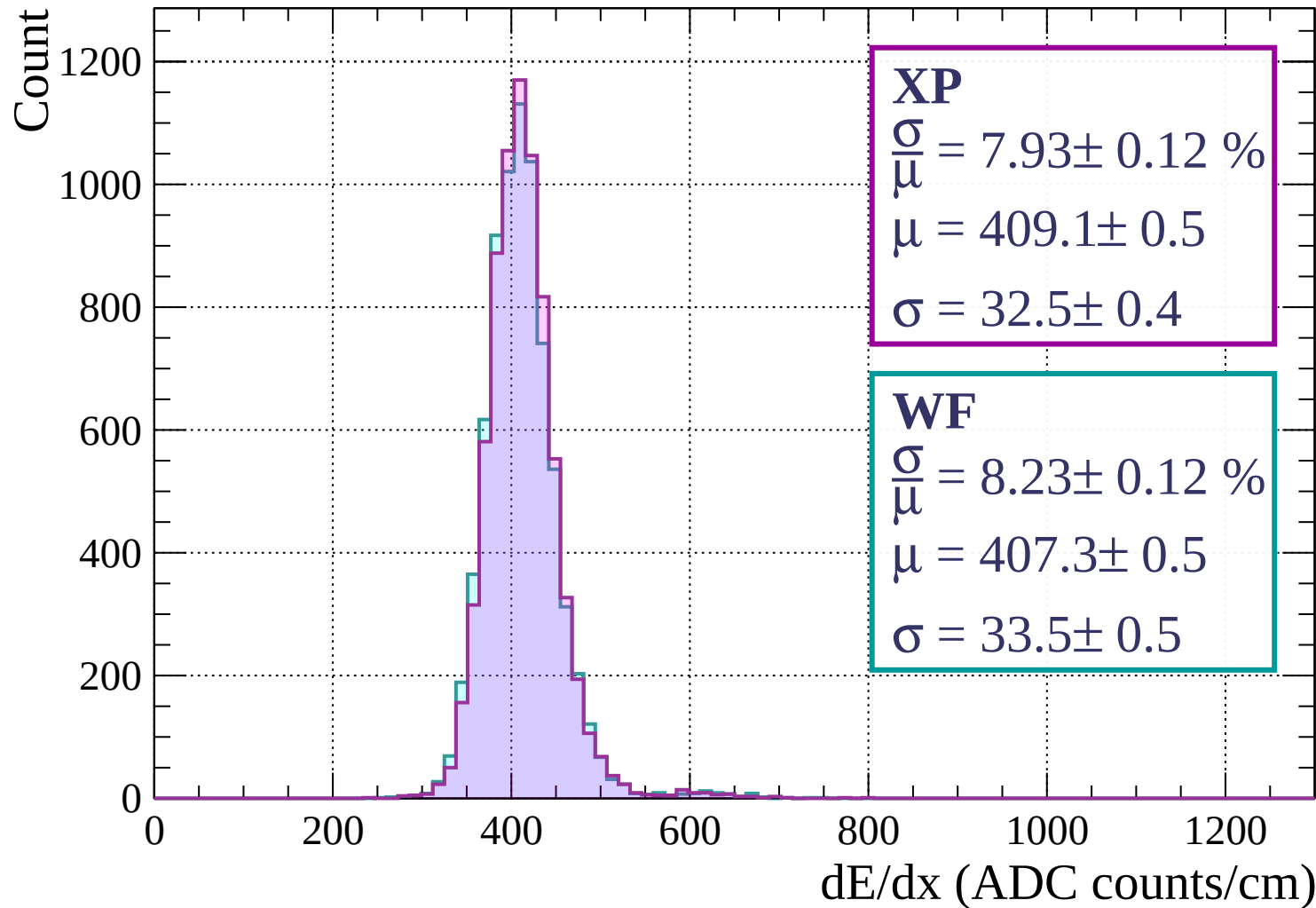
Energy loss | $80 < p < 160$



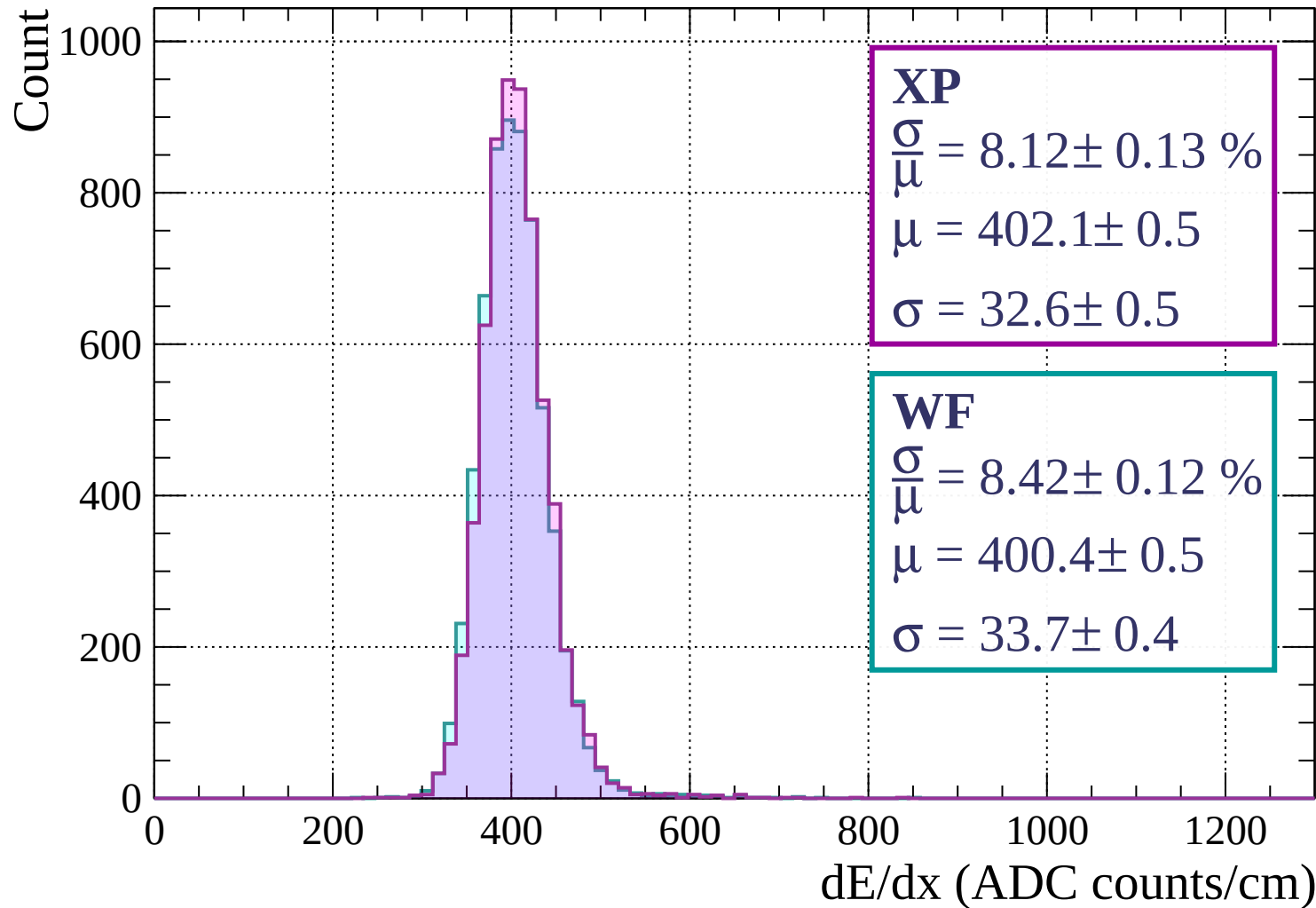
Energy loss | $160 < p < 240$



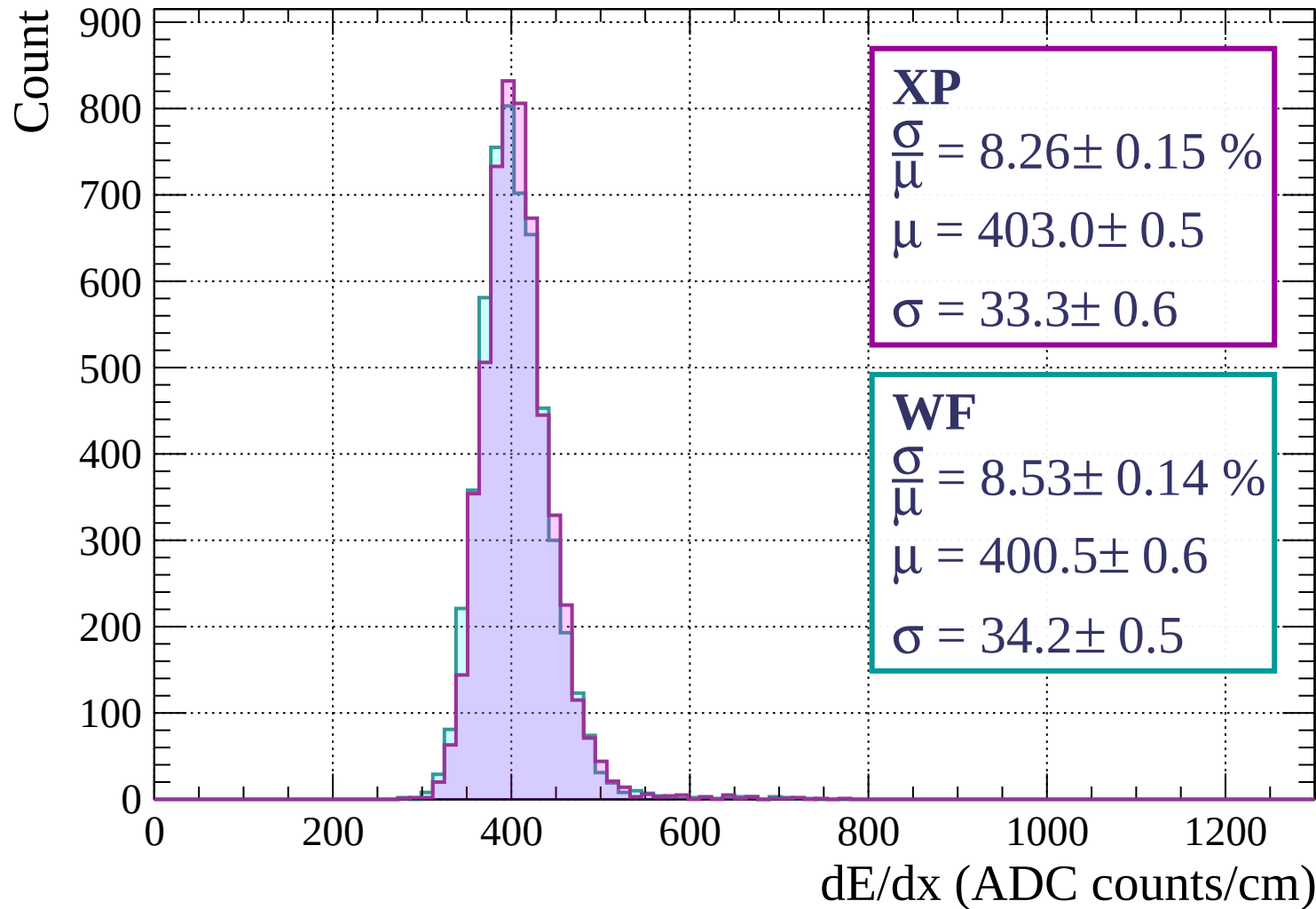
Energy loss | $240 < p < 320$



Energy loss | $320 < p < 400$

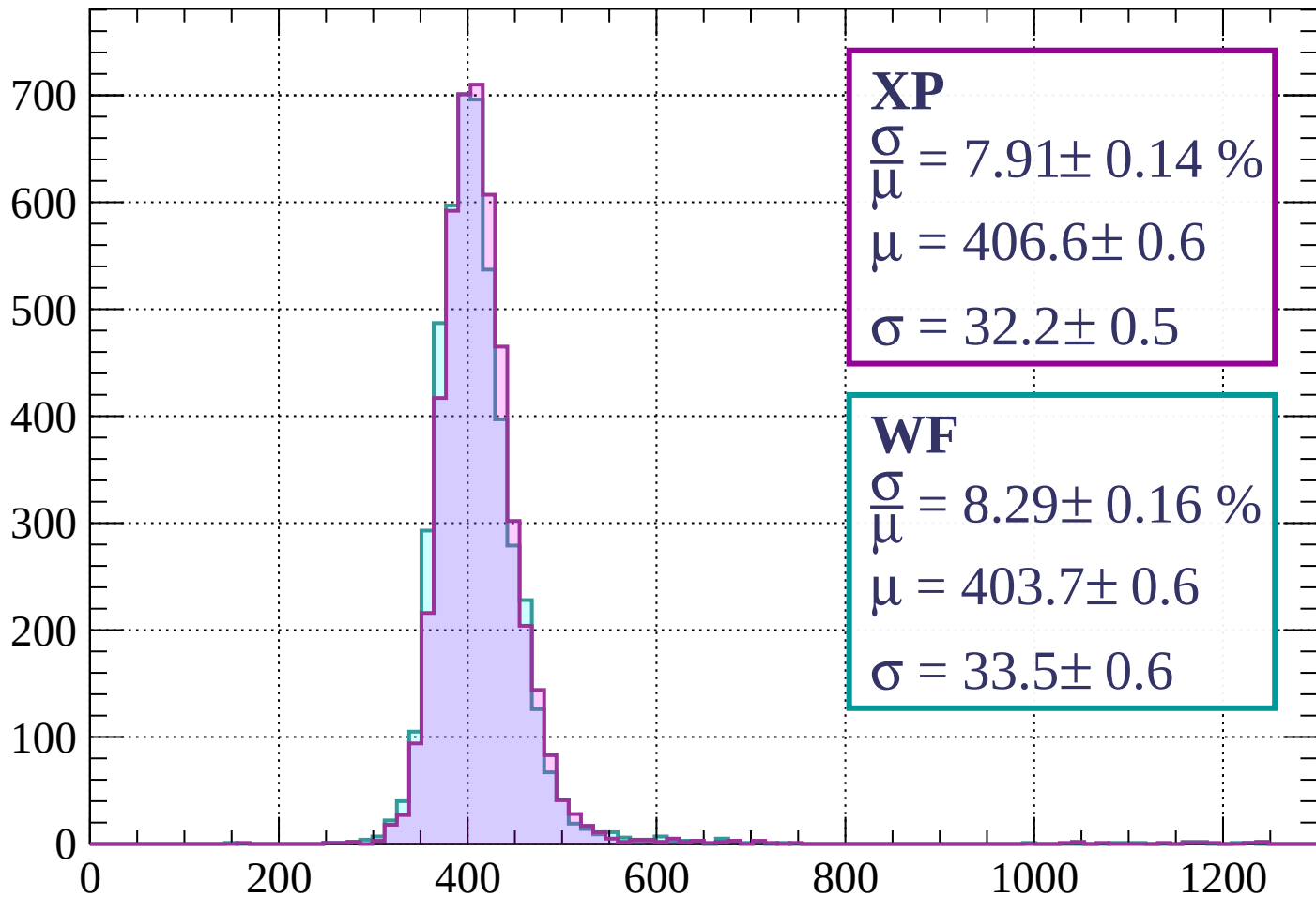


Energy loss | $400 < p < 480$



Energy loss | $480 < p < 560$

Count



XP

$$\frac{\sigma}{\mu} = 7.91 \pm 0.14 \%$$

$$\mu = 406.6 \pm 0.6$$

$$\sigma = 32.2 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 8.29 \pm 0.16 \%$$

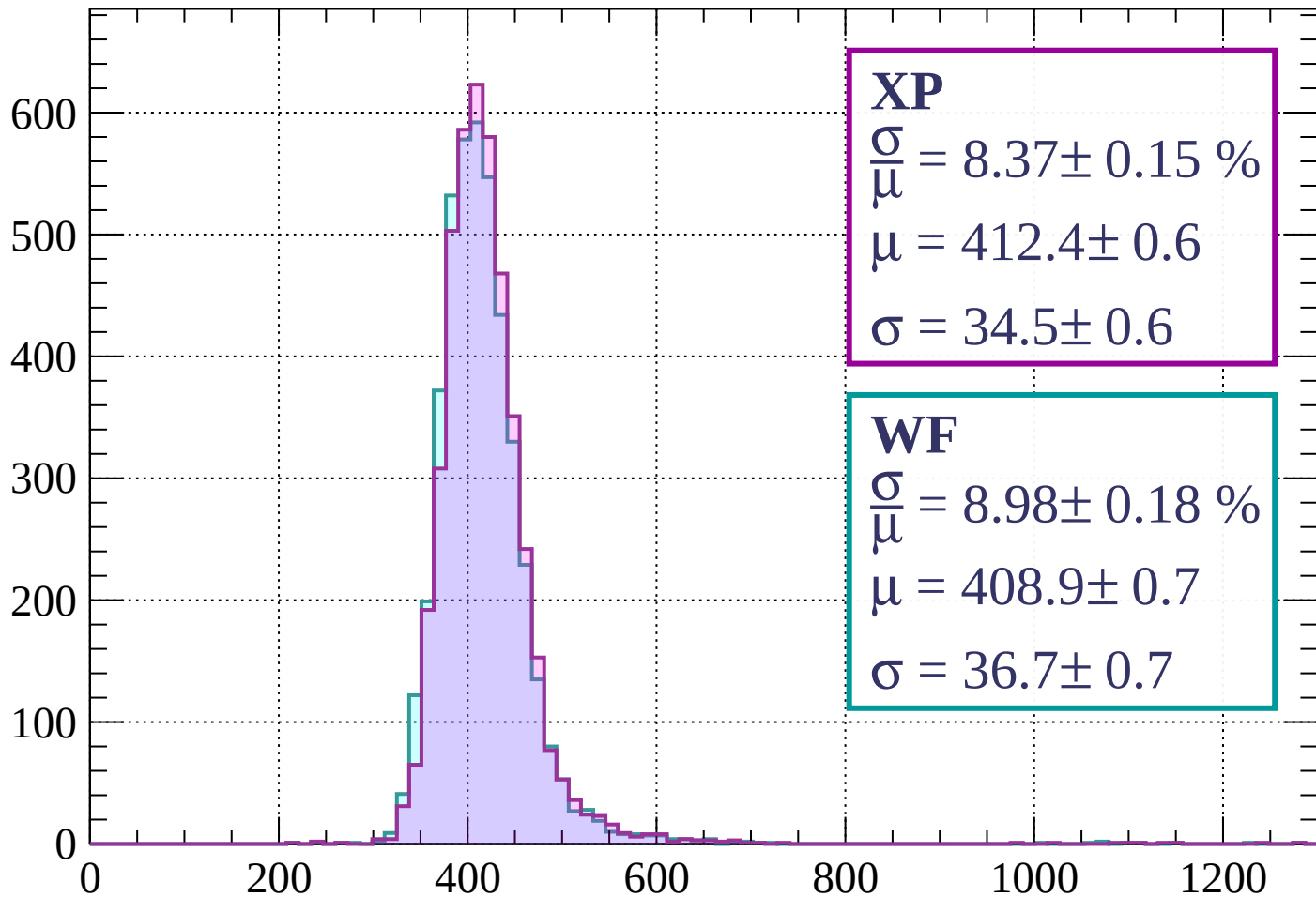
$$\mu = 403.7 \pm 0.6$$

$$\sigma = 33.5 \pm 0.6$$

dE/dx (ADC counts/cm)

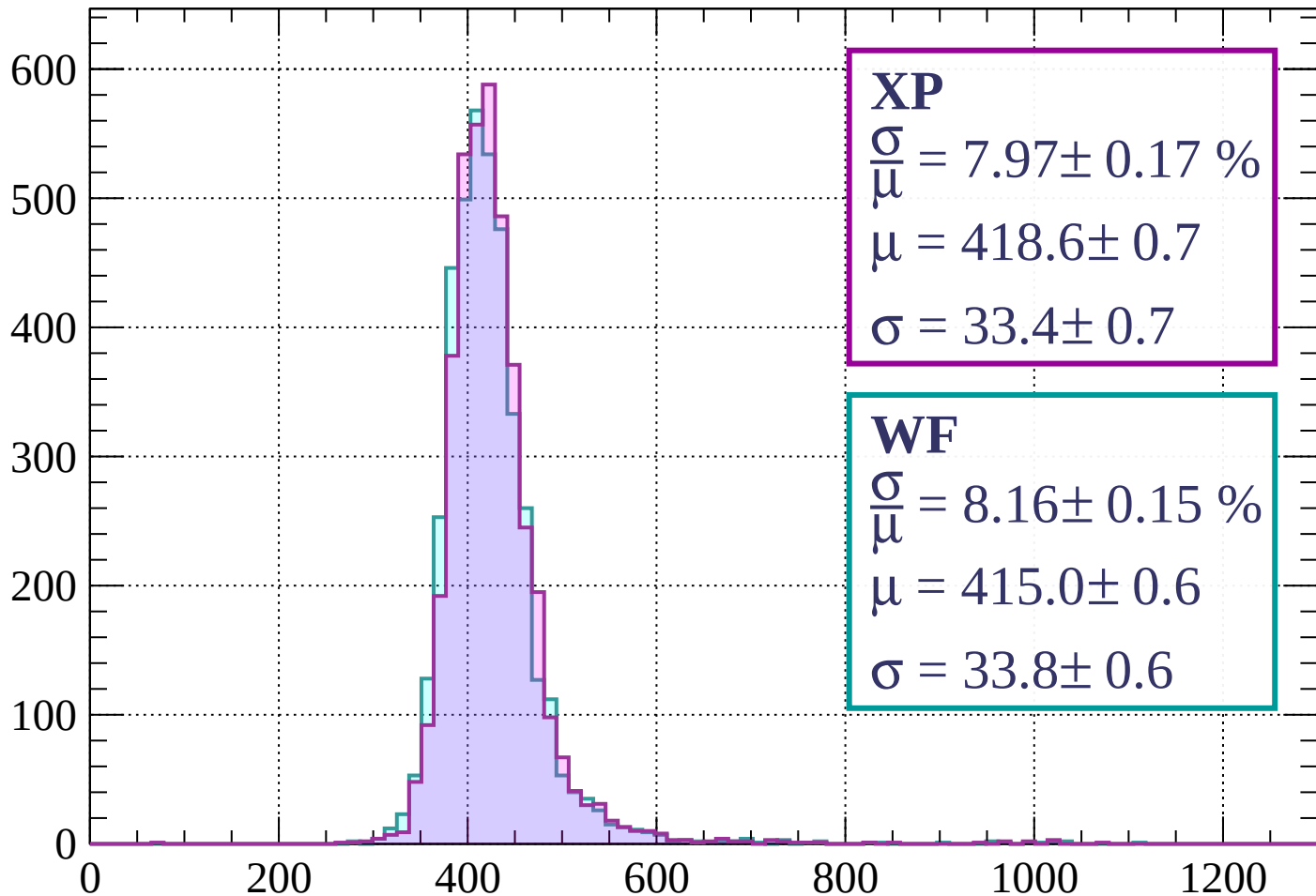
Energy loss | $560 < p < 640$

Count



Energy loss | $640 < p < 720$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 7.97 \pm 0.17 \%$$

$$\mu = 418.6 \pm 0.7$$

$$\sigma = 33.4 \pm 0.7$$

WF

$$\frac{\sigma}{\bar{\mu}} = 8.16 \pm 0.15 \%$$

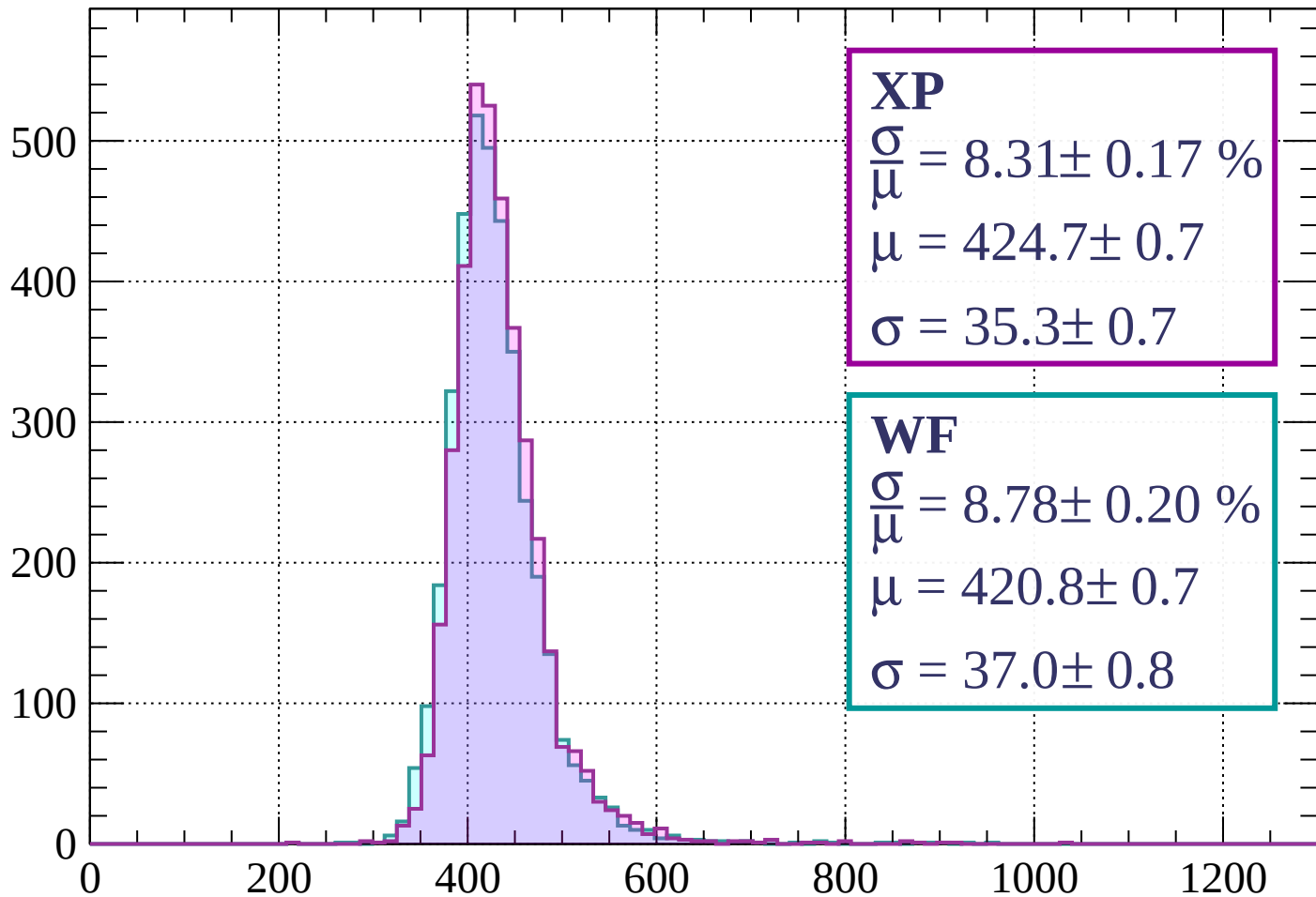
$$\mu = 415.0 \pm 0.6$$

$$\sigma = 33.8 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $720 < p < 800$

Count



XP

$$\frac{\sigma}{\mu} = 8.31 \pm 0.17 \%$$

$$\mu = 424.7 \pm 0.7$$

$$\sigma = 35.3 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 8.78 \pm 0.20 \%$$

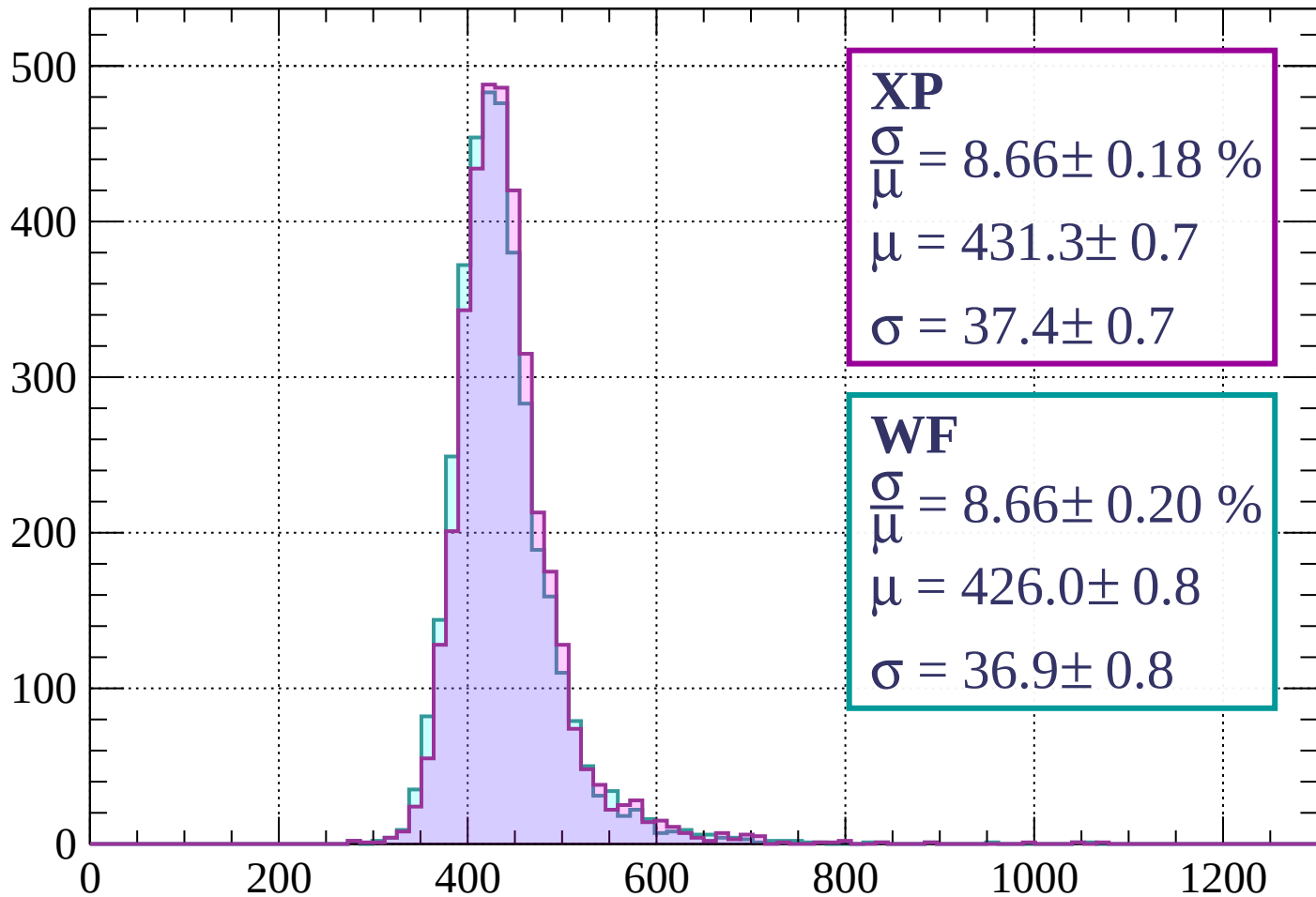
$$\mu = 420.8 \pm 0.7$$

$$\sigma = 37.0 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $800 < p < 880$

Count



XP

$$\frac{\sigma}{\mu} = 8.66 \pm 0.18 \%$$

$$\mu = 431.3 \pm 0.7$$

$$\sigma = 37.4 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 8.66 \pm 0.20 \%$$

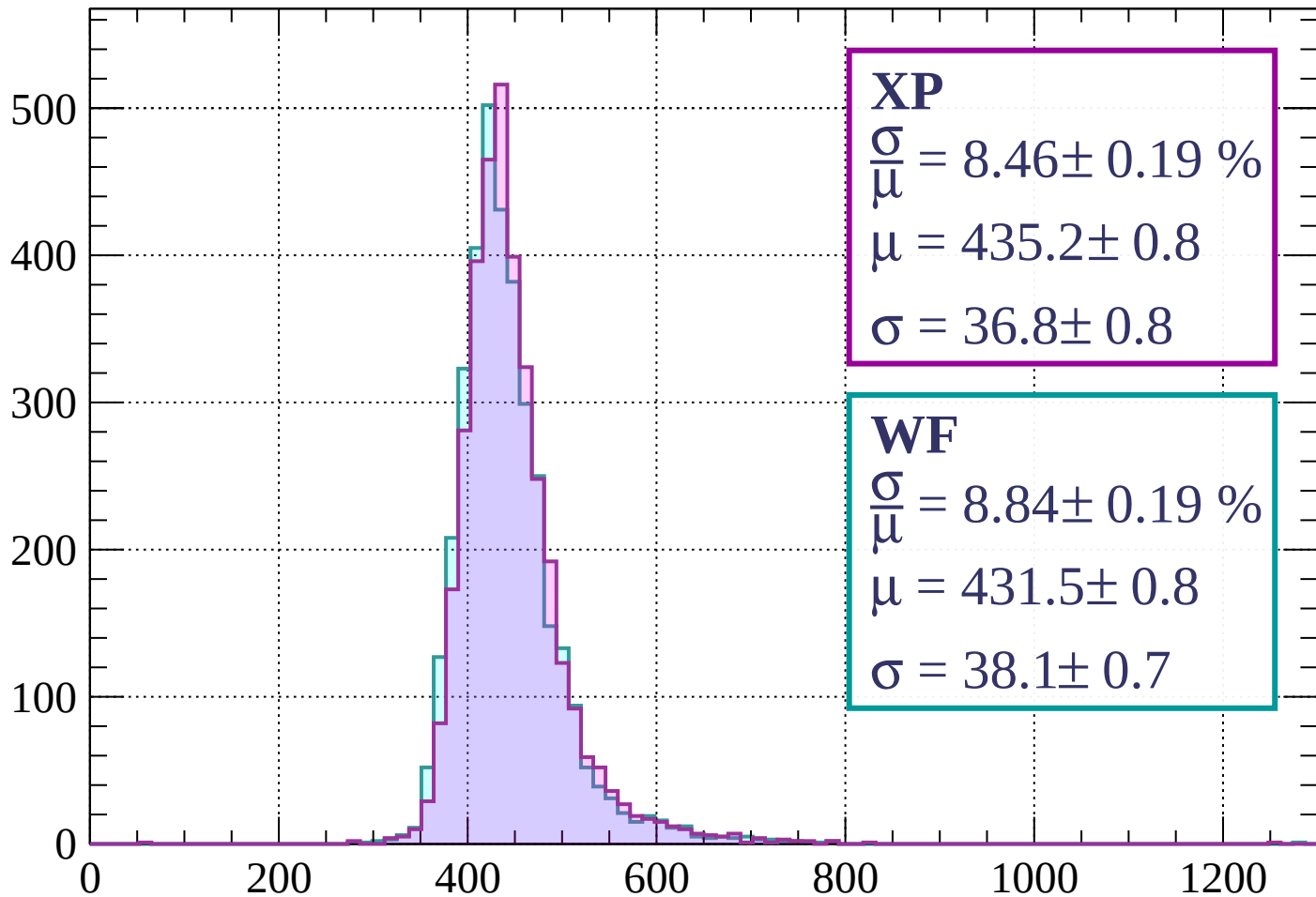
$$\mu = 426.0 \pm 0.8$$

$$\sigma = 36.9 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $880 < p < 960$

Count



XP

$$\frac{\sigma}{\mu} = 8.46 \pm 0.19 \%$$

$$\mu = 435.2 \pm 0.8$$

$$\sigma = 36.8 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 8.84 \pm 0.19 \%$$

$$\mu = 431.5 \pm 0.8$$

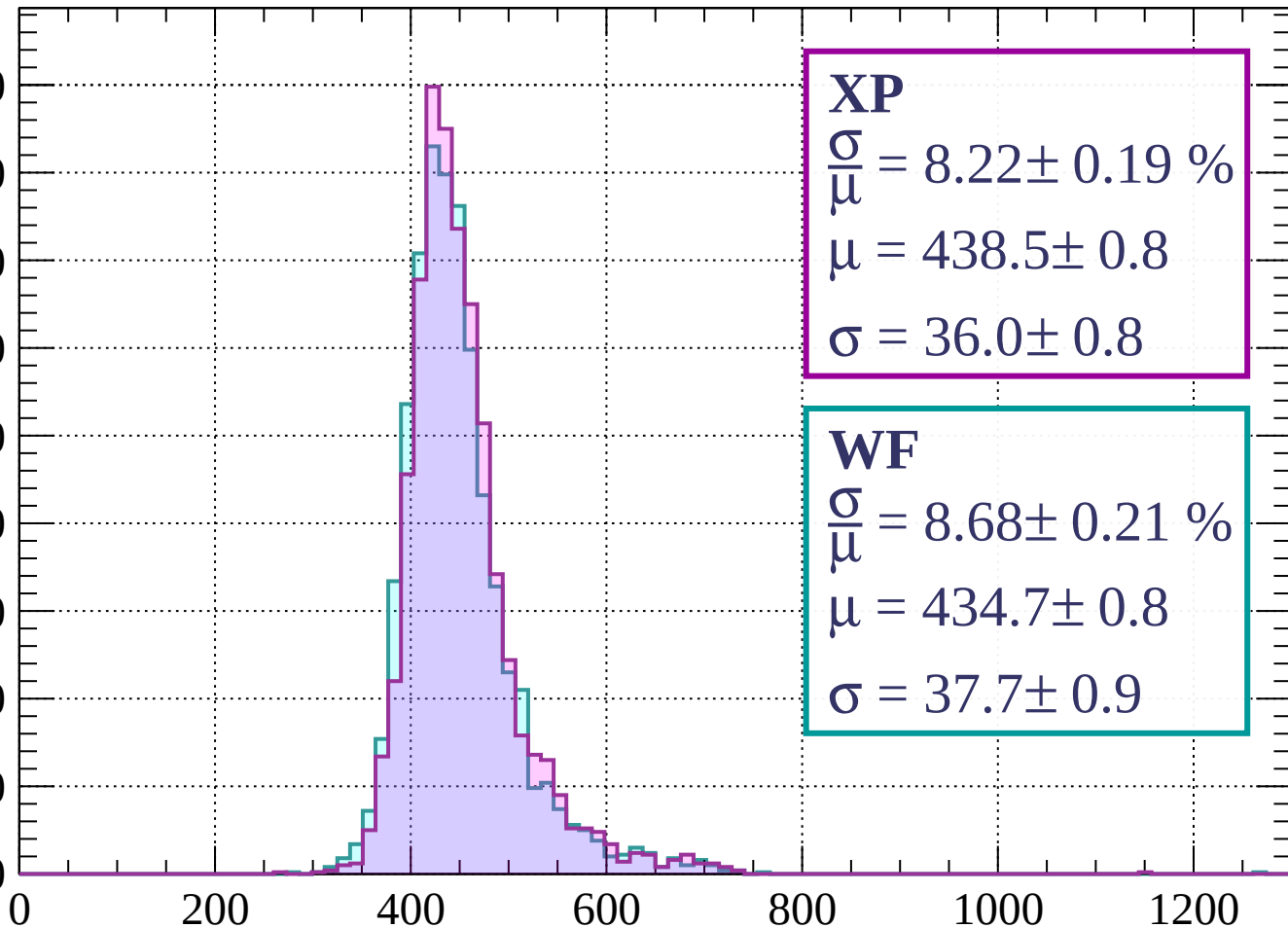
$$\sigma = 38.1 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $960 < p < 1040$

Count

450
400
350
300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 8.22 \pm 0.19 \%$$

$$\mu = 438.5 \pm 0.8$$

$$\sigma = 36.0 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 8.68 \pm 0.21 \%$$

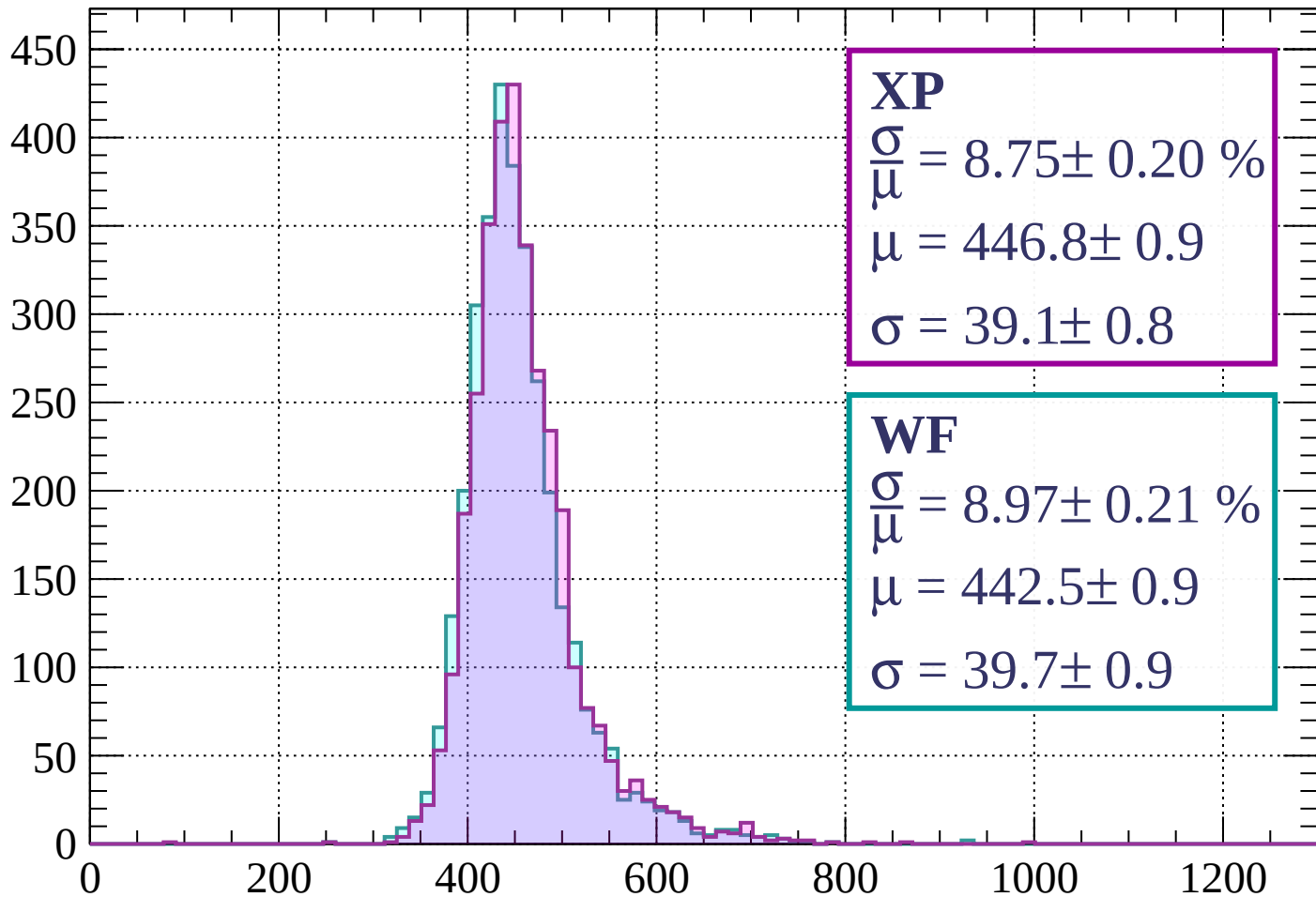
$$\mu = 434.7 \pm 0.8$$

$$\sigma = 37.7 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $1040 < p < 1120$

Count



XP

$$\frac{\sigma}{\mu} = 8.75 \pm 0.20 \%$$

$$\mu = 446.8 \pm 0.9$$

$$\sigma = 39.1 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 8.97 \pm 0.21 \%$$

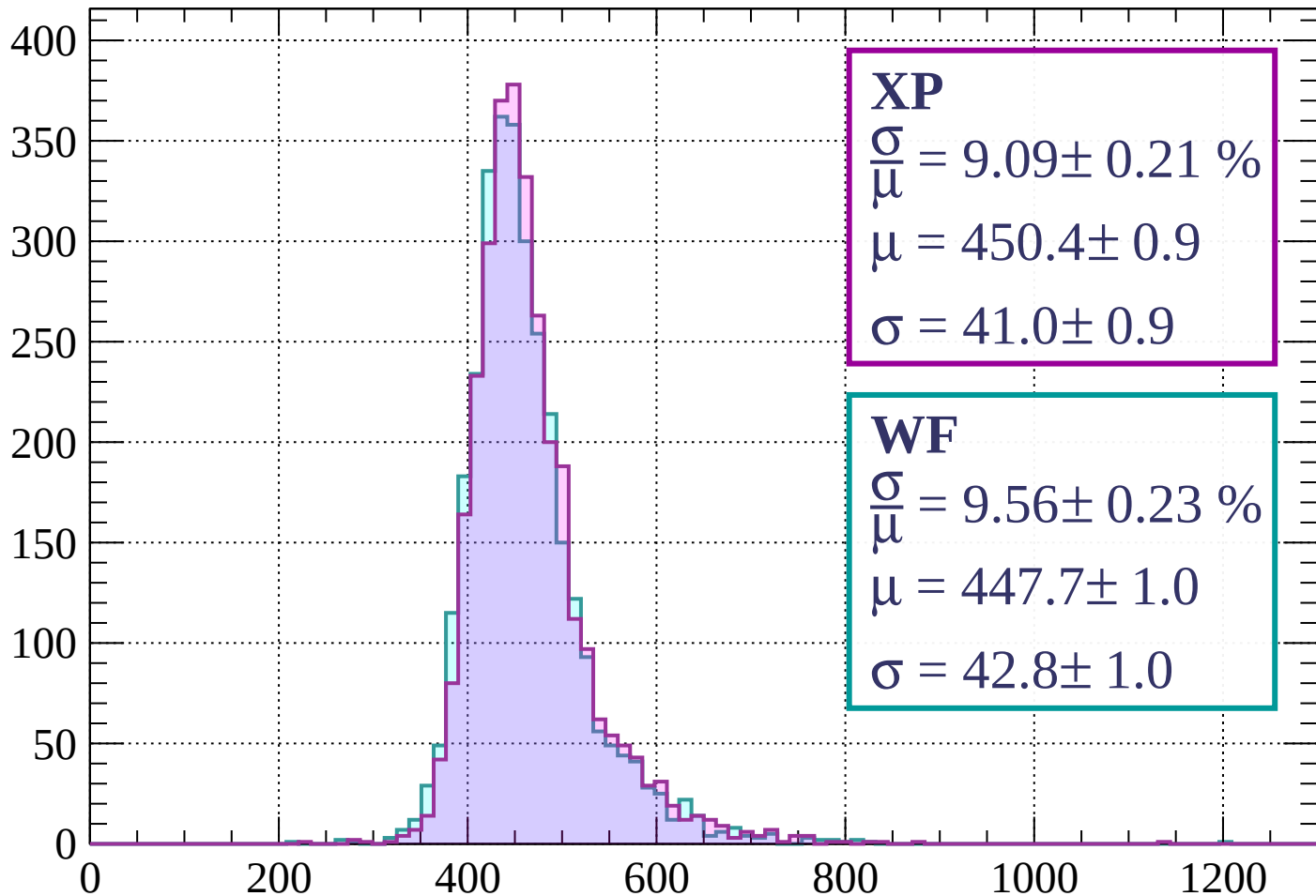
$$\mu = 442.5 \pm 0.9$$

$$\sigma = 39.7 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $1120 < p < 1200$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 9.09 \pm 0.21 \%$$

$$\mu = 450.4 \pm 0.9$$

$$\sigma = 41.0 \pm 0.9$$

WF

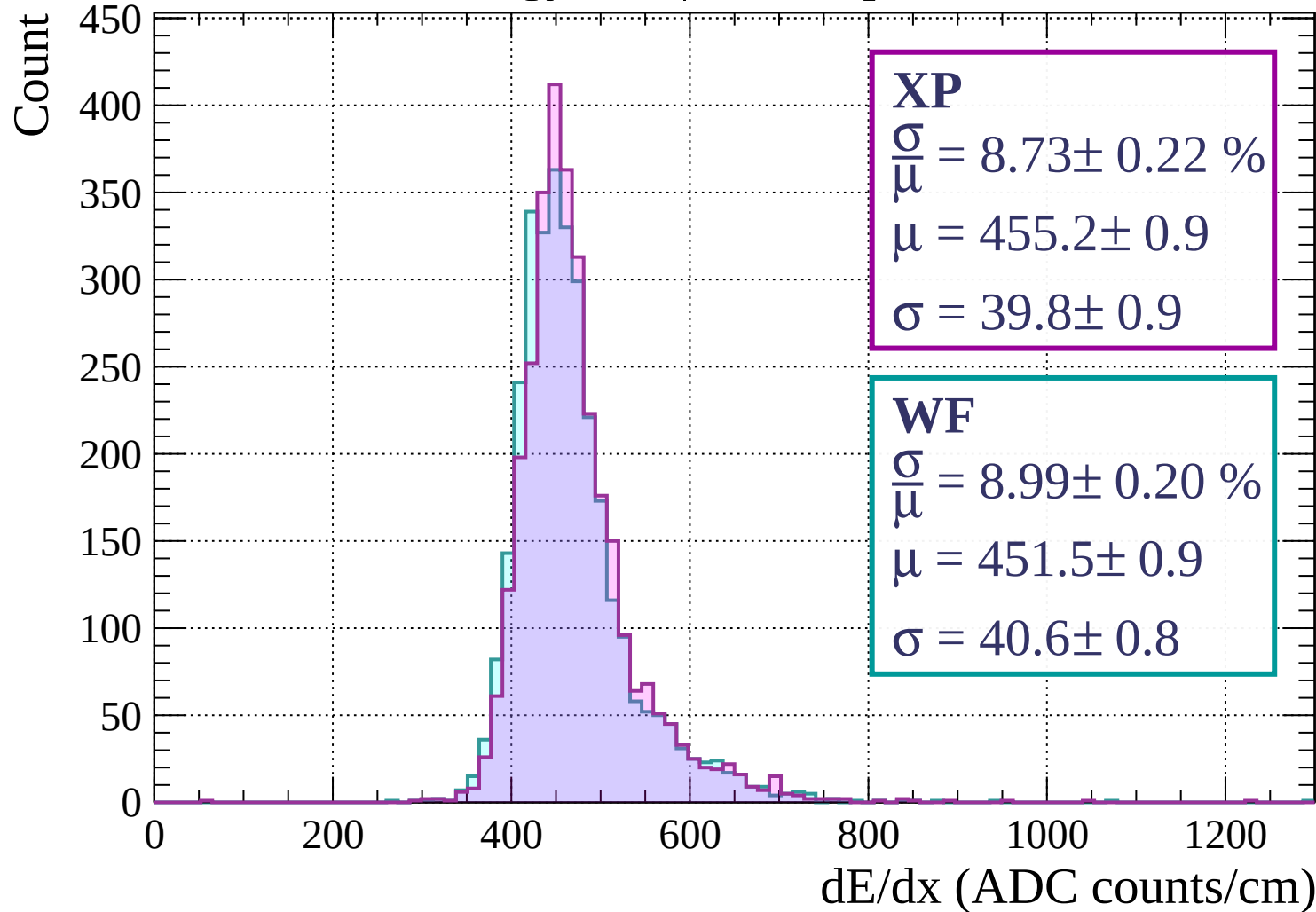
$$\frac{\sigma}{\bar{\mu}} = 9.56 \pm 0.23 \%$$

$$\mu = 447.7 \pm 1.0$$

$$\sigma = 42.8 \pm 1.0$$

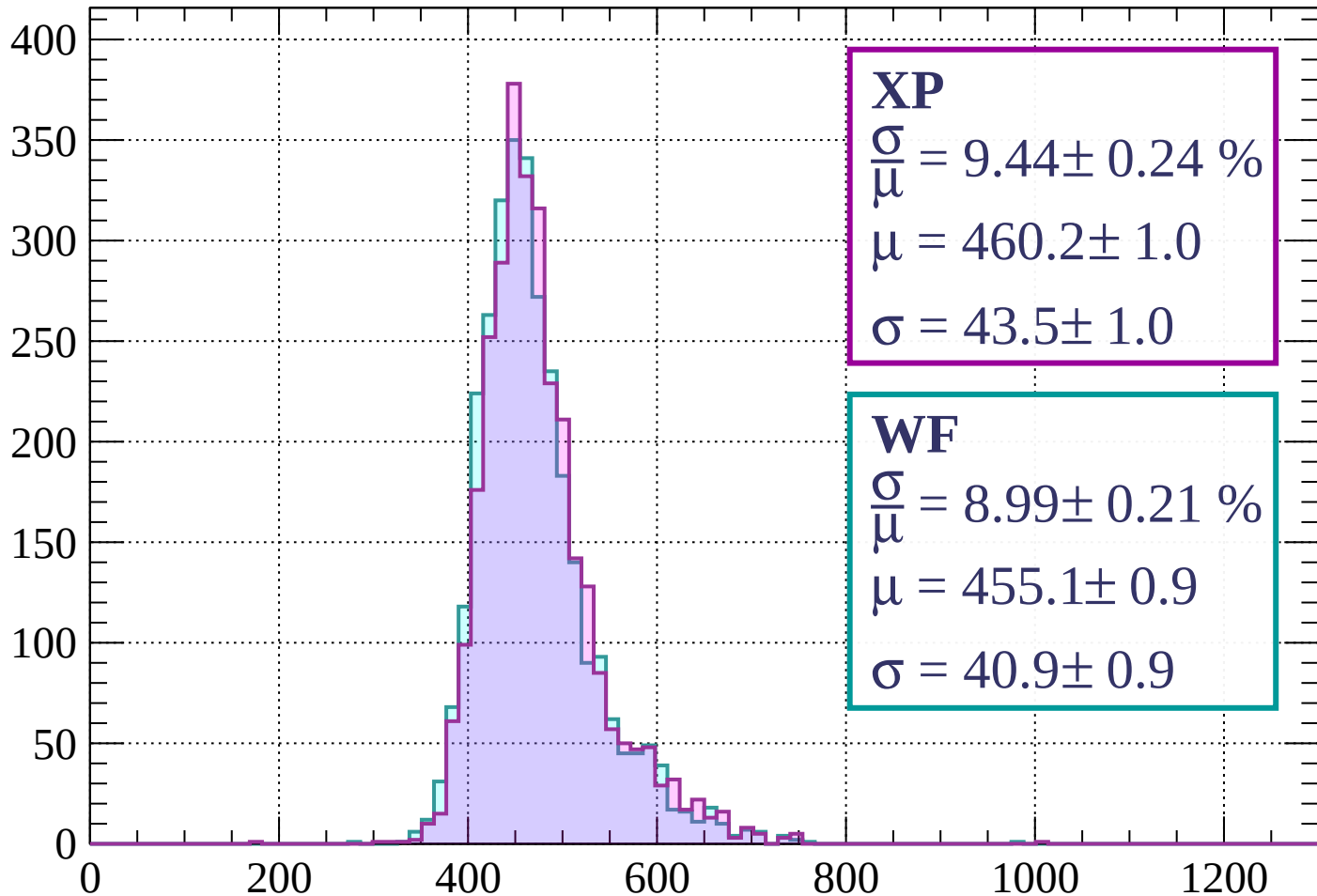
dE/dx (ADC counts/cm)

Energy loss | $1200 < p < 1280$



Energy loss | $1280 < p < 1360$

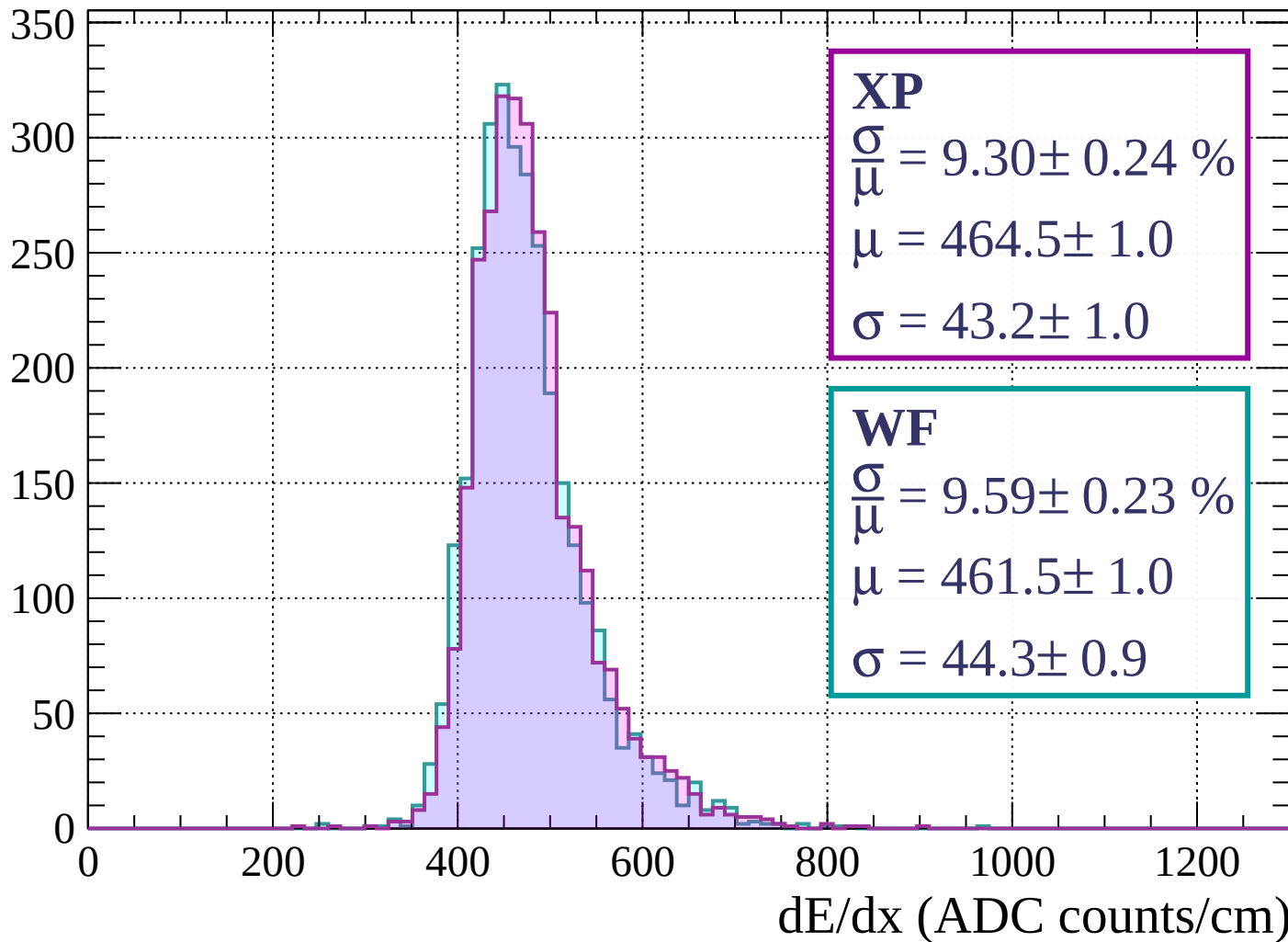
Count



dE/dx (ADC counts/cm)

Energy loss | $1360 < p < 1440$

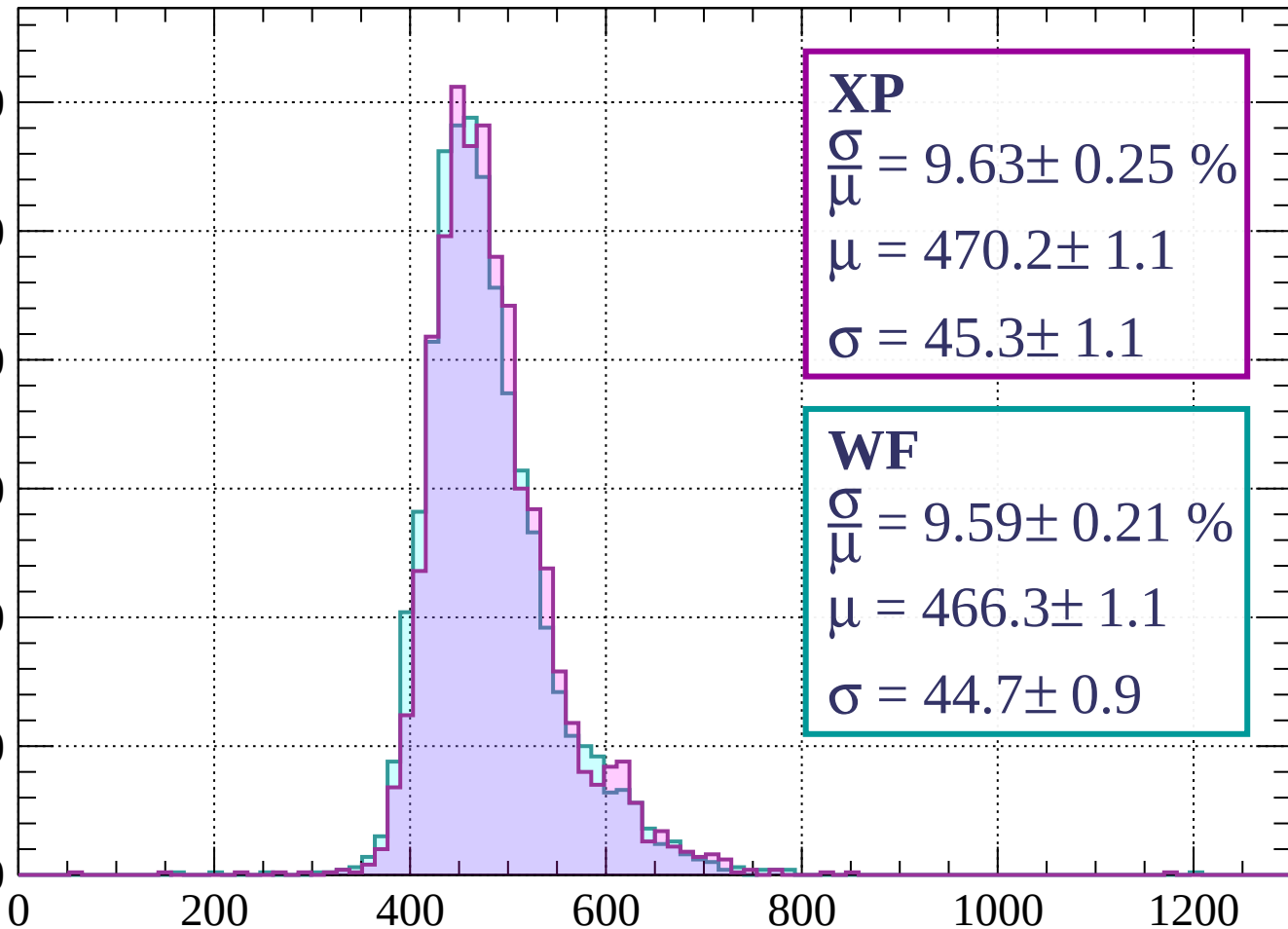
Count



Energy loss | $1440 < p < 1520$

Count

300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 9.63 \pm 0.25 \%$$

$$\mu = 470.2 \pm 1.1$$

$$\sigma = 45.3 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 9.59 \pm 0.21 \%$$

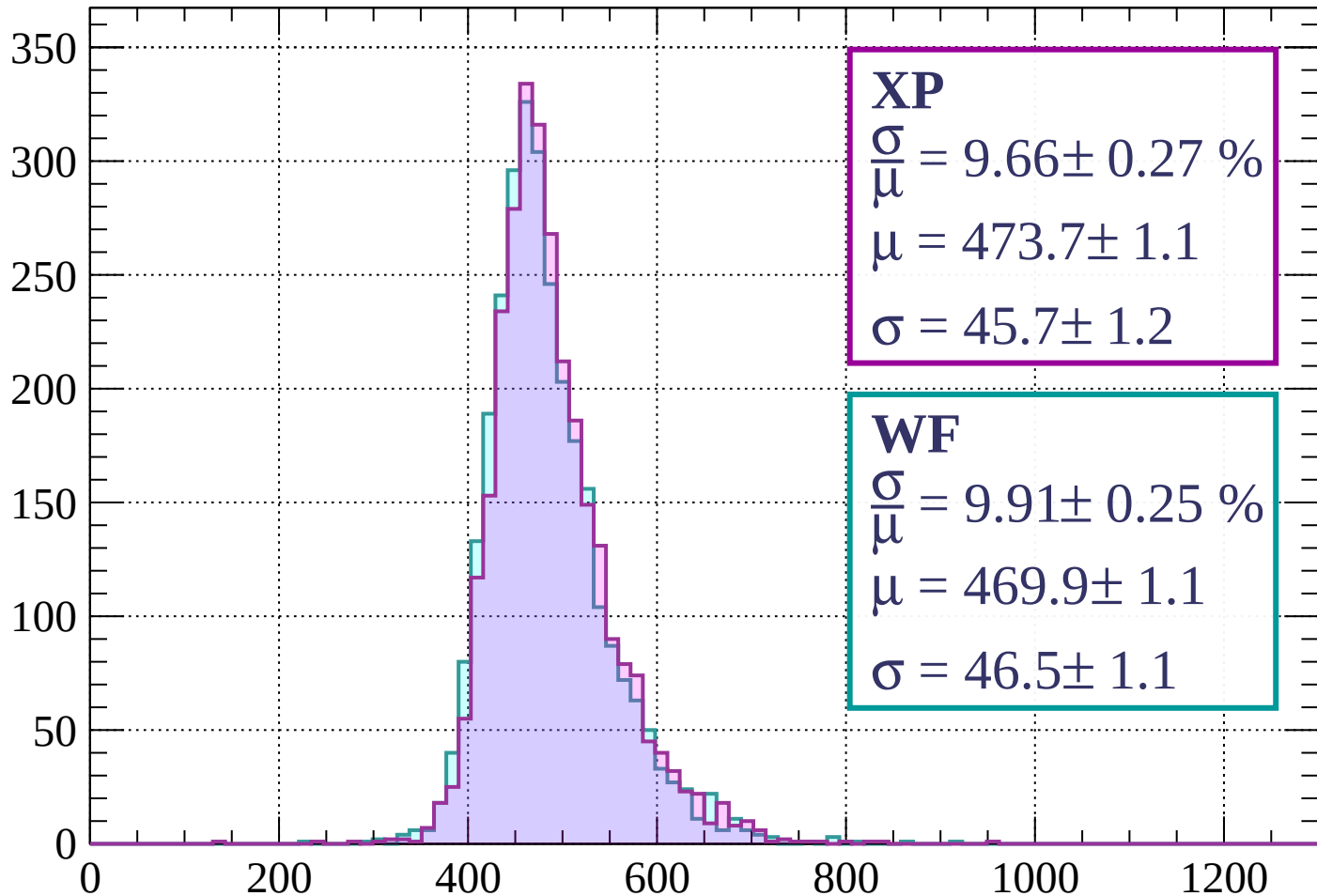
$$\mu = 466.3 \pm 1.1$$

$$\sigma = 44.7 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $1520 < p < 1600$

Count



XP

$$\frac{\sigma}{\mu} = 9.66 \pm 0.27 \%$$

$$\mu = 473.7 \pm 1.1$$

$$\sigma = 45.7 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 9.91 \pm 0.25 \%$$

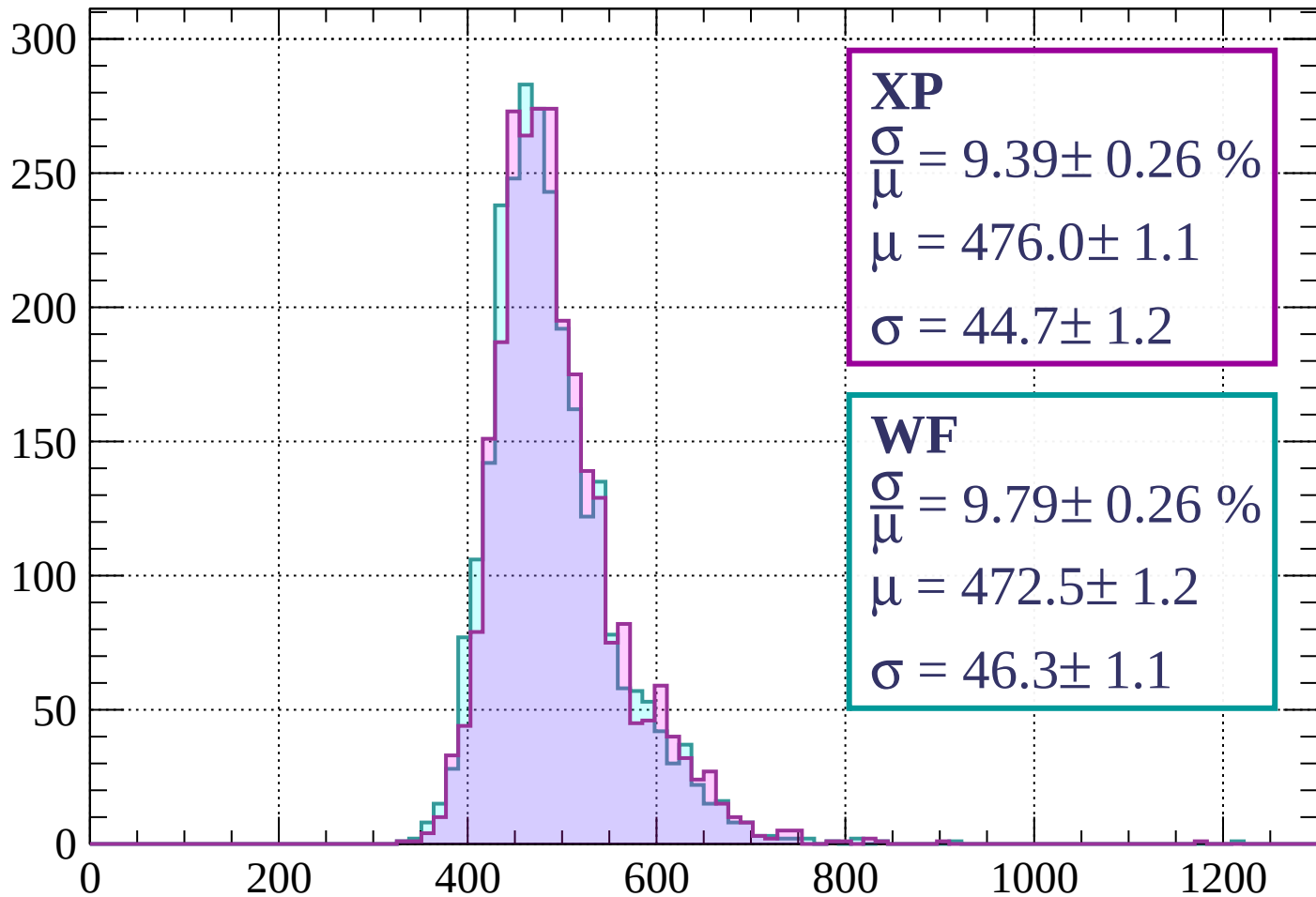
$$\mu = 469.9 \pm 1.1$$

$$\sigma = 46.5 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | $1600 < p < 1680$

Count



XP

$$\frac{\sigma}{\mu} = 9.39 \pm 0.26 \%$$

$$\mu = 476.0 \pm 1.1$$

$$\sigma = 44.7 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 9.79 \pm 0.26 \%$$

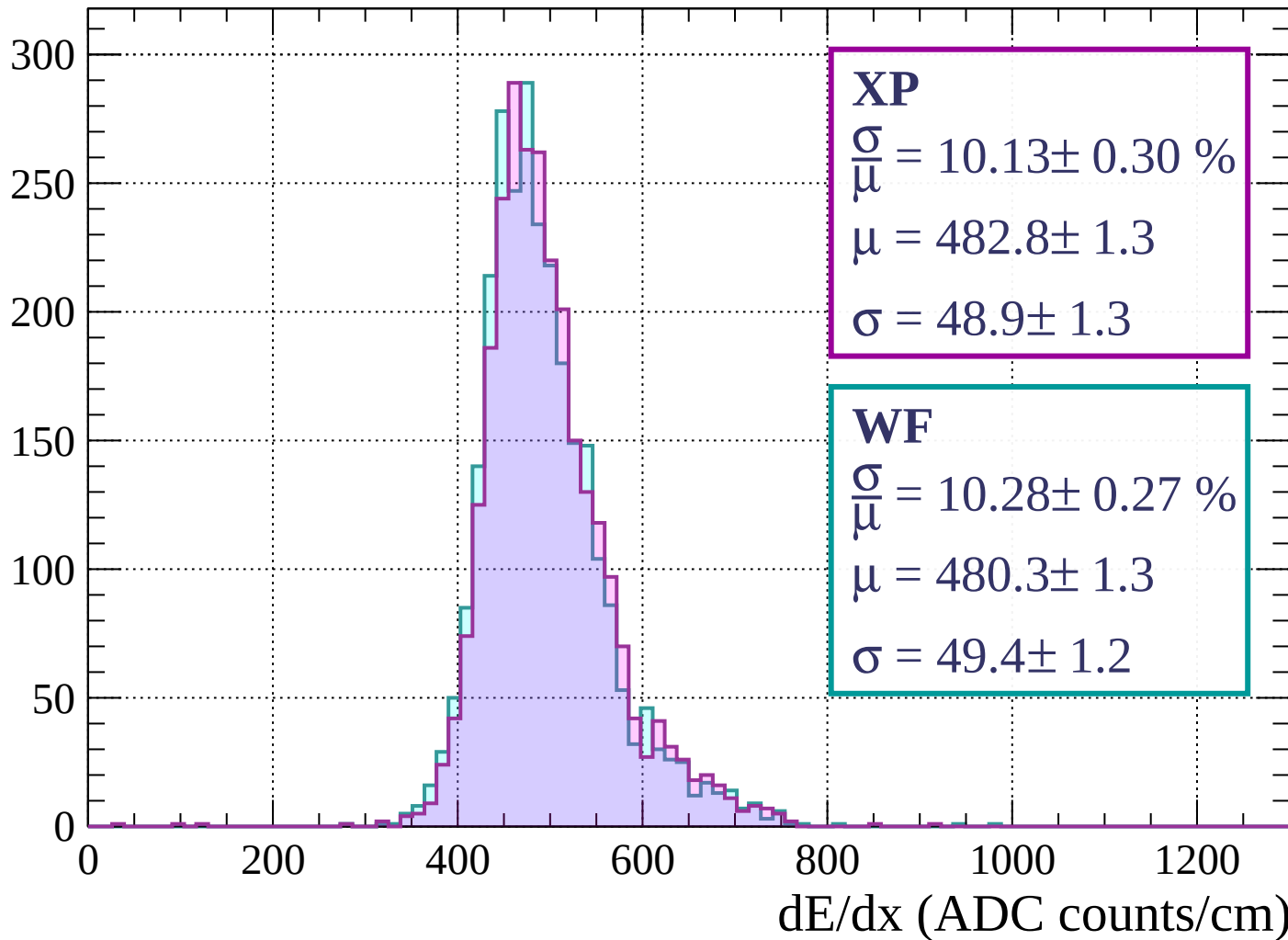
$$\mu = 472.5 \pm 1.2$$

$$\sigma = 46.3 \pm 1.1$$

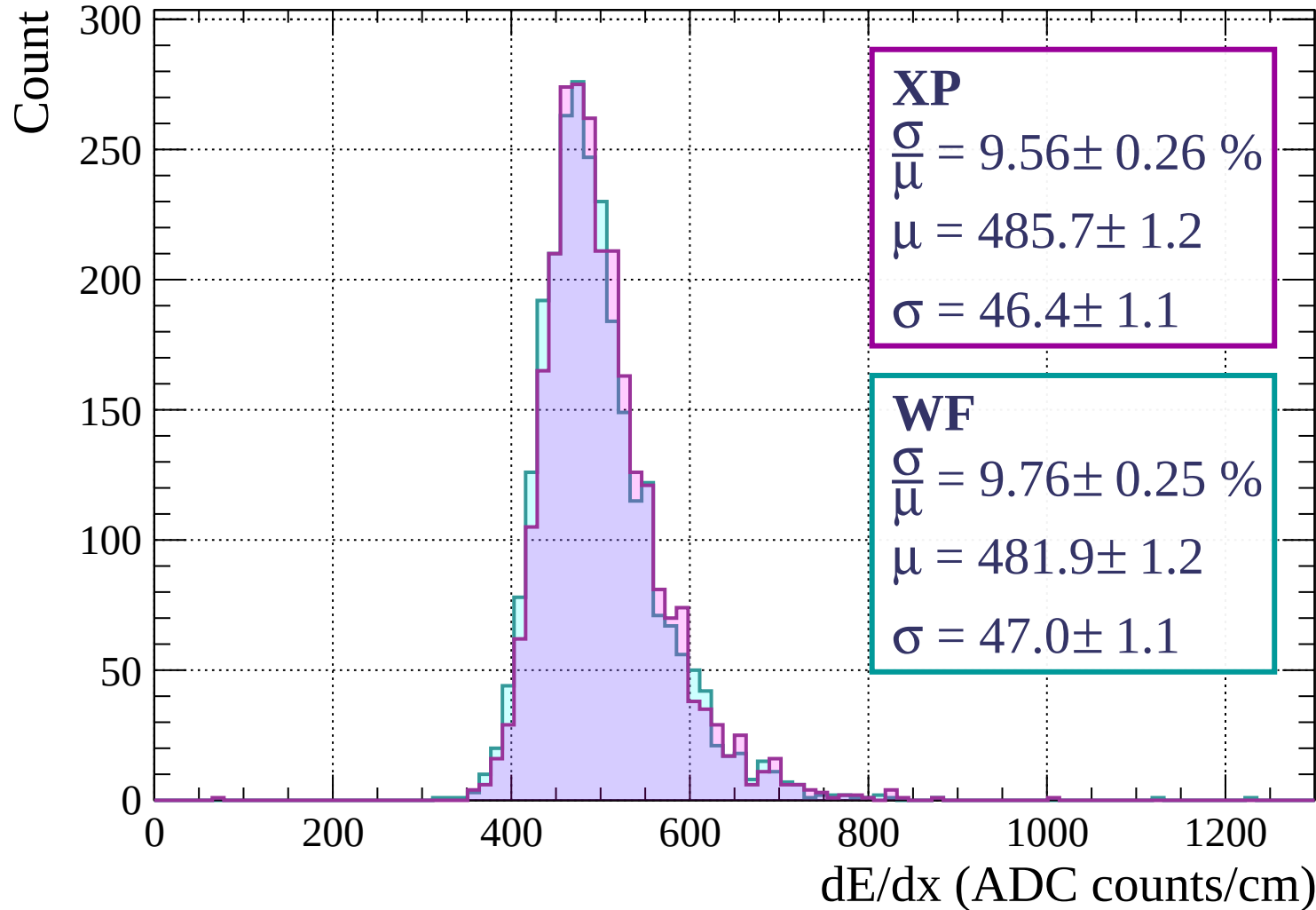
dE/dx (ADC counts/cm)

Energy loss | $1680 < p < 1760$

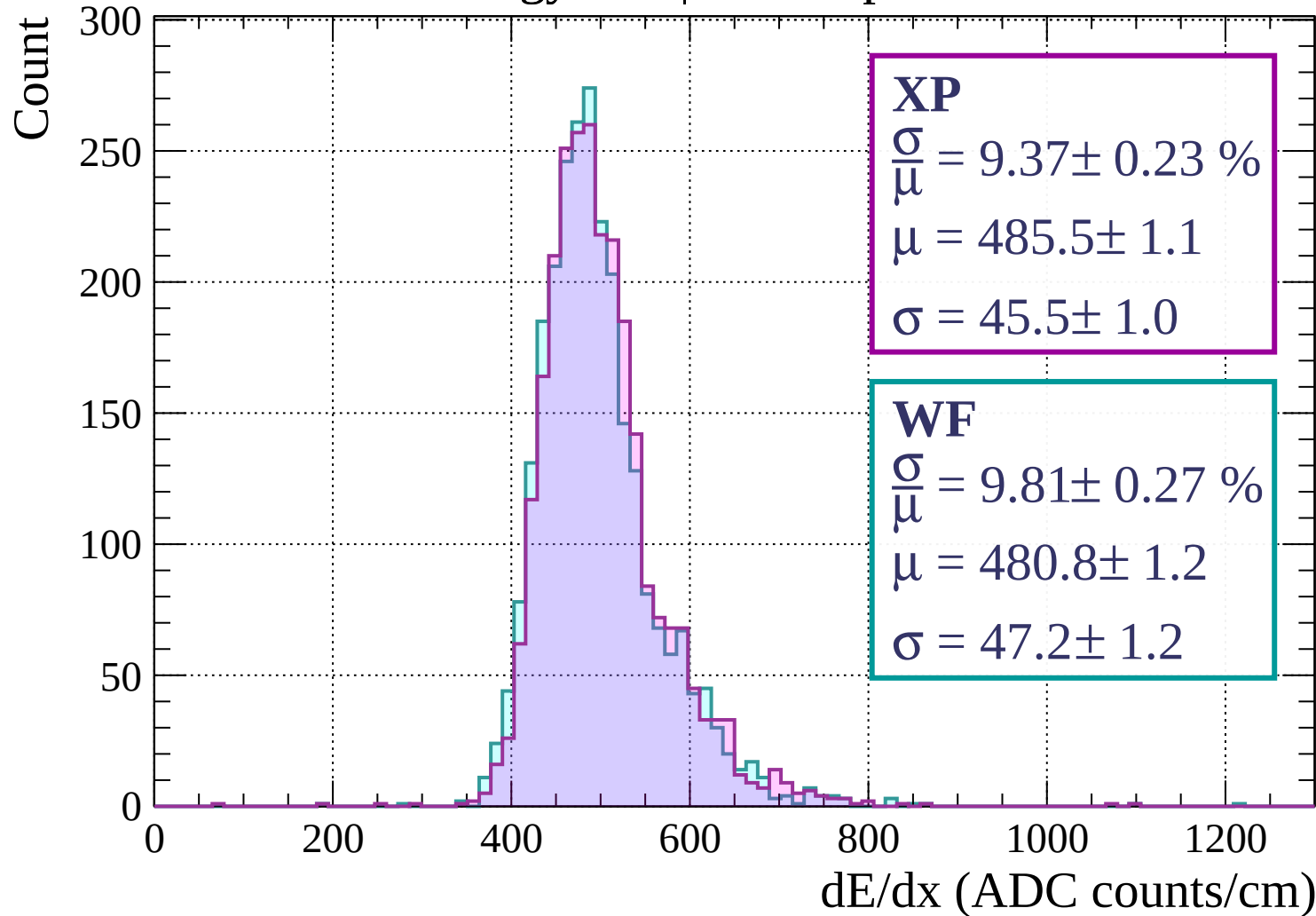
Count



Energy loss | $1760 < p < 1840$

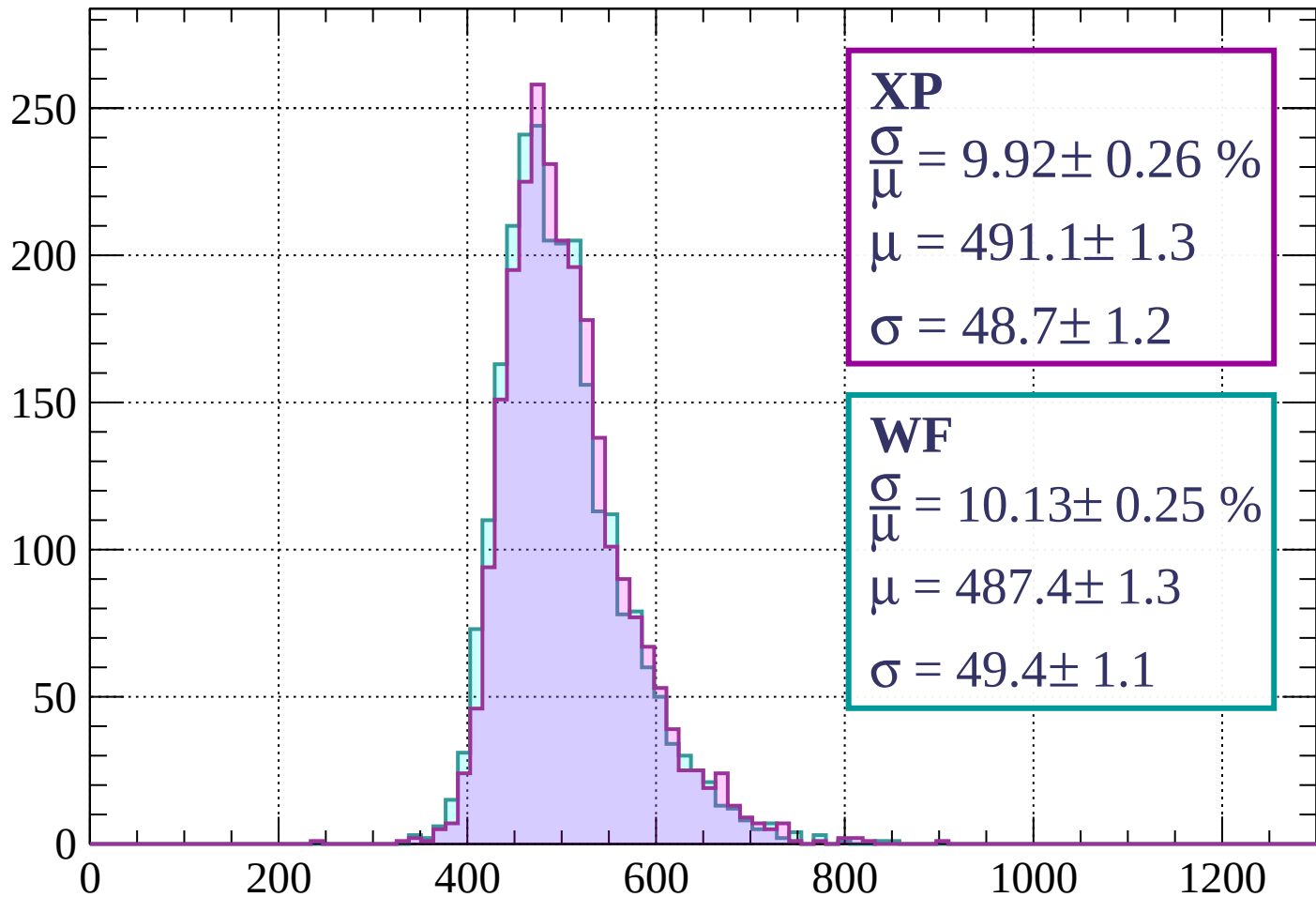


Energy loss | $1840 < p < 1920$



Energy loss | $1920 < p < 2000$

Count



XP

$$\frac{\sigma}{\mu} = 9.92 \pm 0.26 \%$$

$$\mu = 491.1 \pm 1.3$$

$$\sigma = 48.7 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 10.13 \pm 0.25 \%$$

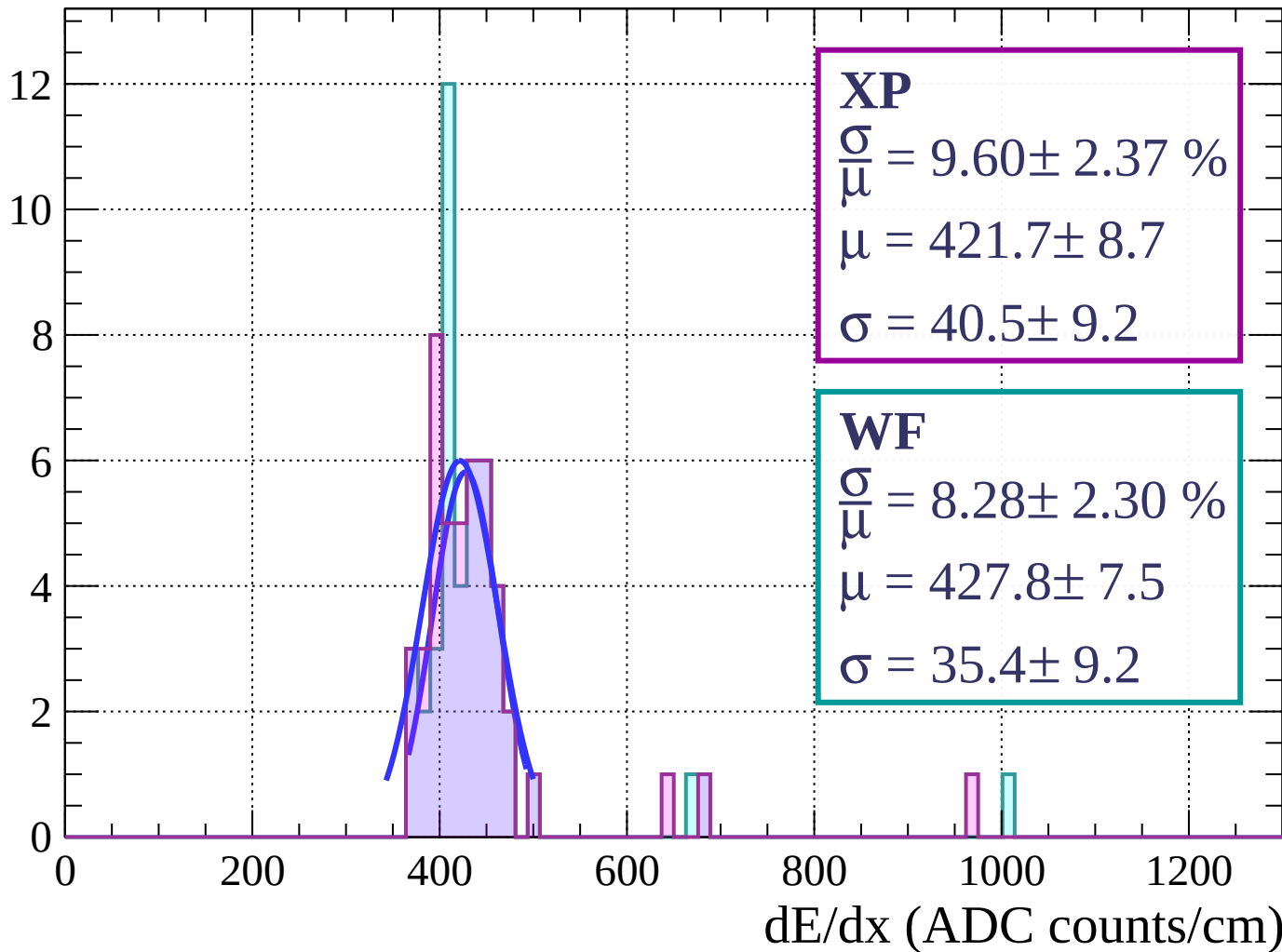
$$\mu = 487.4 \pm 1.3$$

$$\sigma = 49.4 \pm 1.1$$

dE/dx (ADC counts/cm)

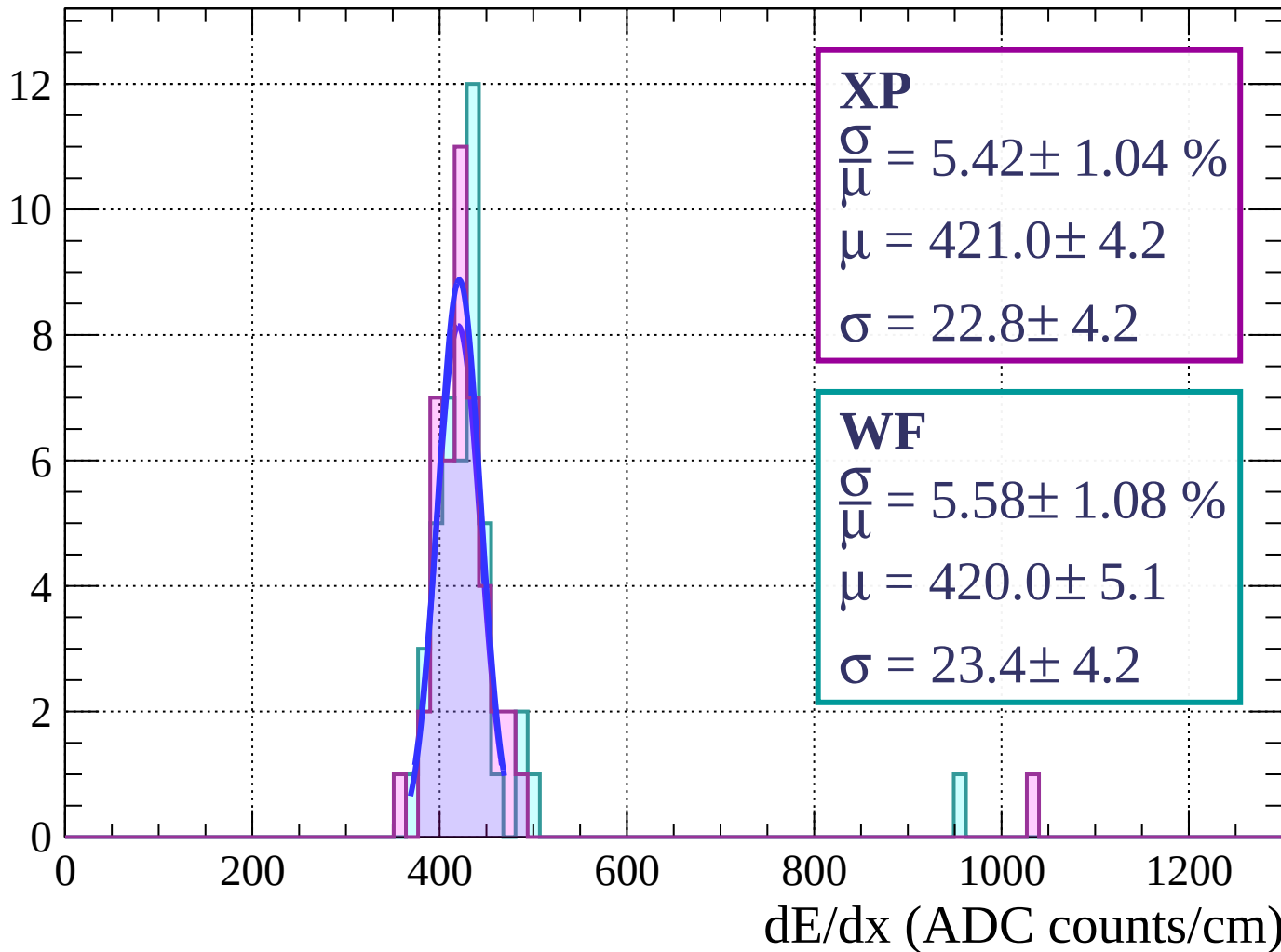
Energy loss | $0 < \phi < 3$

Count



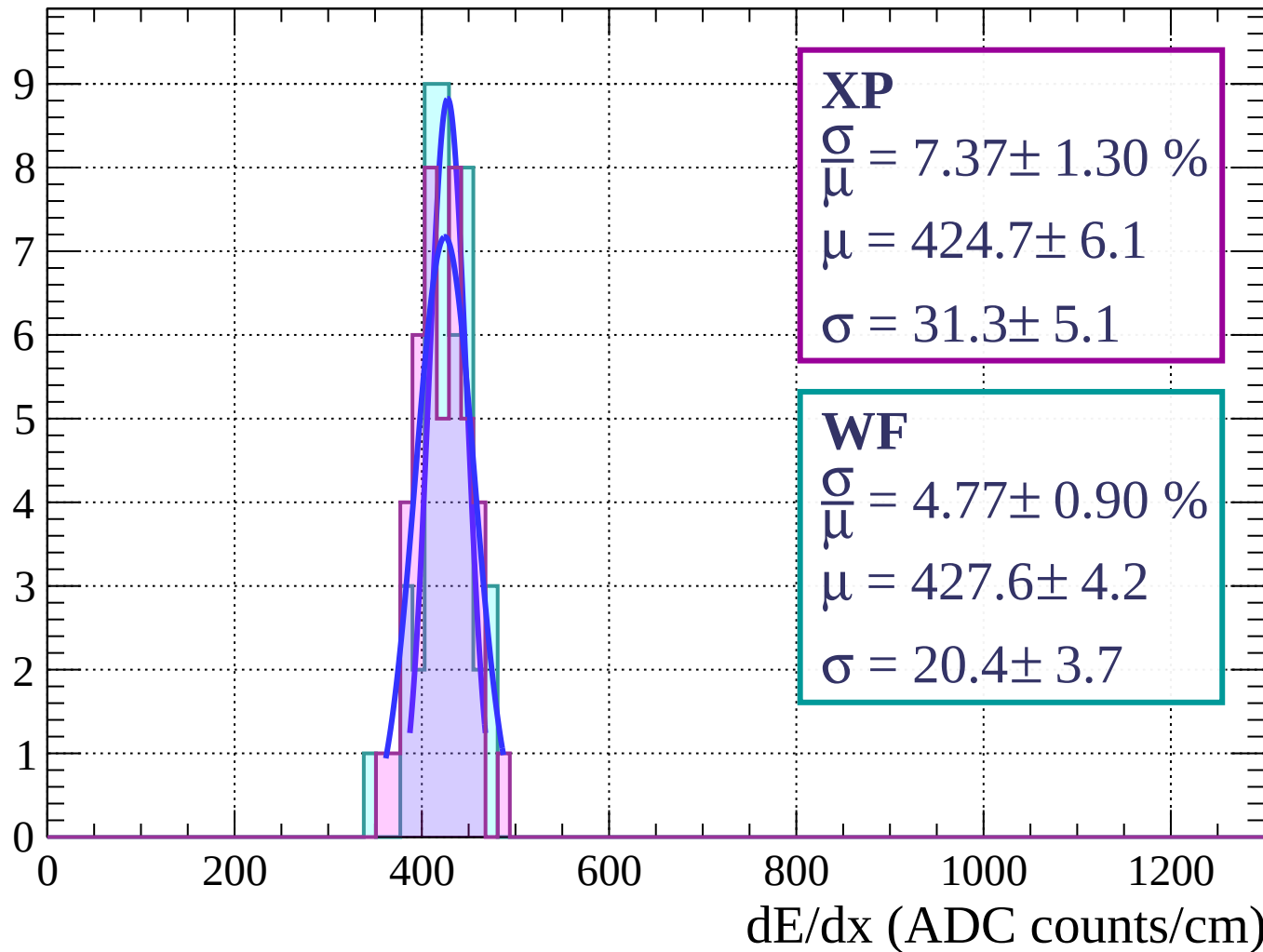
Energy loss | $3 < \phi < 6$

Count



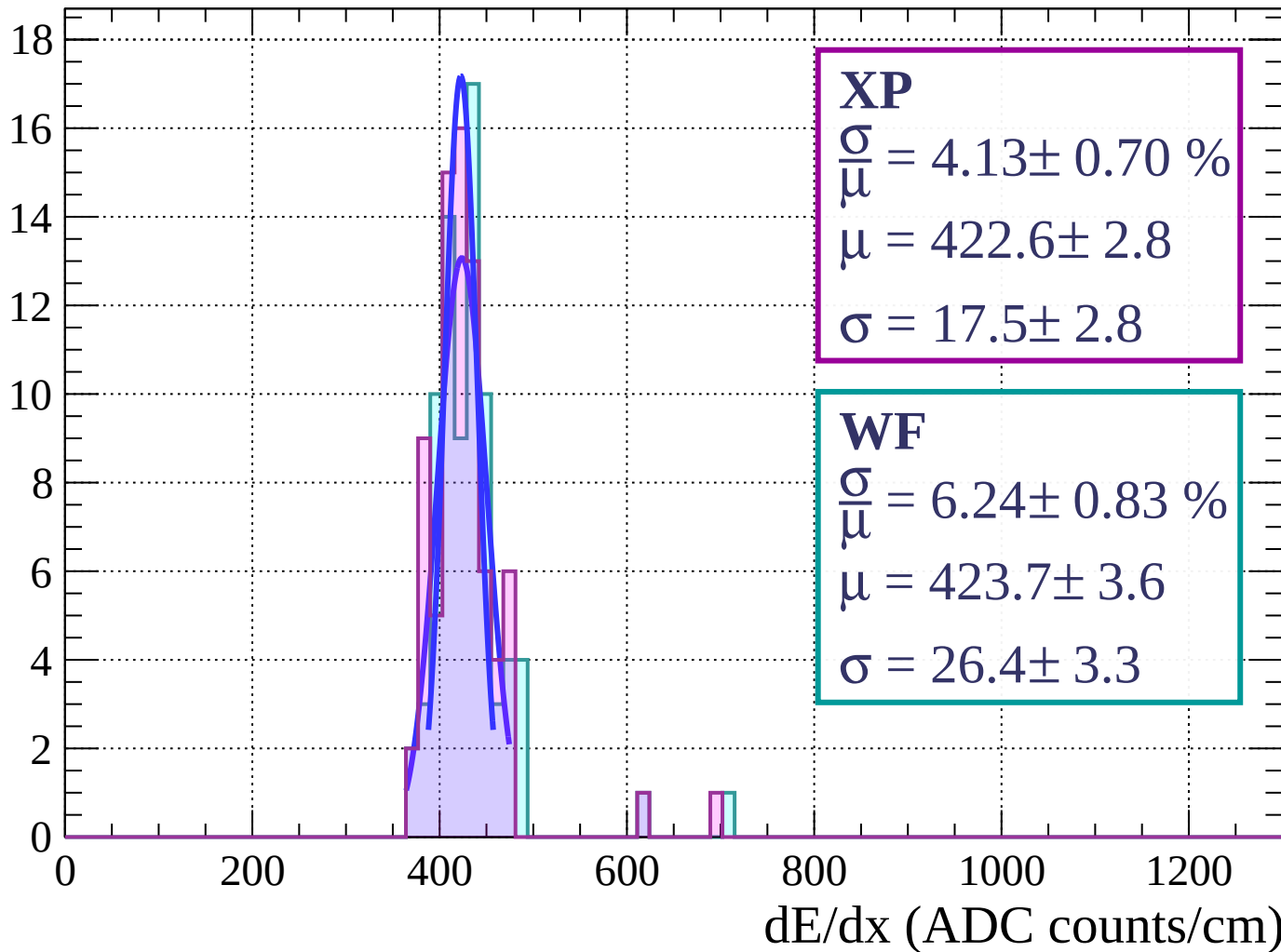
Energy loss | $6 < \phi < 9$

Count



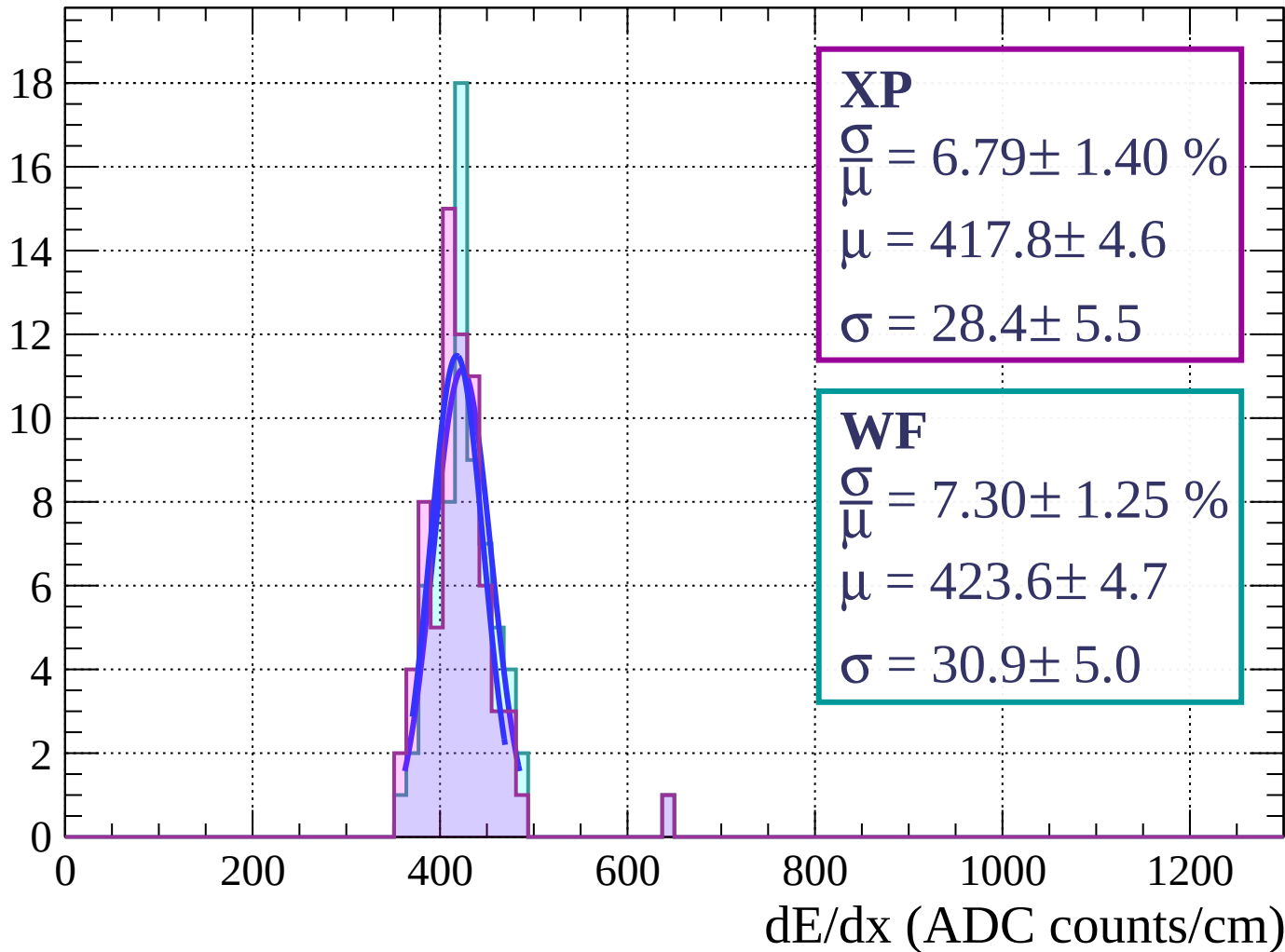
Energy loss | $9 < \phi < 12$

Count



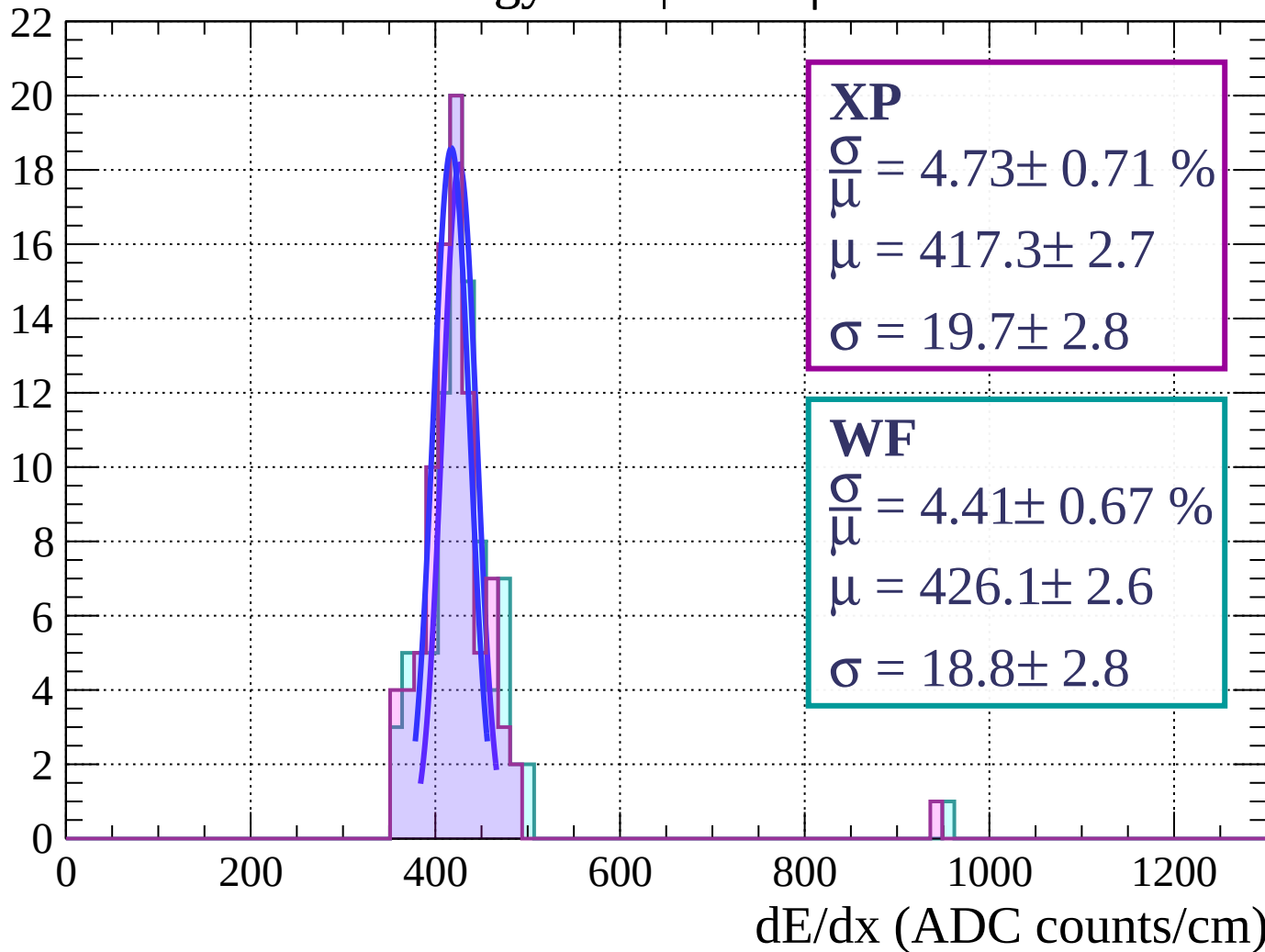
Energy loss | $12 < \phi < 15$

Count



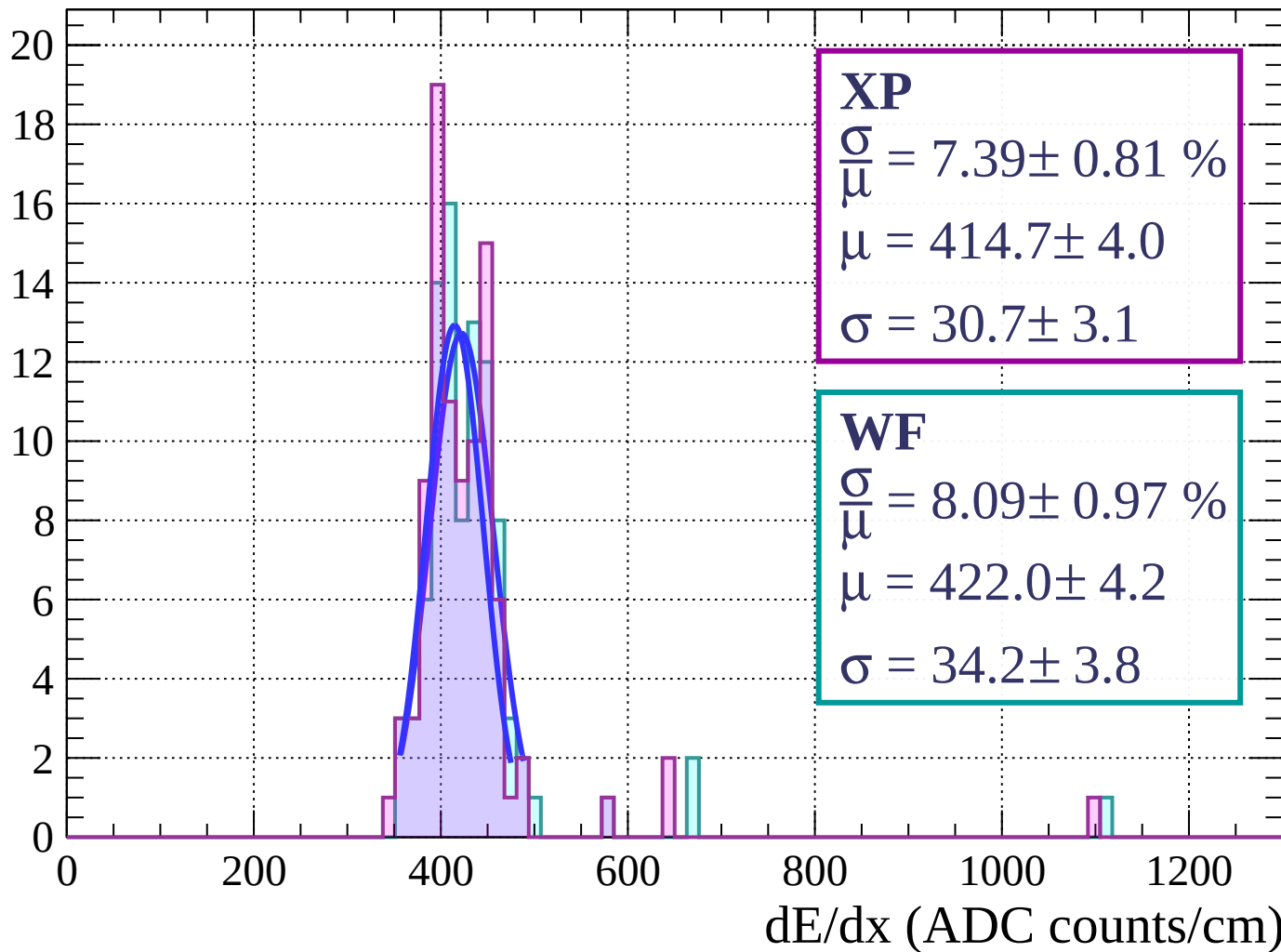
Energy loss | $15 < \phi < 18$

Count



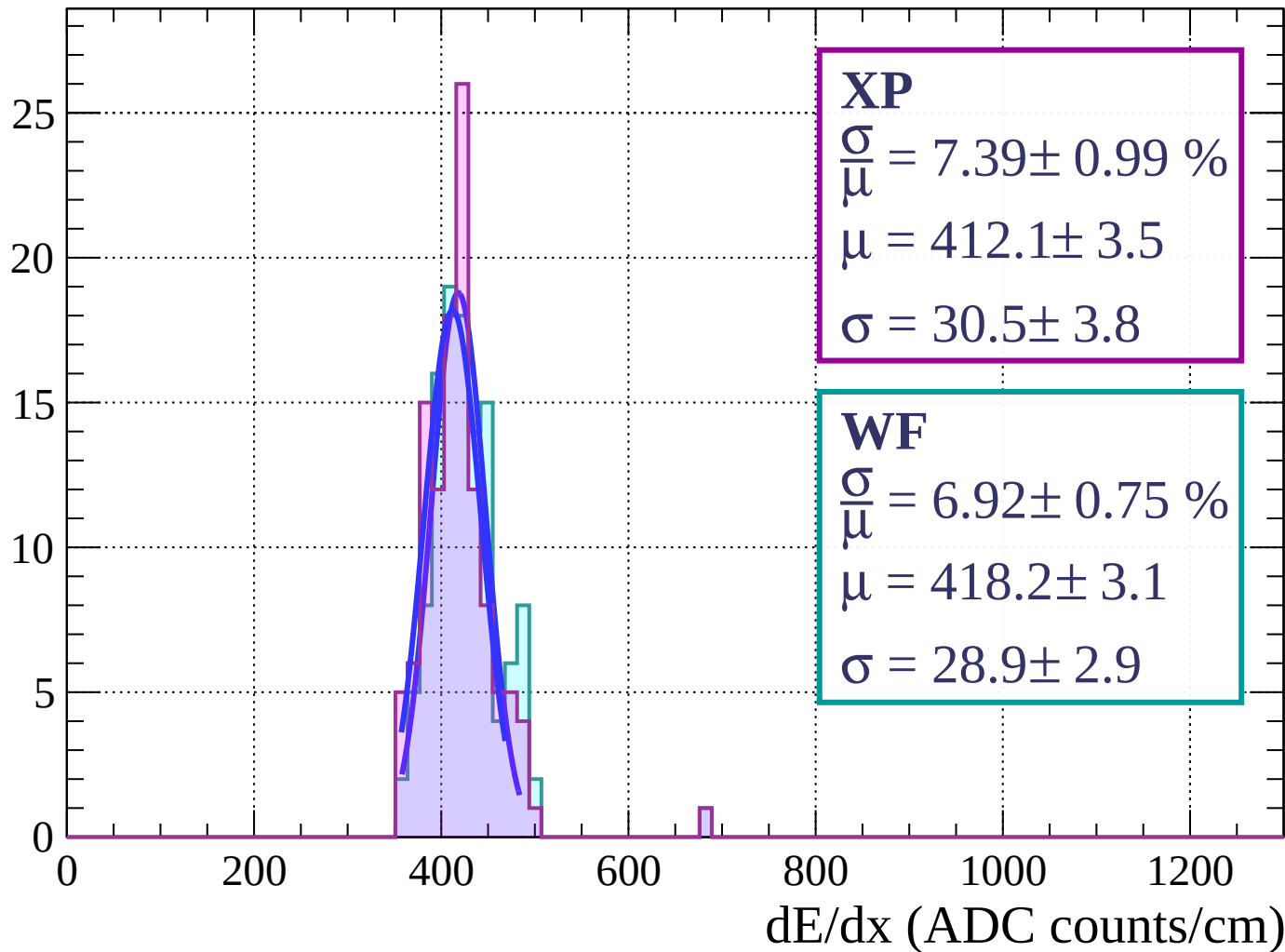
Energy loss | $18 < \phi < 21$

Count



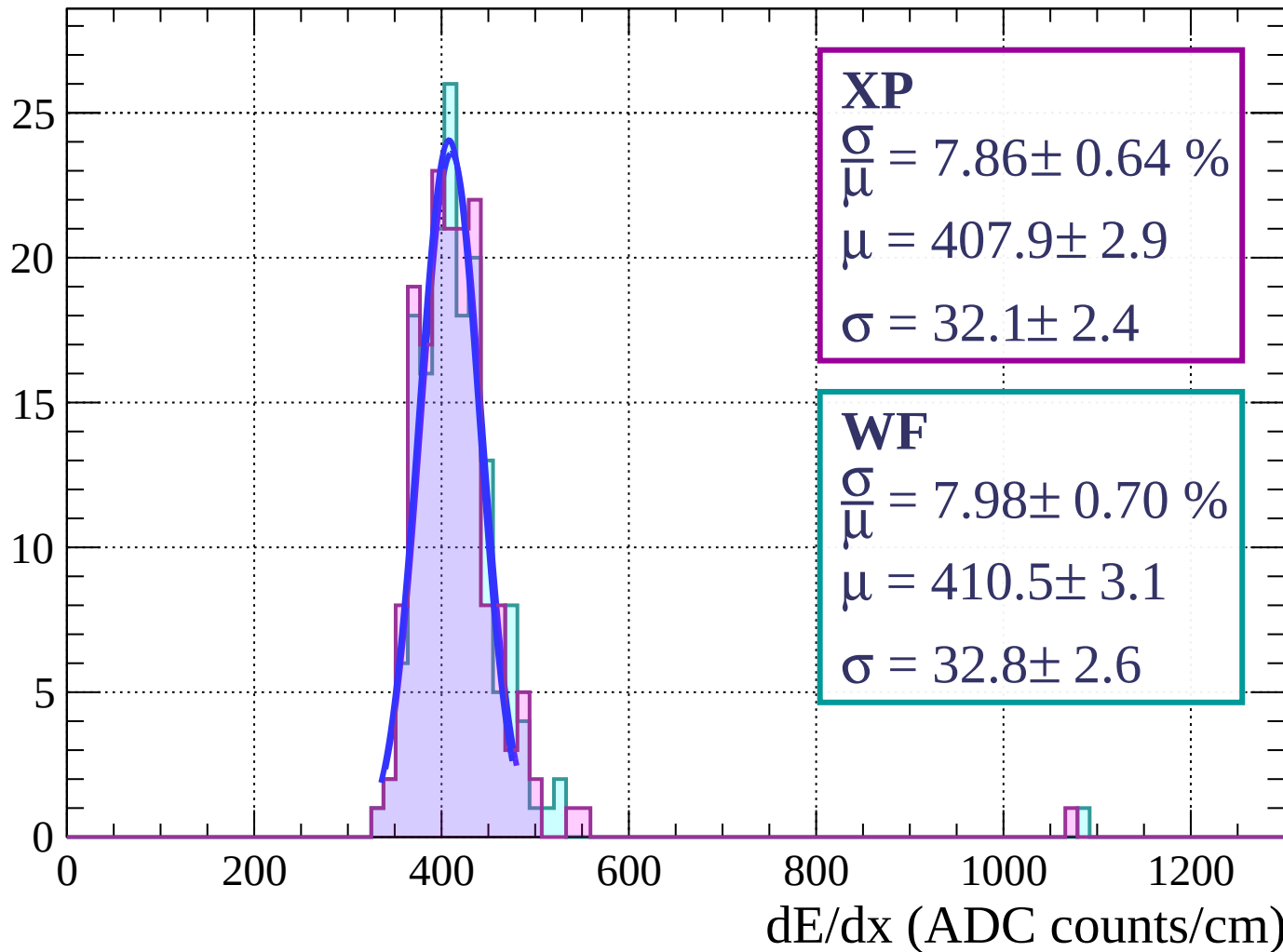
Energy loss | $21 < \phi < 24$

Count



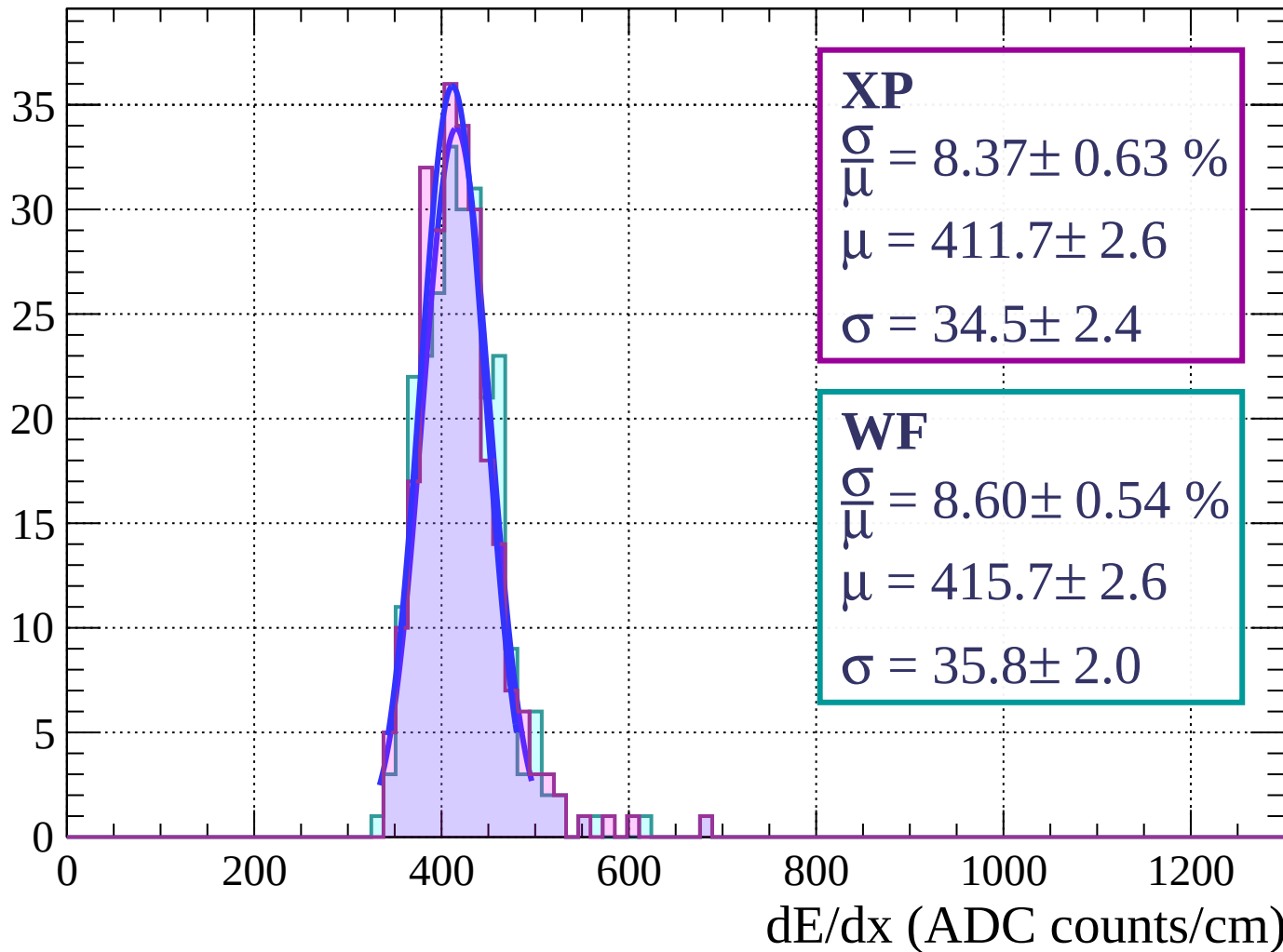
Energy loss | $24 < \phi < 27$

Count



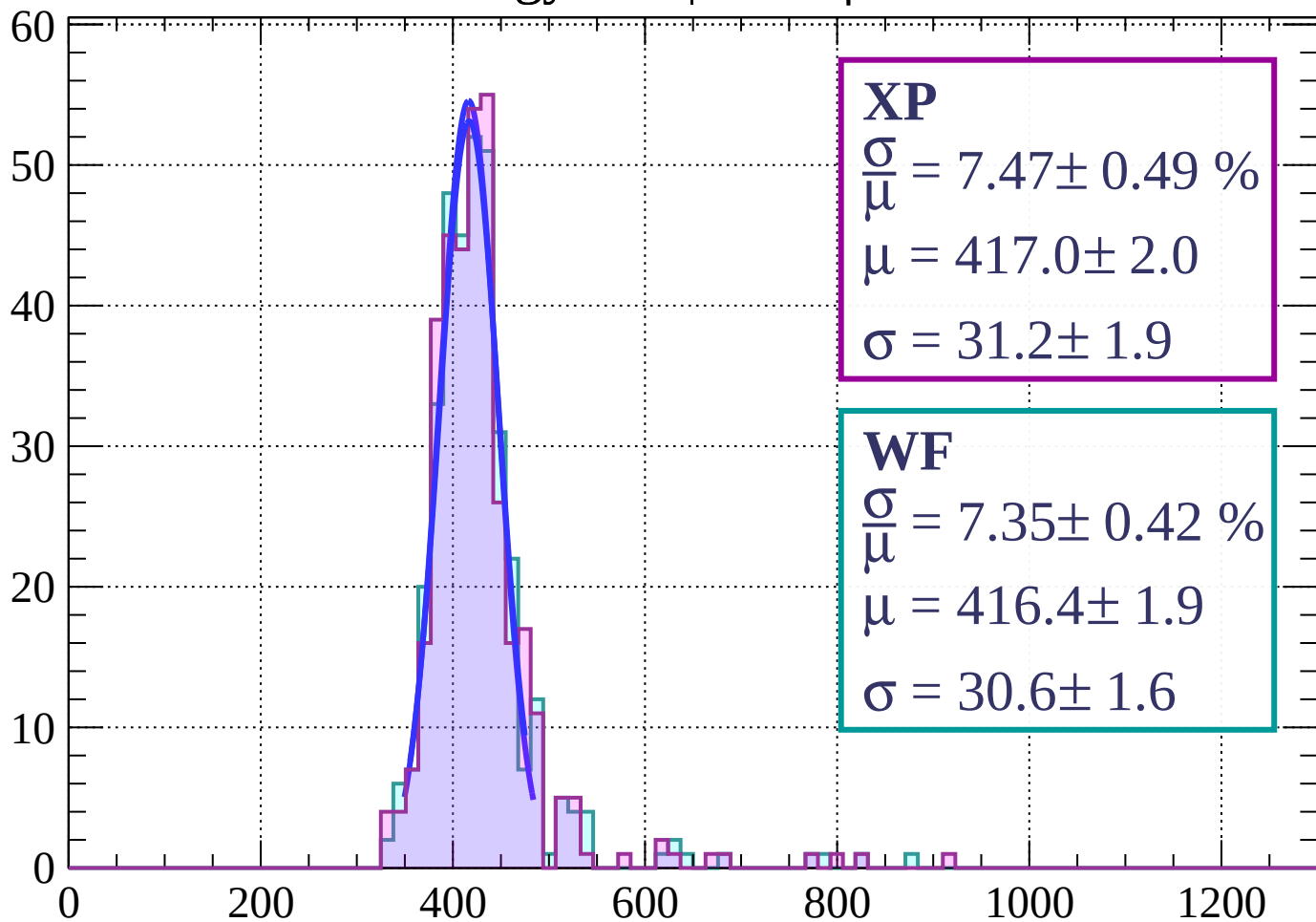
Energy loss | $27 < \phi < 30$

Count



Energy loss | $30 < \phi < 33$

Count



XP

$$\frac{\sigma}{\mu} = 7.47 \pm 0.49 \%$$

$$\mu = 417.0 \pm 2.0$$

$$\sigma = 31.2 \pm 1.9$$

WF

$$\frac{\sigma}{\mu} = 7.35 \pm 0.42 \%$$

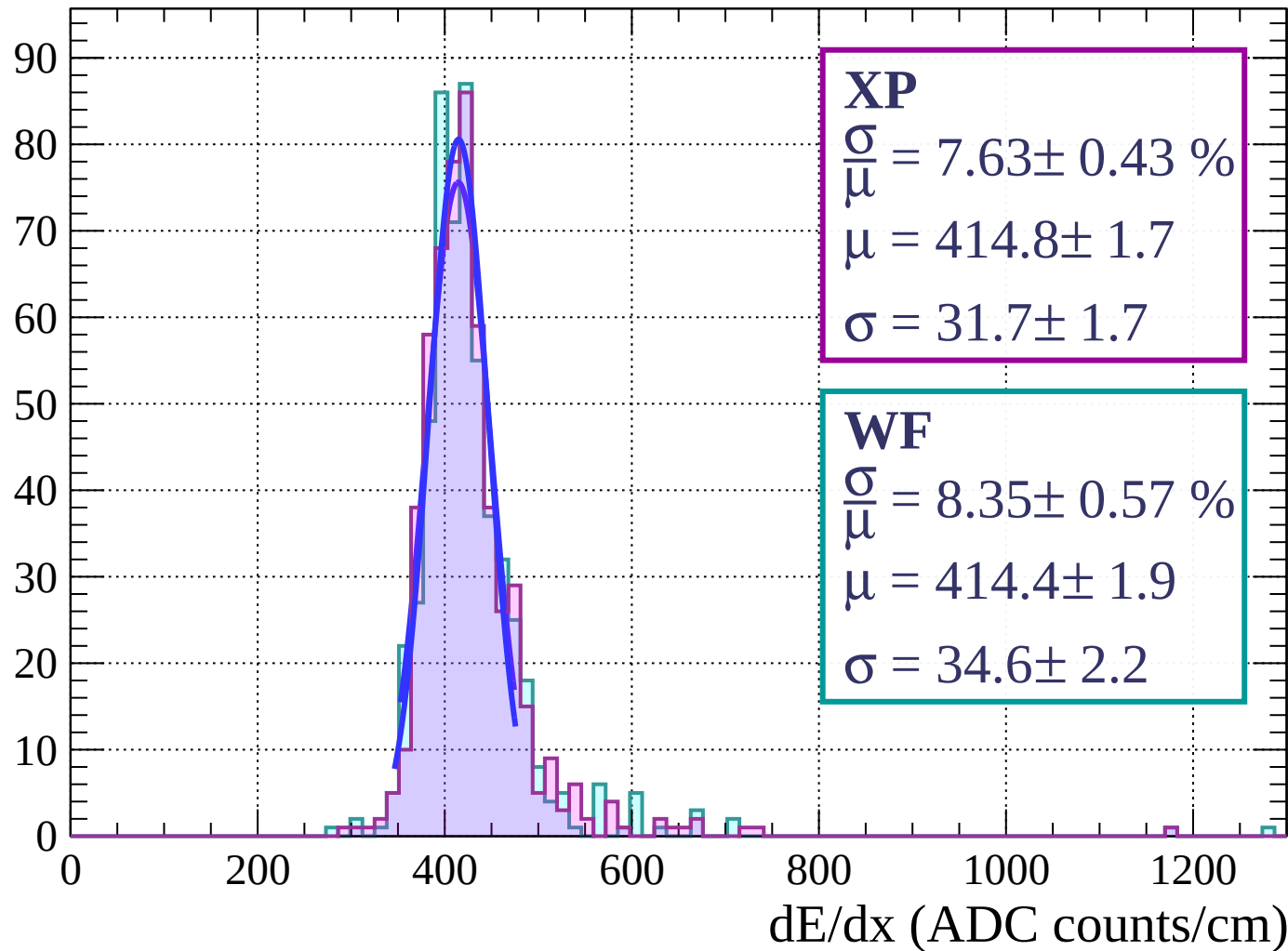
$$\mu = 416.4 \pm 1.9$$

$$\sigma = 30.6 \pm 1.6$$

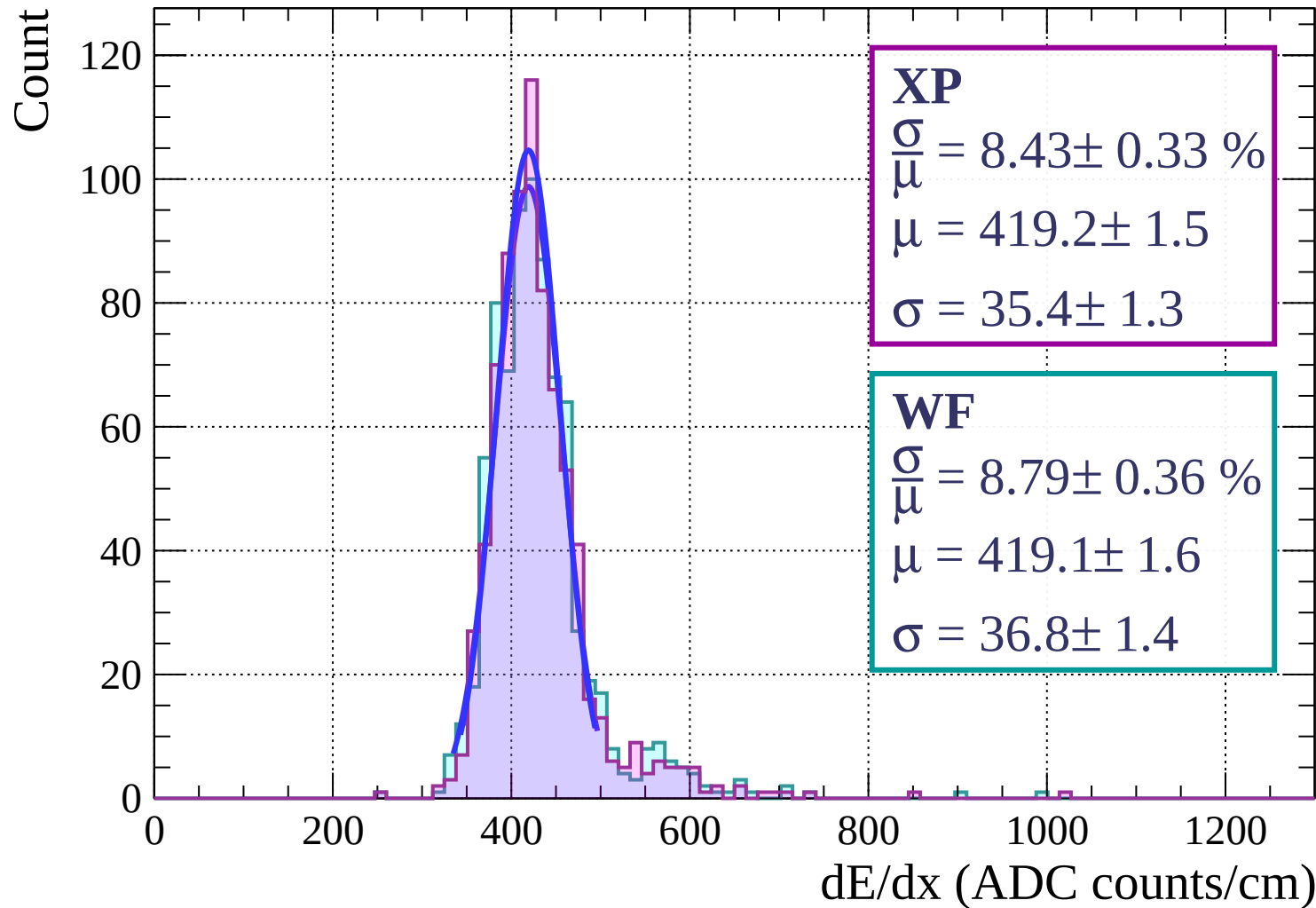
dE/dx (ADC counts/cm)

Energy loss | $33 < \phi < 36$

Count

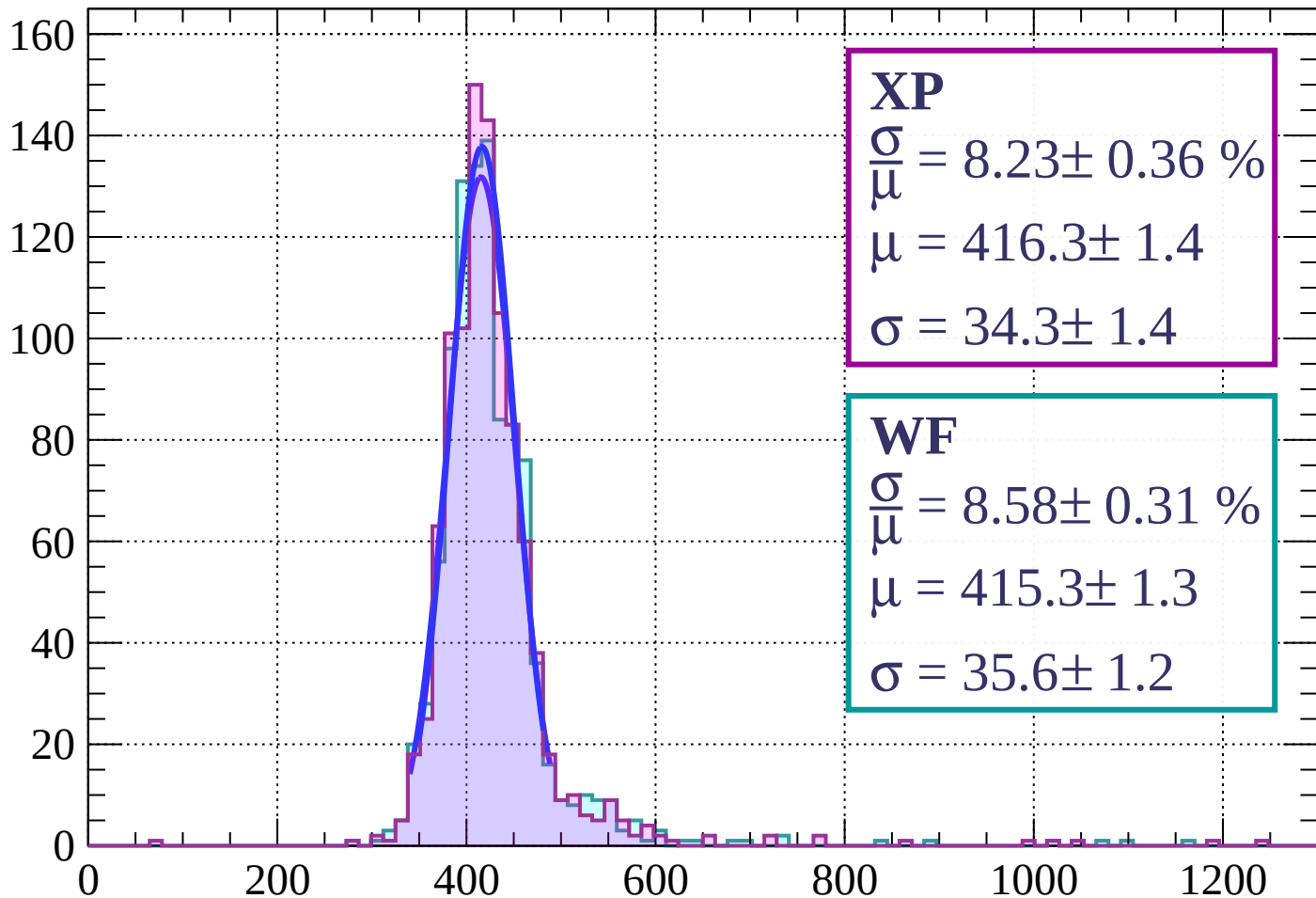


Energy loss | $36 < \phi < 39$



Energy loss | $39 < \phi < 42$

Count



XP

$$\frac{\sigma}{\mu} = 8.23 \pm 0.36 \%$$

$$\mu = 416.3 \pm 1.4$$

$$\sigma = 34.3 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 8.58 \pm 0.31 \%$$

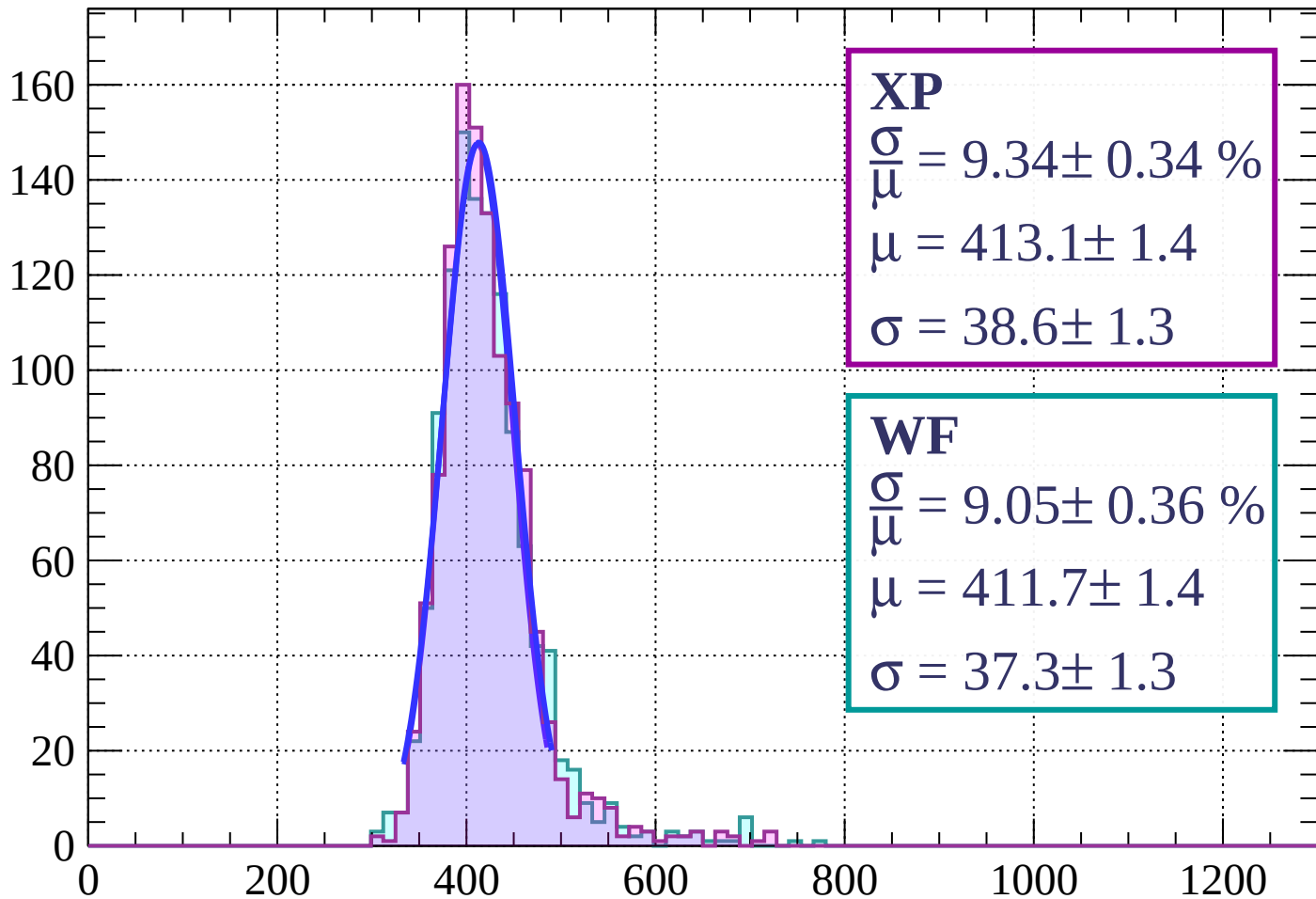
$$\mu = 415.3 \pm 1.3$$

$$\sigma = 35.6 \pm 1.2$$

dE/dx (ADC counts/cm)

Energy loss | $42 < \phi < 45$

Count



XP

$$\frac{\sigma}{\mu} = 9.34 \pm 0.34 \%$$

$$\mu = 413.1 \pm 1.4$$

$$\sigma = 38.6 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 9.05 \pm 0.36 \%$$

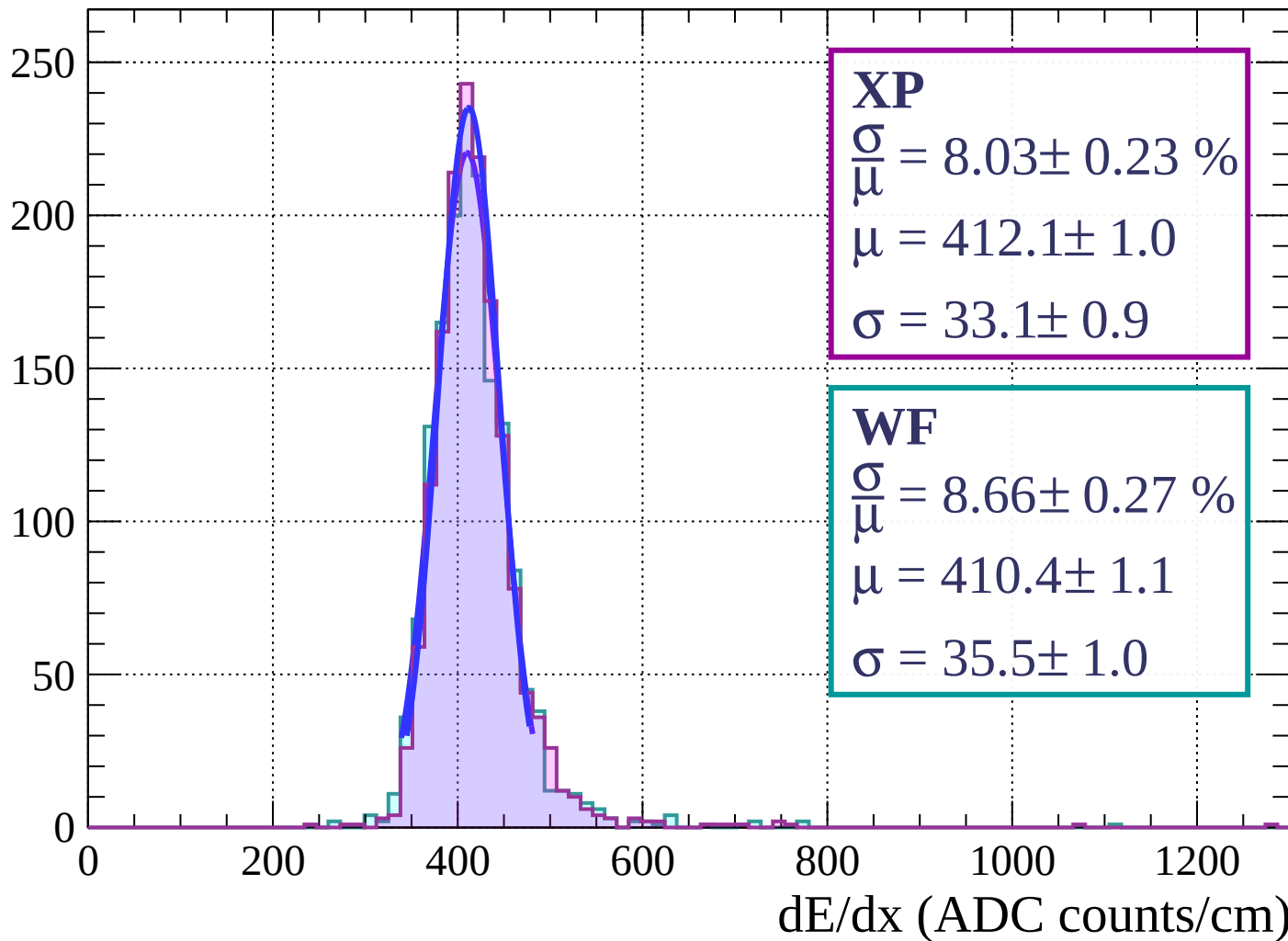
$$\mu = 411.7 \pm 1.4$$

$$\sigma = 37.3 \pm 1.3$$

dE/dx (ADC counts/cm)

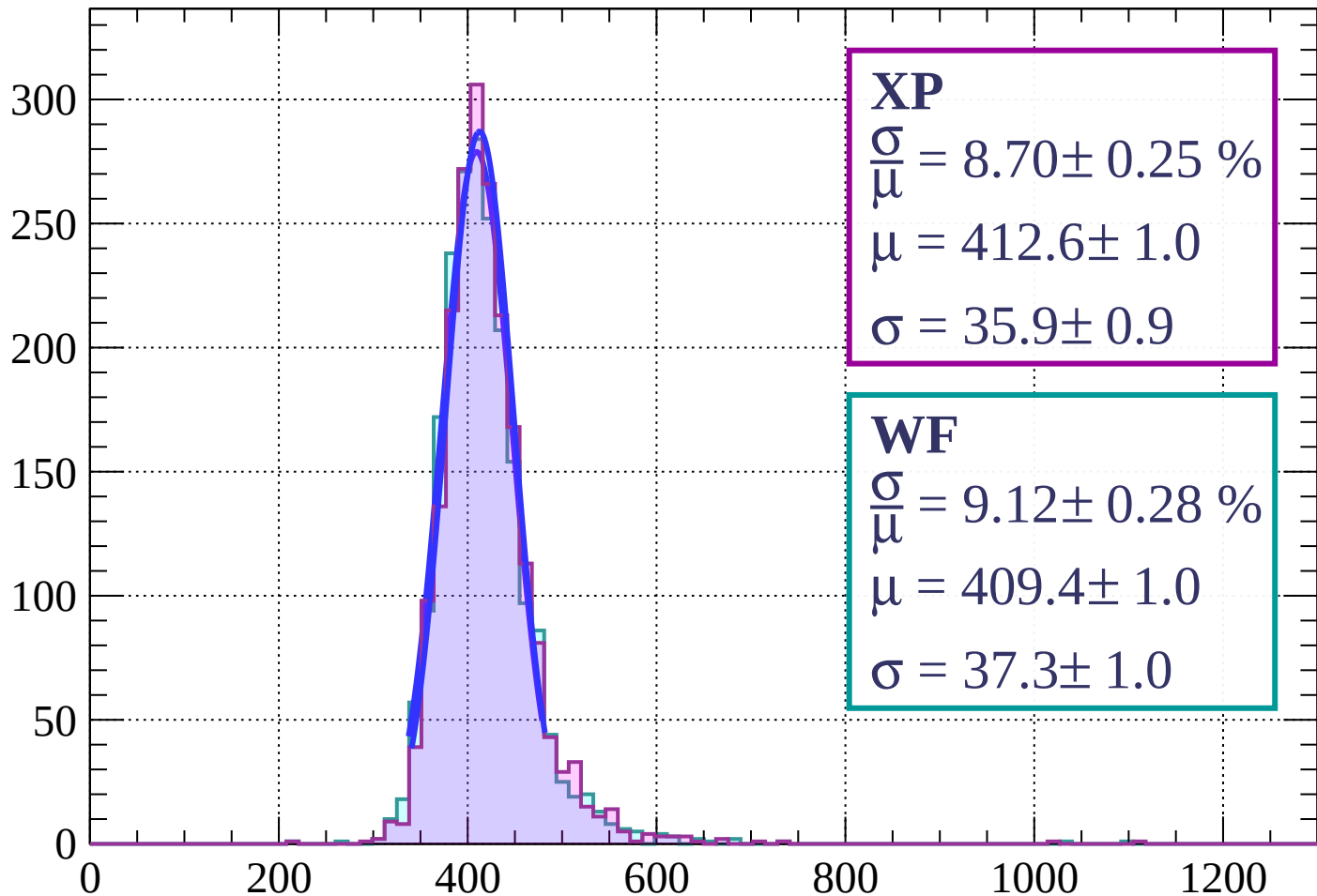
Energy loss | $45 < \phi < 48$

Count



Energy loss | $48 < \phi < 51$

Count



XP

$$\frac{\sigma}{\mu} = 8.70 \pm 0.25 \%$$

$$\mu = 412.6 \pm 1.0$$

$$\sigma = 35.9 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 9.12 \pm 0.28 \%$$

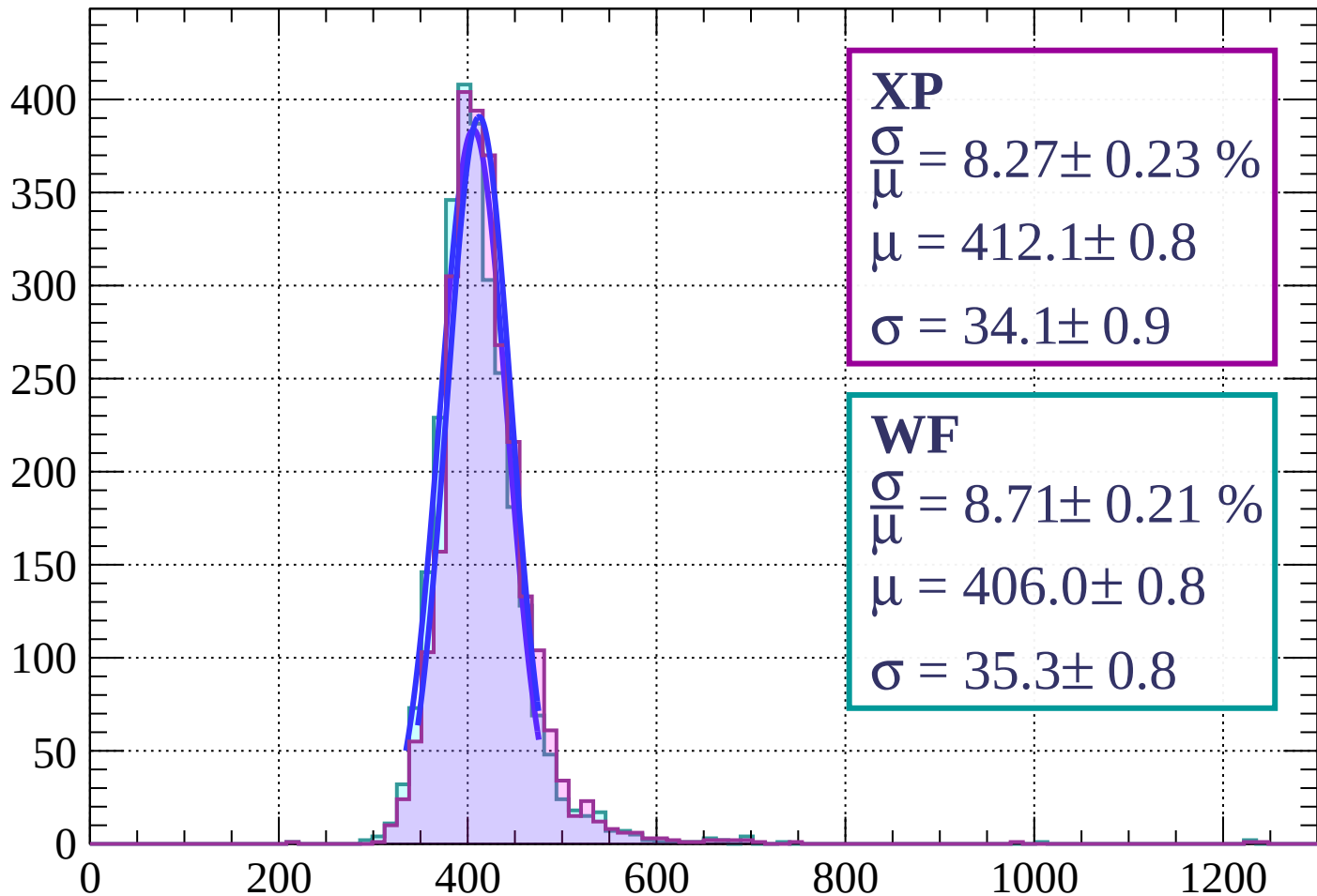
$$\mu = 409.4 \pm 1.0$$

$$\sigma = 37.3 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $51 < \phi < 54$

Count



XP

$$\frac{\sigma}{\mu} = 8.27 \pm 0.23 \%$$

$$\mu = 412.1 \pm 0.8$$

$$\sigma = 34.1 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 8.71 \pm 0.21 \%$$

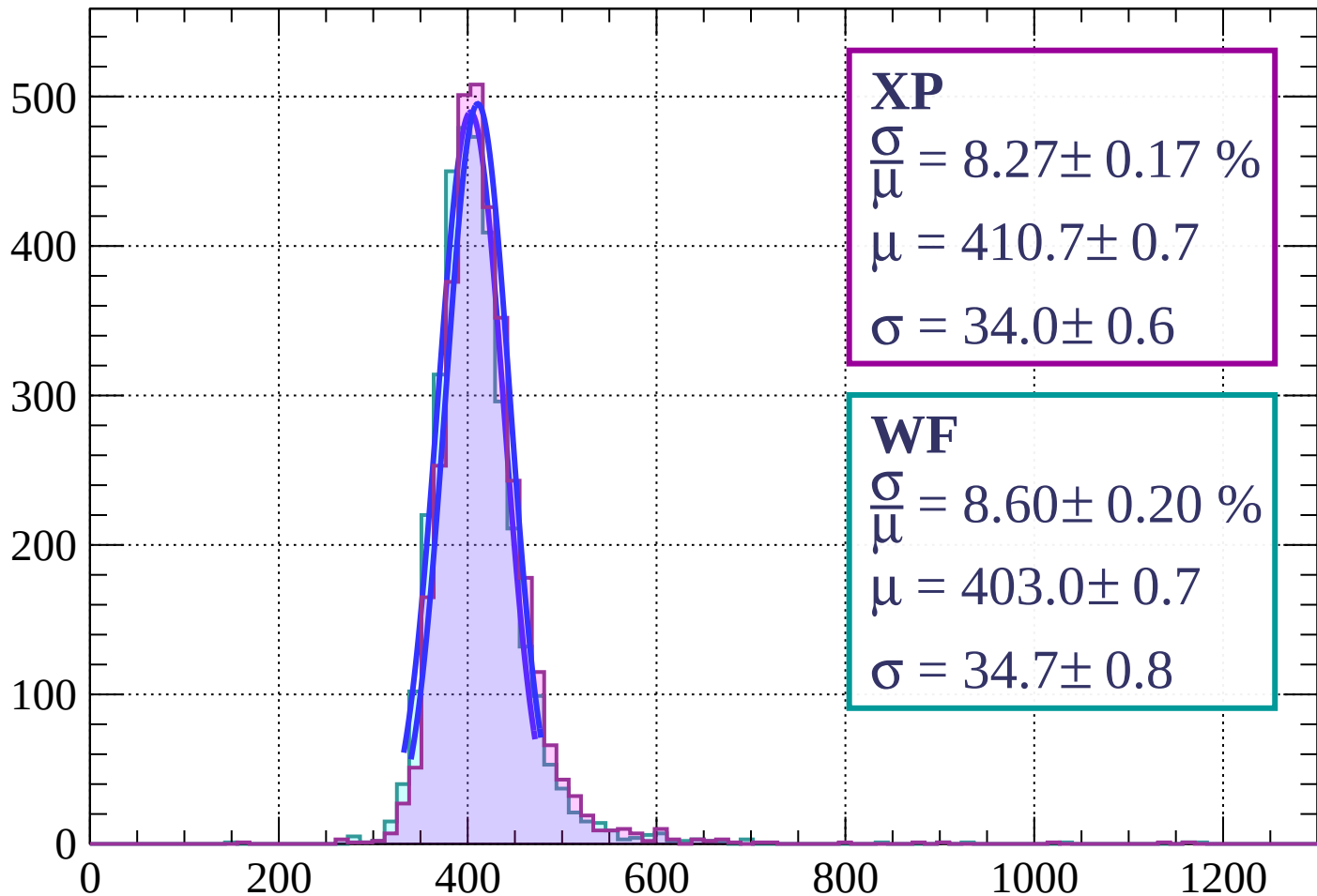
$$\mu = 406.0 \pm 0.8$$

$$\sigma = 35.3 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $54 < \phi < 57$

Count



XP

$$\frac{\sigma}{\mu} = 8.27 \pm 0.17 \%$$

$$\mu = 410.7 \pm 0.7$$

$$\sigma = 34.0 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 8.60 \pm 0.20 \%$$

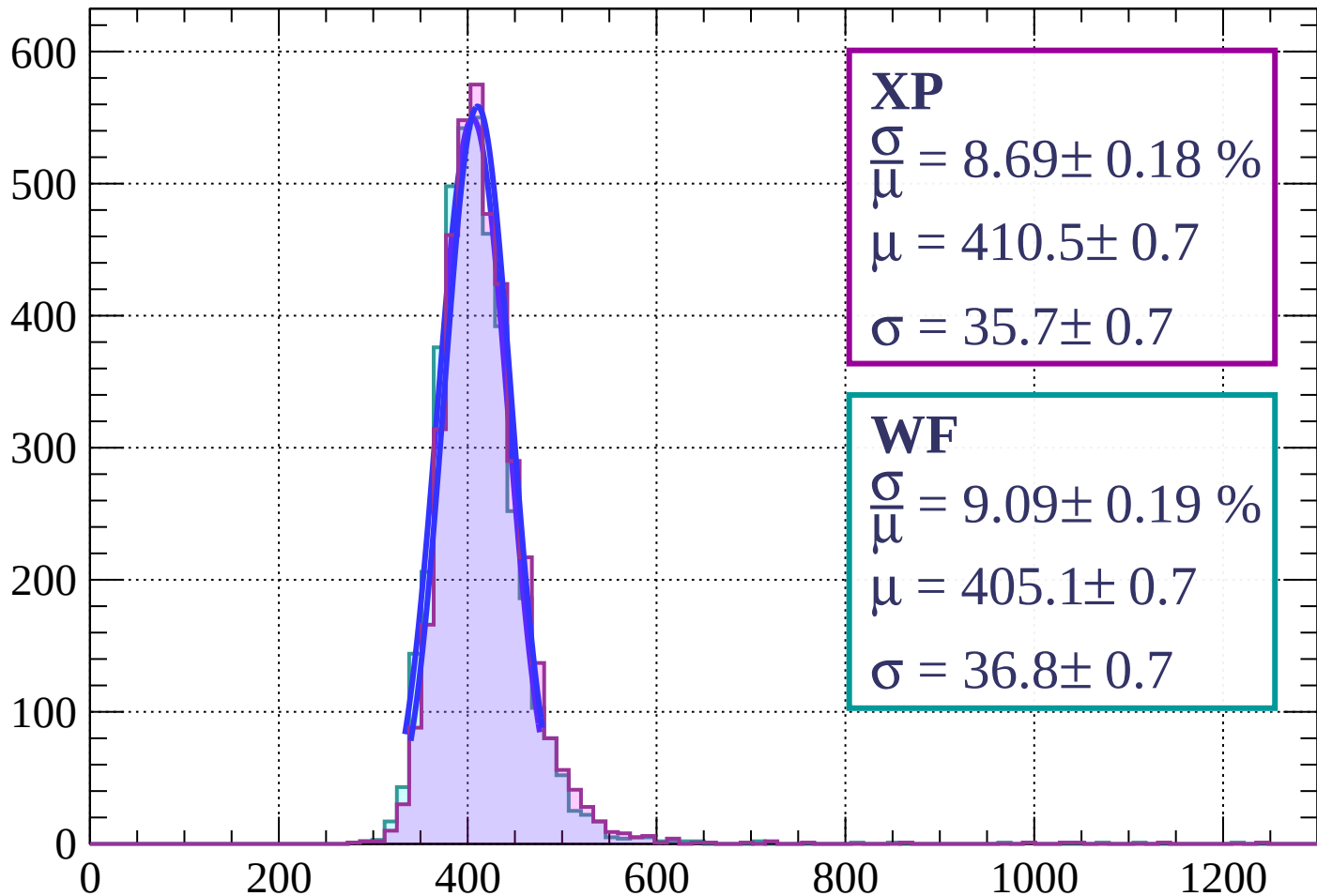
$$\mu = 403.0 \pm 0.7$$

$$\sigma = 34.7 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $57 < \phi < 60$

Count



XP

$$\frac{\sigma}{\mu} = 8.69 \pm 0.18 \%$$

$$\mu = 410.5 \pm 0.7$$

$$\sigma = 35.7 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 9.09 \pm 0.19 \%$$

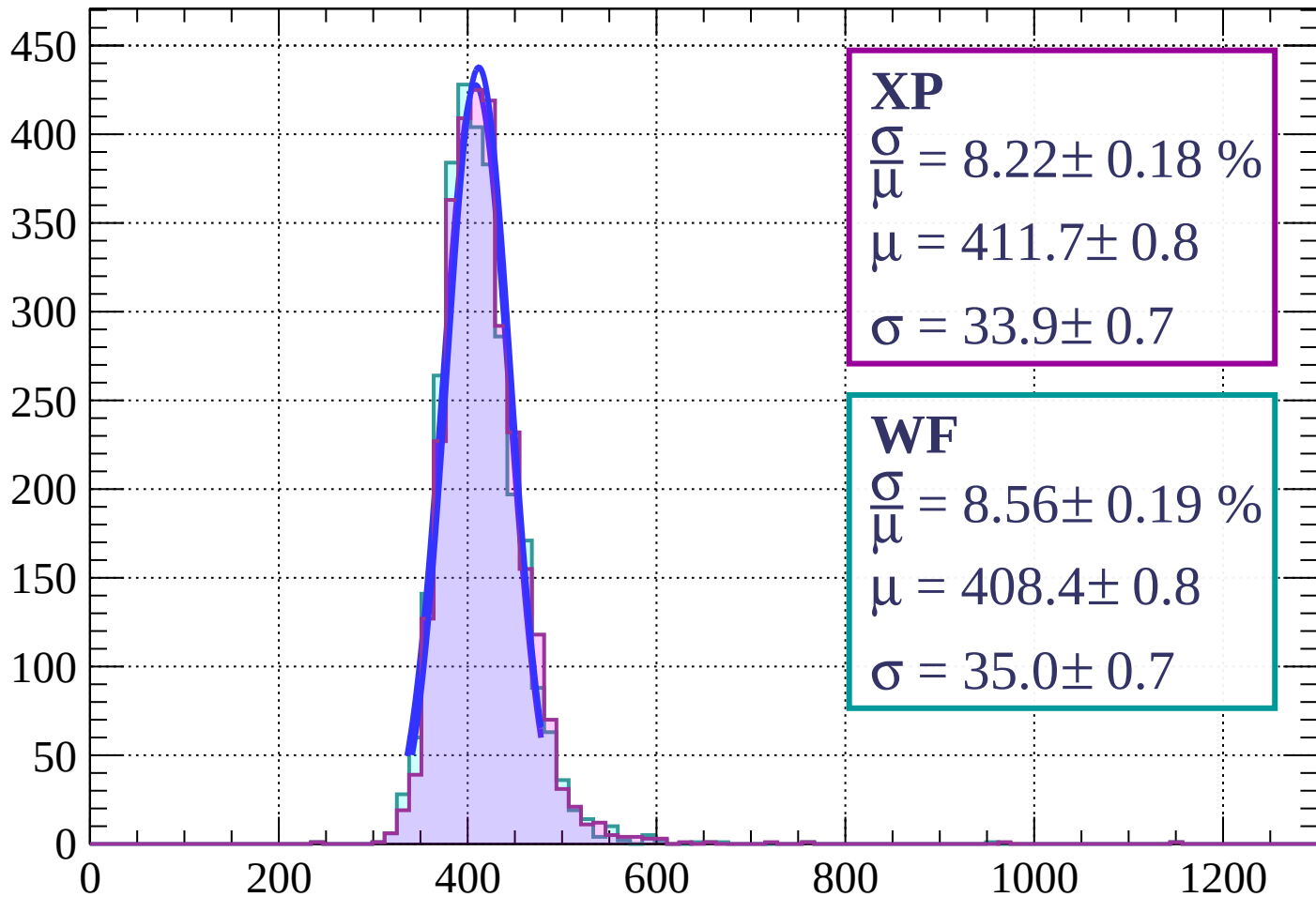
$$\mu = 405.1 \pm 0.7$$

$$\sigma = 36.8 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $60 < \phi < 63$

Count



XP

$$\frac{\sigma}{\mu} = 8.22 \pm 0.18 \%$$

$$\mu = 411.7 \pm 0.8$$

$$\sigma = 33.9 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 8.56 \pm 0.19 \%$$

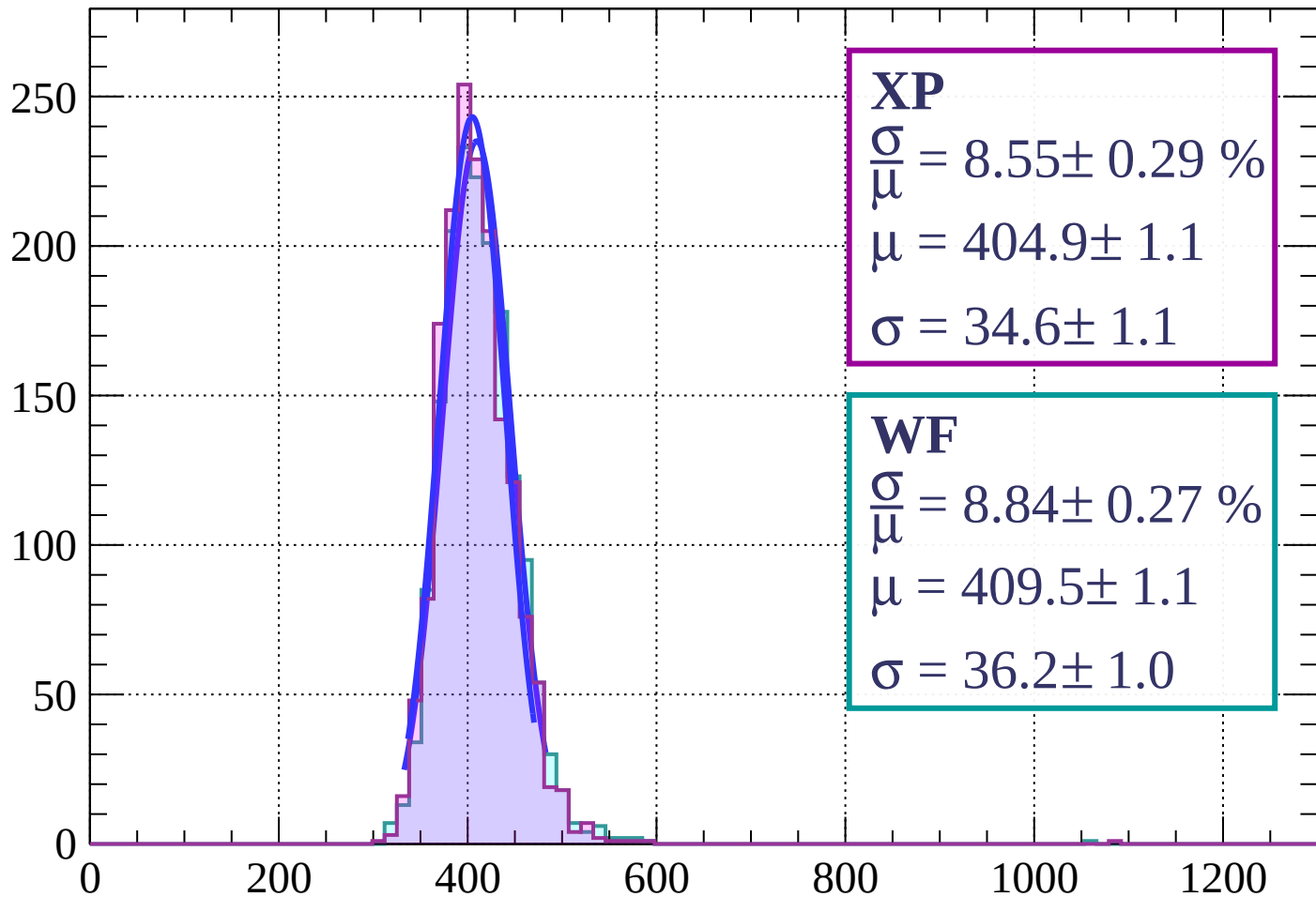
$$\mu = 408.4 \pm 0.8$$

$$\sigma = 35.0 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $63 < \phi < 66$

Count



XP

$$\frac{\sigma}{\mu} = 8.55 \pm 0.29 \%$$

$$\mu = 404.9 \pm 1.1$$

$$\sigma = 34.6 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 8.84 \pm 0.27 \%$$

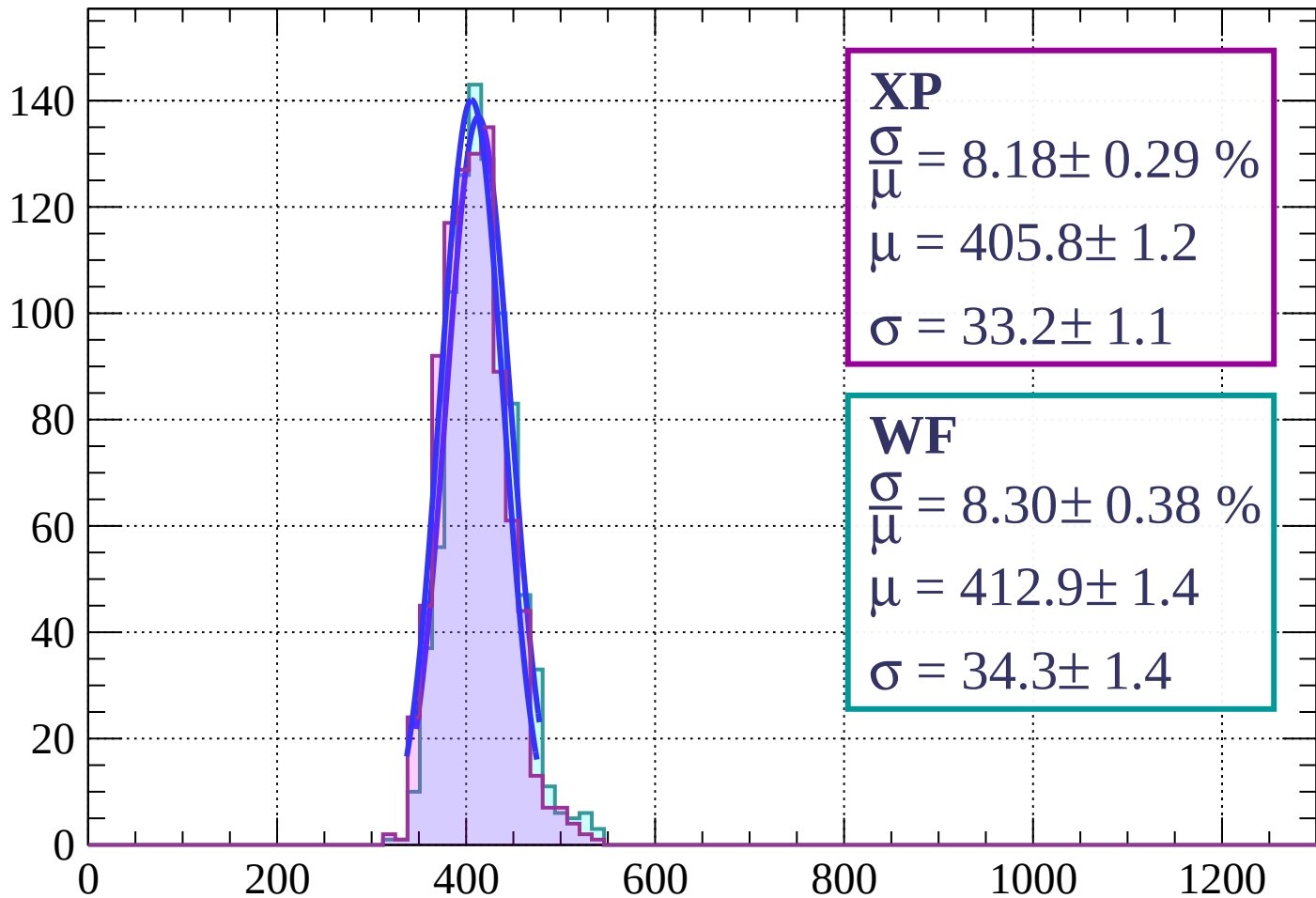
$$\mu = 409.5 \pm 1.1$$

$$\sigma = 36.2 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $66 < \phi < 69$

Count



XP

$$\frac{\sigma}{\mu} = 8.18 \pm 0.29 \%$$

$$\mu = 405.8 \pm 1.2$$

$$\sigma = 33.2 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 8.30 \pm 0.38 \%$$

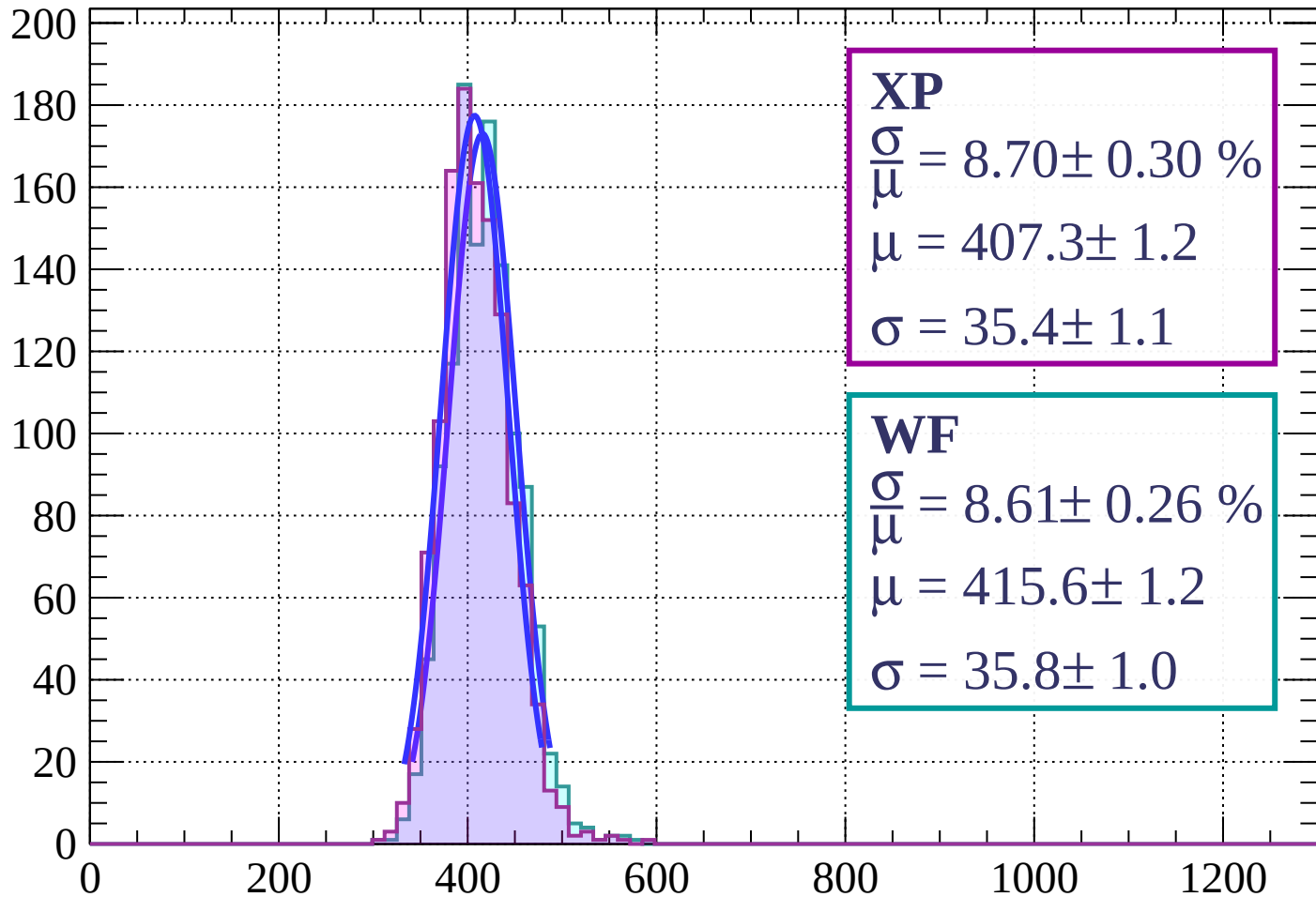
$$\mu = 412.9 \pm 1.4$$

$$\sigma = 34.3 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss | $69 < \phi < 72$

Count



XP

$$\frac{\sigma}{\mu} = 8.70 \pm 0.30 \%$$

$$\mu = 407.3 \pm 1.2$$

$$\sigma = 35.4 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 8.61 \pm 0.26 \%$$

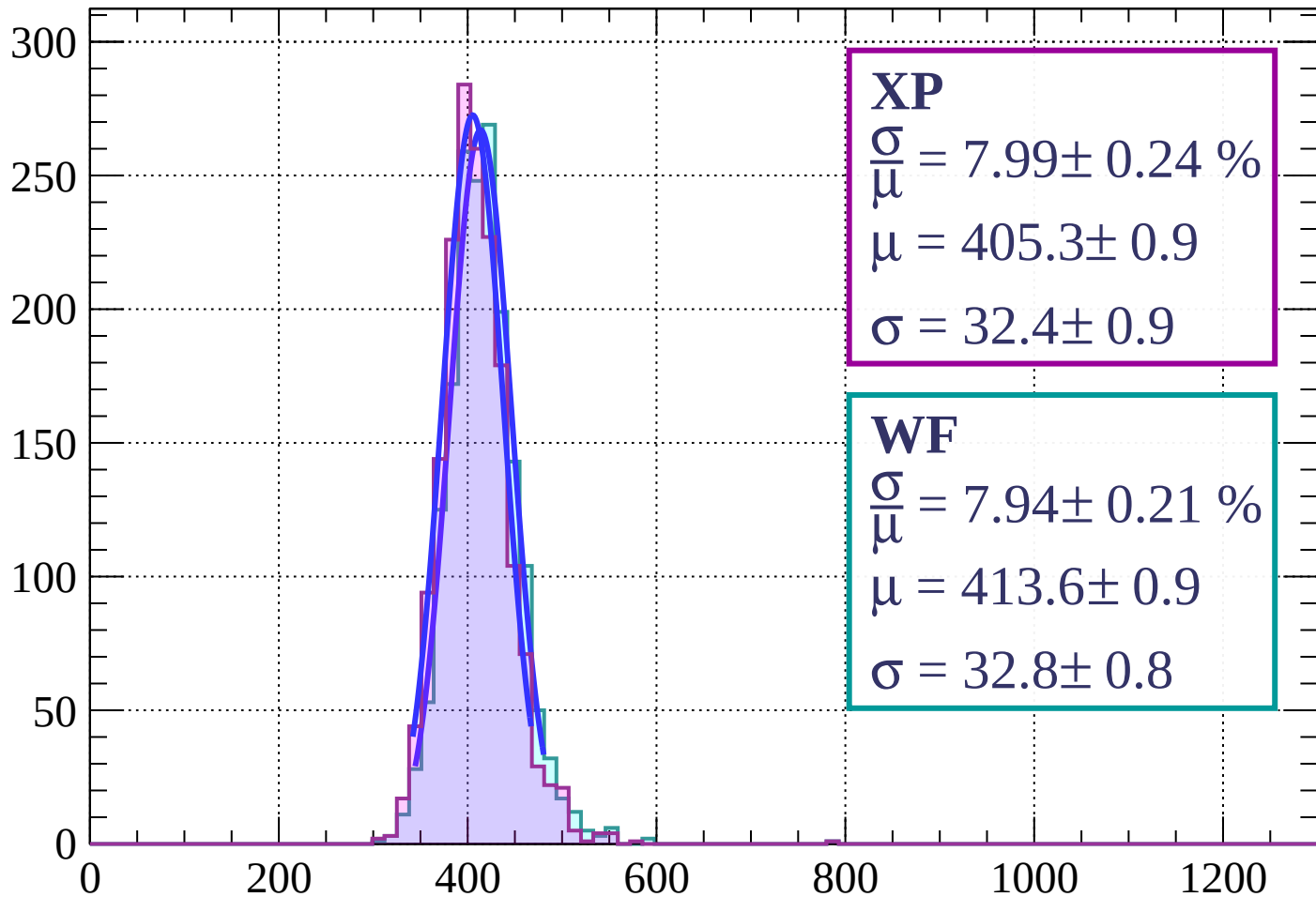
$$\mu = 415.6 \pm 1.2$$

$$\sigma = 35.8 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $72 < \phi < 75$

Count



XP

$$\frac{\sigma}{\mu} = 7.99 \pm 0.24 \%$$

$$\mu = 405.3 \pm 0.9$$

$$\sigma = 32.4 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 7.94 \pm 0.21 \%$$

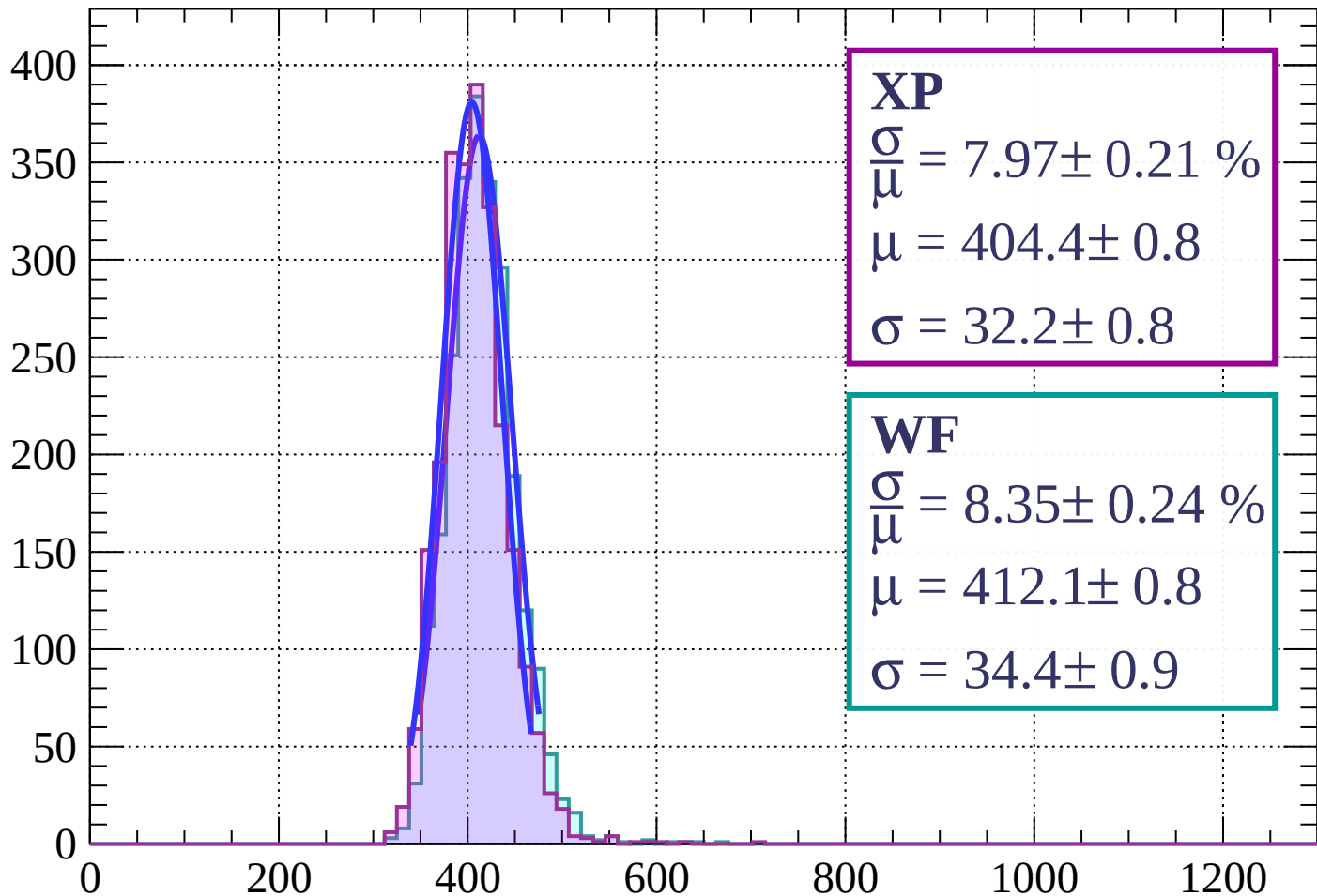
$$\mu = 413.6 \pm 0.9$$

$$\sigma = 32.8 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $75 < \phi < 78$

Count



XP

$$\frac{\sigma}{\mu} = 7.97 \pm 0.21 \%$$

$$\mu = 404.4 \pm 0.8$$

$$\sigma = 32.2 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 8.35 \pm 0.24 \%$$

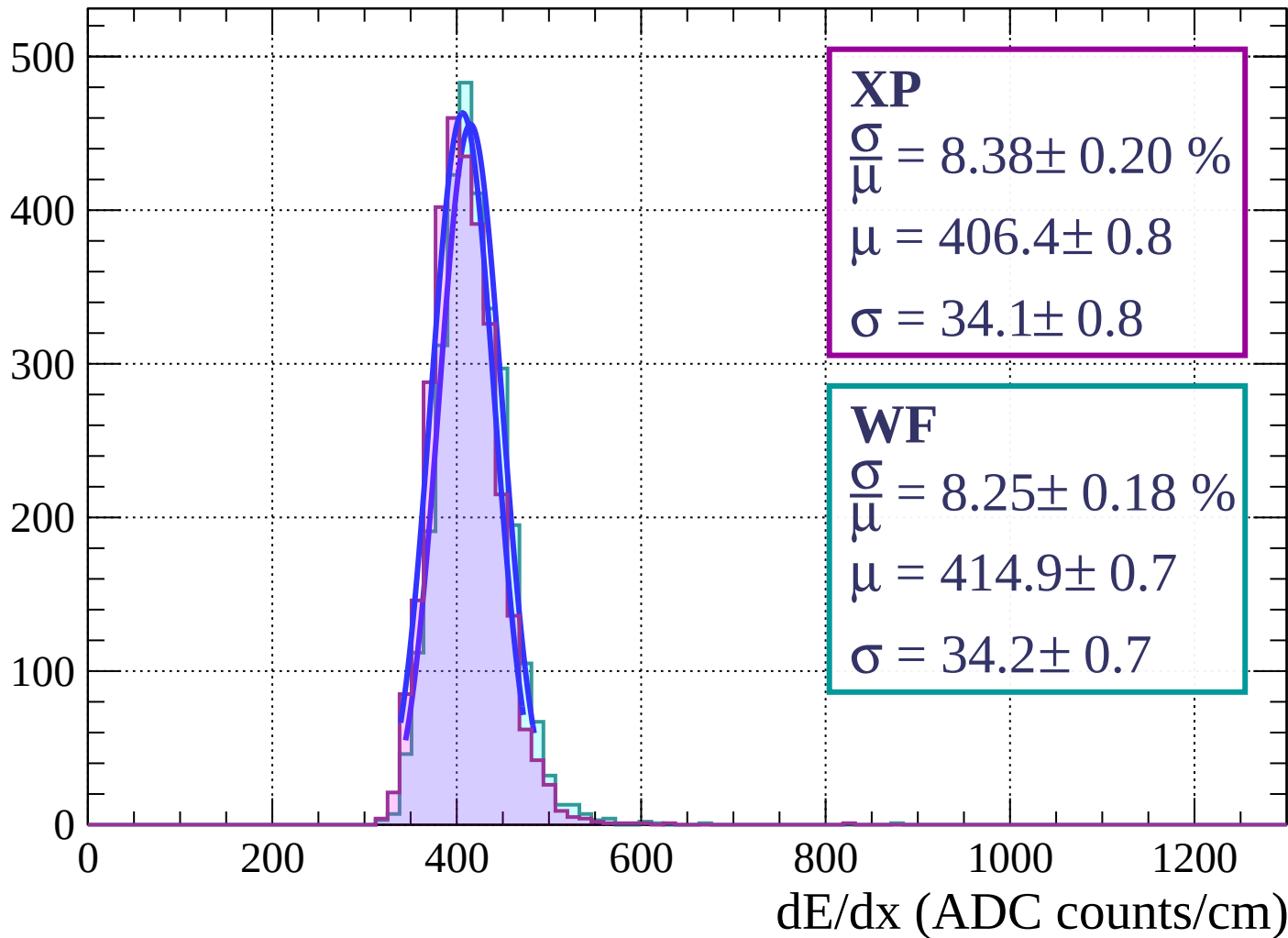
$$\mu = 412.1 \pm 0.8$$

$$\sigma = 34.4 \pm 0.9$$

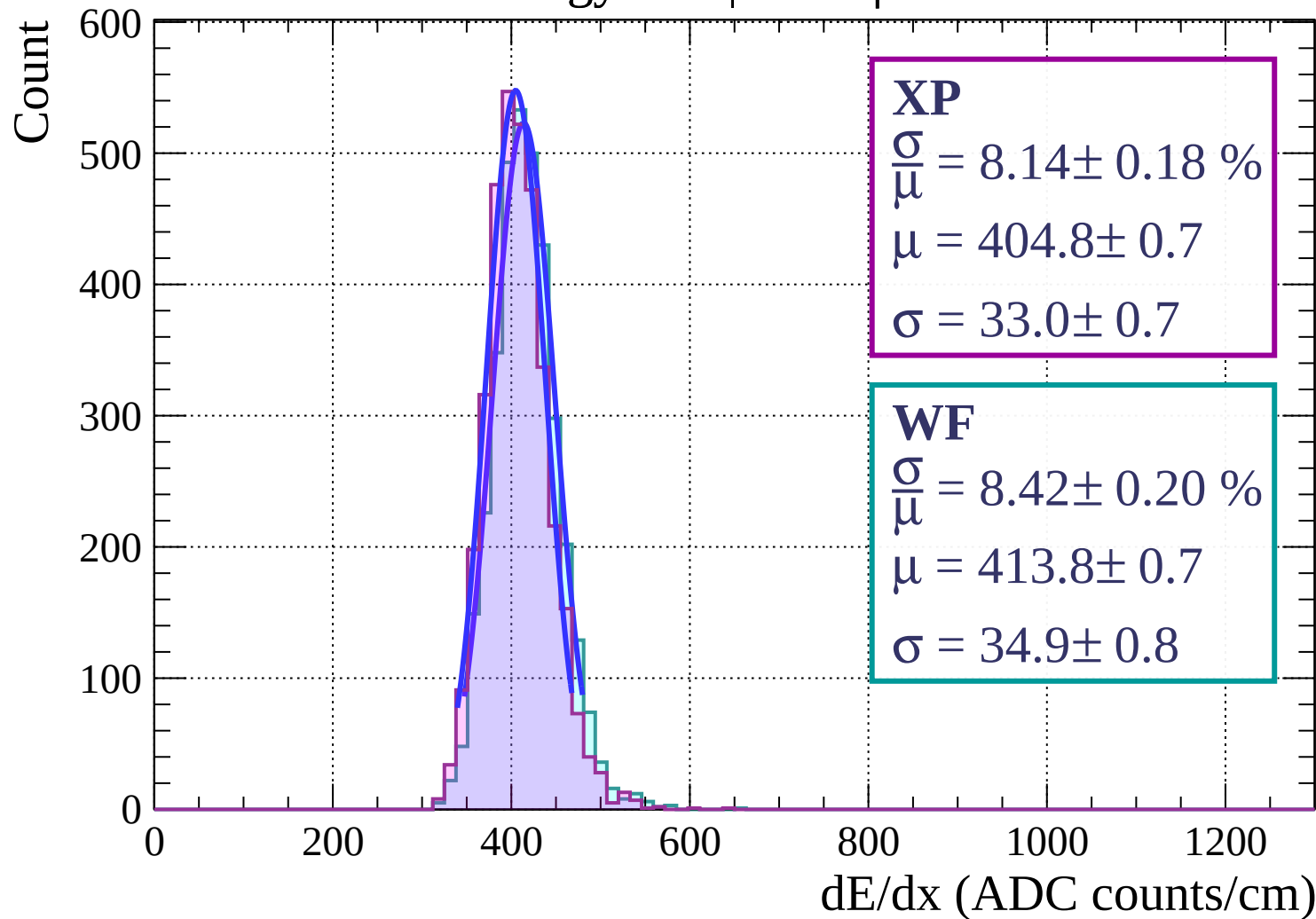
dE/dx (ADC counts/cm)

Energy loss | $78 < \phi < 81$

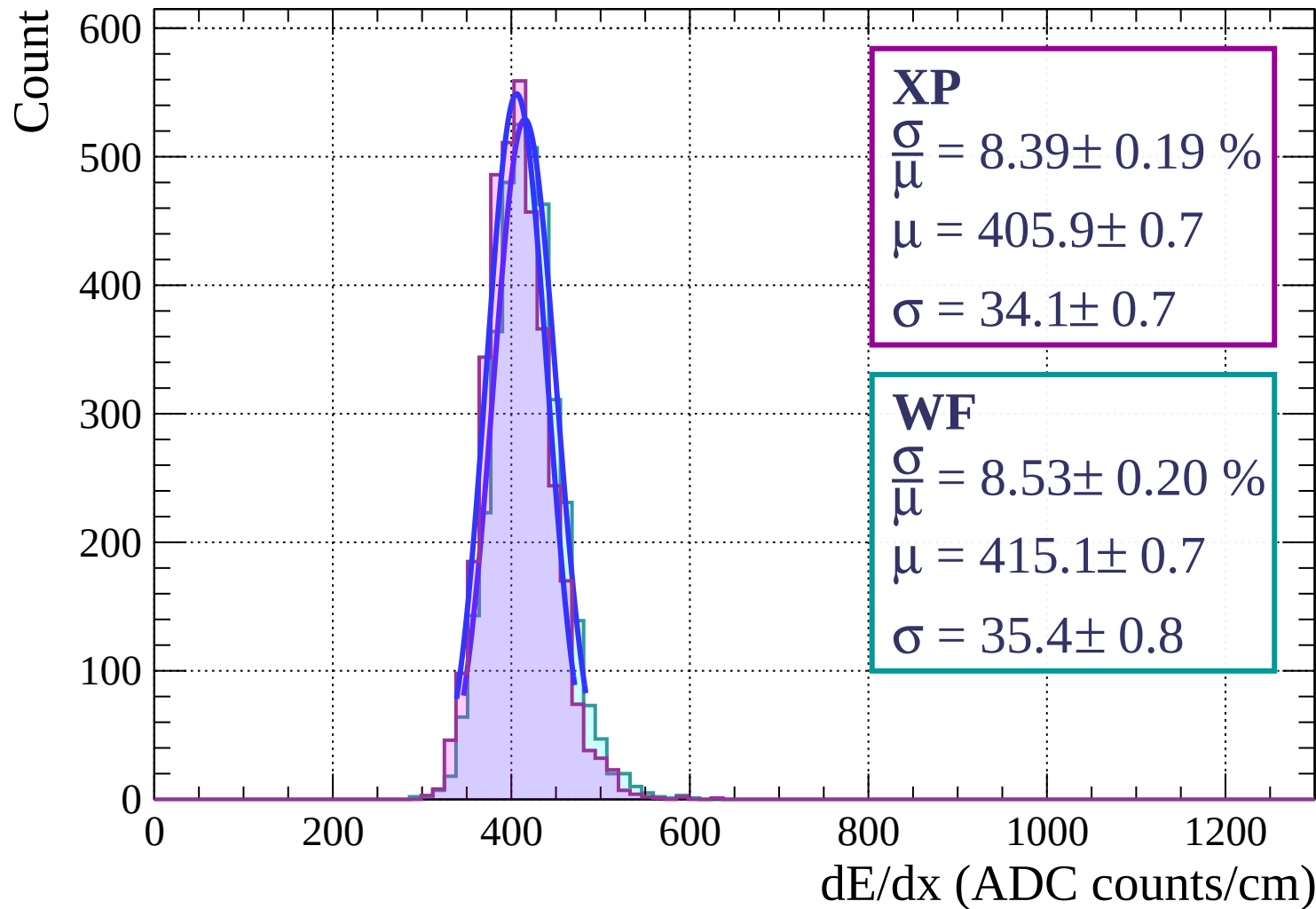
Count



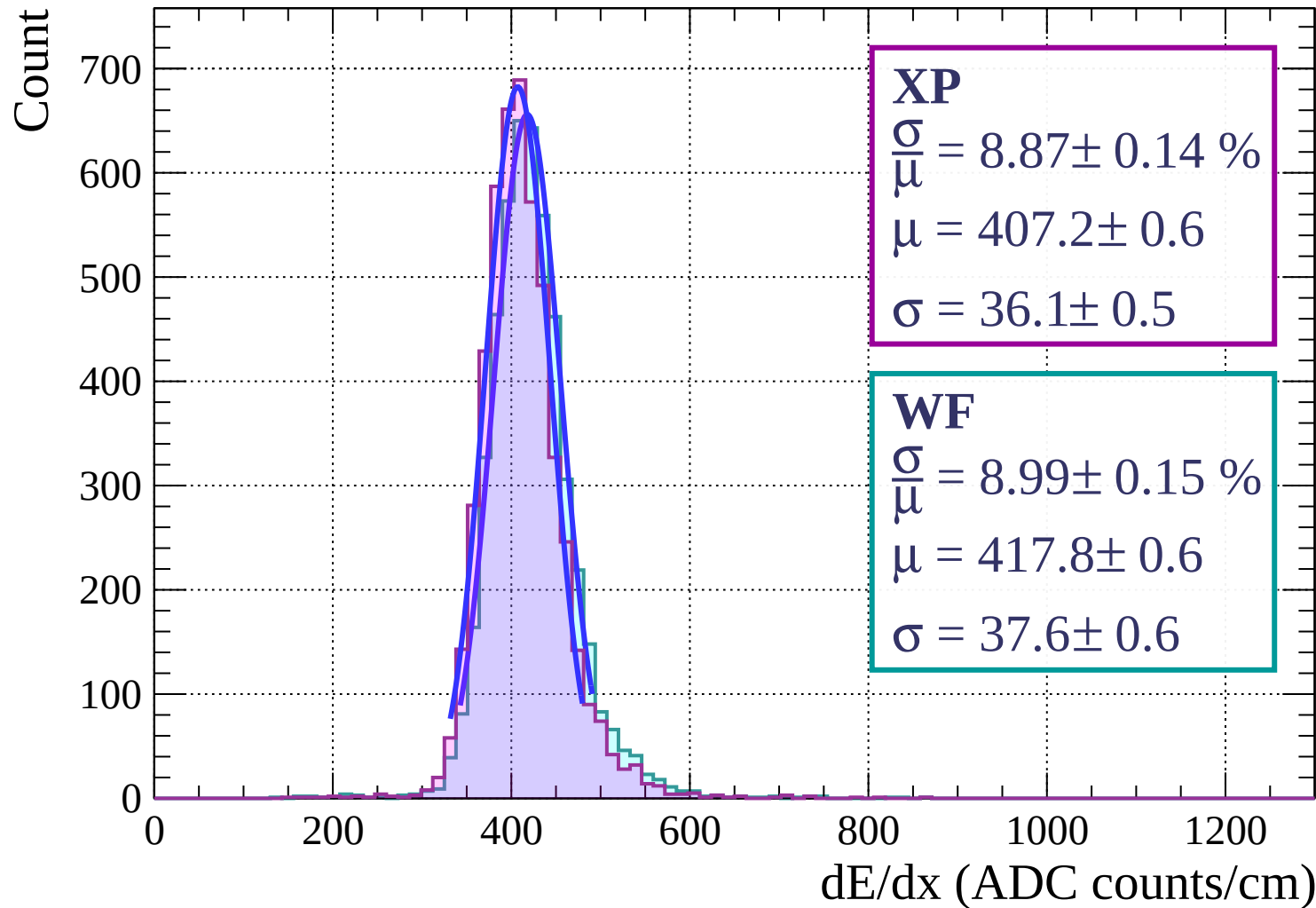
Energy loss | $81 < \phi < 84$



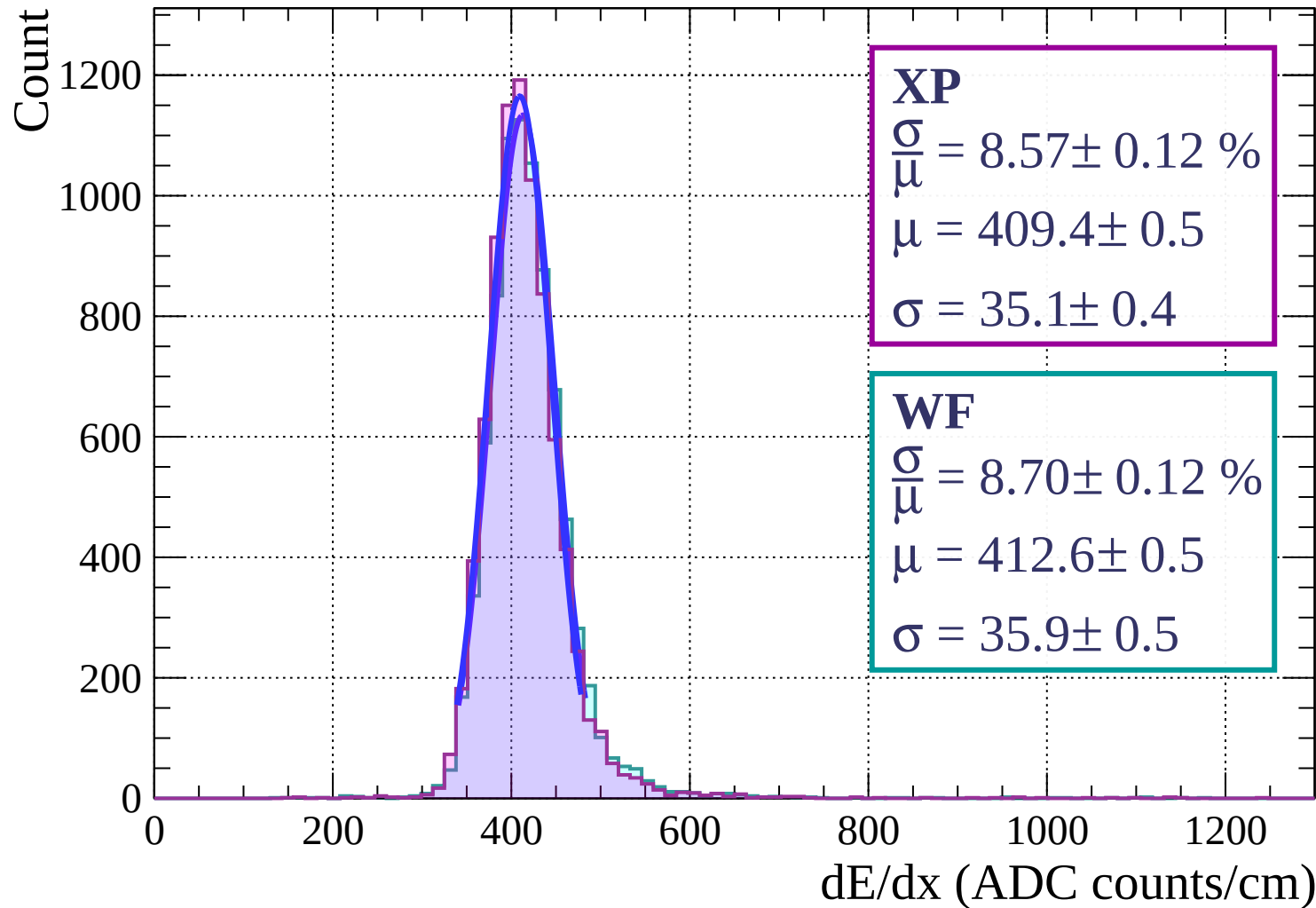
Energy loss | $84 < \phi < 87$



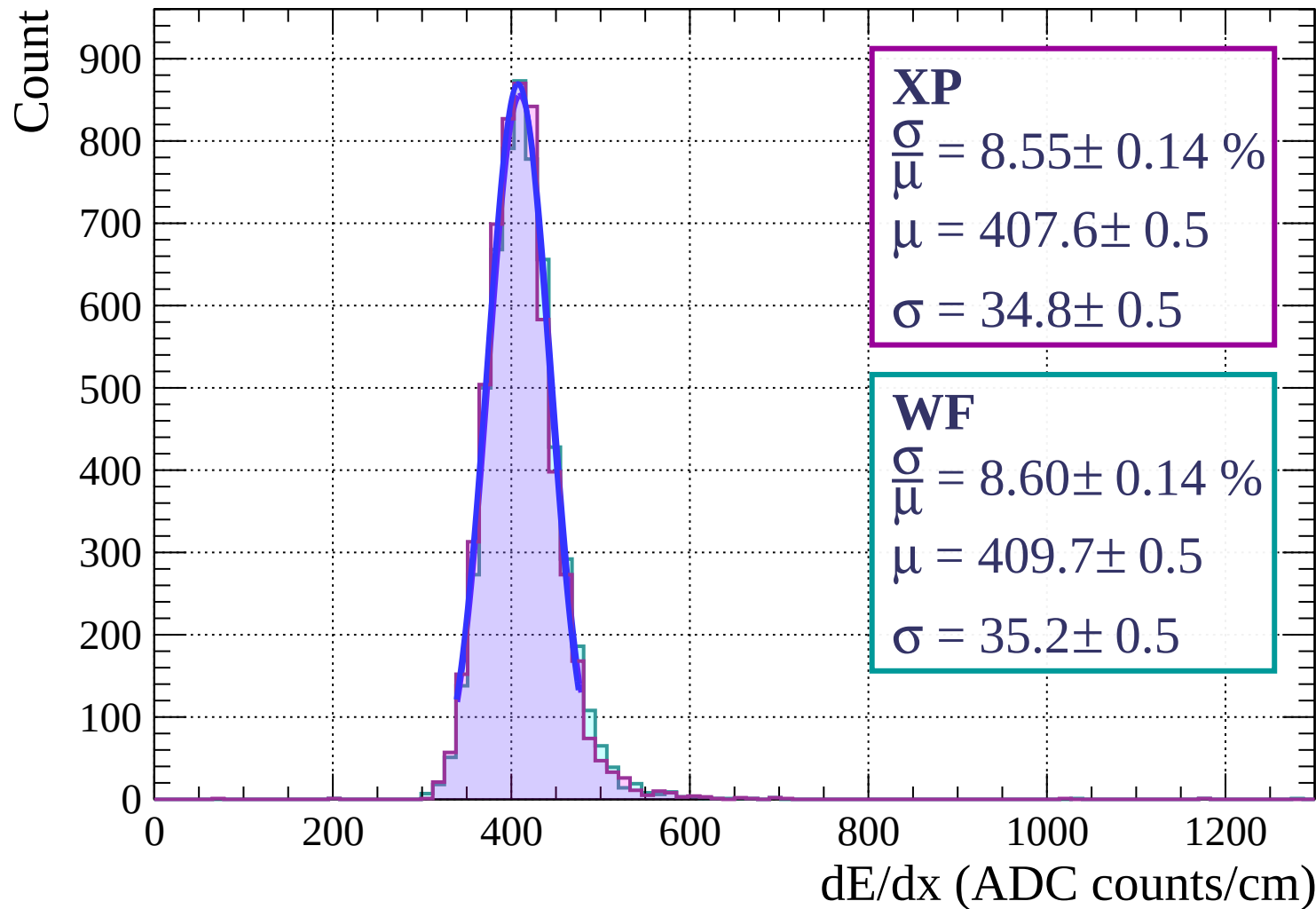
Energy loss | $87 < \phi < 90$



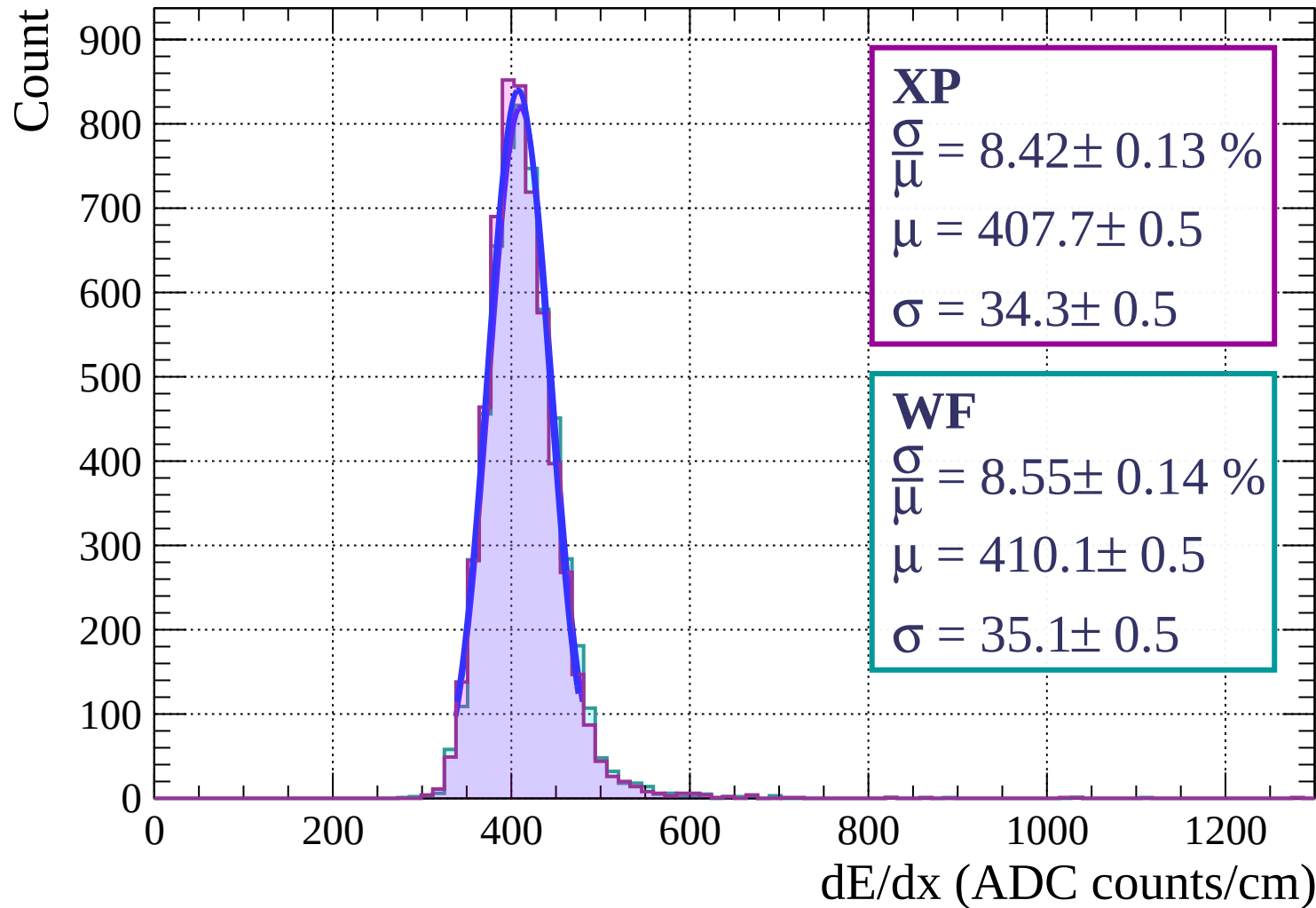
Energy loss | $0 < \theta < 3$



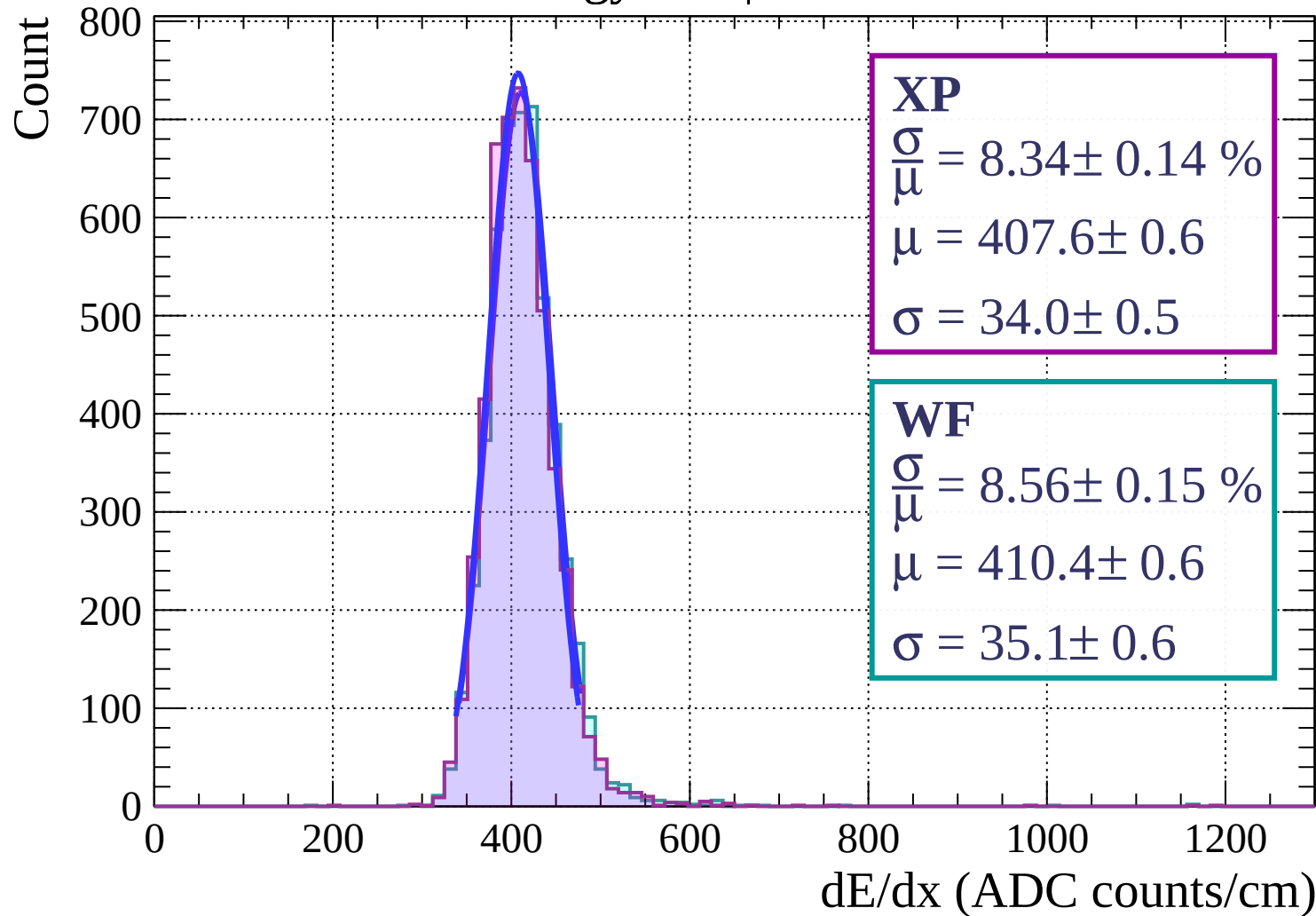
Energy loss | $3 < \theta < 6$



Energy loss | $6 < \theta < 9$

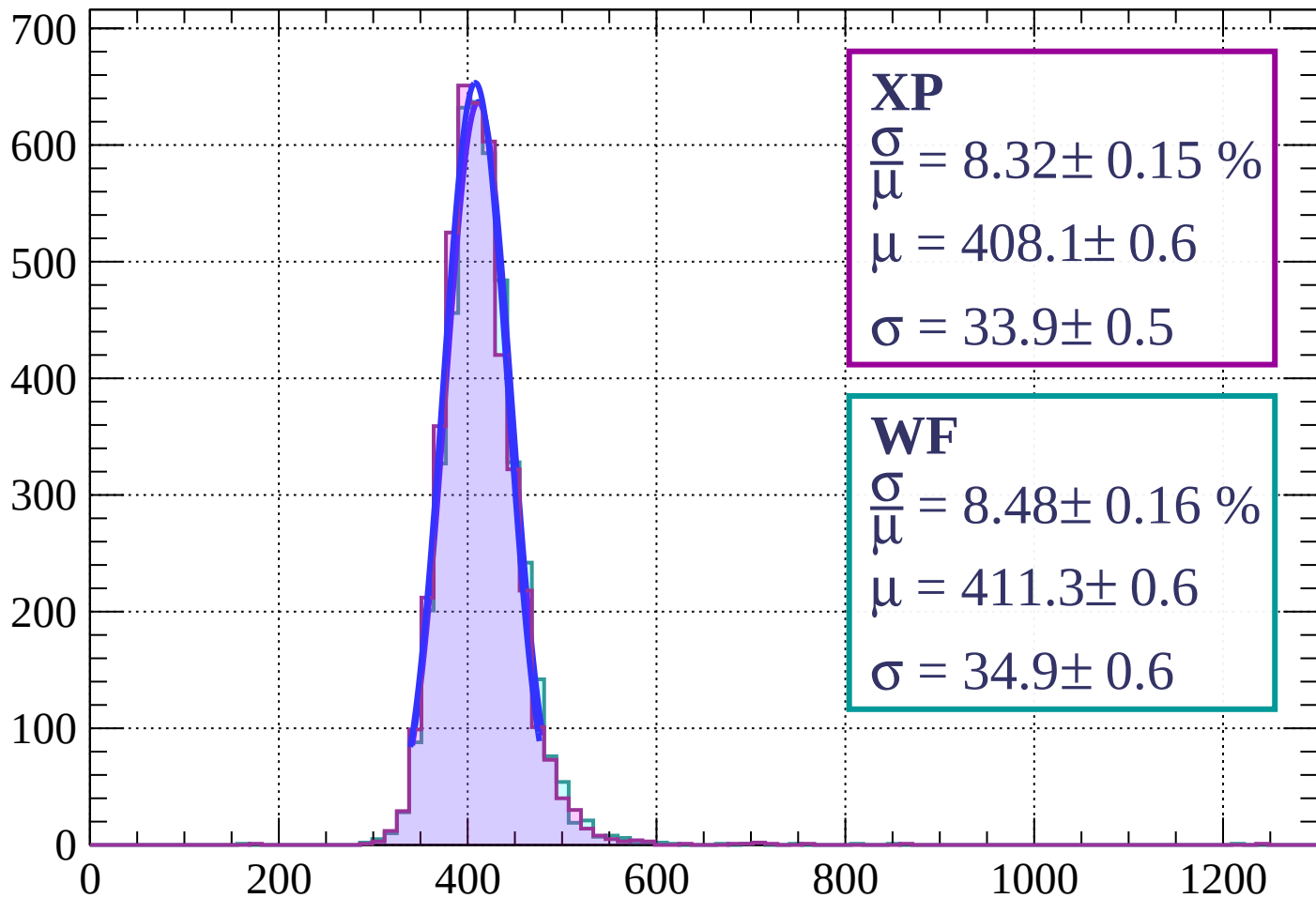


Energy loss | $9 < \theta < 12$



Energy loss | $12 < \theta < 15$

Count



XP

$$\frac{\sigma}{\mu} = 8.32 \pm 0.15 \%$$

$$\mu = 408.1 \pm 0.6$$

$$\sigma = 33.9 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 8.48 \pm 0.16 \%$$

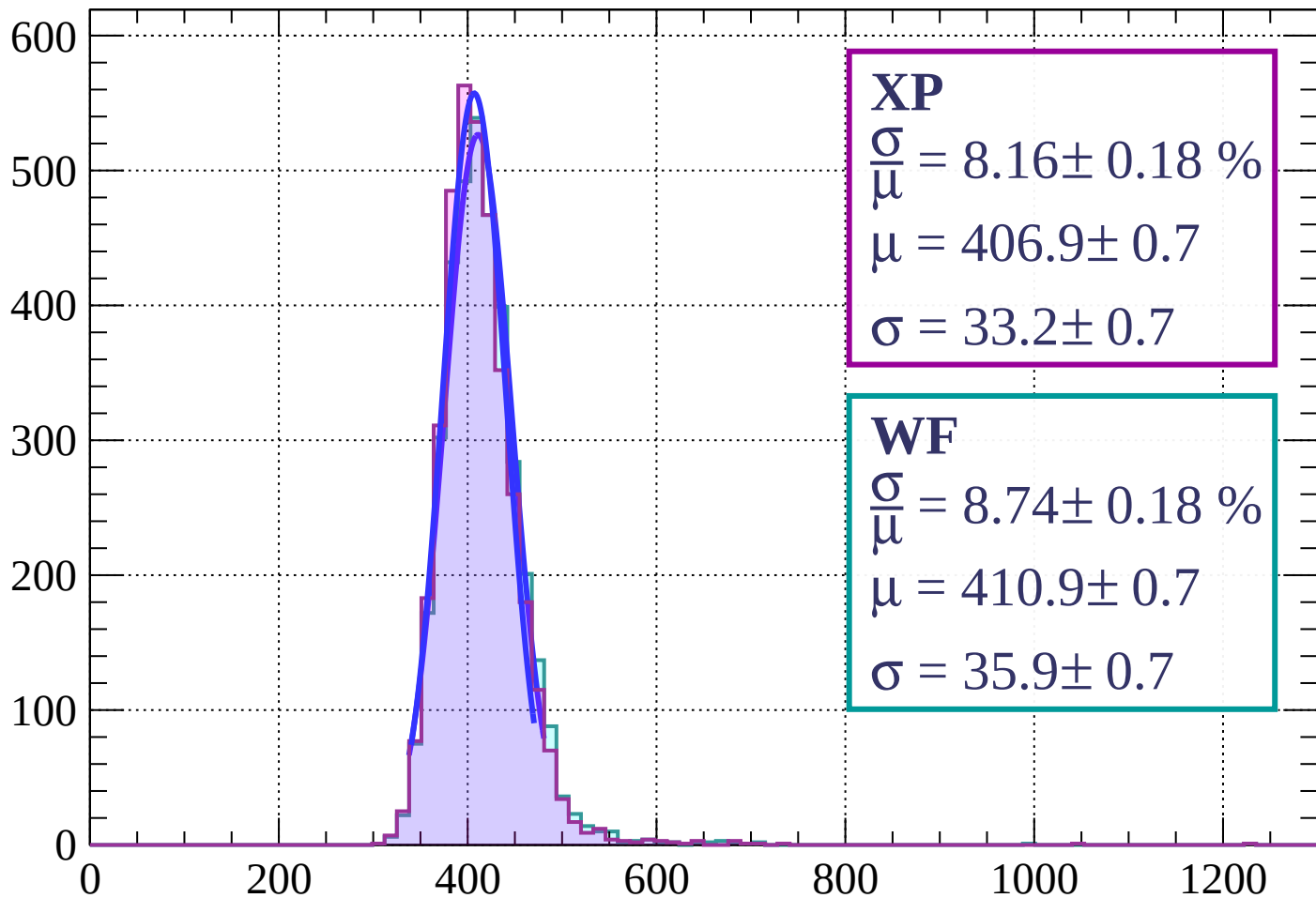
$$\mu = 411.3 \pm 0.6$$

$$\sigma = 34.9 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $15 < \theta < 18$

Count



XP

$$\frac{\sigma}{\mu} = 8.16 \pm 0.18 \%$$

$$\mu = 406.9 \pm 0.7$$

$$\sigma = 33.2 \pm 0.7$$

WF

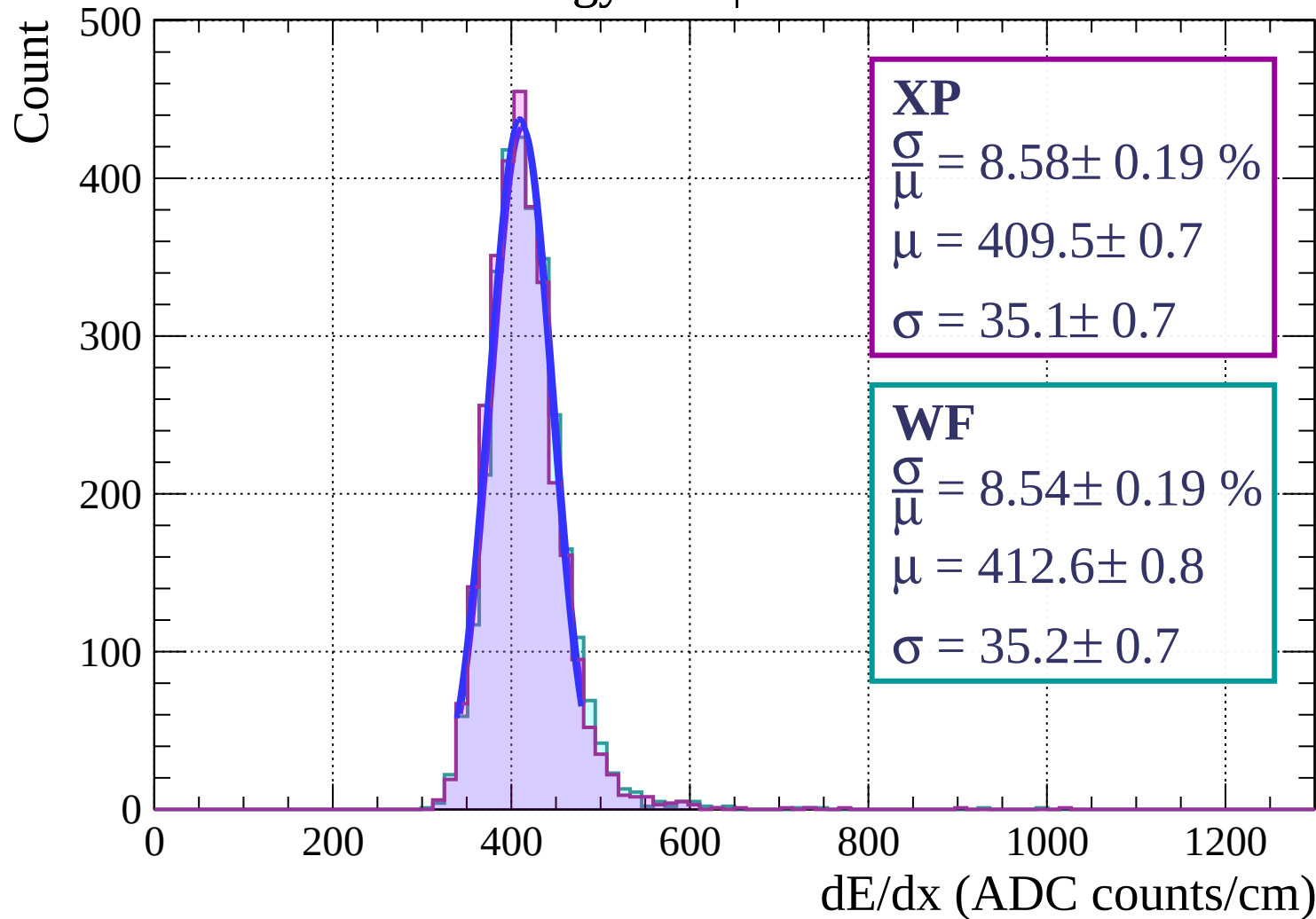
$$\frac{\sigma}{\mu} = 8.74 \pm 0.18 \%$$

$$\mu = 410.9 \pm 0.7$$

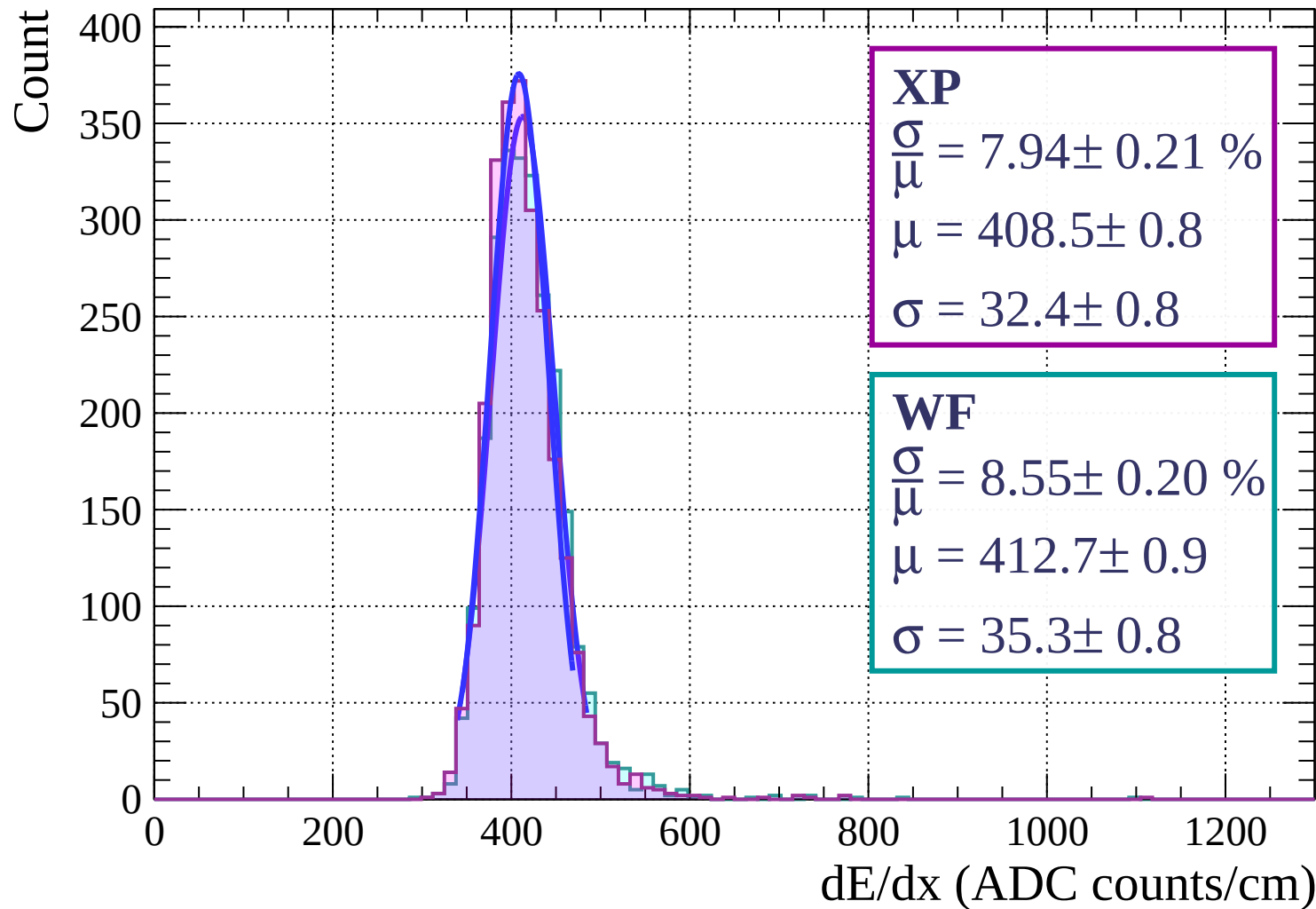
$$\sigma = 35.9 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $18 < \theta < 21$

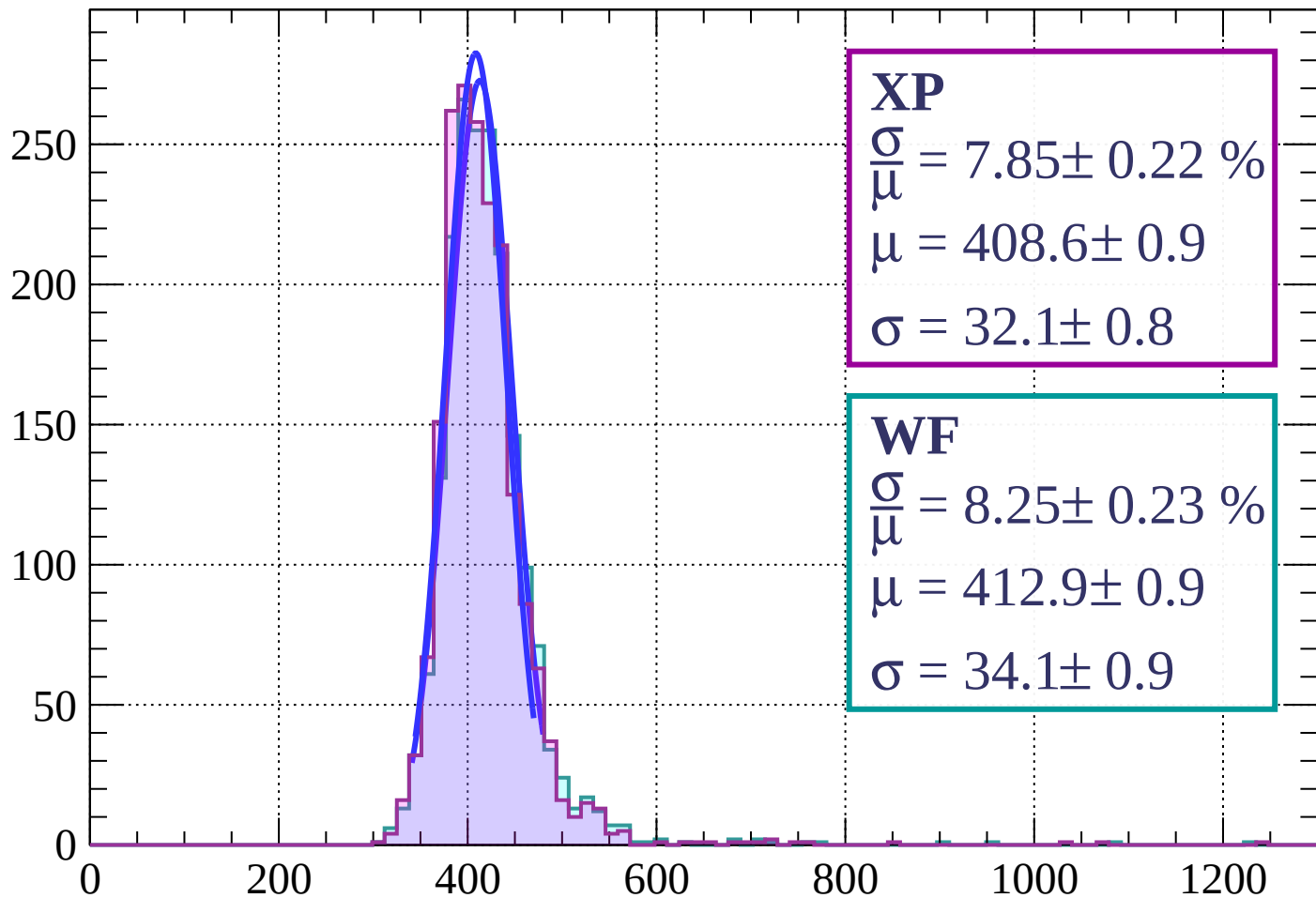


Energy loss | $21 < \theta < 24$



Energy loss | $24 < \theta < 27$

Count



XP

$$\frac{\sigma}{\mu} = 7.85 \pm 0.22 \%$$

$$\mu = 408.6 \pm 0.9$$

$$\sigma = 32.1 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 8.25 \pm 0.23 \%$$

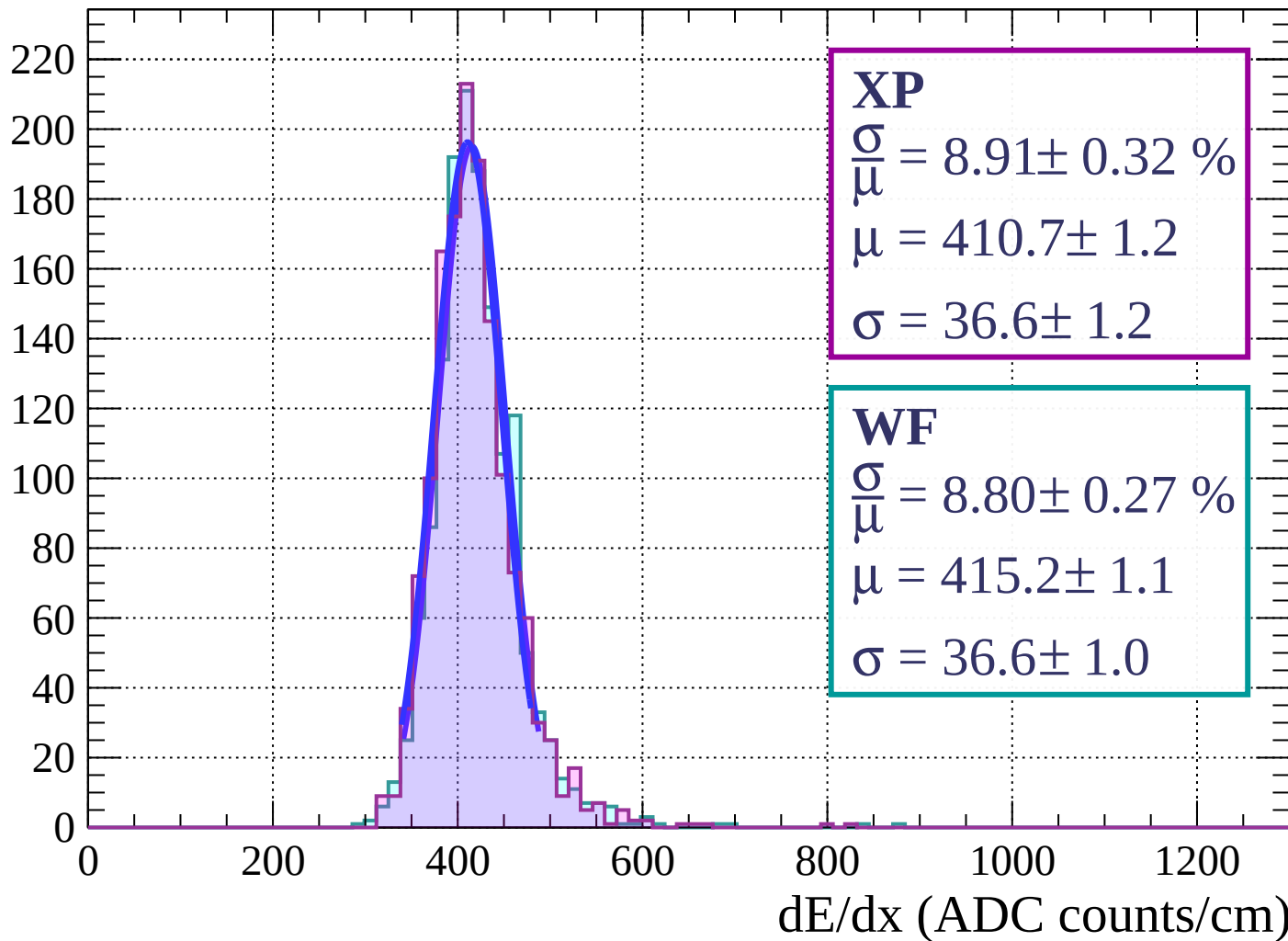
$$\mu = 412.9 \pm 0.9$$

$$\sigma = 34.1 \pm 0.9$$

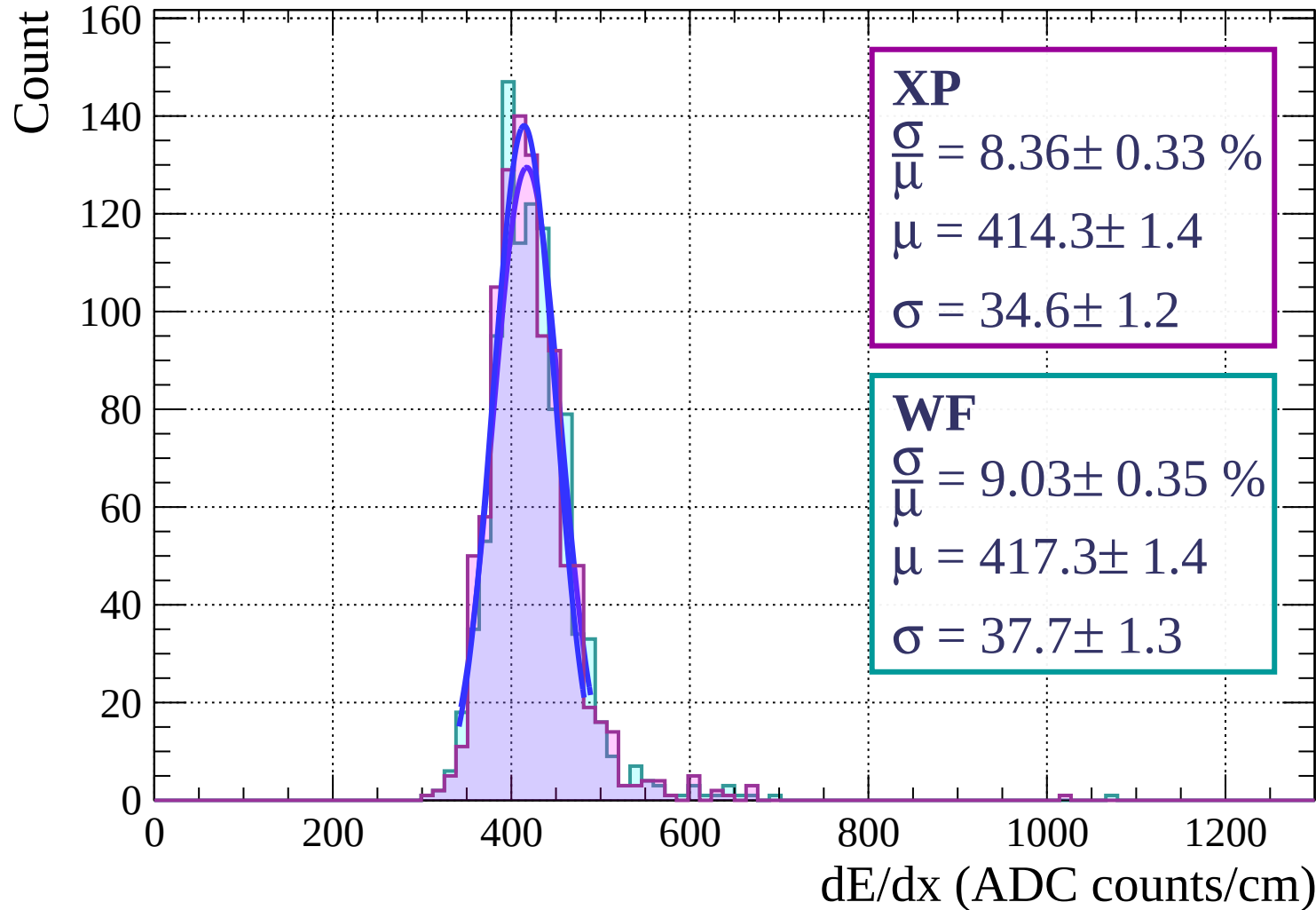
dE/dx (ADC counts/cm)

Energy loss | $27 < \theta < 30$

Count

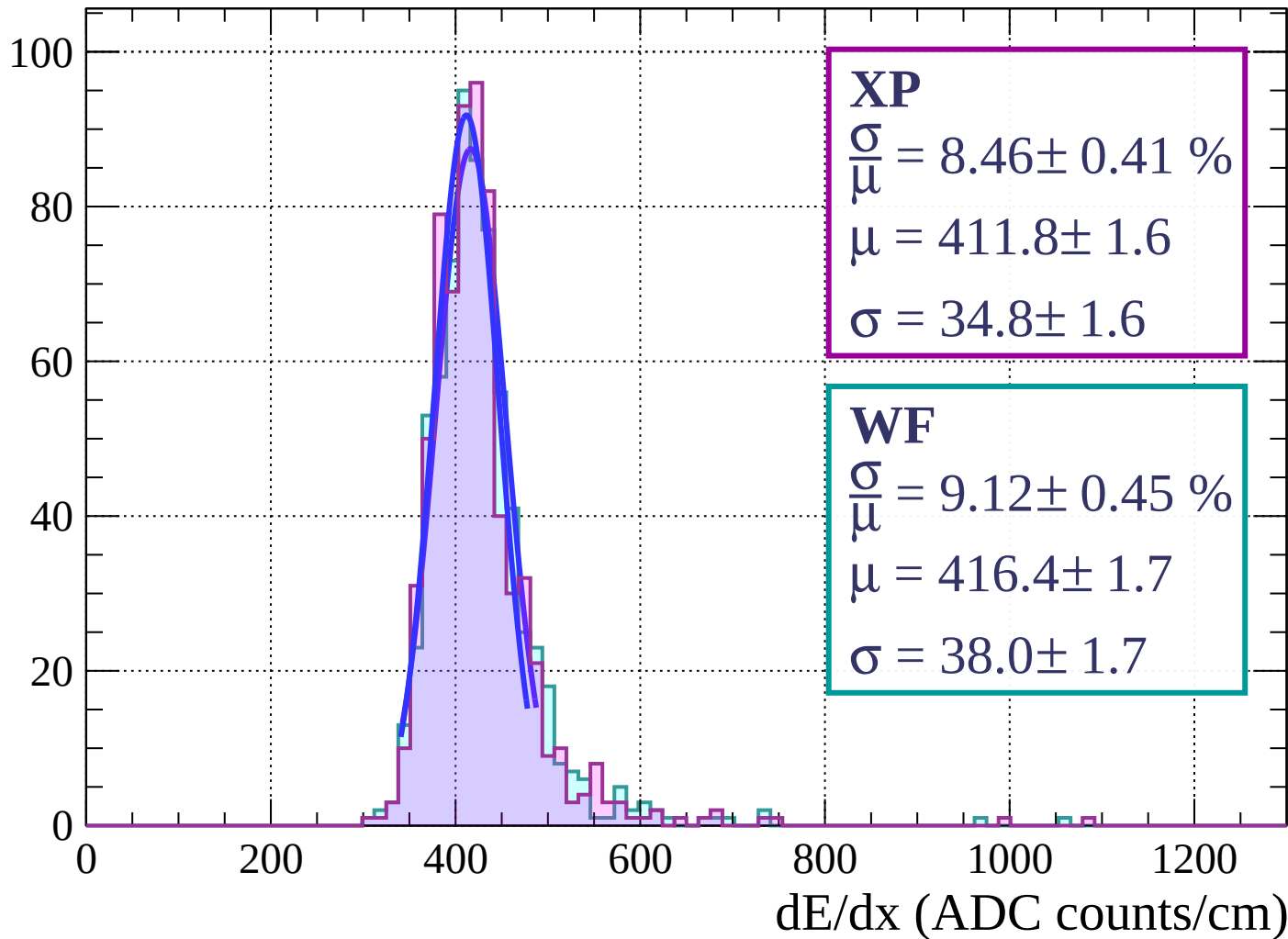


Energy loss | $30 < \theta < 33$



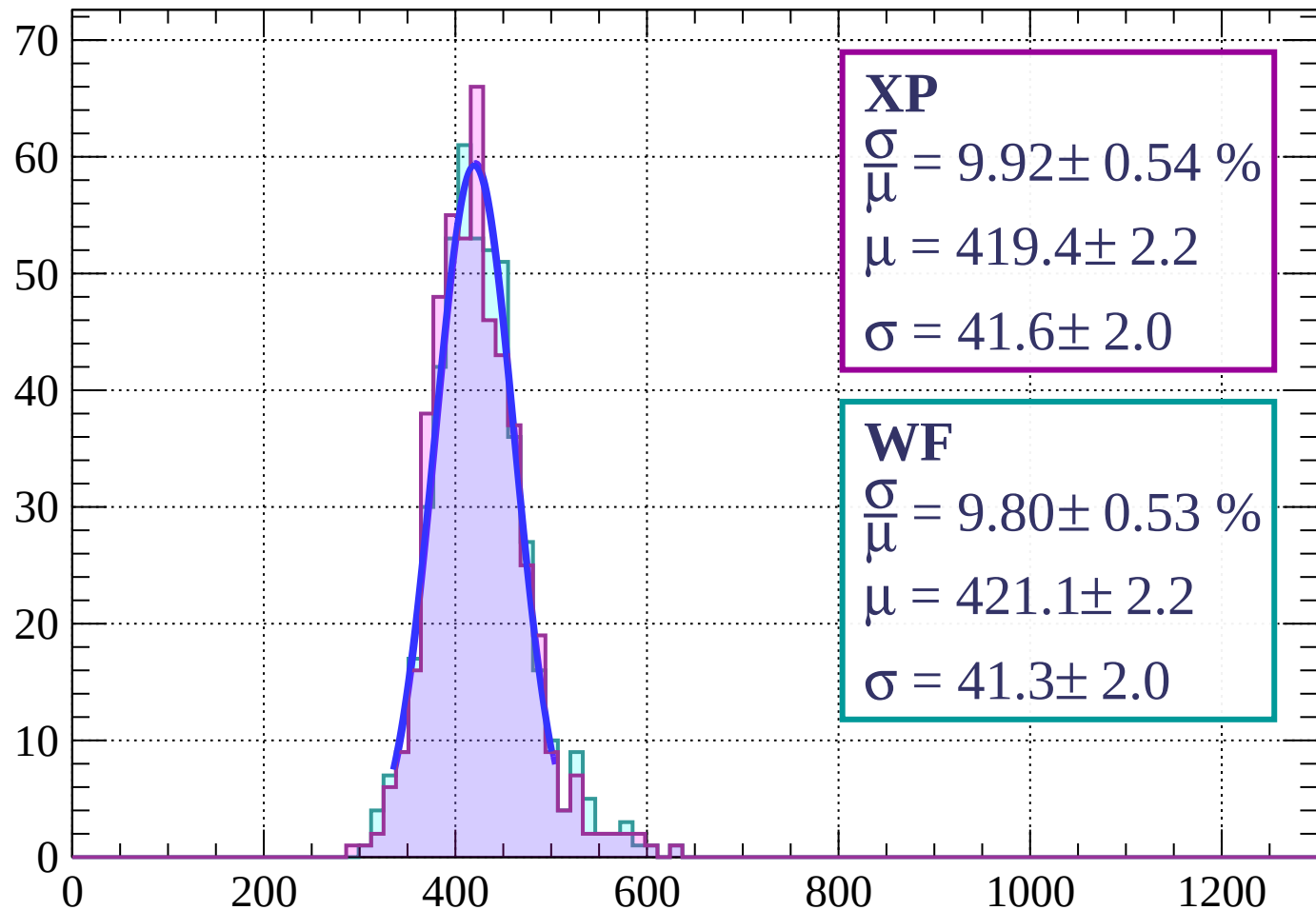
Energy loss | $33 < \theta < 36$

Count



Energy loss | $36 < \theta < 39$

Count



XP

$$\frac{\sigma}{\mu} = 9.92 \pm 0.54 \%$$

$$\mu = 419.4 \pm 2.2$$

$$\sigma = 41.6 \pm 2.0$$

WF

$$\frac{\sigma}{\mu} = 9.80 \pm 0.53 \%$$

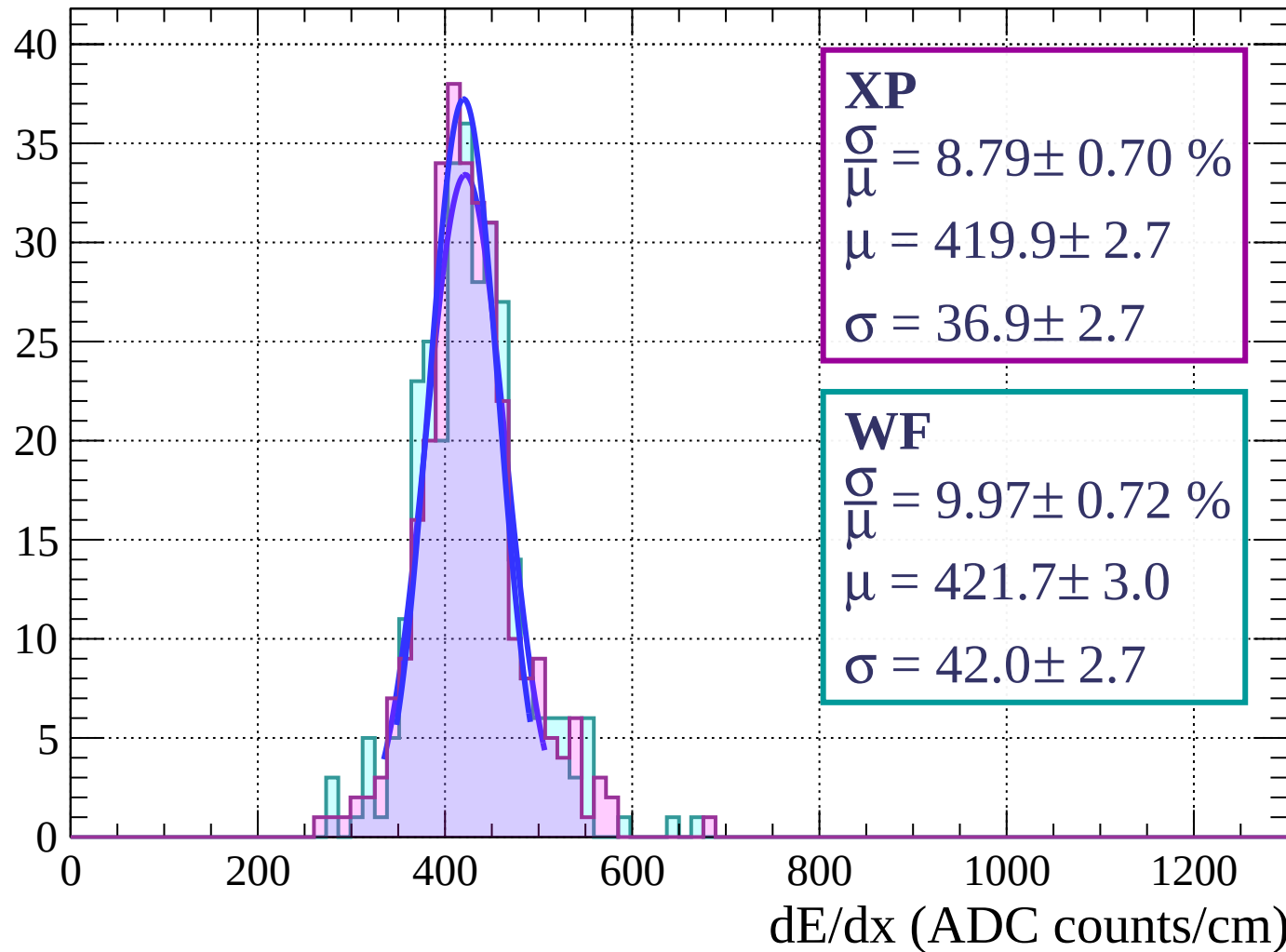
$$\mu = 421.1 \pm 2.2$$

$$\sigma = 41.3 \pm 2.0$$

dE/dx (ADC counts/cm)

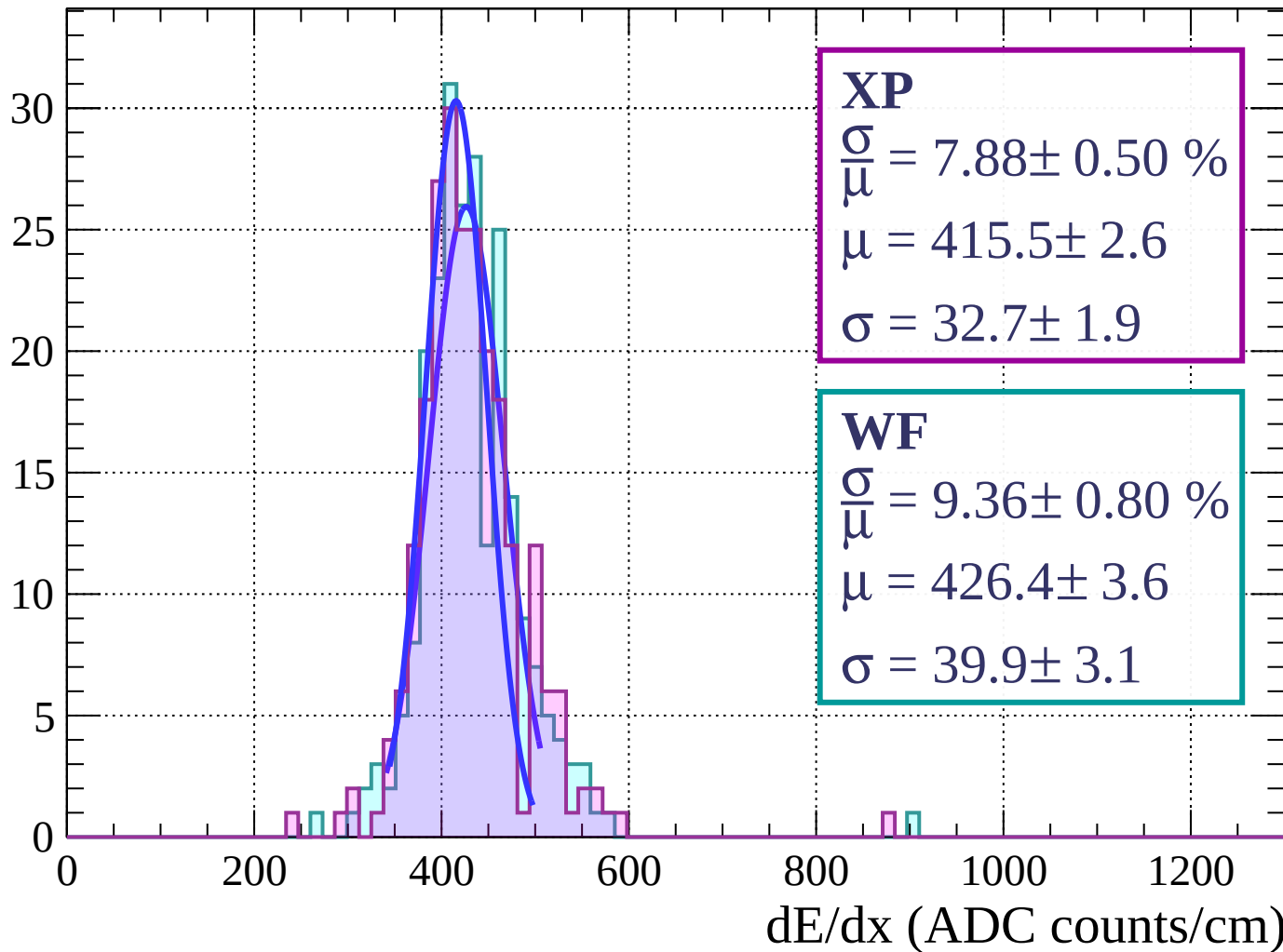
Energy loss | $39 < \theta < 42$

Count



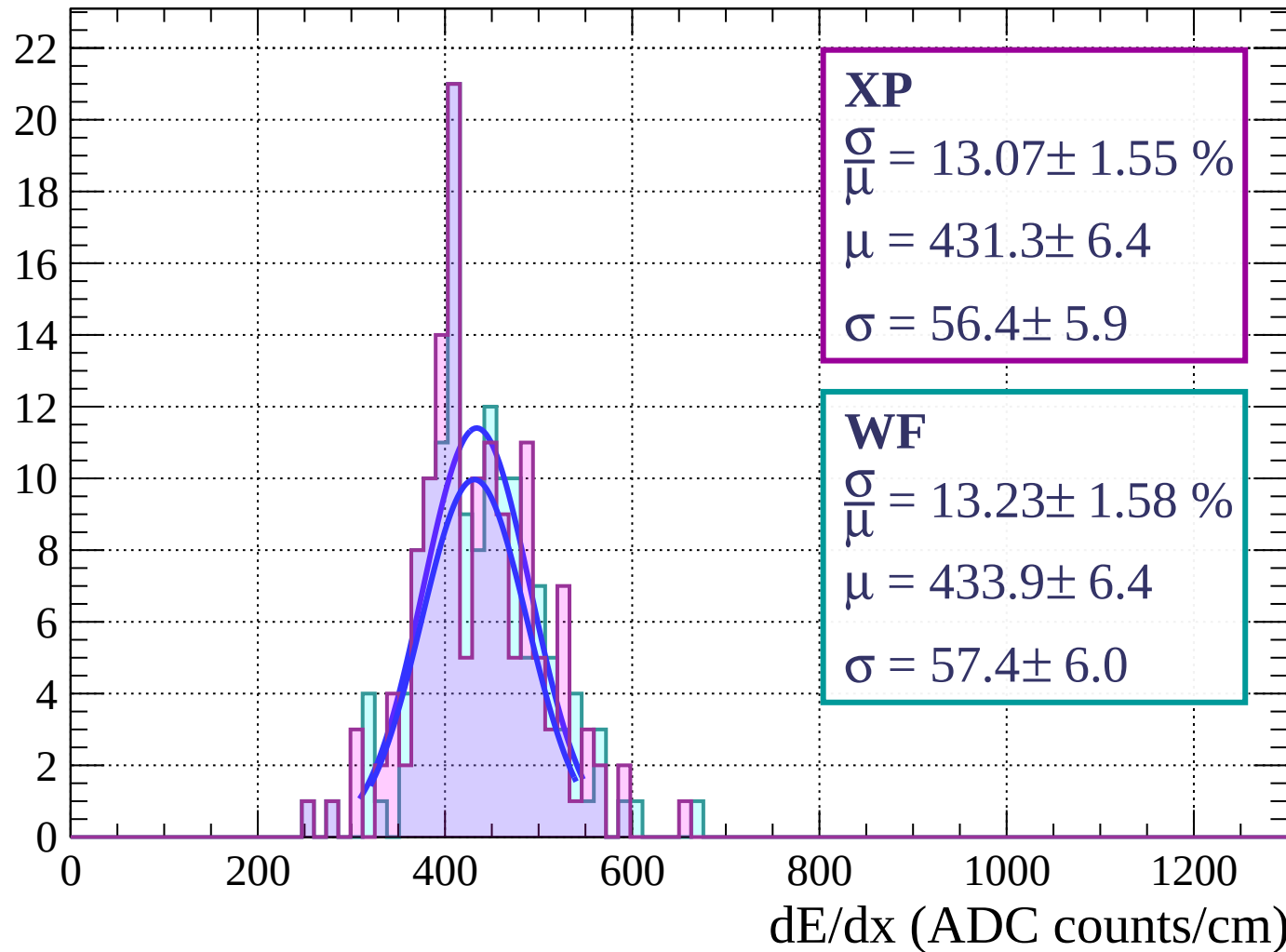
Energy loss | $42 < \theta < 45$

Count



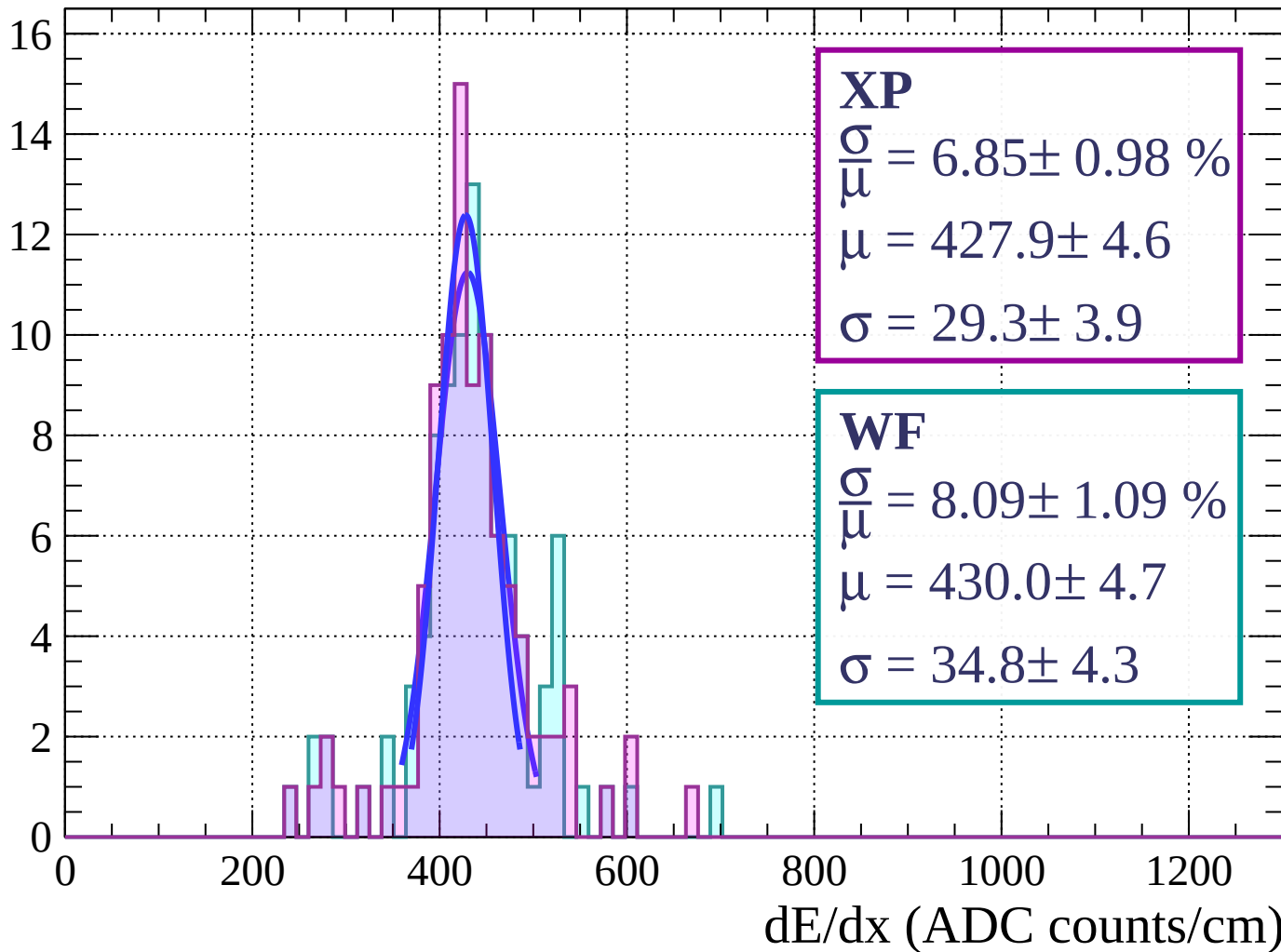
Energy loss | $45 < \theta < 48$

Count



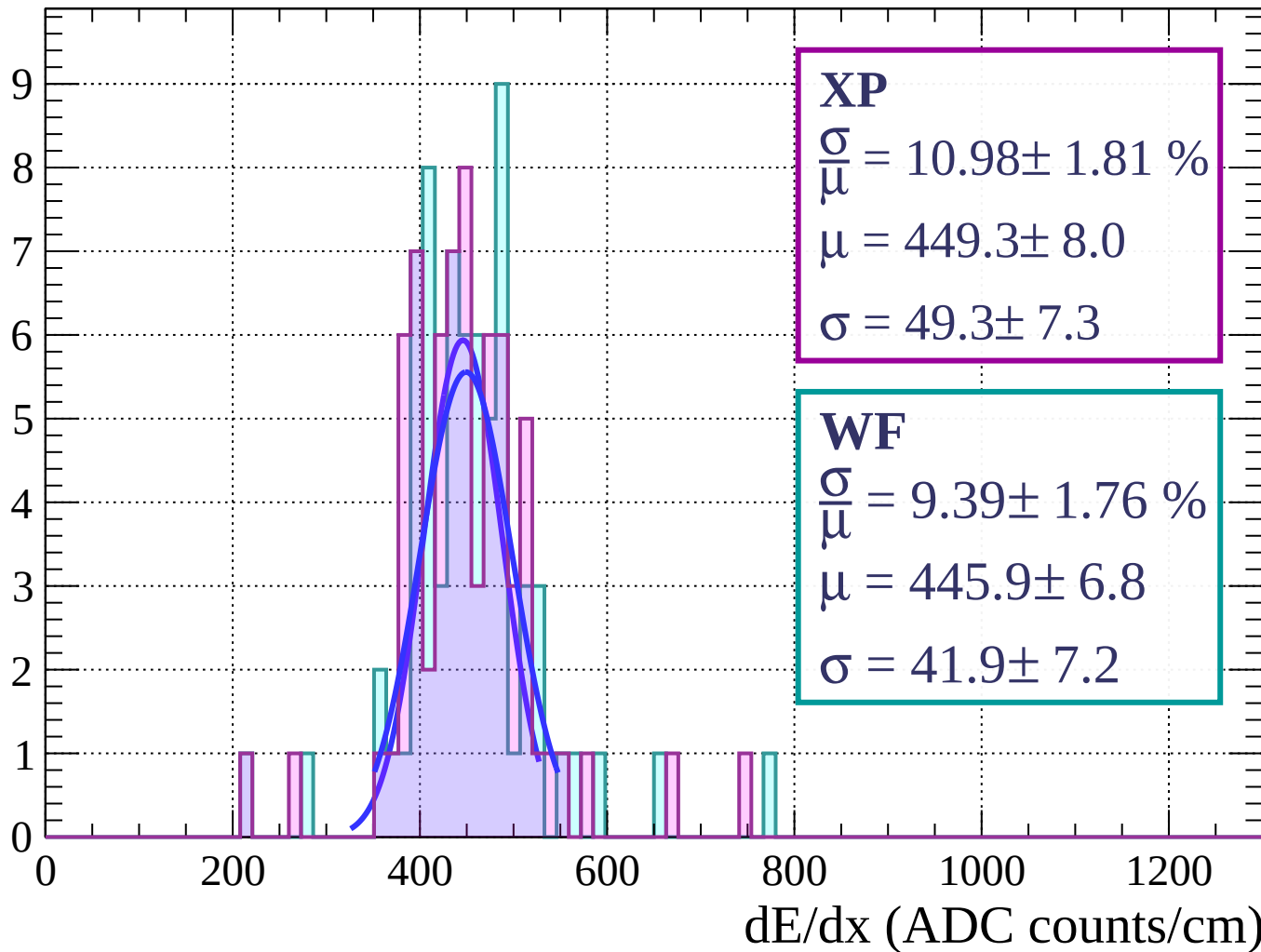
Energy loss | $48 < \theta < 51$

Count



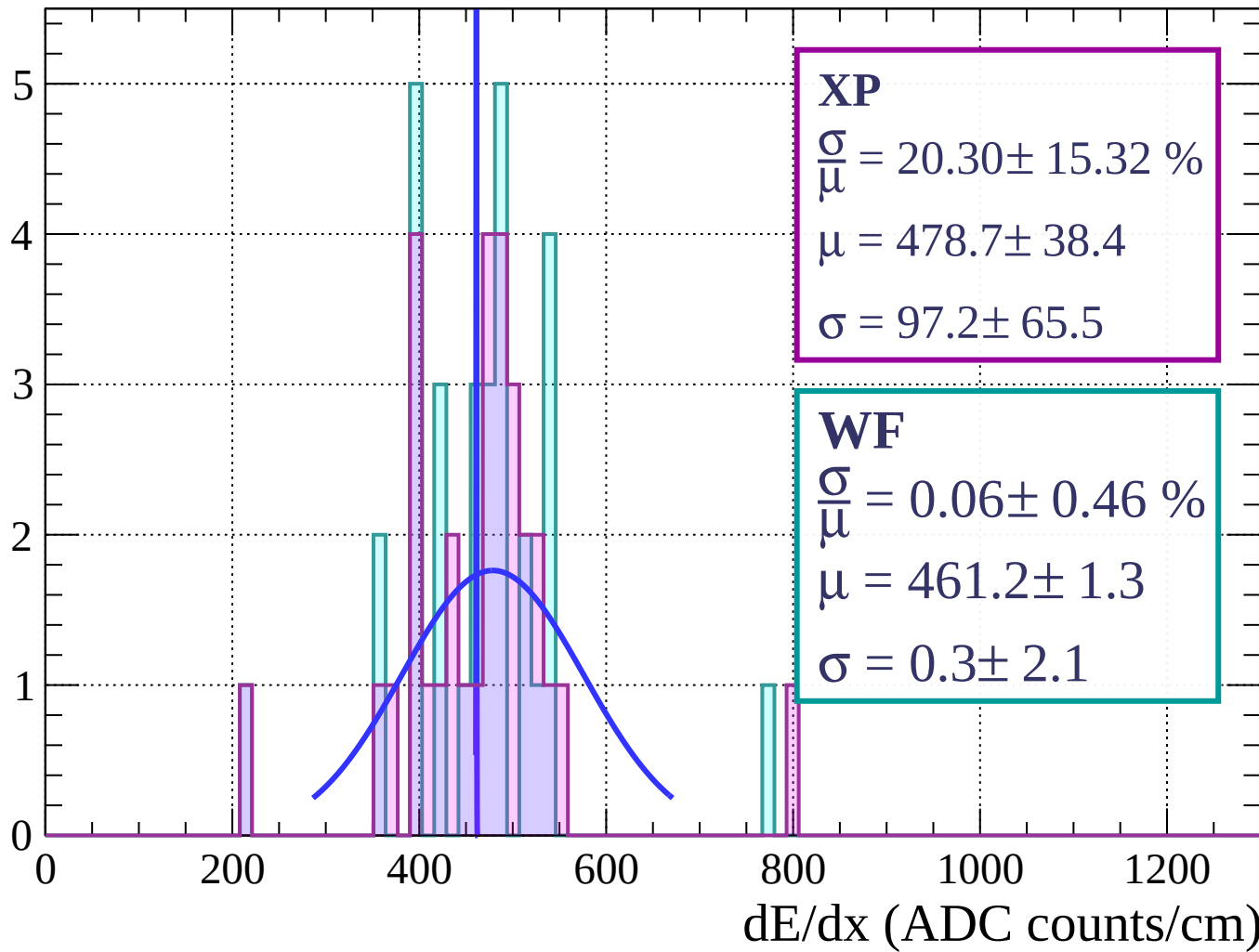
Energy loss | $51 < \theta < 54$

Count



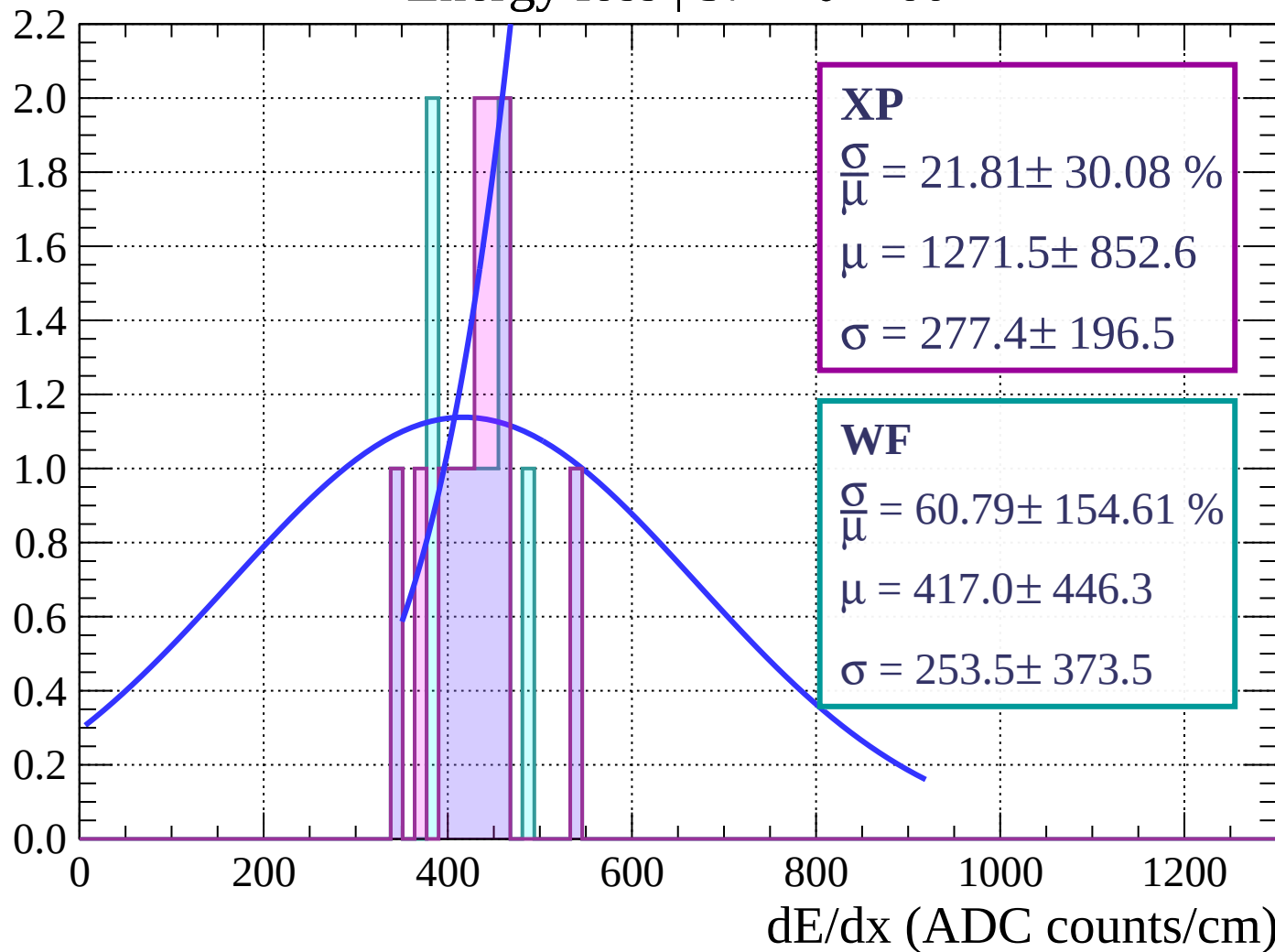
Energy loss | $54 < \theta < 57$

Count



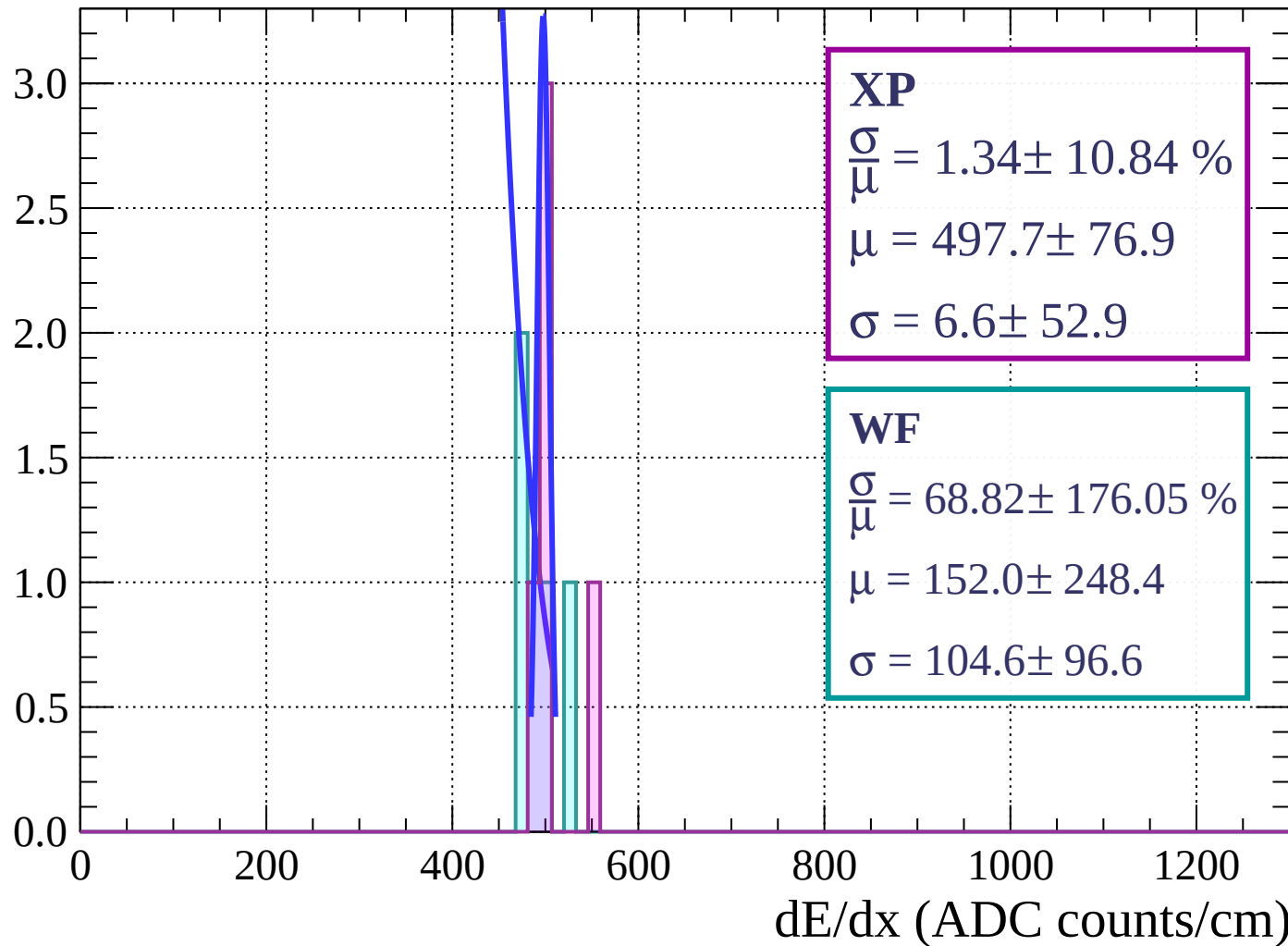
Energy loss | $57 < \theta < 60$

Count



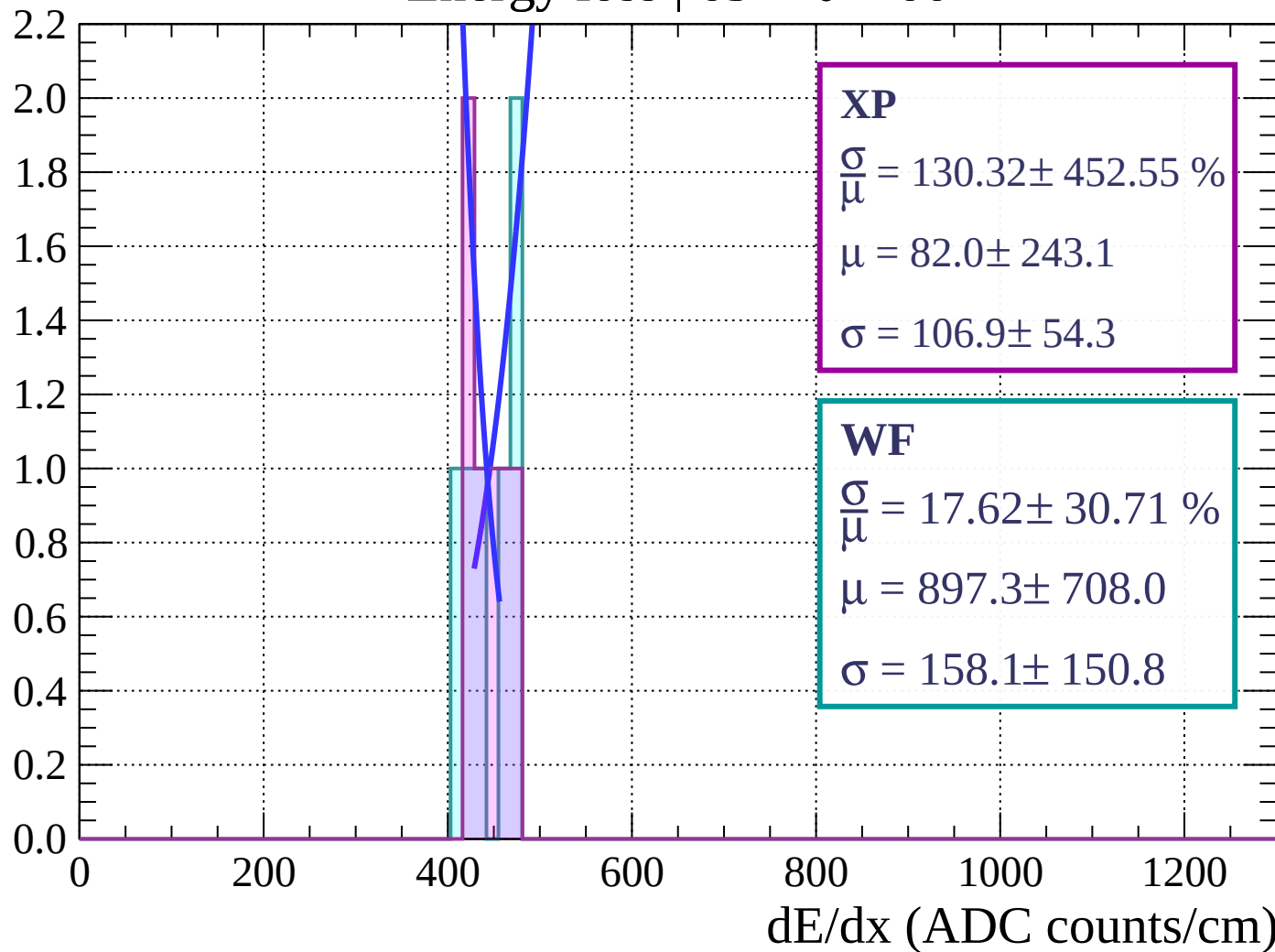
Energy loss | $60 < \theta < 63$

Count

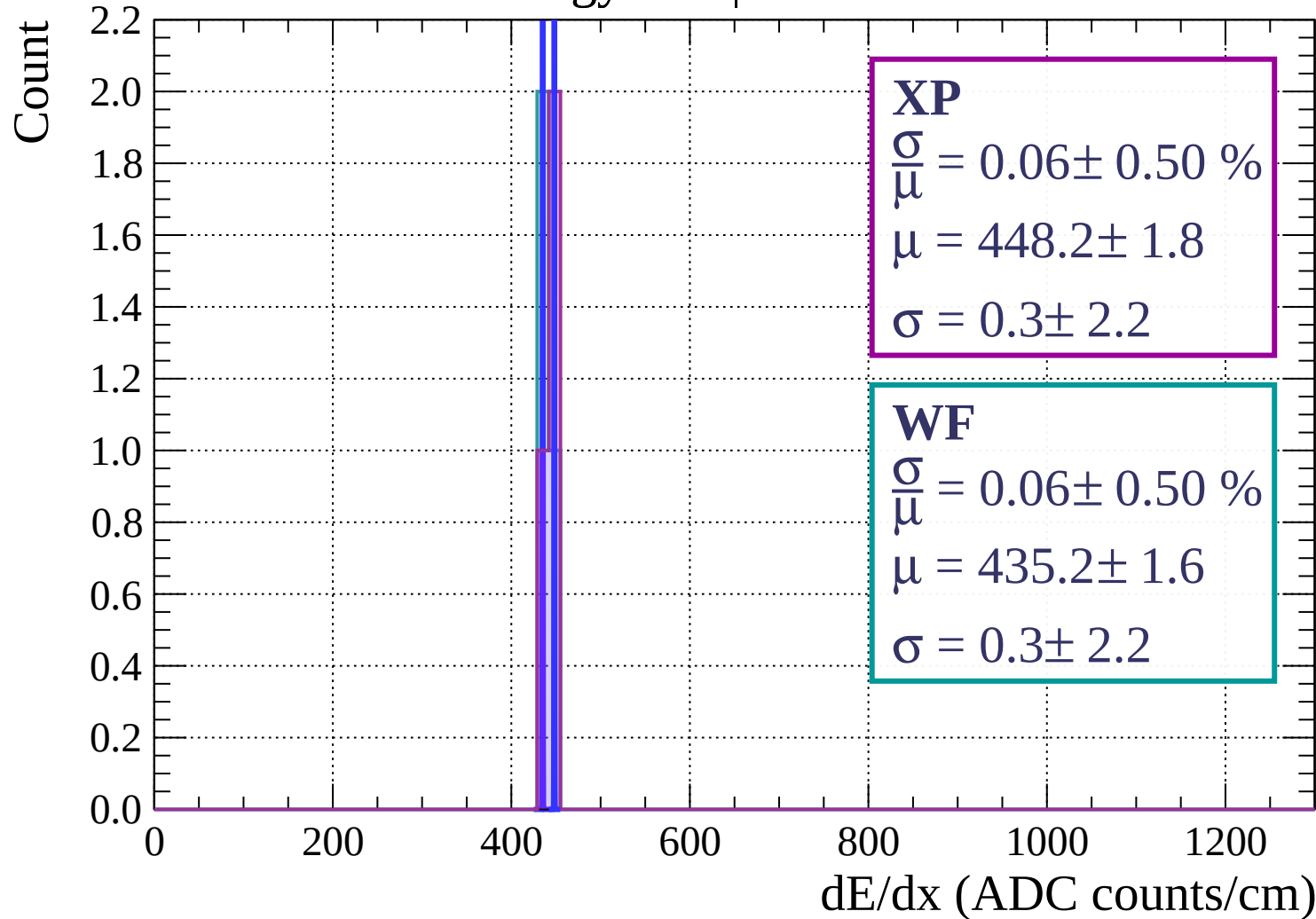


Energy loss | $63 < \theta < 66$

Count

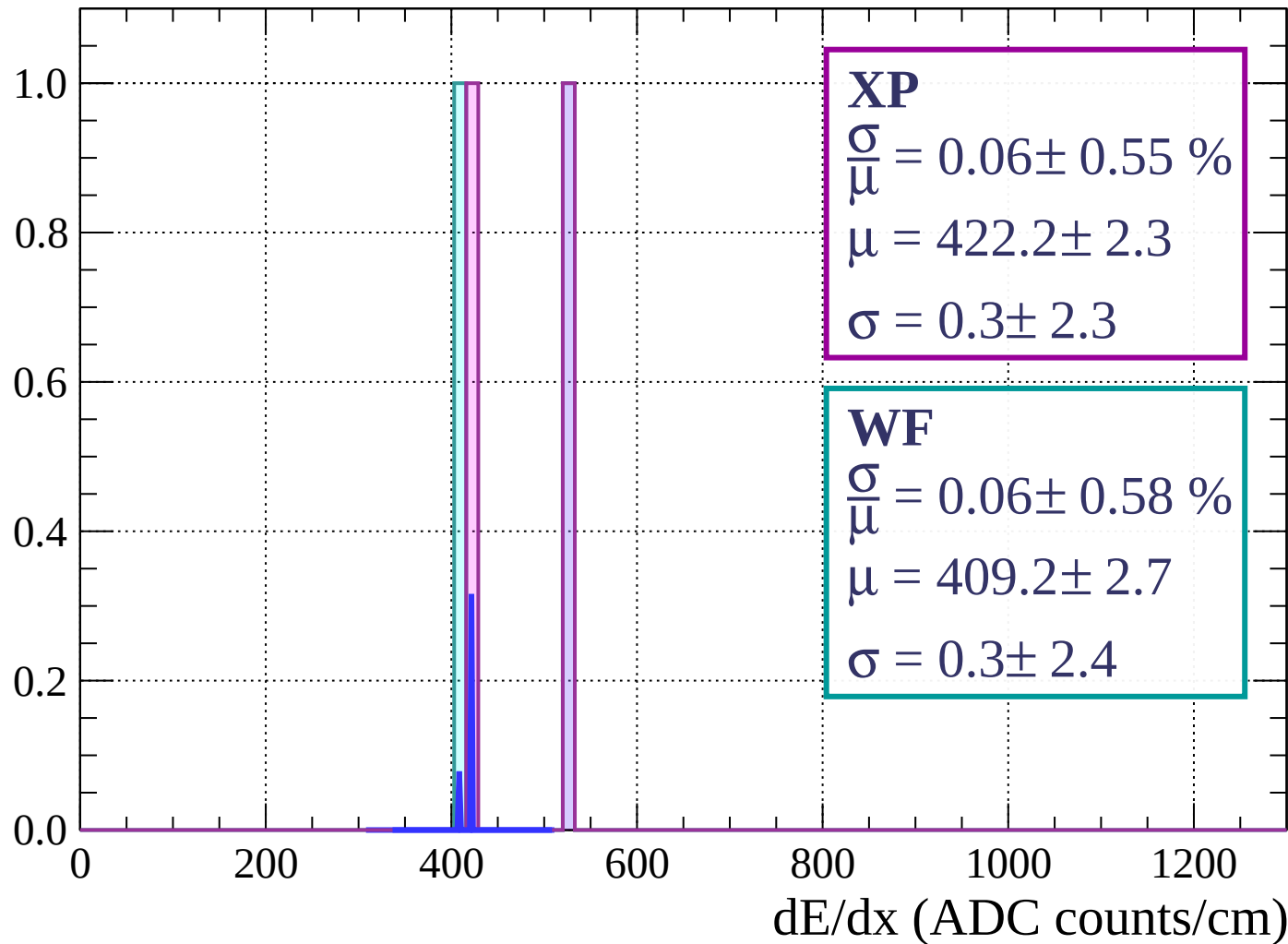


Energy loss | $66 < \theta < 69$



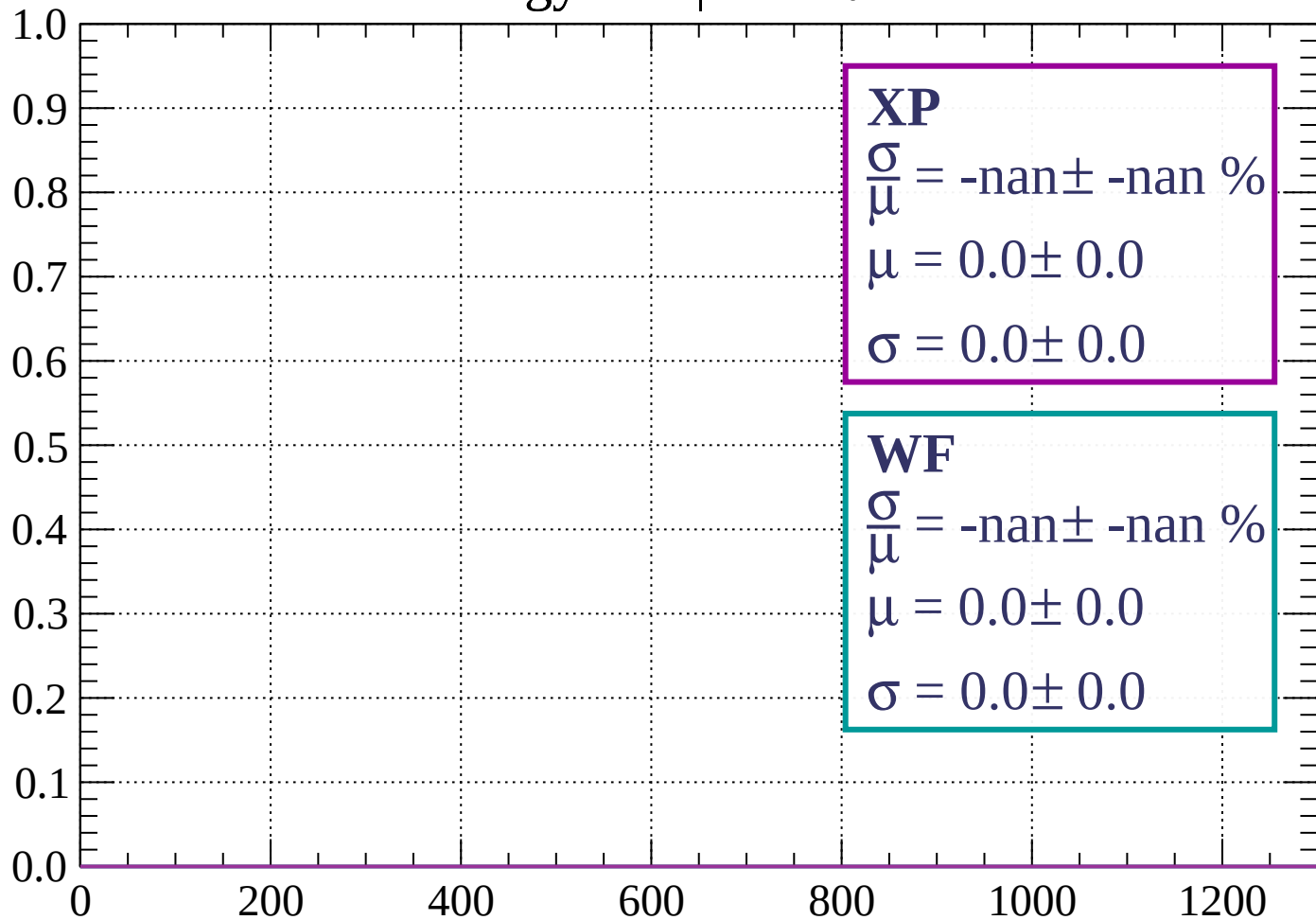
Energy loss | $69 < \theta < 72$

Count



Energy loss | $72 < \theta < 75$

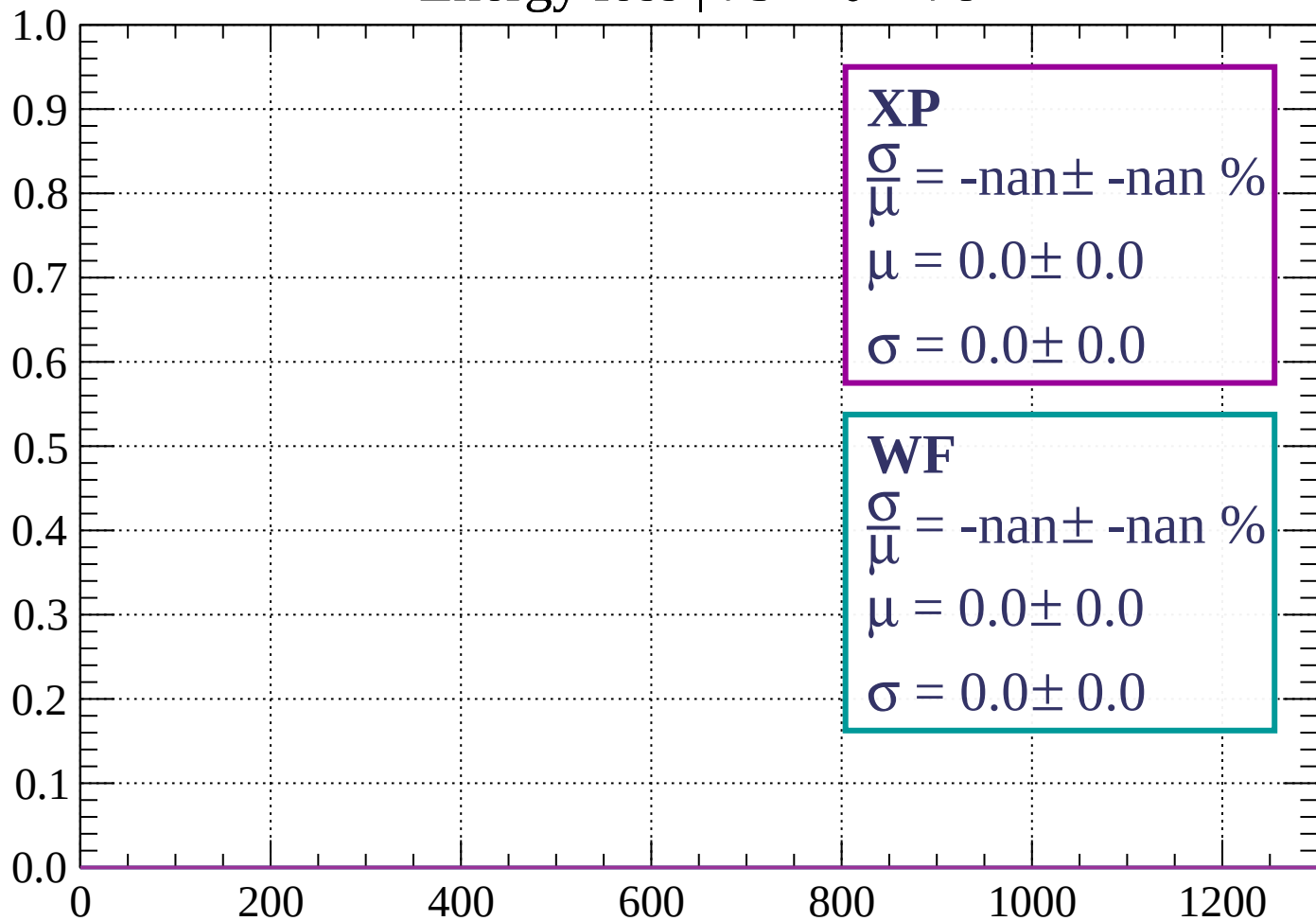
Count



dE/dx (ADC counts/cm)

Energy loss | $75 < \theta < 78$

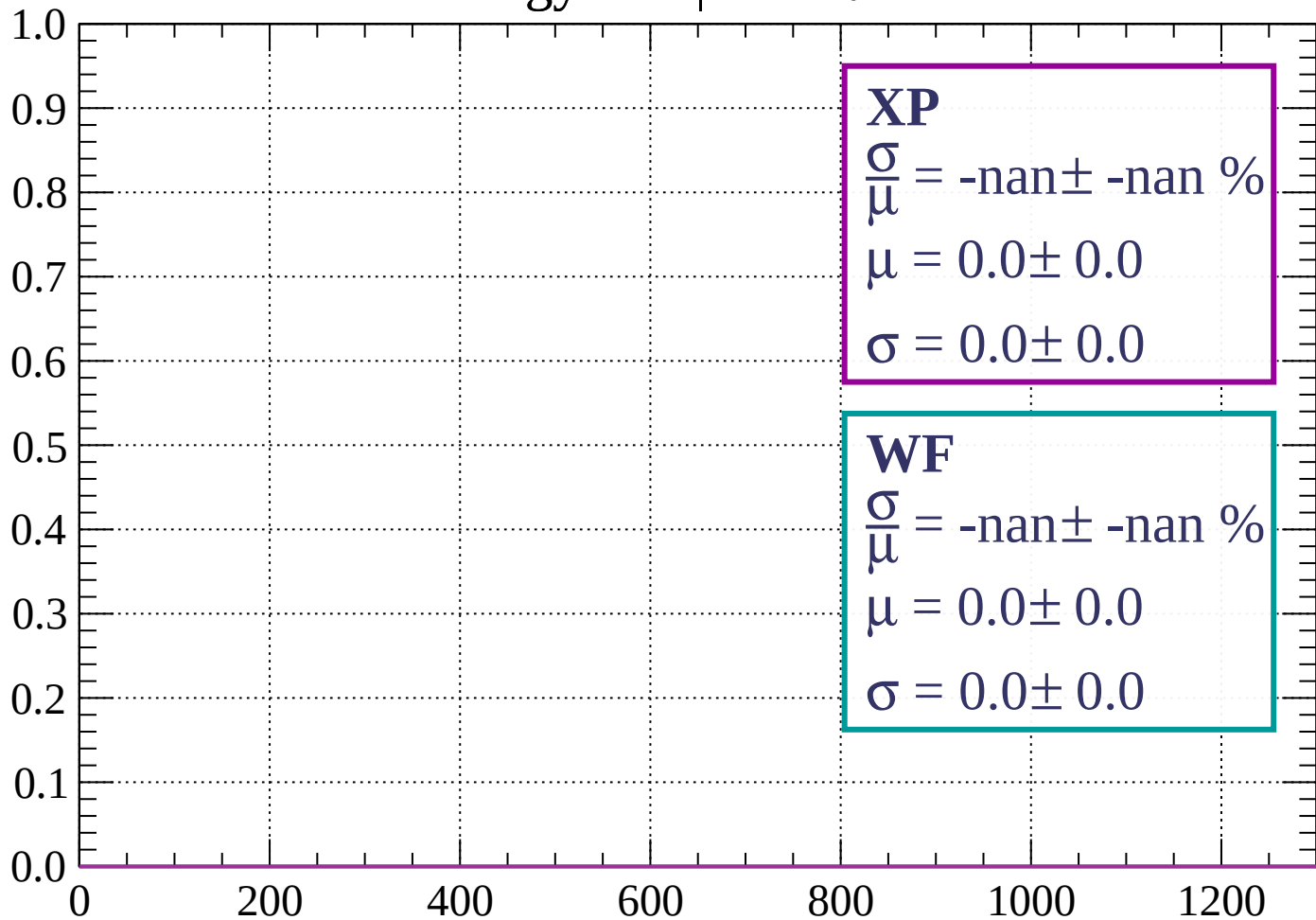
Count



dE/dx (ADC counts/cm)

Energy loss | $78 < \theta < 81$

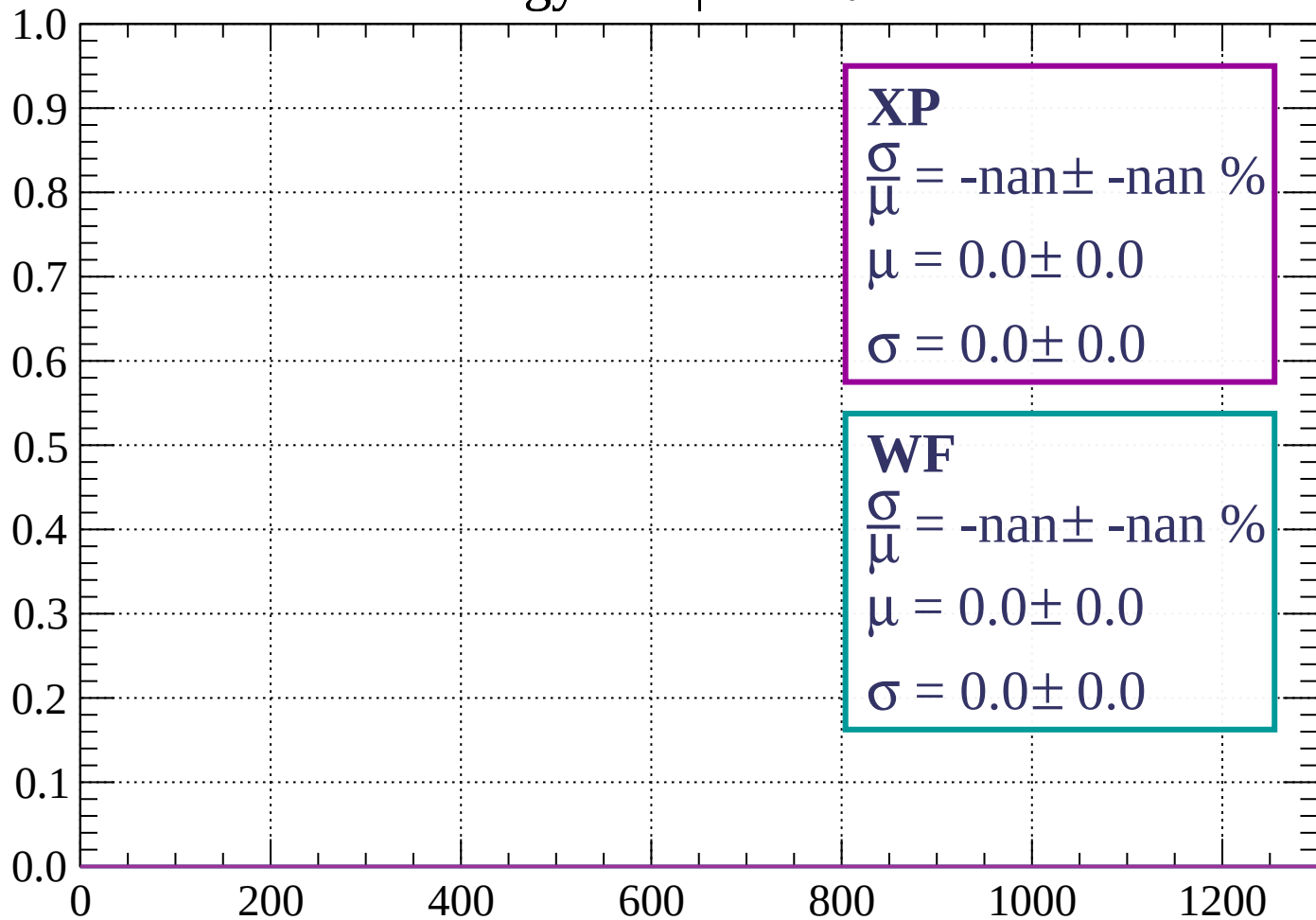
Count



dE/dx (ADC counts/cm)

Energy loss | $81 < \theta < 84$

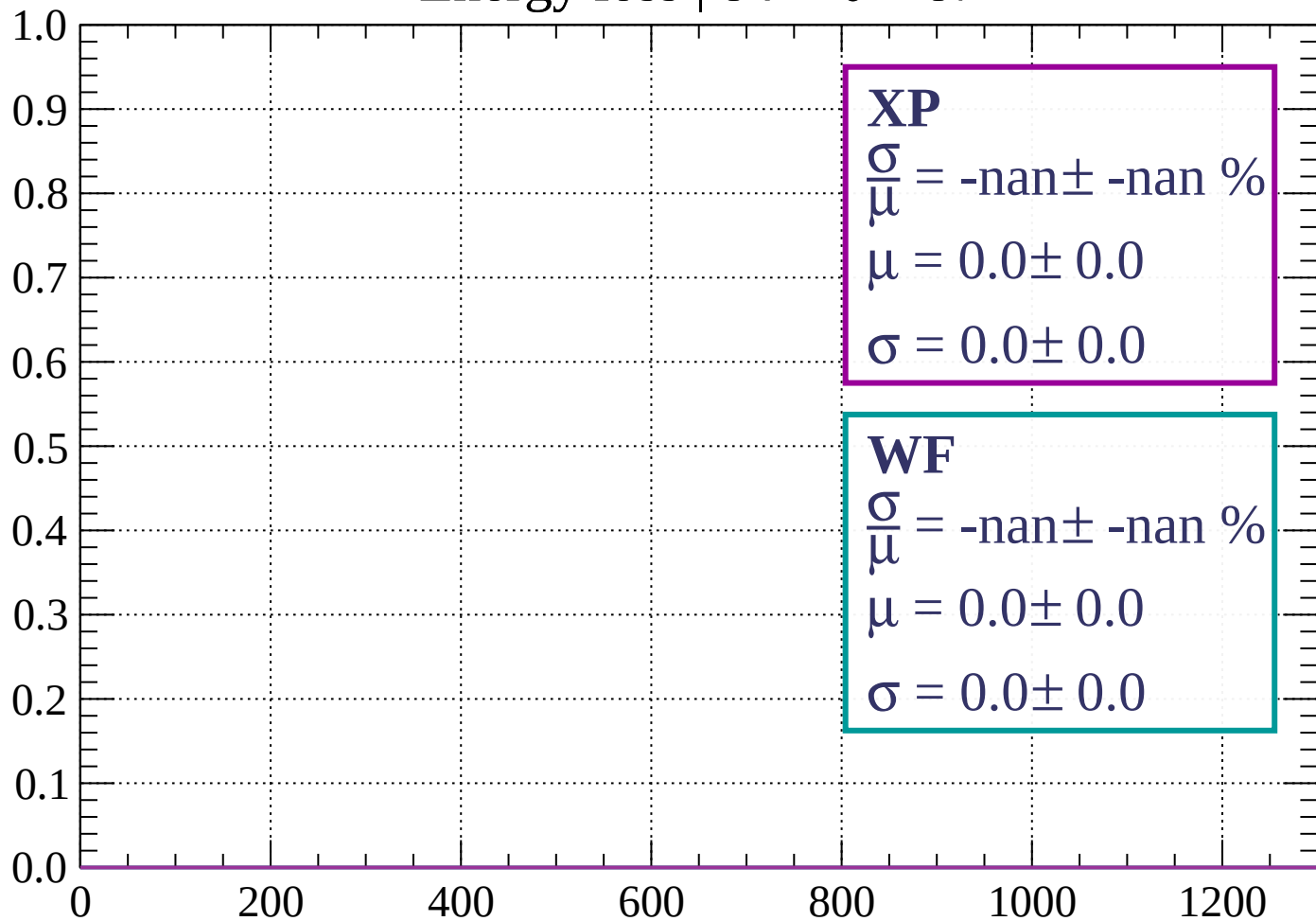
Count



dE/dx (ADC counts/cm)

Energy loss | $84 < \theta < 87$

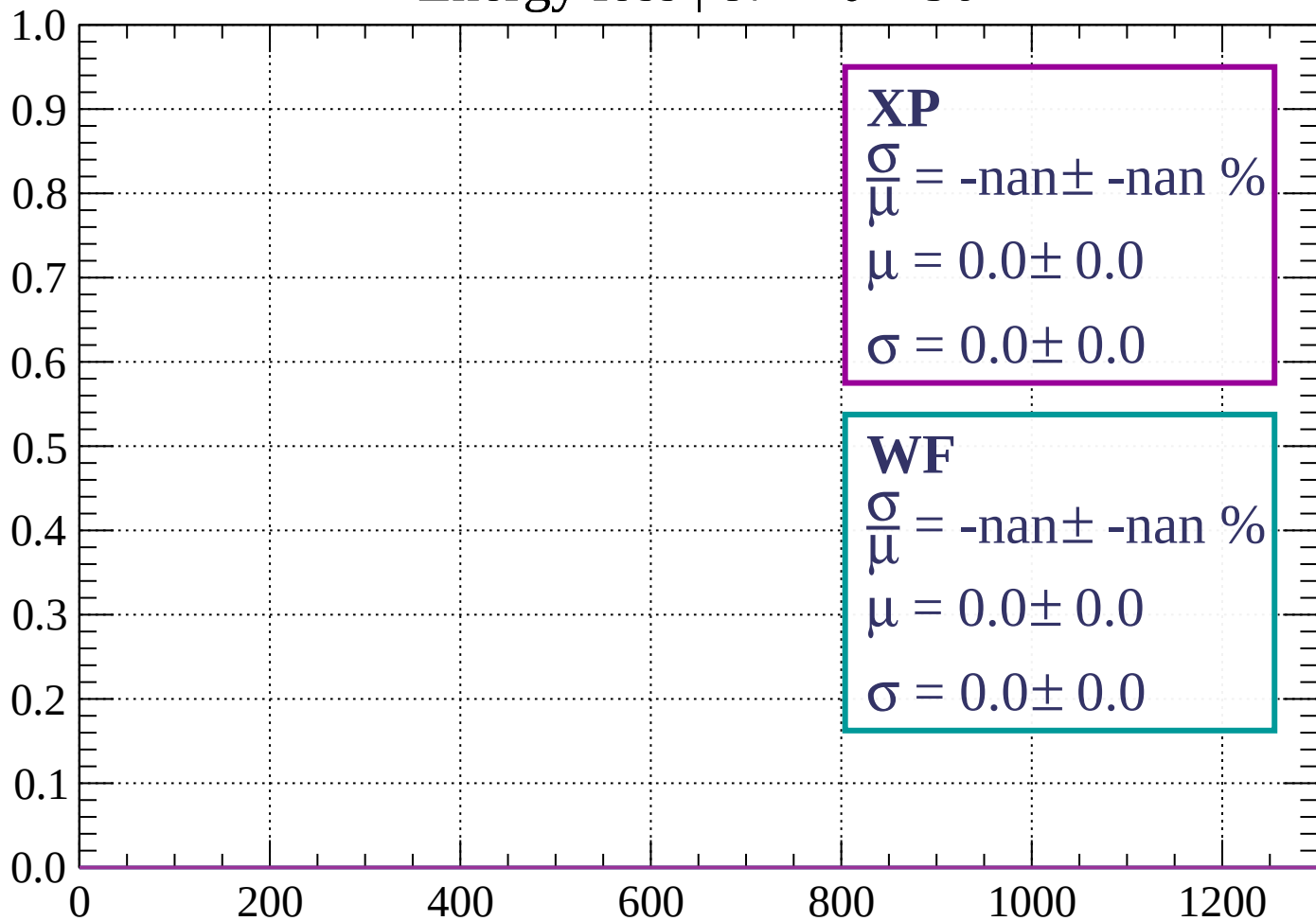
Count



dE/dx (ADC counts/cm)

Energy loss | $87 < \theta < 90$

Count



dE/dx (ADC counts/cm)

